



UNIVERSITY OF SAN CARLOS
SCHOOL OF ENGINEERING

PHYSICALLY CONSTRAINED MACHINE
LEARNING SURROGATES OF ACTIVATED
SLUDGE SYSTEMS FOR PROCESS OPTIMIZATION
DISSERTATION

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Cebu City, Philippines, 2026
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Declaration

I hereby declare that the work presented in the dissertation entitled “Physically Constrained Machine Learning Surrogates of Activated Sludge Systems for Process Optimization” is original and the result of my own research. Formulations and ideas taken from other sources are cited as such.

Cebu City, Philippines, 2026

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Abstract

Activated sludge process optimization requires rapid predictions that preserve material balances and non-negative concentrations. This dissertation integrates a comparison of thirteen reactor predictors, an interpretable second-order regression, and a connected plant optimization study. Mechanistic simulations provide component-resolved reference states. Projection adjusts predicted concentrations to satisfy the specified physical constraints. All 766,350 assessed reactor projection instances passed the mass conservation and non-negativity checks. At the largest design size, projection reduced aggregate component error for five of the thirteen predictors. The structured regression had the lowest mean composite error at the smallest design size of 500 cases, while the multilayer perceptron had the lowest error at the largest size. The structured regression’s named operating, loading, interaction, and coupling terms supported direct examination of the fitted response. The complete projection procedure returned physically checked states in all 2000 fixed-partition cases. In the connected plant, joint projection linked the mixer, reactor train, clarifier outlets, and solids inventory. It reduced aggregate holdout normalized root mean squared error by 8.19% across 1845 accepted states and improved 30 of 32 location–composite scores. Across ten influent scenarios, mean optimization times were 1.726 s for the connected surrogate and 5.236 s for the direct mechanistic route. Independent mechanistic evaluation gave lower objectives for the direct decisions in the nominal case and all ten scenarios. Together, these findings establish physical enforcement, predictive agreement, model interpretation, and operating quality as complementary parts of surrogate assessment. The integrated framework supports rapid candidate generation and component-resolved verification for activated sludge process optimization. Future research can develop adaptive sampling, uncertainty-aware prediction, dynamic treatment models, and validation with plant measurements.

Keywords. Activated sludge process, machine learning surrogate, wastewater treatment, mass conservation, non-negativity, interpretable regression, operating optimization.

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Principal Symbols

Symbol	Meaning
F, P, K	Numbers of component states, reaction processes, and independent stoichiometric invariants
ν	Stoichiometric matrix with processes in rows and components in columns
ρ	Vector of mechanistic process rates
A	Full-row-rank stoichiometric invariant operator
V_0, Z	Orthonormal bases of the stoichiometric right null space and the null space of A
c_{in}, c_{out}	Influent state and mechanistic reference effluent state
$c_{raw}, \tilde{c}$	Raw component prediction and projected component prediction
I_{comp}	Fixed component-to-reported-composite map
D_i	Hydraulic dilution rate for the stated reactor and flow basis
ω, e_{SO}	Net dissolved-oxygen transfer rate and dissolved-oxygen coordinate selector
u	Standalone-reactor operating input vector
ϕ, B, Γ, R	Second-order feature vector, driver coefficients, learned component coupling, and coupled-system matrix
$\hat{C}, \Phi$	Auxiliary fitted component states and training feature matrix
N	Number of observations in the stated training-size design
n_r, L	Numbers of biological reactor stages and clarifier layers
ϑ	Connected-plant control vector
q_P, q_C, q_E, q_U	Flow ratios for the reactor train, clarifier feed, overflow, and underflow relative to fresh flow
m, c_i, c_E, c_U	Mixer, reactor-stage, overflow, and underflow concentrations
g_E, g_U	Flow-weighted clarifier outlet component responses
χ, M_{cl}	Joint plant response and clarifier solids inventory
D_χ	Positive scaling matrix for the plant response
J	Dimensionless engineering objective

Symbols specific to a formulation are defined with their equations. A numerical

coordinate is interpreted using the unit of its corresponding component. The plant projection uses u locally for its standardized response projection.

Chapter 1

Introduction and Research Purpose

1.1 Background and rationale

1.1.1 Wastewater treatment as an engineering need

Wastewater treatment removes pollutants or changes them into forms that reduce their effects before water is discharged or reused. Wastewater carries organic matter, nutrients, and suspended particles from communities and industries. Influent is the water entering a treatment system. Effluent is the water leaving it. Their composition determines the treatment required and the quality achieved. Madhav et al. (2020) describe the sources and environmental effects of water pollutants. Lin et al. (2022) examine their different health consequences. These concerns establish why treatment decisions require more than a single measure of effluent quality.

The need is relevant to the Philippines, where water management connects treatment infrastructure to institutions, financing, and local conditions. Tabios III (2020) discusses the physical setting and governance of Philippine water resources. Damalerio et al. (2022) identify technical, financial, social, and institutional barriers to wastewater management in developing countries and relate their discussion to Philippine technology selection. Their work supports a practical motivation for methods that help engineers compare treatment alternatives transparently. A faster calculation cannot resolve infrastructure or governance barriers by itself. It can support the technical assessment that informs planning and operation.

Treatment must also fit the resources available to the community or operator. The guide edited by Ulrich et al. (2009) links decentralized treatment to technical requirements and local implementation conditions. Jafarinejad (2020) places energy efficiency, cost, reliability, and sustainability within the same treatment-design problem. These sources explain why engineering relevance concerns the balance among treatment requirements and resource use. It does not follow from prediction accuracy alone.

Biological wastewater treatment uses microorganisms to transform organic matter and nutrients. In activated sludge treatment, microorganisms grow as suspended

biomass in tanks containing wastewater. A secondary clarifier separates much of this biomass from the treated water by settling. Part of the settled sludge returns to the biological tanks, and another part is withdrawn to control solids accumulation (Henze et al., 2008; Takács et al., 1991).

Organic substrate is material that microorganisms use for energy or growth. It can be oxidized or incorporated into biomass. Nitrogen can move among ammonium, nitrite, nitrate, biomass, and nitrogen gas. Nitrification oxidizes ammonium to nitrite and nitrate. Denitrification converts oxidized nitrogen to nitrogen gas under suitable conditions. Phosphorus can move among dissolved, stored, and particulate forms. Duan et al. (2022) describe biological nitrogen pathways, while Wentzel et al. (1992) explain the connections among nitrogen and phosphorus removal. These interactions require treatment analysis to track several material quantities together.

The activated sludge model (ASM) family models these changes through component balances and process rates (Henze et al., 1987, 1999, 2000). A component is a material category tracked by the model, such as dissolved ammonium, soluble organic substrate, or a biomass group. It may combine several chemical species with a common process role. A component state is the collection of their concentrations at a specified location. Process rates describe how quickly biological or chemical transformations consume or produce these components (Henze et al., 2000).

Reported water quality quantities combine selected components. Chemical oxygen demand (COD) describes oxygen-demand equivalents associated with oxidizable material. Total nitrogen (TN) and total phosphorus (TP) summarize nitrogen and phosphorus on the adopted reporting basis. Total suspended solids (TSS) measures suspended particulate material (Henze et al., 2000, 2008). Such combined quantities are called composites in this dissertation. They do not identify a unique component state. Two effluents can have the same TN while containing different proportions of ammonium and nitrate. The same total can therefore describe different treatment conditions.

Operating choices influence the entire treatment response. Hydraulic retention time (HRT) is reactor volume divided by the flow rate specified for that reactor. It describes a nominal liquid residence time. Aeration transfers oxygen into the wastewater for biological reactions. Recycle returns liquid or settled sludge to an earlier treatment location. Wasting withdraws sludge from the plant to control the solids retained (Henze et al., 2008). An operating decision consequently affects several components and locations at once. These relationships matter when predicted effluent quality is used to compare candidate settings.

1.1.2 From process modeling to repeated operating calculations

Mechanistic models calculate treatment responses from equations for biological conversion, chemical reactions, liquid flow, and solids settling. Their inputs include influent concentrations and operating settings. Their outputs include component concentrations at the treatment locations. An inventory is the amount of material contained within a treatment unit. A steady-state calculation describes conditions at which modeled inventories no longer change with time under fixed inputs. Marais and Ekama (1976) establish steady-state activated sludge analysis, and Hauduc et al. (2013) review the process knowledge and limitations of activated sludge modeling. Each calculation remains conditional on its component definitions, parameter values, and system boundary. The system boundary identifies the units and material streams included in the balance.

Simulation supports design and operating analysis because it permits engineers to examine conditions that have not been directly observed. Calise et al. (2020) address model validation for an activated sludge process. Rivas et al. (2008) connect model-based calculations to treatment-plant design. These applications show how a process model becomes a decision tool. A candidate operating setting must be evaluated, its treatment response interpreted, and its feasibility compared with the intended engineering requirements.

Several routes can support these comparisons. An engineer can examine settings through repeated simulation and judgment. Numerical optimization searches for settings that improve a stated objective while satisfying specified limits. For activated sludge operation, the objective can combine effluent quality and resource use. A statistical surrogate is a model fitted to approximate the responses of a more detailed model at lower evaluation cost (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019). In this research, the surrogate predicts component concentrations from influent conditions and operating settings. A feasible setting satisfies the imposed limits. It is not necessarily the best setting. Nonlinear reaction and recycle relationships can create several locally favorable settings, so termination of a numerical search does not establish the best setting over the entire allowed domain (Buzzi-Ferraris & Manenti, 2013; de Araujo et al., 2013).

Aeration illustrates the need for repeated calculations. More oxygen supply can change substrate removal and nitrification, but it also changes a resource demand. de Araujo et al. (2013) examine operating sensitivities in an activated sludge optimization problem. Gu et al. (2023) review aeration optimization and control strategies. Choosing a setting requires assessment of the joint treatment response and the declared priorities. The preferred setting cannot be determined from an aeration variable or one effluent quantity alone.

Statistical surrogates learn from paired influent and operating inputs and the corre-

sponding mechanistic treatment responses (Fang et al., 2011; Selerio, 2026). Training is the fitting of model parameters to these pairs. A training sample here is one accepted steady-state simulation, rather than a measurement from an operating plant. Durkin et al. (2024) use surrogates in wastewater resource-recovery optimization. Sundui et al. (2021) review biological wastewater applications of machine learning. These developments motivate faster activated sludge calculations while retaining checks on the physical meaning of the predicted component state.

1.1.3 Why accurate prediction is insufficient

A prediction can be close to a mechanistic reference while violating a physical requirement. Consider a hypothetical effluent ammonium concentration of 0.2 g N m^{-3} and a prediction of -0.1 g N m^{-3} . Its absolute error is only 0.3 g N m^{-3} , but a negative ammonium concentration is physically invalid. Larger substrate or biomass concentrations can dominate an average error score and conceal this problem. A small average error can also conceal a violation of mass conservation distributed across several components (Hansen et al., 2024; Kircher & Votsmeier, 2025).

Mass conservation and non-negativity therefore need separate checks on the same predicted effluent. Mass conservation requires the material entering, leaving, accumulating, and reacting within the system boundary to agree under the adopted component accounting. Stoichiometry specifies the relative amounts consumed and produced by each modeled reaction. A stoichiometric invariant is a weighted combination of components that those reactions do not change (Henze et al., 2000; Sturm & Wexler, 2020, 2022). Boundary inputs and outputs must still be included when such a combination is used to check a reactor balance (Takács & Vanrolleghem, 2006).

Non-negativity requires each concentration to be zero or greater. Setting a negative ammonium prediction to zero can disturb the nitrogen balance. Enforcing only a balance can leave one of its component concentrations negative (Harder et al., 2022; Kircher & Votsmeier, 2025).

Projection adjusts a completed component prediction to satisfy a specified set of constraints. In this research, those constraints include mass conservation and non-negativity, as imposed jointly in reactive-chemistry and activated sludge surrogates (Kircher & Votsmeier, 2025; Selerio, 2026). The feasible set contains the concentration states that satisfy the imposed constraints. The projection criterion specifies how changes from the raw prediction are measured. The raw prediction is the output before this adjustment, and the projected prediction is the output afterward. Kircher and Votsmeier (2025) develop projection for simultaneous mass conservation and non-negativity in mechanistic kinetics surrogates. The method can be applied to different fitted models. Its guarantee concerns the imposed constraints, so it does not establish that the adjusted ammonium, substrate, or biomass concentrations reproduce every reaction rate or improve every prediction error (Selerio, 2026; Sturm & Silva, 2025).

Validation checks predictions against reference responses excluded from the fit being assessed. Internal validation supports model selection, while a held-out assessment set evaluates the selected procedure (Hastie et al., 2009; Kaufman et al., 2012). In this research, these checks assess agreement with simulated activated sludge responses. They do not establish agreement with a measured plant (Machado et al., 2014; Newhart et al., 2019).

Influent concentrations and selected aeration settings are available before a treatment response is calculated. The resulting effluent ammonium concentration is unknown at that point and cannot be supplied as an ordinary predictive input. Information leakage occurs when unavailable outputs or reserved assessment information influence fitting or model selection (Kaufman et al., 2012; Newhart et al., 2019). Keeping that information separate makes the assessment more credible. Extrapolation, which means prediction beyond the sampled input range, still requires separate assessment.

Interpretability concerns whether the dependence of a prediction on its inputs can be examined. A structured regression separates fitted terms for operation, influent composition, curvature, and component coupling (Selerio, 2026). Such an explicit model supports examination of fitted relationships (Brunton et al., 2016; Rudin, 2019). Curvature allows the predicted response to change at different rates as an input changes. Influent loading is the material supplied to the system through the incoming concentrations and flow. An interaction allows an aeration response to depend on that loading. Component coupling expresses fitted dependence among outputs such as substrate and biomass concentrations. These terms describe statistical relationships and need not identify biological mechanisms uniquely.

Projection can change both a concentration and its sensitivity to an operating input (Selerio, 2026). Interpretation therefore follows the raw regression and the final projected effluent.

Operating optimization adds a further concern. A search may select an aeration or recycle setting with favorable predicted effluent quality where the surrogate is inaccurate. This can occur even when average held-out error is small. Alexandrov et al. (1998) develop methods for checking and managing model approximations during optimization. Pedrozo et al. (2025) compare surrogate optimization with reference process calculations. Their work supports an independent mechanistic calculation at the selected controls. Mass conservation and non-negativity remain necessary properties of a component prediction. Decision quality also depends on the treatment response and resource demand calculated at the selected setting.

1.1.4 Rationale for the research

The research is motivated by the need for a fast treatment prediction that remains understandable and usable as a component state. Existing literature supplies methods for statistical approximation, projection, and process optimization (Bhosekar &

Ierapetritou, 2018; Durkin et al., 2024; Kircher & Votsmeier, 2025). The engineering problem is to connect these methods while preserving the different questions each one answers. Prediction error describes agreement with a reference. Mass conservation and non-negativity describe specified physical requirements. Interpretability concerns the basis of the returned response. Decision quality concerns the operating choice evaluated with the reference model.

This distinction matters at both reactor and plant scale. For one reactor, projection can impose mass conservation and non-negativity on the full component prediction (Selerio, 2026). For a connected plant, mixing, reactor, clarifier, recycle, and inventory relationships must also agree across the returned response (Alex et al., 2008; Jeppsson & Diehl, 1996). A physically acceptable state at one unit does not establish consistency of its connections to other units.

The research therefore uses a common mechanistic foundation to examine three linked capabilities. The first is statistical approximation and projection of the complete reactor component response. The second is an interpretable second-order regression with a defined projection procedure. The third is a connected plant surrogate that supports operating analysis and independent evaluation of selected controls. These capabilities address a progression from prediction to physical constraints and then to engineering decisions.

The research is intended to support more transparent treatment analysis. Researchers can distinguish prediction error from the effect of projection. Engineers can inspect the returned component state and evaluate a selected operating setting with the original process model. A valid projection establishes mass conservation and non-negativity under its stated assumptions (Kircher & Votsmeier, 2025; Selerio, 2026). Accuracy, computational cost, and decision quality remain separate assessment questions.

1.2 Statement of the problem

The central problem is how to use machine learning surrogates for activated sludge analysis when accurate approximation alone is insufficient for a usable engineering response. The research addresses four connected limitations.

1.2.1 Uncertain agreement with the complete component response

An error score for effluent TN alone does not establish accurate ammonium, nitrite, nitrate, and biomass predictions. Component errors can compensate in the total, and errors in small concentrations can be hidden (Henze et al., 2000; Selerio, 2026). Accuracy can also depend on the number of reactor cases used for training and on whether

an influent or operating setting lies within the sampled ranges (Li et al., 2024; Viering & Loog, 2023). A common assessment of all component concentrations is needed to distinguish these effects while keeping the activated sludge prediction target fixed.

1.2.2 Mass conservation and non-negativity are not established by training alone

Ordinary regression minimizes prediction error without requiring every new activated sludge effluent to satisfy mass conservation or concentration bounds. A residual is the numerical discrepancy in a balance equation. A physical residual penalty adds a cost for that discrepancy during fitting. A finite penalty can reduce violations without eliminating them at every new influent and operating setting (Hansen et al., 2024; Raissi et al., 2019). Projection and independent numerical checks provide one way to assess both physical requirements on the returned component state (Kircher & Votsmeier, 2025; Selerio, 2026).

1.2.3 A visible regression structure can still be difficult to interpret

A regression coefficient is the fitted weight of an input or input product. Its meaning depends on its units and its place in the prediction calculation. For example, a coefficient multiplying aeration and influent substrate describes a different relationship from a coefficient multiplying aeration alone. Component coupling and projection can further change the final predicted ammonium or biomass response (Selerio, 2026). Interpretation must therefore follow the complete calculation. Fitting and validation must also use only the information available for the intended activated sludge prediction (Kaufman et al., 2012).

1.2.4 Unit predictions do not establish the quality of plant decisions

Separately predicted activated sludge unit responses can disagree at their connections. The predicted clarifier underflow, for example, may imply a return-sludge contribution different from the one predicted at the mixer. Consistent reactor and clarifier connections are required by the component-transport and plant models (Alex et al., 2008; Jeppsson & Diehl, 1996). Joint projection can impose specified plant balances. Satisfaction of these selected relations does not establish agreement with every mechanistic equation, which remains outside the stated projection guarantee (Kircher & Votsmeier, 2025; Selerio, 2026). An operating search can still select a setting affected by these approximation errors. Selected aeration, recycle, and wasting controls therefore require independent mechanistic evaluation under the same treatment and resource

priorities. This assessment follows the distinction between approximation quality and optimization behavior developed in surrogate optimization research (Alexandrov et al., 1998; Pedrozo et al., 2025).

1.3 Research questions and objectives

The general objective is to develop a unified framework for activated sludge surrogate prediction, projection, interpretation, and operating assessment. The framework treats mass conservation and non-negativity as explicit requirements and keeps predictive error and decision quality as separate endpoints.

The first question asks how accurately different statistical learners approximate the complete reactor component response as training data change and inputs move beyond the sampled domain. Its objective is to establish a benchmark, which is a defined simulation task for comparing models. The comparison uses shared component targets, data partitions, and evaluation scales.

The second question asks how projection affects mass conservation, non-negativity, error, displacement, and computational cost for the same raw predictions. Its objective is to compare raw and projected states in paired assessments. The comparison identifies the effect of projection without attributing it to a different fitted learner.

The third question asks how an explicit second-order regression and learned component coupling can support interpretable activated sludge prediction with mass conservation and non-negativity. A second-order regression includes individual inputs, their squares, and pairwise products. Its driver is the fitted expression evaluated before component coupling is applied. The objective is to formulate and examine this model, its fitting problem, and its projection procedure. Interpretation distinguishes the driver, the raw component predictions, and the final projected effluent. Detailed formulations are developed in Chapter 5.

The fourth question asks how joint prediction and projection can support operating choices for a connected activated sludge plant. Its objective is to impose the declared unit connections, formulate common engineering priorities, and independently evaluate selected controls. Surrogate-assisted search is compared with direct mechanistic optimization on the original reference basis.

1.4 Significance of the study

1.4.1 Scientific significance

The scientific value is a clear connection between predicted activated sludge responses and the component accounting of the mechanistic model. Stoichiometry defines the relative component changes in each reaction. The reactor and plant boundaries

define the material streams included in mass conservation. Non-negativity applies to each modeled concentration, including dissolved nutrients and biomass. These explicit requirements make the physical meaning of a predicted treatment state available for examination.

This contribution also clarifies the limit of a physical guarantee. A state that satisfies mass conservation and non-negativity need not satisfy the full nonlinear kinetic equations. Likewise, a reported nutrient total need not be identical to an inventory used for mass conservation. Identifying these distinctions supports a more precise interpretation of scientific machine learning in biological wastewater treatment (Henze et al., 2000; Takács & Vanrolleghem, 2006).

1.4.2 Methodological significance

For researchers comparing activated sludge surrogates, the study provides separate evidence for prediction and projection. Models within each assessment receive common influent and operating inputs and predict the same component concentrations. Preprocessing transforms data into the coordinates used for fitting. Tuning selects model settings, called hyperparameters, using internal validation. These settings control model structure or fitting and differ from the coefficients estimated during training. Preprocessing and tuning use only the relevant training information to avoid information leakage (Kaufman et al., 2012). Paired raw and projected effluents identify which differences arise from projection. The reactor and plant assessments retain their specified validation designs, so their scores answer related questions without establishing one shared model ranking.

The design reports errors for individual component concentrations and for the combined water quality quantities. It also reports mass conservation residuals, negative concentrations, projection displacement, and computational effort. Projection displacement measures how much the concentration vector changes during enforcement. These measures distinguish an accurate treatment prediction from one that requires substantial adjustment to satisfy the imposed balances. Their joint reporting makes the limits of an average effluent error score visible, as illustrated by component-level projection and activated sludge assessments (Selerio, 2026; Sturm & Silva, 2025).

The structured regression provides an inspectable activated sludge model. Its named terms and component coupling permit direct coefficient and sensitivity calculations. Sensitivity measures the local change in a predicted concentration when an input such as aeration changes. Projection is analyzed as another stage of the returned effluent calculation. This supports examination of the treatment response without treating fitted coefficients as causal biological parameters (Brunton et al., 2016; Selerio, 2026).

1.4.3 Engineering and practical significance

Engineers need to compare operating conditions while maintaining a consistent account of material across the treatment plant (Alex et al., 2008; Henze et al., 2008). The connected surrogate models mixer, reactor, clarifier-outlet, and inventory responses together. Joint projection addresses contradictions that separate unit predictions can leave unresolved. The resulting numerical checks provide a basis for detecting specified mass conservation and non-negativity violations before a prediction enters an operating search.

Independent mechanistic evaluation gives the operating assessment a practical reference, consistent with reference-model checks used in surrogate optimization (Alexandrov et al., 1998; Pedrozo et al., 2025). It determines treatment responses at the controls actually selected, checks the original engineering requirements, and distinguishes available decisions from failed or ineligible cases. This is useful for assessing whether a faster search supports the intended decision. It does not establish lower operating costs or better field treatment before those outcomes are evaluated.

The framework is also relevant to technical capacity in wastewater planning. Philippine and other resource-limited settings require decisions that connect treatment needs with the resources and conditions of implementation (Damalerio et al., 2022; Tabios III, 2020; Ulrich et al., 2009). A transparent, computationally manageable assessment can support such decisions. Plant calibration, reliable measurements, operating expertise, and institutional capacity remain necessary for practical application.

1.4.4 Significance for further research

The formulations provide a defined starting point for examining uncertainty, dynamic activated sludge operation, and other biological treatment configurations. Their value is the separation of physical constraints, statistical agreement, interpretation, and decision evidence. An extension must define its own modeled components and boundary streams. For example, a dynamic treatment model must account for changing reactor and clarifier inventories (Alex et al., 2008; Takács et al., 1991). It cannot inherit the steady-state reactor or plant guarantees solely because it uses the same learning method.

1.5 Contributions of the research

The first contribution is a common reactor assessment that retains the complete component target. It examines learning behavior under specified data availability and input-domain conditions. Component and composite scores have distinct roles, so a favorable reported total cannot conceal every defect in the component state.

The second contribution is a paired projection assessment for mass conservation and

non-negativity. It adopts the projection principles of Kircher and Votsmeier (2025) and applies a common physical requirement across different learners. Mathematical feasibility, projection displacement, prediction error, and cost are evaluated separately. The contribution concerns their organized use in activated sludge analysis and does not claim invention of the underlying projection method.

The third contribution is an interpretable second-order regression with learned component coupling and checked projection. The explicit feature groups support structural inspection of the activated sludge regression (Selerio, 2026). The fitting formulation distinguishes soft physical penalties from the projection used to impose mass conservation and non-negativity on the returned state. Interpretation follows the complete response calculation rather than an isolated coefficient block.

The fourth contribution is a connected plant formulation with joint projection and independent operating verification. It links unit responses through the declared mixing, reaction-inventory, clarification, and stored-solids relations. Selected controls are then assessed with the original mechanistic equations and a common objective. This connects the physical meaning of a surrogate state to the quality of the decision made with it.

Table 1.1 makes the problem–contribution relationship explicit. The evidence column identifies how each contribution is assessed. It does not report an empirical outcome.

Table 1.1: Research problems, methodological contributions, and assessment evidence.

Research problem	Contribution	Assessment evidence
Complete response is hidden by a few reported totals	Common component-level reactor benchmark	Held-out component and composite errors, learning curves, and beyond-domain assessment
Training does not establish mass conservation and non-negativity	Common projection with paired raw and projected assessment	Equality residuals, concentration checks, error differences, displacement, and cost
Model terms do not directly explain the final response	Explicit second-order regression and component coupling with checked projection	Named coefficient groups, full response sensitivities, and stage-specific diagnostics
Consistent predictions do not establish useful operating decisions	Joint plant projection and original-model verification of selected controls	Unit and inventory checks, common reference objectives, engineering eligibility, and failure accounting

These contributions are mathematical and methodological. Their significance is the capacity to address the stated limitations through a coherent assessment. Empirical superiority over another learner or optimization route is not presumed.

1.6 Research framework

The research begins with a mechanistic activated sludge model and specified ranges of influent concentrations and operating settings. These ranges form the input domain. Each simulated case pairs those inputs with steady-state component concentrations. The pairs support statistical fitting. Reaction stoichiometry and boundary flows supply the mass conservation relations (Henze et al., 2000; Sturm & Wexler, 2022). Non-negativity supplies the concentration requirement. Projection adjusts each raw effluent prediction to satisfy these requirements, following the joint enforcement approaches of Kircher and Votsmeier (2025) and Selerio (2026).

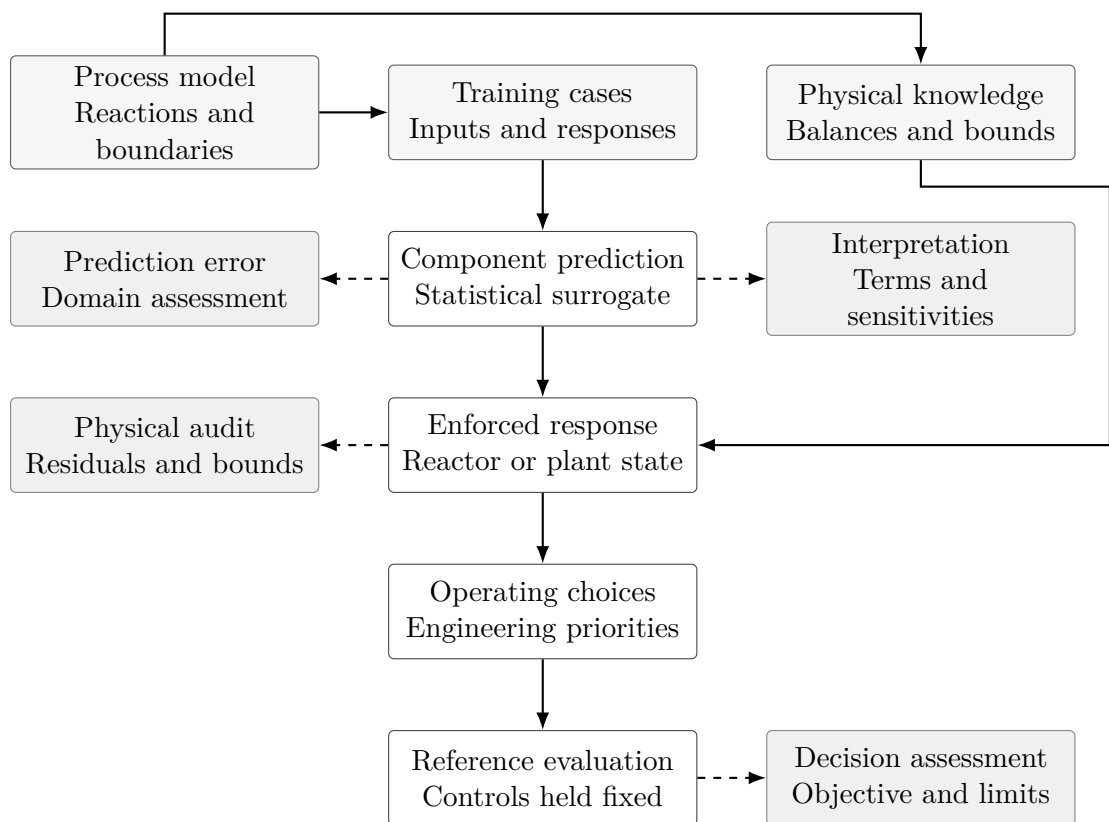
The logic extends to interpretation and operating analysis. A structured predictor exposes terms and sensitivities. A connected plant formulation adds relationships among unit responses and stored solids. An operating search selects controls, and independent mechanistic evaluation establishes their reference treatment response. Each step addresses a specific claim. The evidence required for one claim does not substitute for the evidence required for another.

Figure 1.1 links the information used to construct the prediction procedure to the evidence used to assess its engineering value. The central sequence concerns prediction, projection, operating choice, and reference evaluation. The side boxes retain the different assessment questions.

The supervised targets are the mechanistic component concentrations paired with known influent and operating inputs. Physical knowledge supplies the requirements used by projection. Error assessment compares predicted ammonium, substrate, biomass, and other concentrations with those targets. Physical checks assess mass conservation, non-negativity, and the additional plant constraints. Structural inspection examines the fitted treatment model. Decision assessment calculates the mechanistic treatment response at the selected controls. The reactor and connected plant use this logic with separately specified predictors and physical boundaries.

1.7 Scope and limitations

The study concerns simulated steady-state activated sludge responses. It includes a standalone reactor and a connected plant with biological reactors, secondary clarification, and recycle streams. The mechanistic equations supply reference responses under fixed process definitions and parameter values. The reactor and connected plant have separately specified operating domains.



Solid arrows connect modeling steps. Dashed arrows identify assessment.

Figure 1.1: Conceptual research logic linking process knowledge, component prediction, projection, interpretation, and independent assessment of operating choices.

The state includes organic, nitrogen, phosphorus, oxygen, alkalinity, biomass, and solids coordinates within the adopted model basis. Mass conservation is imposed through that basis and the stated system boundary. Non-negativity is imposed on the represented concentrations. The guarantees do not establish complete elemental accounting for unrepresented species, full kinetic feasibility of every projected state, or dynamic stability beyond the specified checks. These limits follow from the adopted component boundary and the scope of the imposed constraints (Selerio, 2026; Takács & Vanrolleghem, 2006).

Agreement with simulated responses establishes agreement with the selected mechanistic reference. It does not establish performance at an operating treatment plant. Calibration uncertainty, measurement error, microbial adaptation, and unrepresented reactions require additional field evidence (Machado et al., 2014; Newhart et al., 2019). Reliable calibration and process understanding remain essential (Meijer, 2004; Petersen et al., 2003).

Exploratory simulation informed the connected plant assessment design. Its separate holdout is a reserved set of simulated plant cases. It supports descriptive assessment conditional on those design choices, which is termed post-selection assessment. It does not provide independent external confirmation. Fitting and tuning still exclude the holdout responses. Independent mechanistic evaluation checks the selected controls, but it does not remove this statistical limitation.

Attached-growth treatment, spatial transport, resource-recovery networks, and dynamic control provide relevant background and possible extensions. They are not additional evaluated case studies. The connected operating problem uses the declared topology and local numerical assessment. It does not certify a global optimum or a new network configuration.

Numerical examples and conceptual figures explain the mathematical operations and are explicitly hypothetical. They do not report treatment, prediction, optimization, or timing findings. Empirical results are reported separately in the reactor projection, structured-regression, and connected-plant assessments.

Chapter 2

Review of Related Literature

Activated sludge surrogate modeling draws on several bodies of research. Mechanistic models define the treatment processes and component balances to be approximated. Data-driven studies show how treatment quantities can be estimated from measurements or simulations. Physical enforcement methods restrict predictions to specified balances and concentration bounds. Interpretable modeling explains the fitted relationships, while optimization studies examine their use in operating decisions. This chapter reviews the contributions of these studies and the questions that remain when their methods are combined for activated sludge analysis.

2.1 Mechanistic foundations of activated sludge modeling

2.1.1 Development of component and nutrient models

Activated sludge modeling developed from the need to relate operating conditions to organic matter removal, oxygen demand, and sludge production. Marais and Ekama (1976) formulated steady-state activated sludge analysis around substrate utilization and biomass. Dold et al. (1981) developed a general model that linked several biological processes within a common treatment calculation. These contributions established a basis for calculating treatment responses from material balances and biological rates.

Henze et al. (1987) presented the single-sludge framework associated with Activated Sludge Model No. 1 (ASM1). The model distinguishes substrate, biomass, and nitrogen components so that carbon oxidation, nitrification, and denitrification can be calculated together. Later models addressed processes that this framework did not include explicitly. Henze et al. (1999) extended phosphorus-removal modeling to include the denitrifying activity of phosphorus-accumulating organisms. Gujer et al. (1999) introduced Activated Sludge Model No. 3 (ASM3), which includes organic substrate storage and uses endogenous respiration to model biomass decay.

These developments addressed different biological modeling needs. The phosphorus-removal extension connects nitrate use to phosphate behavior. The storage-based model separates substrate uptake from its later use by biomass. The collected formulations in Henze et al. (2000) provide a common reference for these component definitions and their associated reaction terms. Their contribution to surrogate modeling is a physically defined prediction target. Effluent COD or TN can summarize treatment quality, while the component concentrations describe the material forms underlying those totals.

Hauduc et al. (2013) compared seven activated sludge models by examining their treatment of hydrolysis, fermentation, microbial growth and decay, and biological storage. Their review identifies conceptual differences between the models and relates those differences to the underlying process knowledge. It shows that model selection depends on the treatment process and the question being studied. For a surrogate comparison, this means that all candidate predictors need to approximate the same selected process model. Otherwise, differences in prediction can arise from different biological assumptions as well as different statistical methods.

2.1.2 Biological dependence and calibration uncertainty

Nutrient-removal research explains why operating inputs and influent composition must be considered together. Wentzel et al. (1992) examined the interactions among nitrification, denitrification, and excess phosphorus removal. Guerrero et al. (2011) studied competition for organic substrate in a pilot system intended for simultaneous nitrogen and phosphorus removal. Their experiments found that the nature of the supplied carbon source affected the ability to maintain phosphorus removal when nitrate entered the initially anaerobic zone. The study therefore addresses a specific biological interaction between substrate composition, recycle conditions, and nutrient removal.

For statistical modeling, these findings support retaining both operating conditions and component-resolved influent information. A model based only on total organic loading can omit differences between carbon sources that have different biological roles. A model that treats aeration or recycle independently of loading can also overlook the conditions under which those actions affect nutrient removal. This is a reason to examine interactions in a treatment surrogate. It is not evidence that a fitted interaction coefficient identifies a microbial mechanism by itself.

Calibration research addresses another part of the modeling problem. Petersen et al. (2003) reviewed experimental designs for estimating activated sludge parameters. Vanrolleghem et al. (1999) examined the use of respirometry, which measures oxygen consumption, to estimate model parameters and component combinations. These methods help connect the internal quantities of a process model to observations. They also show why the information supplied by an experiment must match the parameter being estimated.

Machado et al. (2014) calibrated a phosphorus-removal model using data from a full-scale plant in Manresa, Spain. They assessed parameter confidence intervals and examined how influent and operating variables affected the model fit. They also varied those inputs while retaining default kinetic and stoichiometric parameters to evaluate output sensitivity. The study addresses the influence of input uncertainty alongside uncertainty in biological parameters. Its relevance to surrogate modeling is that simulated targets depend on both sets of assumptions. Agreement with those targets assesses the surrogate's approximation, while agreement with an operating plant requires calibration and observational evidence.

The mechanistic literature thus establishes meaningful components, reaction relationships, and methods for calibration. The unresolved question for this dissertation is how to approximate those defined treatment responses efficiently while retaining their material accounting. The surrogate is evaluated against an adopted mechanistic reference. It does not replace the separate task of calibrating that reference to a particular plant.

2.1.3 Clarification and connected plant balances

Biological reactors do not determine the final discharge independently of solids separation. Takács et al. (1991) developed a clarification and thickening model based on solids fluxes and balances around layers of a one-dimensional settler. The model calculates the solids profile and the overflow and underflow concentrations under steady and changing conditions. Härtel and Pöpel (1992) also modeled secondary clarification with sludge thickening. These studies provide mechanistic descriptions of how suspended material is retained, concentrated, and discharged.

Jeppsson and Diehl (1996) examined the propagation of biological components through the secondary clarifier. They identified difficulties in coupling a biological reactor model to a settler model because the two models describe particulate material differently. Their evaluation of component-transport algorithms focused particularly on the composition of sludge returned to the reactor. This study directly establishes why predicting aggregate solids alone is insufficient for every connected component calculation.

Alex et al. (2008) addressed the need for comparable plant assessments through Benchmark Simulation Model No. 1. The benchmark connects five biological compartments, a secondary settler, internal recycle, return sludge, and wasting. It also specifies influent conditions, control settings, and assessment criteria. Its contribution is a common process and evaluation setting in which alternative operating strategies can be compared. It makes the connection between biological treatment and separation explicit.

Together, these studies establish the plant relationships that a connected surrogate must approximate. A clarifier prediction affects both the discharge and the recycled

biomass entering the biological tanks. Stored solids also contribute to the plant inventory used in solids-retention calculations. A single-reactor surrogate therefore addresses only part of the plant response. Extending physical enforcement to a connected plant requires consistent component flows at shared streams and an explicit account of retained solids.

2.2 Data-driven prediction of wastewater treatment responses

2.2.1 Soft sensors and reported treatment quantities

Data-driven treatment studies developed partly because plants collect measurements that are difficult to interpret jointly. Newhart et al. (2019) reviewed methods for fault detection, variable prediction, and advanced process control. They identify missing observations, outliers, correlated variables, and changing process behavior as important features of plant data. Sundui et al. (2021) reviewed machine learning applications across biological wastewater treatment. These reviews establish several useful prediction tasks and explain the need to match a method to its data and intended application.

A soft sensor estimates a quantity that is difficult or slow to measure from other available observations. Ching et al. (2022) developed a soft sensor for five-day biochemical oxygen demand using extreme gradient boosting (XGBoost). They evaluated influent and effluent estimates at two treatment plants and used supporting measurements selected from variables such as COD, solids, nutrients, and simpler sensor readings. The model estimated a wide range of oxygen-demand values and could accommodate missing supporting measurements. The study addresses the practical need for faster monitoring of a selected water quality quantity.

Y. Wang et al. (2025) compared a mechanistic activated sludge model with six machine learning models for effluent TN prediction using one year of high-resolution municipal plant data. Their results showed that the mechanistic model captured the direction of TN variation, while the learned models achieved greater predictive accuracy in the assessed data. They also used feature-attribution analysis to identify influential inputs. This provides evidence that statistical modeling and model inspection can improve a defined effluent-prediction task.

Ekinici et al. (2023) addressed waste-sludge prediction using 208 daily observations from an advanced biological treatment plant in Turkey. Their inputs included flow, polymer dosage, and the amounts of solids and pollutants removed. They compared learning methods and reported improved prediction after feature selection. The study adds a useful solids-management target to the quality indicators considered in other work. Its measured removal inputs support that estimation task. A prediction made

before evaluating a proposed operating setting would require a different set of available inputs.

The common achievement of these studies is useful estimation of selected treatment quantities from available plant information. Their results support monitoring, diagnosis, and particular operating analyses. The remaining question here concerns a different output requirement. Predicting all modeled effluent components is needed when the analysis must check mass conservation, non-negativity, and the distribution of material among biological forms. Accuracy for TN, oxygen demand, or sludge production supplies part of that evidence, while complete component prediction requires its own assessment.

2.2.2 Component estimation and simulation-based surrogates

Component-estimation studies move closer to the information required by mechanistic models. P. Wang et al. (2024) used a backpropagation neural network to estimate five COD-related component quantities from measured influent COD. They first characterized the components in an operating plant and then evaluated the neural predictions against those measurements. Their reported prediction accuracy exceeded 80% for the five assessed components. The study addresses the difficulty of repeatedly measuring component allocations needed as model inputs.

This contribution differs from predicting the effluent component state at a proposed operating setting. Influent fractionation estimates how the incoming organic material is divided among model categories. An effluent surrogate estimates the result of biological conversion under specified controls. The two tasks can complement one another, but they have different reference outputs and assessment requirements. The component estimates of P. Wang et al. (2024) therefore support improved influent characterization without establishing the complete treatment mapping investigated here.

Simulation-based studies address that mapping more directly. Fang et al. (2011) combined an extended activated sludge model with a support vector surrogate and a genetic optimization search for a full-scale nutrient-removal system. Mechanistic simulations supplied the relationship between operating conditions and effluent quality learned by the surrogate. Their optimization suggested that anoxic volume and internal recycle could be reduced while retaining the specified discharge quality. The study is an early example of combining process simulation, statistical approximation, and operating search.

He et al. (2023) developed a Kriging surrogate for greenhouse-gas emissions from papermaking wastewater treatment. Kriging uses statistical relationships among sampled input conditions to estimate responses at other inputs. They used steady-state process simulations to calculate emissions and fitted the surrogate to those outputs. The reported coefficient of determination exceeded 0.96, and the model supported analysis and optimization of controllable treatment variables. Their work addresses

the cost of repeated evaluation of an environmental objective. Its learned target is the emissions measure, rather than the complete component response needed for the mass conservation checks of this dissertation.

Selerio (2026) addressed complete single-reactor component prediction through invariant-constrained second-order regression. The study used 10,000 simulated steady-state cases with 20 effluent components and compared the structured regression with nine statistical predictors. Its evaluation included training-size variation, reported water quality errors, and physical violation diagnostics. This provides a closer precedent for the present component-level task than the selected-indicator studies. Its physical enforcement and interpretation are examined in the following sections.

Table 2.1 summarizes the progression from monitoring to process approximation. The distinctions concern the purpose of the studies and the information their assessments establish. They identify the additional evidence required for a complete activated sludge response without treating a different prediction task as a methodological failure.

Table 2.1: Treatment prediction studies and the scope of their evidence.

Study and task	Contribution established	Additional question for this dissertation
Ching et al. (2022) Oxygen-demand soft sensing	Estimation across two plants with support for missing sensor observations.	Can a predictor estimate the complete effluent state from influent and proposed operating inputs?
Y. Wang et al. (2025) Effluent TN prediction	Comparison of six learned models with a mechanistic model and inspection of influential inputs.	Are the individual nitrogen and other component concentrations accurate and physically consistent?
P. Wang et al. (2024) Influent component estimation	Neural estimation of five COD-related quantities used in process modeling.	How accurately is the resulting effluent component state predicted after treatment?
Fang et al. (2011) Simulation-based operating search	Mechanistic simulation combined with a support vector surrogate and optimization.	Can connected component predictions be constrained and checked at the selected controls?
Selerio (2026) Complete reactor prediction	Structured prediction of 20 components with physical checks and training-size assessment.	How do common projection and plant connections affect prediction and operating assessment?

2.2.3 Model comparison, data availability, and extrapolation

The methods used in treatment prediction provide several ways to approximate non-linear responses. Breiman (2001) developed random forests by averaging randomized

trees. Chen and Guestrin (2016) developed XGBoost, while Ke et al. (2017) introduced the light gradient boosting machine (LightGBM). Smola and Schölkopf (2004) described support vector regression, which uses regularized fitting and kernel-based similarity. These methodological studies explain the learners used in wastewater applications. Their contributions support a comparator set with different fitting structures, while treatment studies establish which structures perform well for a particular process target.

Inspectable linear and nonlinear models provide another part of that comparison. Wold et al. (2001) described partial least squares regression, which uses latent components to summarize related input and output variation. These statistical components are weighted combinations of variables rather than physical wastewater constituents. Woo et al. (2009) applied a kernel extension to estimate COD, TN, and cyanide concentrations in industrial cokes wastewater treatment. It captured nonlinear process relationships and outperformed the assessed linear and other nonlinear partial least squares formulations. This establishes the benefit of the extension for that application. Ordinary partial least squares remains a different comparator for the present component target.

The tabular-model comparison of Shwartz-Ziv and Armon (2022) and the neural-model survey of Borisov et al. (2024) provide further reasons to compare learning families rather than select one solely from its complexity. This is relevant to activated sludge data, where a row contains operating values and influent concentrations. However, the applicable model ranking must come from the shared treatment target and assessment design. The broader methodological literature guides comparator selection, while the treatment benchmark supplies process-specific evidence.

Training-size assessment addresses the cost of obtaining that evidence. Viering and Loog (2023) reviewed learning curves and found that their shapes depend on the learner, data, and task. A learning curve relates prediction performance to the number of training observations. Selerio (2026) used such an assessment for its reactor surrogates. This addresses data efficiency within its stated input range. A broader comparison still needs to determine how the complete component errors and physical adjustments change together as simulated training cases become scarce.

Assessment also depends on information separation. Kaufman et al. (2012) defined information leakage and described how information unavailable for the intended prediction can enter model development. Applied to the treatment tasks reviewed above, this means that current effluent measurements can support a monitoring estimate but cannot automatically serve as inputs to a prediction made before treatment is evaluated. Model tuning and scaling also need to exclude the observations reserved for assessment. These are conditions for interpreting an error comparison. They do not imply that every study using downstream measurements has handled its data incorrectly.

The remaining comparative question combines model choice, training support, and

operating domain. The reviewed indicator and component studies demonstrate useful prediction within their assessed conditions. A common complete-state comparison is needed to assess several learners under the same input and output definitions. It must also distinguish prediction within the training ranges from extrapolation beyond them. This motivates evaluating component error and physical validity together as influent and operating conditions change.

2.3 Methods for imposing physical constraints on predictions

2.3.1 Physical guidance during fitting

Physical constraints have been introduced into learning to address predictions that disagree with governing equations. Karniadakis et al. (2021) reviewed approaches that combine physical knowledge with machine learning. One influential approach is the residual-based fitting of Raissi et al. (2019), which adds discrepancies in differential equations to the training objective. This makes equation disagreement costly alongside prediction error. The method provides a way to use governing equations when response observations are limited.

Li et al. (2024) examined this approach for water-treatment unit dynamics, including an activated sludge reactor. They compared physics-informed neural models with conventional learned models under limited data and different loading conditions. The physically informed models showed stronger predictive capability and generalization in those assessed systems. Their study therefore provides direct wastewater evidence that process equations can improve learning when data are scarce.

The distinction between this contribution and output enforcement follows from the fitting problem. A finite residual penalty balances equation agreement with other terms in the objective. It does not require every new predicted concentration vector to satisfy a balance equality exactly. Thus, the demonstrated learning benefit of Li et al. (2024) supports physical guidance during fitting, while sample-specific mass conservation and non-negativity require an additional construction or check. These are complementary requirements in activated sludge modeling.

2.3.2 Conservation through model construction

Other studies address conservation by changing how outputs are calculated. Beucler et al. (2021) introduced neural architectures that encode analytic constraints and compared them with constraints imposed through a loss function. In their climate-convection application, the architectural approach enforced the selected conservation laws to machine precision without degrading overall predictive performance. The study

establishes that appropriately constructed outputs can satisfy known equalities directly.

Sturm and Wexler (2020) developed a conservation framework in which learned transfers are mapped to changes in chemical species through balanced process relationships. Sturm and Wexler (2022) then placed the stoichiometric mapping in a fixed neural layer for a photochemical surrogate. The later construction avoided the need to reconstruct all reaction transfers as separate training targets and retained atom conservation in the output changes. These studies address the coupling of species that independent output regression leaves unspecified.

Their relevance to activated sludge modeling lies in the shared use of stoichiometry. A balanced mapping can preserve the component combinations implied by the adopted reaction network. Concentration bounds remain a separate condition because a balanced change can consume more of a component than is present. This follows from the difference between a reaction-induced change and the final concentration. The construction must be assessed for the quantity it constrains.

Harder et al. (2022) explored both physical penalties and constrained output completion in an aerosol microphysics surrogate. Their assessment treated mass conservation and positive mass as explicit modeling concerns. It illustrates why conservation and concentration bounds need to be examined together when outputs are completed or adjusted. For the activated sludge task, the required state must satisfy both conditions after the complete prediction procedure has been applied.

2.3.3 Projection after statistical prediction

Projection provides an alternative that can be applied after a model has produced its output. Sturm and Silva (2025) developed a species-weighted least-squares correction for atom conservation in an atmospheric chemistry surrogate. The weights accounted for the magnitude and uncertainty of individual species changes. In their assessment, unweighted correction could damage prediction of a small radical, while weighting retained its accuracy and slightly improved overall agreement. Their results establish that the projection metric affects prediction quality even when the enforced conservation relation is unchanged.

The corresponding activated sludge issue is variation in concentration scale across components. A numerical adjustment that is small for abundant substrate can be large relative to a residual nutrient concentration. The study of Sturm and Silva (2025) therefore supports examining adjustment weights and component errors together. Its equality correction addresses atom accounting, while a simultaneous concentration-bound construction is needed when non-negativity is also required.

Kircher and Votsmeier (2025) addressed that joint requirement through a positivity-preserving projection and linear interpolation backtracking. They evaluated atmospheric chemistry, heterogeneous catalysis, and synthetic reaction networks. The corrected predictions satisfied atom balances and remained positive in the tested systems

without reducing overall prediction accuracy. This study establishes a method for imposing both requirements across different underlying surrogates. The remaining activated sludge question concerns its behavior under wastewater component units, reactor boundaries, training scarcity, and extrapolated operating inputs.

Selerio (2026) supplied a direct activated sludge example. Its structured regression used physical penalties during fitting, followed by a check of each raw prediction and projection onto the mass conservation equalities. A linear program, which optimizes a linear objective under linear constraints, was used when non-negativity still failed. The published assessment reported no final mass conservation or non-negativity violations. The raw regression did not satisfy both requirements, so the final enforcement procedure was responsible for the physical result. This resolves the question of whether a structured single-reactor surrogate can return a component state with both imposed properties.

The same study also reported an accuracy trade-off relative to some unconstrained comparators. Consequently, its result establishes physical reliability for the returned state without making that state the most accurate prediction under every reported score. The next comparative question is how a common projection changes different learners when their raw predictions are held fixed. A paired comparison of raw and projected states can identify that effect across components and operating conditions.

Figure 2.1 brings together the approaches described above. Fitting penalties, constrained output construction, and final-state projection address physical consistency at different stages. The distinction follows from the methods of the cited studies and explains which part of a predictor is responsible for a stated constraint.

2.3.4 Constraint scope and recent extensions

More recent work connects physical enforcement to uncertainty and model capacity. Hansen et al. (2024) used integral conservation relations to adjust a probabilistic prediction, accounting for both predicted means and uncertainty. Their framework enforced conservation exactly when the imposed balance was treated as having zero uncertainty and was evaluated on a family of transport equations. Utkarsh et al. (2025) trained a probabilistic predictor together with a differentiable projection and evaluated it on differential-equation and time-series problems. A differentiable calculation provides derivatives that can be used during fitting. These studies address uncertainty as part of constrained prediction, rather than only correcting a single concentration estimate.

Min et al. (2024) developed neural output layers for input-dependent linear and convex constraints and established approximation results under the stated mathematical conditions. A convex feasible set contains the straight line joining any two feasible states. Their work addresses the concern that hard constraints, which require outputs to satisfy the imposed conditions, may restrict model capacity. Valente et al. (2025) examined projection onto physical manifolds, which are sets of outputs satisfying se-

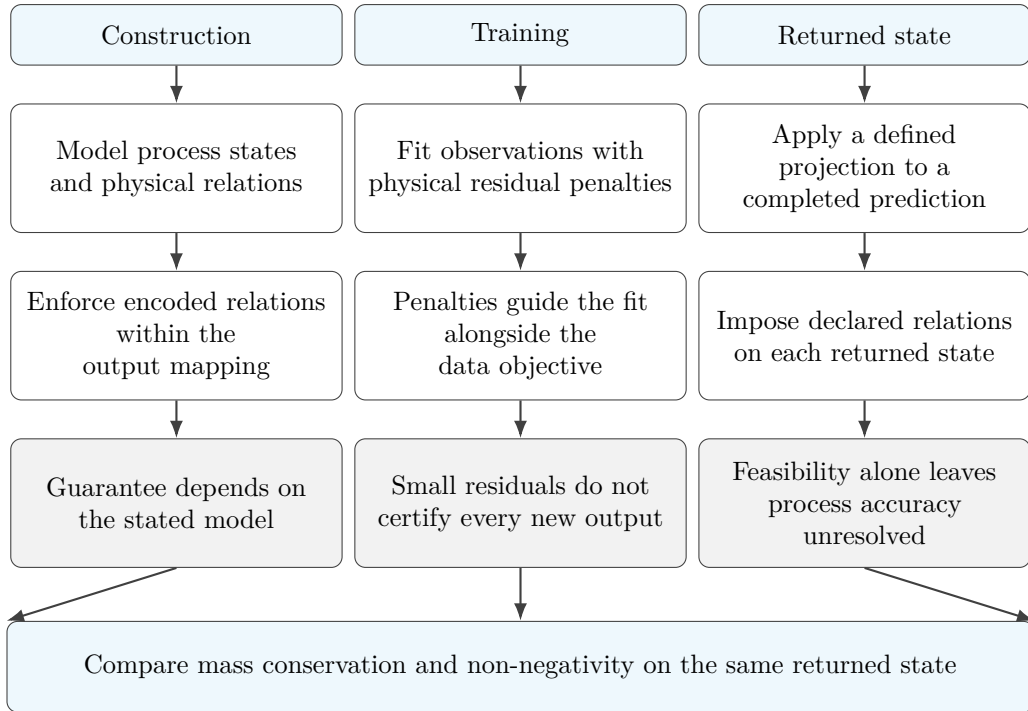


Figure 2.1: Stages at which the reviewed methods introduce physical information. Fitting penalties guide equation agreement. Output construction and projection impose the relations encoded in their formulations. Simultaneous mass conservation and non-negativity require an assessment of the final component state.

lected physical laws. Their spring-mass and reactive-plasma examples extend output correction to nonlinear physical relations. These contributions broaden the available constraint methods. Their evaluated problems and mathematical conditions must still be matched to a wastewater application.

Taken together, the physical-learning literature has already addressed conservation through several constructions and joint positivity through specific methods. The gap for this dissertation is therefore not the absence of physical enforcement. It is the need to assess its accuracy, adjustment size, and computational cost for a common activated sludge component task and then extend the stated accounting conditions to connected plant outputs. The adopted constraints remain narrower than the full biological and settling equations.

2.4 Interpretable modeling of treatment responses

2.4.1 Inspection of inputs and fitted relationships

Interpretability research addresses the need to examine why a model predicts a particular response. Rudin (2019) argued for using directly inspectable models in consequential applications when such models are available. In wastewater treatment,

this need appears when an engineer must understand how influent composition or an operating setting relates to predicted quality. The argument supports inspection of the model used for the decision. It does not establish the biological meaning of every fitted coefficient.

The treatment studies reviewed earlier use different forms of inspection. Y. Wang et al. (2025) combined TN prediction with feature-attribution analysis to identify important inputs. Ekinici et al. (2023) used feature selection and attribution in sludge prediction. These approaches address which observed variables contribute to a specified statistical output. They can guide process investigation and reduce unnecessary inputs. Their explanations concern the fitted indicator or sludge model rather than all component balances of a mechanistic treatment system.

Explicit equations offer a complementary approach. Nazlabadi et al. (2021) reviewed response-surface methods in biological wastewater treatment, where polynomial terms describe operating effects and interactions over the assessed range. Brunton et al. (2016) developed sparse equation discovery from candidate functions for nonlinear dynamical systems. The latter is a methodological precedent for inspecting named terms, while response-surface applications provide the treatment context. Together, they support visible algebraic relationships as a useful form of model inspection.

For activated sludge analysis, an aeration term, an influent substrate term, and their product have different interpretations. The product permits the fitted aeration response to depend on the substrate condition. Such terms help describe a treatment response within the fitted domain. Establishing a causal biological parameter would require a different identification problem and evidence about the underlying mechanism. This distinction is particularly relevant when inputs are correlated.

2.4.2 Structured regression and the final projected state

Selerio (2026) addressed the need for an inspectable complete-component surrogate by separating operating effects, influent effects, curvature, interactions, and learned dependence among effluent components. Its explicit second-order terms allow the fitted input relationships to be examined in physical coordinates. Component coupling then contributes to the complete raw prediction, and a fixed composition map produces the reported water quality quantities. The study therefore provides a direct treatment model that combines visible statistical structure with final physical enforcement.

The publication also identifies limits on that interpretation. Its displayed coefficient relationships describe the raw regression before the final constraint correction. The mass conservation projection and linear-program stages can change the returned state. A concentration bound becomes active when a predicted component reaches the permitted boundary, and that change can affect local sensitivity. The study explicitly distinguishes its coefficient displays from derivatives of the final projected output. It also recognizes that different fitted coefficient combinations can produce similar predic-

tions and identifies coefficient stability across resampled data as a further diagnostic.

Shen et al. (2023) reviewed differentiable modeling that connects learned relationships with process equations. Their geoscience examples show how combined models can support interpretation while retaining physical knowledge, and they identify verification of the physical significance of outputs as a continuing need. This perspective supports examining the complete calculation in a treatment surrogate. The specific distinction between raw coefficients and final constrained sensitivities, however, is supplied directly by the activated sludge study of Selerio (2026).

The remaining interpretive question is thus an extension of an established contribution. An explicit raw equation already makes its terms visible. A fuller assessment must connect those terms to component coupling, reporting, and the constraints that determine the final effluent. It must also examine whether the inferred response patterns remain stable when the training cases change. This motivates interpretation of the complete prediction procedure while retaining the distinction between statistical association and biological mechanism.

2.5 Surrogate modeling for operating decisions

2.5.1 Mechanistic and learned routes to operating optimization

Activated sludge operating optimization predates the recent use of machine learning surrogates. de Araujo et al. (2013) optimized a steady-state process model and used sensitivity analysis to select variables for economic control. Their analysis linked active effluent constraints and remaining operating degrees of freedom to the control structure. Nazif et al. (2023) calibrated a treatment model to a plant in Tehran, grouped influents by their quantitative and qualitative characteristics, and optimized operating settings for each group. These studies establish mechanistic routes that connect treatment constraints, influent conditions, and operating choices.

Learned models provide another route when repeated response evaluation is costly. The integrated approach of Fang et al. (2011) used mechanistic simulations to train a support vector surrogate before optimizing the nutrient-removal system. Kusiak and Wei (2013) fitted neural models for airflow, effluent carbonaceous biochemical oxygen demand, and TSS from industrial data. They then used a multiobjective particle-swarm search to examine dissolved-oxygen settings. These studies demonstrate operating searches supported by learned treatment relationships. Their targets determine which aspects of the operating decision are evaluated.

Dynamic control studies provide evidence at a different time scale. Bernardelli et al. (2020) developed a neuro-fuzzy predictive controller for aeration and field-tested it at a large municipal plant. They reported improved effluent quality and energy savings

in that application. Croll et al. (2023) reviewed reinforcement learning for wastewater control and discussed the approaches and challenges involved. These studies establish that learning can participate in changing treatment operation, while the present dissertation concerns fixed-input steady-state operating assessment. Their dynamic results are relevant background, but they do not supply the evaluation of that steady-state task.

The operating literature therefore addresses both process-based and learned decision routes. The question for a new surrogate is how its selected settings compare on a common engineering basis. This requires the same treatment objectives, operating bounds, and reference calculations. It also requires distinguishing the cost of developing a learned model from the cost of using it during a search.

2.5.2 Integrated resource recovery and connected responses

Network and resource-recovery studies extend the decision beyond one biological unit. Padrón-Páez et al. (2020) examined sustainable treatment design through multiple objectives, while Aboagye et al. (2021) combined wastewater-network design, optimization, and sustainability assessment. Their contributions show how treatment and resource priorities can be evaluated together when several process choices are available. They establish a broader systems context for the operating objective of a connected plant.

Durkin et al. (2024) developed surrogate-based process synthesis for resource recovery from brewery wastewater. Their framework used high-fidelity simulations, regression surrogates, and classification models for process feasibility. The application considered biogas recovery, recoverable nutrient quality, and aeration demand, together with uncertainty in wastewater composition. It demonstrates that surrogates can support integrated decisions about several resource pathways while accounting for simulation feasibility.

This work addresses a network-design problem with alternative process configurations. The connected operating task here retains a specified activated sludge topology and predicts the mixer, reactor stages, clarifier outlets, and stored solids jointly. The additional requirement is consistency of those predicted component quantities at fixed physical connections. A classification of simulator feasibility and an explicitly imposed component balance answer different questions. The contribution of Durkin et al. (2024) motivates integrated decision analysis, while the reactor and clarifier literature defines the balances needed for the present extension.

The single-reactor result of Selerio (2026) supplies physical enforcement at one unit. The connected mechanistic benchmark of Alex et al. (2008) supplies a precedent for comparing whole-plant responses and controls. Combining these lines of work leads to a specific remaining question. Can a statistical plant response preserve declared unit connections and solids inventories while supporting useful operating choices? This

requires assessment beyond single-reactor feasibility and beyond the accuracy of one operating objective.

2.5.3 Approximation error at selected operating conditions

Surrogate optimization research addresses the reliability of the response used during search. Bhosekar and Ierapetritou (2018) reviewed surrogate modeling for feasibility analysis and optimization. McBride and Sundmacher (2019) reviewed surrogate construction from experiments and simulations for process-system decisions. Both place model development within a larger optimization problem. Their relevance is that a surrogate must approximate the quantities and operating region used by the decision, not only reproduce an average response over its training design.

Alexandrov et al. (1998) developed a trust-region framework that manages approximation models during optimization. The method uses reference information to assess a local approximation and control the region in which it is used. This is a methodological response to inaccurate surrogate behavior during search. For treatment analysis, it provides a reason to check the approximation near the controls actually selected rather than rely entirely on a prediction score averaged over other cases.

Pedrozo et al. (2025) compared five surrogate approaches for an optimization problem involving pooling of carbon-dioxide streams. They assessed predictive accuracy, computational effort, and optimization behavior separately. Their comparison of one-shot optimization with a trust-region filter strategy showed that the models differed in convergence and reliability during search. This is process-systems evidence that a favorable fitting result and a favorable optimization result are distinct outcomes. Its evaluated system is carbon capture, so the corresponding activated sludge decision still requires a treatment-specific comparison.

Mass conservation and non-negativity exclude certain impossible component states, but several feasible states can have different nitrogen distributions or biomass concentrations. An optimizer can therefore select a physically admissible prediction that remains inaccurate. The physical-enforcement studies establish the relevant constraints, while approximation-management studies establish the need for checking the response used during search. Together, they motivate independent mechanistic evaluation of the selected aeration, recycle, and wasting settings.

The remaining decision gap concerns the combined evidence for a connected activated sludge surrogate. Its unit balances and stored solids need to be consistent, its selected settings need to meet the original treatment requirements, and its objective needs to be evaluated on the same mechanistic basis as the direct comparator. Search time and verification effort must also be reported on a defined basis. This makes decision quality and computational benefit testable without assuming that constraint satisfaction alone establishes either outcome.

2.6 Synthesis of established contributions and remaining gaps

2.6.1 What the reviewed studies have addressed

The literature establishes a clear progression. Mechanistic activated sludge models define components, reaction rates, separation, and recirculation. Calibration studies connect those quantities to plant observations. Soft sensors and indicator models improve estimation of selected treatment responses. Simulation-based surrogates extend learned prediction to operating calculations. Physical-learning studies then address equation guidance and explicit constraint enforcement. Interpretable regression and surrogate optimization connect these predictions to inspection and decision use.

Several requirements have already been addressed directly. Kircher and Votsmeier (2025) established joint atom balance and positivity for specified reactive-chemistry surrogates. Selerio (2026) established final mass conservation and non-negativity for a structured activated sludge reactor surrogate and identified the distinction between raw and projected interpretation. Durkin et al. (2024) demonstrated integrated wastewater resource-recovery optimization using surrogates. These achievements define the starting point for the dissertation rather than requirements that the literature has entirely overlooked.

The remaining gaps arise when these contributions are brought into the same complete-component activated sludge assessment. Table 2.2 identifies the studies that address each need and the more specific question that remains. The questions concern the reviewed evidence and the declared reactor and plant setting. They do not claim that the underlying methods have never been developed.

2.6.2 The four linked questions of the dissertation

The first gap concerns comparative prediction of the complete reactor response. The indicator studies establish useful outputs, and the structured-regression study extends prediction to all modeled components. The remaining assessment requires a shared component target across learning families and a clear account of how errors change with training size and inputs beyond the sampled range. This extends established prediction work to a common comparison rather than treating an indicator score as evidence about every component.

The second gap concerns the consequences of physical enforcement. Joint mass conservation and concentration bounds have already been established by specific methods. What remains is a paired assessment of their effect on the same raw predictions across several activated sludge learners. Component error, residuals, negative concentrations, adjustment size, and cost answer different questions. Their joint evaluation identifies the value and limits of enforcement without assuming an accuracy improvement.

Table 2.2: Established studies and the remaining questions addressed by the dissertation.

Research need	Studies addressing the need	Remaining question
Complete treatment prediction	Y. Wang et al. (2025) assesses TN prediction. P. Wang et al. (2024) estimates influent components. Selerio (2026) predicts complete reactor components.	How do several learners compare on one complete-state target under training scarcity and extrapolation?
Physical validity of predictions	Beucler et al. (2021) and Sturm and Wexler (2022) impose conservation. Kircher and Votsmeier (2025) and Selerio (2026) address conservation and concentration bounds together.	How does common enforcement affect component accuracy, adjustment size, and cost across learners?
Interpretation of treatment responses	Y. Wang et al. (2025) inspects input contributions. Selerio (2026) exposes regression terms and distinguishes raw from final interpretation.	How do coupling, reporting, and active constraints affect sensitivities and stability of the explanation?
Connected operating decisions	Fang et al. (2011) and Durkin et al. (2024) use surrogates in treatment decisions. Alex et al. (2008) defines a connected plant benchmark.	Can joint component enforcement support settings that remain credible under independent plant evaluation?

The third gap concerns interpretation of the returned treatment state. The structured reactor regression already supplies visible terms and identifies the limits of raw coefficient inspection. The further question is how operating and influent sensitivities pass through coupling, composite reporting, and projection, including changes when concentration bounds become active. Examining stability across training partitions adds evidence about the reproducibility of those fitted relationships.

The fourth gap concerns connected prediction and operating assessment. Existing mechanistic plant models establish the shared streams and solids storage, while surrogate applications establish learned operating and resource-recovery decisions. The remaining requirement here is a joint constrained component response for the specified plant and an independent calculation of treatment outcomes at its selected controls. Comparison with direct mechanistic optimization then assesses the decision and computational cost under common priorities.

Figure 2.2 shows why these questions are linked. Physical enforcement acts on the concentrations being scored and interpreted. Those same concentrations enter the operating objective and constraints. A change in projection can therefore affect component errors, local sensitivities, and selected settings. The integrated assessment must follow that shared response through each stage.

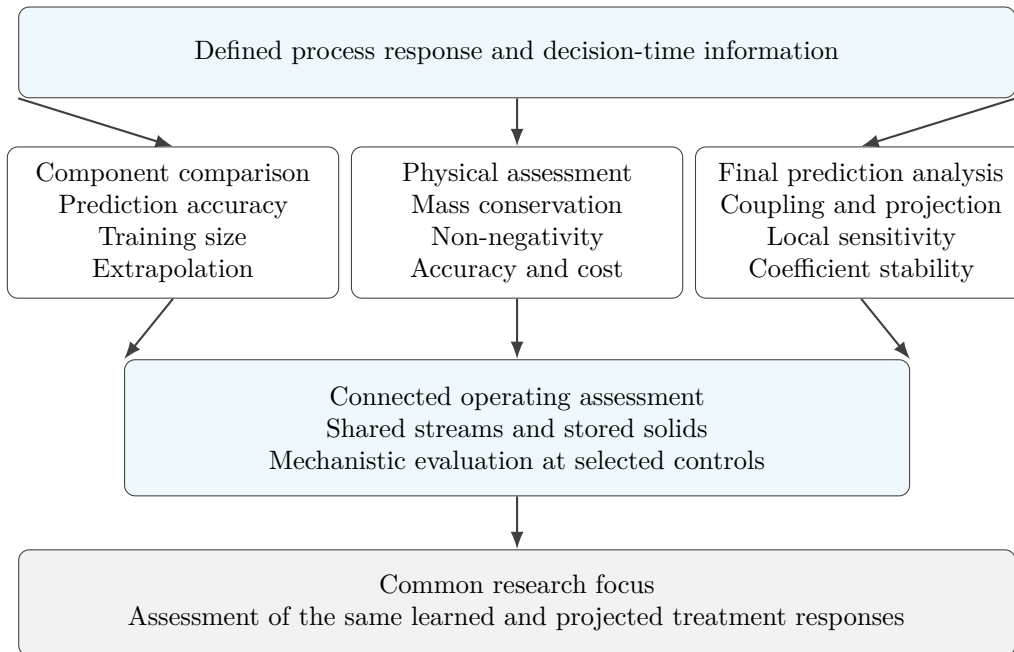


Figure 2.2: Links among the remaining comparative, physical, interpretive, and operating questions. Each question concerns the same defined activated sludge response. Established methods address parts of the problem, while their combined use requires common component and decision assessment.

The literature therefore supports the dissertation’s progression from a defined reactor response to connected operating decisions. Chapters 3 and 4 develop the common

component foundation and paired prediction assessment. Chapter 5 develops the structured regression and follows interpretation through physical enforcement. Chapter 6 extends the response to connected units and independently assesses selected controls. Chapters 4–6 each present the rationale, theory, methodology, results, and discussion of their assessment. Chapter 7 synthesizes the findings and develops directions for future research.

Chapter 3

Mechanistic Foundations of Physically Admissible Prediction

3.1 The component state as the object of prediction

An activated sludge reactor changes the form and distribution of material. Organic substrate can become biomass, ammonium can become nitrite and nitrate, and soluble phosphate can enter biological storage or a precipitate. A description of treatment therefore needs more than one aggregate effluent concentration. The ASM framework separates material into components and connects their changes through biological and chemical processes (Henze et al., 2000). This separation provides the physical meaning of a surrogate prediction. The surrogate estimates a complete component state from the influent state and the operating conditions.

The mechanistic model is Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN). It combines the phosphorus-removal framework of Henze et al. (2000) with the nitrogen-transformation formulation used by Guerrero et al. (2011). The state contains 20 components and the reaction description contains 28 processes. Ten components are dissolved and ten are particulate. Stoichiometry records the relative component amounts consumed and produced by these processes. Kinetics specifies their rates as functions of the reactor state and fixed parameters. The surrogate approximates the resulting steady-state concentrations while projection imposes selected accounting constraints.

3.1.1 Reading component vectors and matrix operations

A concentration is first a scalar, which is one numerical value with a specified unit. For example, 5 g N m^{-3} means 5 grams of nitrogen per cubic metre. Its numerical value is also 5 mg N L^{-1} because both the mass and volume units change by a factor of one thousand. Oxygen, COD, nitrogen, phosphorus, solids, and alkalinity have different reporting bases. A unit written as g COD m^{-3} refers to oxygen-demand

equivalents rather than grams of one named organic molecule. These bases determine which coefficients can be used when quantities are combined.

A vector collects related scalar values in a fixed order. A three-component column vector and its transpose can be written as

$$c = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}, \quad c^\top = \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix}.$$

The superscript $\top$ means transpose. It changes a column into a row, or interchanges the rows and columns of a matrix. It does not change the component values. The notation $c \in \mathbb{R}^3$ says that the vector has three real-valued coordinates. The notation $c \geq 0$ means that c_1 , c_2 , and c_3 are each non-negative. It does not mean only that their sum is non-negative.

A strictly positive coordinate satisfies $c_j > 0$, whereas non-negativity permits $c_j = 0$. The distinction matters when a logarithm or a reciprocal is evaluated. Such an operation can require a strictly positive argument even though a physical concentration may legitimately be zero.

Subscripts identify the coordinate being discussed. The symbol c_j means component j . When several observations are involved, c_{ij} means component j in observation i . The labels *in* and *out* identify the side of the reactor boundary. Thus $c_{out,ij}$ is the effluent concentration of component j in observation i . Parenthetical superscripts used for a surrogate or prediction type are identifiers rather than powers.

A matrix arranges coefficients in rows and columns. If a matrix has two rows and three columns, its dimension is 2×3 . A matrix product is defined when the number of columns of the first factor equals the number of rows of the second factor. Each output entry is obtained by multiplying corresponding entries in a row and a column and adding the products. For hypothetical numerical values,

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 1(3) + 2(4) + 0(5) \\ 0(3) + 1(4) + 1(5) \end{bmatrix} = \begin{bmatrix} 11 \\ 9 \end{bmatrix}.$$

The product has two coordinates because the matrix has two rows. In a physical calculation, each coefficient also converts the unit of its associated input into the required output unit. The numerical example explains the operation without assigning these coefficients to an activated sludge process.

A row-column product gives a weighted sum. For a coefficient column a , the scalar product $a^\top c$ means $a_1 c_1 + a_2 c_2 + a_3 c_3$. The summation notation $\sum_{j=1}^3 a_j c_j$ writes the same operation by identifying its index, starting value, and ending value. An identity matrix, denoted I , leaves a vector unchanged. A zero matrix has every entry equal to zero. These operations allow many component equations to be recorded in one expression while preserving the scalar accounting behind them.

The magnitude of a vector is described by a norm. The Euclidean norm is the square root of the sum of squared coordinates. A hypothetical vector $(3, -4)^\top$ has Euclidean norm $\sqrt{3^2 + (-4)^2} = 5$. The sum of absolute coordinates gives its absolute-sum norm, $|3| + |-4| = 7$. Squaring a norm and squaring each coordinate are therefore different steps that must be applied in the stated order. When coordinates have different physical bases, a numerical norm is meaningful only with the adopted unit convention or an explicit standardization.

The 20-component state uses the same column-vector convention. Its fixed ordering is

$$c = (S_O, S_F, S_A, S_{NH4}, S_{NO2}, S_{NO3}, S_{N2}, S_{PO4}, S_I, S_{ALK}, X_I, X_S, X_H, X_{PAO}, X_{PP}, X_{PHA}, X_{AOB}, X_{NOB}, X_{MeP}, X_{MeOH})^\top. \quad (3.1)$$

Every concentration is represented by its numerical value in the native unit assigned to that component. A vector of these values combines several constituent bases. An organic component expressed as g COD m⁻³ is not expressed on the same material basis as ammonium expressed as g N m⁻³. The coefficients of every physical operator must account for this distinction.

Table 3.1: Component meanings, native units, and fixed coefficients for the four reported water quality quantities. Each coefficient converts the native component basis to the corresponding reported basis.

Component	Meaning	Native unit	COD	TN	TP	TSS
S_O	Dissolved oxygen	g O ₂ m ⁻³	0	0	0	0
S_F	Fermentable substrate	g COD m ⁻³	1	0	0	0
S_A	Acetate and fermentation products	g COD m ⁻³	1	0	0	0
S_{NH4}	Ammonium nitrogen	g N m ⁻³	0	1	0	0
S_{NO2}	Nitrite nitrogen	g N m ⁻³	0	1	0	0
S_{NO3}	Nitrate nitrogen	g N m ⁻³	0	1	0	0
S_{N2}	Dissolved dinitrogen	g N m ⁻³	0	0	0	0
S_{PO4}	Soluble phosphate	g P m ⁻³	0	0	1	0
S_I	Soluble inert organic matter	g COD m ⁻³	1	0.01	0	0
S_{ALK}	Alkalinity	mol m ⁻³	0	0	0	0
X_I	Particulate inert organic matter	g COD m ⁻³	1	0.02	0.01	0.75
X_S	Slowly biodegradable substrate	g COD m ⁻³	1	0	0	0.75
X_H	Heterotrophic biomass	g COD m ⁻³	1	0.07	0.02	0.90
X_{PAO}	Phosphate-accumulating biomass	g COD m ⁻³	1	0.07	0.02	0.90
X_{PP}	Stored polyphosphate	g P m ⁻³	0	0	1	3.23
X_{PHA}	Stored polyhydroxyalkanoates	g COD m ⁻³	1	0	0	0.60
X_{AOB}	Ammonia-oxidizing biomass	g COD m ⁻³	1	0.07	0.02	0.90
X_{NOB}	Nitrite-oxidizing biomass	g COD m ⁻³	1	0.07	0.02	0.90
X_{MeP}	Precipitated metal phosphate	g TSS m ⁻³	0	0	1/4.87	1
X_{MeOH}	Metal hydroxide	g TSS m ⁻³	0	0	0	1

The symbols in Table 3.1 identify state variables. They do not introduce additional process abbreviations. For example, X_{AOB} is the concentration coordinate assigned to ammonia-oxidizing biomass. The numerical constituent contents and solids conversion

factors are fixed assumptions of the adopted reporting map. They are not estimated from the surrogate fit. Their meaning follows the component accounting used in activated sludge modeling (Henze et al., 2000).

The full state preserves distinctions that aggregate quantities cannot recover. Ammonium, nitrite, and nitrate contribute to the same reported nitrogen total, but their conversion involves different microbial populations and oxygen requirements. Dissolved dinitrogen remains a component of the modeled nitrogen inventory even though it has zero weight in reported TN. Similarly, soluble phosphate, stored polyphosphate, and metal phosphate have different process roles even when all contribute to TP. These differences matter when a predicted state is used to interpret an operating decision.

3.2 Biological meaning of the component transformations

3.2.1 Electron donors, electron acceptors, and biomass

Microorganisms use part of a substrate to obtain energy and another part to form cellular material. An electron donor supplies electrons in an oxidation reaction. An electron acceptor receives them in a reduction reaction. Under aerobic conditions, organic material can serve as the donor and oxygen as the acceptor. Under anoxic conditions, some organisms use nitrate or nitrite instead of oxygen. Naming the donor and acceptor helps explain why the same organic substrate can support different treatment pathways (Narayanan & Narayan, 2019; Preena et al., 2021).

The carbon source and the energy source describe different properties. Heterotrophic biomass uses organic carbon to form cells. Chemolithoautotrophic nitrifying biomass obtains energy from inorganic nitrogen oxidation and uses inorganic carbon for cell synthesis. Its growth is therefore not simply another form of organic-substrate removal. The separate biomass coordinates in the adopted model retain this distinction (Guerrero et al., 2011; Henze et al., 2000). Complete ammonia oxidation can also occur within one microbial organism (van Kessel et al., 2015). That biological observation does not replace the separate ammonium- and nitrite-oxidation processes in the adopted model.

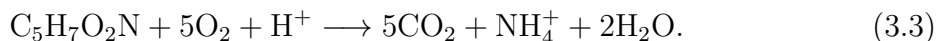
An idealized organic unit, CH_2O , illustrates oxidation without assigning one chemical formula to all wastewater organic matter. Its complete aerobic oxidation is



Both sides contain one carbon atom, two hydrogen atoms, and three oxygen atoms. Using approximate molar masses of 30 g mol^{-1} for the organic unit and 32 g mol^{-1} for oxygen gives $32/30 = 1.067 \text{ g oxygen per g of this idealized material}$. A hypothetical 30 g of this material would require 32 g oxygen for complete oxidation. The example

explains the oxygen-demand basis of organic components. It does not prescribe the oxygen demand of an arbitrary wastewater mixture.

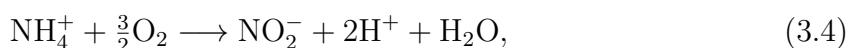
Cell synthesis changes that accounting because part of the carbon and nitrogen enters biomass. The empirical formula $C_5H_7O_2N$ is a useful simplified model of biomass composition, as discussed in the assimilation study of Hoover and Porges (1952). It describes an average composition rather than a single molecular species. For example, an idealized complete oxidation of this biomass with nitrogen released as ammonium can be written as



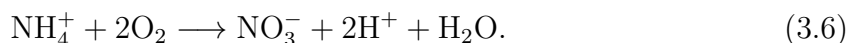
The carbon, hydrogen, oxygen, and nitrogen counts are respectively 5, 8, 12, and 1 on each side. The net charge is +1 on each side. Five moles of oxygen correspond to 160 g, while the empirical biomass unit has an approximate mass of 113 g. The ratio is therefore about 1.416 g oxygen per g of this idealized biomass. These calculations distinguish dry biomass mass from biomass COD. The adopted model uses its specified yields, constituent contents, and lysis processes rather than substituting these teaching reactions for its stoichiometric entries.

3.2.2 Nitrogen transformations and their oxygen requirements

Separating nitrification into two steps makes the oxygen demand and the nitrite intermediate visible. Ignoring cell synthesis for this teaching calculation, the ammonium-oxidation and nitrite-oxidation reactions are



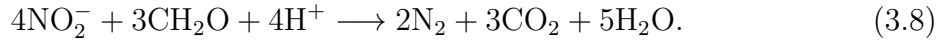
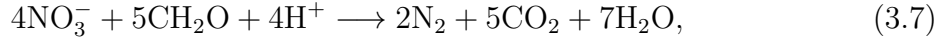
The first reaction preserves one nitrogen atom, four hydrogen atoms, and three oxygen atoms. Its right-hand charge is $-1 + 2 = +1$, which equals its left-hand charge. The second reaction preserves one nitrogen atom, three oxygen atoms, and a net charge of -1 . Adding the reactions and cancelling the intermediate nitrite gives



Each mole of nitrogen has an approximate mass of 14 g. The first step requires $1.5(32)/14 = 3.43$ g oxygen per g nitrogen. The second requires $0.5(32)/14 = 1.14$ g oxygen per g nitrogen. Their sum is 4.57 g oxygen per g nitrogen. A hypothetical conversion of 10 g ammonium nitrogen to nitrate would therefore require about 45.7 g oxygen under these idealized assumptions. Actual process demand also depends on biomass synthesis and the adopted yields. The hydrogen ions in the first step

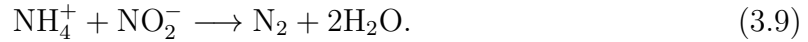
explain why nitrification also draws on the buffering capacity represented by alkalinity (Guerrero et al., 2011; Henze et al., 2000).

Denitrification uses oxidized nitrogen as an electron acceptor and can convert it to dinitrogen. With CH_2O as the illustrative donor and with cell synthesis omitted, two balanced net reactions are



For the nitrate reaction, both sides contain four nitrogen atoms, five carbon atoms, fourteen hydrogen atoms, and seventeen oxygen atoms. Its left-hand charge is $-4 + 4 = 0$, matching the uncharged products. The nitrite reaction can be checked in the same way. The donor requirements are $5/4$ and $3/4$ moles of the illustrative organic unit per mole of nitrogen. This difference explains why the pathway through nitrite can require less organic donor than the pathway through nitrate under the same simplified assumptions. Carbon supply and donor identity also affect denitrification kinetics, as reviewed by Lu et al. (2014). A stoichiometric requirement alone does not establish a reaction rate.

Anaerobic ammonium oxidation provides a different nitrogen pathway. An idealized net reaction that omits biomass synthesis and minor products is



It preserves two nitrogen atoms, four hydrogen atoms, two oxygen atoms, and zero net charge. It couples ammonium and nitrite without an organic electron donor in this idealized balance. The cultivation work of Strous et al. (1998) and the pathway studies of Dietl et al. (2015) and Maalcke et al. (2016) explain why it differs biologically from ordinary heterotrophic denitrification. Its application requires its own organisms and operating conditions (Vandekerckhove, 2018; Winkler & Straka, 2019; Xie, 2021). It is included here to clarify the boundary of the nitrogen discussion. It is not an additional process in the adopted 20-component, 28-process reactor model.

3.2.3 From elemental accounting to the adopted state basis

The teaching reactions permit direct checks of atom counts and charge because every displayed species has a chemical formula. An activated sludge state instead combines oxygen-demand equivalents, nitrogen mass, phosphorus mass, solids mass, and alkalinity. A biomass coordinate in g COD m^{-3} cannot be added directly to a nitrogen coordinate in g N m^{-3} to obtain a material total. Its nitrogen-content coefficient must first convert its contribution to the nitrogen basis.

Takács and Vanrolleghem (2006) explain the distinction between balances on lumped wastewater components and complete elemental accounting. A full elemental

balance may require water, carbon dioxide, gas exchange, and other species outside a particular activated sludge state. The mass conservation conditions in this dissertation are established from the adopted stoichiometric matrix and the declared flow and transfer boundary. They certify the quantities represented by that model. They do not imply that every unrepresented chemical species or external flux has been measured. The derivation below identifies the precise quantities that can be imposed without adding an unsupported physical condition.

3.3 Process transformations and their rates

Hydrolysis transfers slowly biodegradable material toward forms that microorganisms can use more readily. Heterotrophic metabolism links substrate consumption to biomass growth and electron-acceptor use. Under aerobic conditions oxygen acts as an electron acceptor. Under anoxic conditions nitrate or nitrite can support the corresponding modeled transformations. Fermentation supplies acetate from suitable organic material. Biomass lysis redistributes material from active populations to substrate and inert pools. Biological phosphorus removal adds storage, growth, phosphate uptake, and release mechanisms. Chemical precipitation and redissolution redistribute phosphorus between soluble and particulate forms (Guerrero et al., 2011; Henze et al., 2000).

Two-step nitrification represents ammonium oxidation to nitrite separately from nitrite oxidation to nitrate. It therefore retains both ammonia-oxidizing and nitrite-oxidizing biomass. A one-step description can reproduce an overall ammonium-to-nitrate conversion while hiding a nitrite intermediate. Retaining the intermediate allows the state to distinguish a low-ammonium effluent that contains nitrite from one in which nitrite has been converted further. The distinction concerns the reaction model rather than an assurance that every nitrogen pathway occurring in a real plant is included. The broader diversity of microbial nitrogen-removal pathways reviewed by Duan et al. (2022) sets that limitation.

The stoichiometric matrix, also called the Petersen matrix, records which components are consumed and produced by each process. A small hypothetical reaction network illustrates the entries before the full model is written. Let three material pools be denoted by A , B , and C . One unit of B contains twice the tracked inventory of one unit of A or C . Consider the transformations



Figure 3.1 connects these two transformations to their component changes. The coefficients below each arrow count changes in the three pools, while the inventory weights explain why both processes preserve the same material quantity.

Two units of A contain the same inventory as one unit of B . That unit of B contains

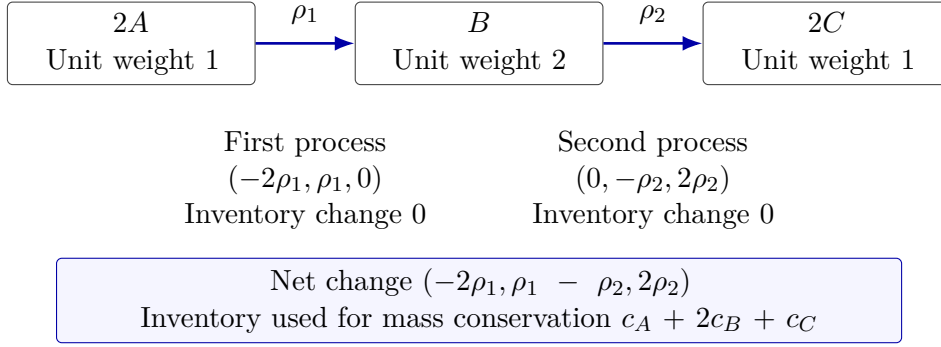


Figure 3.1: Hypothetical reaction accounting for three pools with inventory weights 1, 2, and 1. Each process redistributes the tracked inventory without creating or removing it.

the same inventory as two units of C . The middle pool receives material from the first process and supplies it to the second. Its net concentration change therefore depends on the difference between the two rates, even though each process separately preserves the weighted inventory.

The first process consumes two units of A and produces one unit of B . It has no direct effect on C . The second consumes one unit of B and produces two units of C . If the two process rates are ρ_1 and ρ_2 , their scalar contributions are

$$\begin{aligned}
 \dot{c}_A \Big|_{\text{reaction}} &= -2\rho_1, \\
 \dot{c}_B \Big|_{\text{reaction}} &= \rho_1 - \rho_2, \\
 \dot{c}_C \Big|_{\text{reaction}} &= 2\rho_2.
 \end{aligned}$$

The dot indicates a time derivative. These equations isolate reaction contributions and contain no hydraulic transport. They can be organized into a process-by-component matrix and a rate column,

$$\nu_{\text{toy}} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \end{bmatrix}, \quad \rho_{\text{toy}} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}.$$

Transposing the matrix puts each component's coefficients in one row. Multiplication then recovers all three scalar rates,

$$\nu_{\text{toy}}^\top \rho_{\text{toy}} = \begin{bmatrix} -2 & 0 \\ 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix} = \begin{bmatrix} -2\rho_1 \\ \rho_1 - \rho_2 \\ 2\rho_2 \end{bmatrix}.$$

For hypothetical rates of 3 and 1 in compatible amount-per-time units, the contributions are -6 , 2 , and 2 . The middle pool increases because its production rate exceeds its consumption rate. A positive value in one component does not imply that either process creates the complete tracked inventory.

The full model uses the same convention. Its orientation is fixed as

$$\nu \in \mathbb{R}^{P \times F}, \quad P = 28, \quad F = 20. \quad (3.10)$$

Row p describes process p and column j describes component j . A negative entry ν_{pj} represents consumption and a positive entry represents production. A zero entry means that the process has no direct stoichiometric contribution to that component. Coefficients contain yields and constituent conversions appropriate to the native units.

For component j of the full model, the individual process contributions are $\nu_{1j}\rho_1$, $\nu_{2j}\rho_2$, and so on through $\nu_{28j}\rho_{28}$. Adding them gives the net reaction rate of that component. The expression $\rho(c, u)$ indicates that process rates depend on the current component state and operating inputs. For a rate vector $\rho(c, u) \in \mathbb{R}^{28}$, the reaction contribution to the component balance is

$$r_{\text{reaction}}(c, u) = \nu^\top \rho(c, u), \quad r_{\text{reaction},j}(c, u) = \sum_{p=1}^{28} \nu_{pj} \rho_p(c, u). \quad (3.11)$$

Stoichiometry specifies the proportions of a transformation. Kinetics specifies its speed at a particular state. These are different responsibilities. Increasing the concentration of biomass can change a process rate while leaving its stoichiometric row unchanged. Likewise, changing aeration can alter the oxygen limitation of a process without changing the material accounting of that process.

A simple saturation expression makes this distinction concrete. Suppose, only for illustration, that the substrate concentration and its saturation constant are both 4 g COD m⁻³. Dividing the concentration by their sum gives 4/(4 + 4) = 0.5. The unit cancels because numerator and denominator use the same constituent basis. If a limiting dissolved substrate has concentration S and saturation constant $K_S > 0$, the factor

$$\frac{S}{K_S + S} \quad (3.12)$$

equals one half at $S = K_S$, approaches zero as S approaches zero, and approaches one as S becomes large. A representative growth expression combines a maximum specific growth rate, biomass abundance, and two limitation factors. For hypothetical values $\mu = 2 \text{ d}^{-1}$, $X = 10 \text{ g biomass COD m}^{-3}$, a substrate factor of 0.5, and an oxygen factor of 0.25, the growth contribution is $2(10)(0.5)(0.25) = 2.5 \text{ g biomass COD m}^{-3} \text{ d}^{-1}$. Multiplication by dimensionless limitation factors reduces the unrestricted contribution $\mu X = 20$ to 2.5. A representative growth expression can combine substrate and oxygen limitation with biomass abundance,

$$\rho_{\text{growth}} = \mu X \frac{S}{K_S + S} \frac{S_O}{K_O + S_O}. \quad (3.13)$$

Here μ is a maximum specific growth rate, X is the associated biomass coordinate, and K_O is an oxygen saturation constant. Equation (3.13) illustrates the role of multiplicative limitation in activated sludge kinetics. It does not replace the complete 28-process

rate formulation, whose process-specific yields, inhibition terms, and storage dependencies follow Henze et al. (2000) and Guerrero et al. (2011).

A hypothetical substrate concentration of $K_S/9$ gives a substrate factor of 0.1. Raising the concentration to $9K_S$ gives a factor of 0.9. The growth rate changes by a factor of nine if all other factors stay fixed, although the substrate concentration changes by a factor of 81. This calculation explains why a linear relationship between influent load and effluent state is not generally sufficient. Several limiting factors and competing populations can act together. It also explains why an accurate surrogate needs to represent interactions without confusing them with a new stoichiometric law.

3.4 The steady-state reactor boundary

The single-reactor prediction problem uses a completely mixed reactor at steady state. Perfect mixing makes the reactor concentration equal to the effluent concentration. The standalone boundary contains hydraulic inflow and outflow, the modeled reaction processes, and dissolved-oxygen transfer. It excludes a clarifier, solids return, and other external additions. These exclusions define the balance being enforced. They do not imply that those processes are absent from wastewater treatment plants.

Let c_{in} and c_{out} be the influent and effluent component vectors. Let $D > 0$ be the hydraulic dilution rate. For component j , hydraulic entry contributes $Dc_{in,j}$ and hydraulic exit contributes $-Dc_{out,j}$. Reaction contributes the sum of the 28 process terms. Oxygen transfer contributes only to the oxygen coordinate. The scalar balance is therefore

$$0 = Dc_{in,j} - Dc_{out,j} + \sum_{p=1}^{28} \nu_{pj} \rho_p + \omega(e_{S_O})_j.$$

The selector has first coordinate one because S_O is first in the chosen component order. Its other nineteen coordinates are zero. Thus the oxygen equation contains $+\omega$, while the ammonium equation contains no direct oxygen-transfer term. Writing the oxygen equation separately makes this distinction explicit,

$$0 = D(S_{O,in} - S_{O,out}) + \sum_{p=1}^{28} \nu_{p,S_O} \rho_p + \omega.$$

All terms are numerical rates on the oxygen basis for this row. For another component, the units change to that component's native basis. With $e_{S_O} \in \mathbb{R}^{20}$ selecting dissolved oxygen, the same twenty scalar equations are collected into the steady balance

$$0 = D(c_{in} - c_{out}) + \nu^\top \rho(c_{out}, u) + \omega e_{S_O}, \quad (3.14)$$

Moving the hydraulic difference to the opposite side changes its sign. For one component, this operation gives $D(c_{out,j} - c_{in,j}) = \sum_p \nu_{pj} \rho_p + \omega(e_{S_O})_j$. The vector equation is equivalently

$$D(c_{out} - c_{in}) = \nu^\top \rho(c_{out}, u) + \omega e_{S_O}. \quad (3.15)$$

The hydraulic term has the same native component-unit-per-day basis as the corresponding reaction and transfer terms. Treating the balance as a numerical vector equation requires a fixed unit convention throughout.

The operating inputs are HRT and a dimensionless aeration coordinate u_{air} . HRT specifies the time needed to replace one reactor volume on the stated flow basis. A 12-hour residence time corresponds to two reactor-volume replacements per day because $24/12 = 2$. The dilution rate is consequently an inverse time. Oxygen transfer uses a different inverse-time coefficient. Multiplying that coefficient by the oxygen concentration deficit produces a concentration-per-time rate. The adopted parameterization is

$$D = \frac{24 \text{ h d}^{-1}}{\text{HRT}}, \quad (3.16)$$

$$k_{La} = (2 + 18u_{\text{air}}) \text{ d}^{-1}, \quad (3.17)$$

$$\omega = k_{La}(S_{O,\text{sat}} - S_{O,\text{out}}), \quad S_{O,\text{sat}} = 8.5 \text{ g O}_2 \text{ m}^{-3}. \quad (3.18)$$

The dilution rate connects the time basis of the hydraulic specification to that of the process rates. The oxygen-transfer term depends on both the aeration setting and the difference between saturation and the reactor oxygen concentration.

For a hypothetical HRT of 12 h, Equation (3.16) gives $D = 2 \text{ d}^{-1}$. A component difference of 5 g m^{-3} then corresponds to a net transformation or transfer of $10 \text{ g m}^{-3} \text{ d}^{-1}$ in that coordinate. For a hypothetical aeration setting of 1, the transfer coefficient is 20 d^{-1} . If dissolved oxygen is $2 \text{ g O}_2 \text{ m}^{-3}$, the oxygen-transfer rate is $130 \text{ g O}_2 \text{ m}^{-3} \text{ d}^{-1}$. These calculations illustrate the accounting. They are not simulated treatment outcomes.

Steady state means that each modeled inventory has zero time derivative. It does not establish a unique steady state, dynamic stability, or a field-calibrated description. Sensitivity of operating choices to the process model remains relevant even when a balance closes (de Araujo et al., 2013). In particular, a small numerical residual certifies agreement with the stated equations at the returned state. It cannot certify the completeness of the equations for a real wastewater system.

3.5 From stoichiometry to mass conservation invariants

An invariant describes reaction accounting rather than a concentration that must stay fixed at every treatment location. Ammonium can decrease while nitrate increases during nitrification. A conserved combination accounts for the corresponding material across the included nitrogen pools. Influent, effluent, recycle, and external sources must be included when that combination is used in a reactor or plant balance.

A stoichiometric invariant is a linear combination of component concentrations that the modeled reactions cannot change. In the hypothetical three-pool network, adding the concentration values without weights would count one unit of B as if it contained the same inventory as one unit of A . Its material basis instead requires

$$I_{\text{toy}} = c_A + 2c_B + c_C.$$

Substitution of the three scalar reaction rates shows directly that its derivative is zero,

$$\frac{dI_{\text{toy}}}{dt} = (-2\rho_1) + 2(\rho_1 - \rho_2) + (2\rho_2) = 0.$$

The ρ_1 and ρ_2 terms each cancel separately. The identity therefore holds at every combination of process rates, including the hypothetical pair that gives component changes $(-6, 2, 2)^\top$. Its weighted change is $-6 + 2(2) + 2 = 0$.

The weights can be found instead of guessed. Let an unknown coefficient column be $a = (a_A, a_B, a_C)^\top$. Requiring zero weighted contribution from each process gives

$$\begin{aligned} -2a_A + a_B &= 0, \\ -a_B + 2a_C &= 0. \end{aligned}$$

The first equation gives $a_B = 2a_A$. The second then gives $a_C = a_A$. Choosing $a_A = 1$ produces $(1, 2, 1)^\top$. Choosing $a_A = 3$ produces $(3, 6, 3)^\top$ and the same mass conservation law multiplied by three. These columns lie in the null space of ν_{toy} , which is the set of columns sent to zero by that matrix.

For the full component basis, $\nu a = 0$ imposes the same zero-contribution condition on every modeled process. If $a \in \mathbb{R}^{20}$ satisfies this condition, then

$$a^\top \nu^\top \rho = (\nu a)^\top \rho = 0 \tag{3.19}$$

for every process-rate vector. This identity does not depend on the accuracy of an estimated rate or on the value of an operating input. It follows from the reaction accounting alone. Mass conservation-aware surrogate constructions use this separation to preserve balances without solving all of the original rate equations during inference (Hansen et al., 2024; Sturm & Wexler, 2020, 2022).

The number of independent mass conservation laws follows from the number of independent reaction directions. In the toy matrix, neither reaction row is a multiple of the other. More explicitly, suppose $s(-2, 1, 0) + t(0, -1, 2) = (0, 0, 0)$. The first coordinate gives $s = 0$ and the third gives $t = 0$. There are therefore two independent rows. Their rank is two. Three component coordinates subject to two independent zero-contribution equations leave one free invariant coefficient. This relation is called rank-nullity,

$$\text{number of component coordinates} = \text{rank} + \text{null-space dimension}.$$

The one free coefficient in the toy calculation is the choice of a_A . A basis is a collection of independent vectors from which all vectors in the space can be formed by weighted addition. One basis column suffices for that one-dimensional invariant space.

Singular-value decomposition determines such directions without solving every null-space equation manually. It factors a matrix into orthonormal directions and non-negative singular values. Orthonormal columns have unit Euclidean norm and zero scalar product with each other. A zero singular value marks a direction that the matrix sends to zero. For the toy matrix, multiplying it by its transpose gives

$$\nu_{\text{toy}}\nu_{\text{toy}}^\top = \begin{bmatrix} 5 & -1 \\ -1 & 5 \end{bmatrix}.$$

This square matrix has eigenvalues 4 and 6, so the two nonzero singular values of ν_{toy} are 2 and $\sqrt{6}$. The remaining component direction is the invariant direction. The term eigenvalue means the factor by which a square matrix stretches a corresponding direction. Singular values describe the corresponding non-negative stretching magnitudes for a matrix that can be rectangular.

For the full decomposition, U has dimension 28×28 , Σ has dimension 28×20 , and V has dimension 20×20 . The rectangular Σ contains singular values on its diagonal and zeros elsewhere. Multiplication in the order $U\Sigma V^\top$ recovers a 28×20 matrix. An independent invariant basis is obtained by singular-value decomposition of the 28×20 matrix ν . The factorization is

$$\nu = U\Sigma V^\top, \quad (3.20)$$

In finite numerical arithmetic, an exactly zero singular value may appear as a very small number. The rank calculation therefore uses a threshold rather than testing exact numerical equality with zero. The largest singular value sets the matrix scale. Machine precision sets the scale of rounding error, and the larger matrix dimension accounts for the calculation size. The numerical rank counts singular values exceeding

$$\tau_{\text{rank}} = \max(P, F)\epsilon_{\text{mach}}\|\nu\|_2. \quad (3.21)$$

Here ϵ_{mach} is the machine precision used for the calculation and $\|\nu\|_2$ is the largest singular value. The adopted stoichiometric model has rank 15. The rank-nullity identity therefore gives five independent invariants. This is a structural property of the specified reaction matrix.

The toy invariant column $(1, 2, 1)^\top$ has length $\sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}$. Dividing each entry by that length produces the normalized column

$$V_{0,\text{toy}} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

Its two process dot products are $(-2 + 2 + 0)/\sqrt{6} = 0$ and $(0 - 2 + 2)/\sqrt{6} = 0$. Its transpose gives an invariant row with length one. For the full matrix, five orthonormal

right singular vectors play this role. Let $V_0 \in \mathbb{R}^{20 \times 5}$ contain the right singular vectors spanning $\text{null}(\nu)$. Define

$$A = V_0^\top \in \mathbb{R}^{5 \times 20}. \quad (3.22)$$

The first product below tests every combination of an invariant row and a reaction row. Its (h, p) entry is the scalar sum $\sum_{j=1}^{20} A_{hj} \nu_{pj}$. The second product tests the dot products of invariant rows. A diagonal entry is their squared length and an off-diagonal entry is the dot product of two different rows. Unit lengths and mutual orthogonality therefore give the identity matrix. The resulting operator obeys

$$A\nu^\top = 0, \quad AA^\top = I_5. \quad (3.23)$$

Each row represents an independent component combination defining mass conservation. The orthonormal rows need not correspond individually to named elemental inventories. Their purpose is to provide a numerically specified basis for the same invariant space.

The matrix orientation matters. Because processes are rows of ν , the component combinations defining mass conservation lie in its right null space. Taking the right null space of $\nu^\top$ instead produces vectors in process coordinates. Those vectors describe dependencies among process contributions and have a different interpretation. Dimension checks prevent the two constructions from being confused.

Mass conservation of reactions does not by itself establish mass conservation across an open reactor boundary. For invariant row h , multiply the scalar component balance by A_{hj} and add it over all twenty components. The reaction sum can be grouped by process,

$$\begin{aligned} D \sum_{j=1}^{20} A_{hj} (c_{out,j} - c_{in,j}) &= \sum_{p=1}^{28} \rho_p \sum_{j=1}^{20} A_{hj} \nu_{pj} + \omega \sum_{j=1}^{20} A_{hj} (e_{SO})_j \\ &= 0 + \omega A_{h1}. \end{aligned}$$

The first sum vanishes by the invariant construction. The selector leaves only the oxygen-column coefficient A_{h1} . Reaction cancellation and forcing cancellation are consequently separate checks. Premultiplying Equation (3.15) by A gives

$$DA(c_{out} - c_{in}) = \omega A e_{SO}. \quad (3.24)$$

The aeration forcing must therefore be compatible with the selected invariants. The distinction can be checked in the toy network. Adding one unit to pool A without another change gives a forcing column $(1, 0, 0)^\top$. Its invariant product is $1/\sqrt{6}$, so it changes the tracked inventory. A forcing column $(-2, 1, 0)^\top$ instead gives $(-2 + 2)/\sqrt{6} = 0$ and lies in the reaction-change space. The example concerns an abstract inventory and does not identify either forcing with oxygen transfer.

The range of a matrix is the set of vectors produced by multiplying it by all possible coordinate columns. A forcing in $\text{range}(\nu^\top)$ can be represented as a combination of

reaction-change columns and is annihilated by every invariant row. For the adopted matrix, the oxygen selector lies in $\text{range}(\nu^\top)$. There is a process-coordinate vector $\lambda_O \in \mathbb{R}^{28}$ such that

$$e_{SO} = \nu^\top \lambda_O, \quad Ae_{SO} = A\nu^\top \lambda_O = 0. \quad (3.25)$$

The coordinates in λ_O establish a subspace relation. They do not state that aeration is itself a biological reaction or that these coordinates are non-negative process rates. The scalar consequence is $A_{h1} = 0$ for every invariant row. Dividing $D \sum_j A_{hj}(c_{out,j} - c_{in,j}) = 0$ by the positive dilution rate gives equality of the influent and effluent invariant sums. Since $D > 0$, the invariant boundary relation reduces to

$$Ac_{out} = Ac_{in}. \quad (3.26)$$

The relation is justified jointly by stoichiometry, the forcing direction, and the reactor boundary. It should not be imposed on a system with an unaccounted external phosphorus dose, sludge withdrawal, or another source that changes the selected inventories.

More generally, an additional component source g contributes $\sum_j A_{hj}g_j$ to row h . Dividing this contribution by D converts its concentration-per-time basis into a concentration-equivalent invariant change. An additional component source g gives

$$A(c_{out} - c_{in}) = D^{-1}Ag \quad (3.27)$$

when aeration remains compatible. A hypothetical phosphate addition with a nonzero phosphorus-inventory contribution changes the right-hand side. Forcing equality with the original influent would then remove a real external contribution. This example shows why a physical constraint must be derived from the actual system boundary rather than chosen solely because it improves a statistical score.

The full-precision operator is used for all calculations. Its construction is checked through $A\nu^\top$, $AA^\top - I_5$, and Ae_{SO} . Rounding coefficients for display cannot replace the original operator in a numerical calculation. Any nonsingular row transformation T defines $A' = TA$ and the same equality set. An arbitrary transformation can change the magnitude of a residual. The orthonormal construction fixes the basis and scaling used for the mass conservation diagnostic.

3.6 Reaction progress and the feasible state geometry

The invariant equation can also be understood through reaction progress. In the hypothetical three-pool network, let the first process advance by one unit and the second by one half-unit on a compatible concentration-equivalent basis. The accumulated

component changes are

$$\Delta c_A = -2(1) + 0(0.5) = -2,$$

$$\Delta c_B = 1(1) - 1(0.5) = 0.5,$$

$$\Delta c_C = 0(1) + 2(0.5) = 1.$$

Their weighted sum is $-2 + 2(0.5) + 1 = 0$. Starting from the hypothetical state $(8, 1, 1)^\top$ gives $(6, 1.5, 2)^\top$, and both states have weighted inventory 11. The amount of progress differs from the instantaneous rate. Multiplication by a time scale converts a rate into a concentration-equivalent change.

In a steady reactor balance, the relevant time scale is D^{-1} . Substituting $\omega e_{S_O} = \nu^\top(\omega \lambda_O)$ combines reaction and compatible forcing in the same component-change space. Each component change is then a sum of the form $\sum_p \nu_{pj} r_p$. Rearranging the steady balance and using Equation (3.25) gives

$$c_{out} - c_{in} = \nu^\top r, \quad r = D^{-1}(\rho + \omega \lambda_O). \quad (3.28)$$

The vector r gives one set of process coordinates for the total component change. It is not a unique reconstruction of the biological rates because $\nu^\top$ has rank 15 and 28 columns. It also includes the algebraic term for oxygen transfer. Individual entries of r therefore should not be interpreted as independently measured reaction extents or microbial activity.

A change satisfying mass conservation need not be described in the original process coordinates. In the toy network, the condition on a change v is $v_A + 2v_B + v_C = 0$. Two independent unit-length columns satisfying it are

$$z_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad z_2 = \frac{1}{\sqrt{30}} \begin{bmatrix} 1 \\ 2 \\ -5 \end{bmatrix}.$$

Their invariant dot products are $(-2 + 2 + 0)/\sqrt{5} = 0$ and $(1 + 4 - 5)/\sqrt{30} = 0$. Their mutual dot product is $(-2 + 2 + 0)/\sqrt{150} = 0$. Their squared lengths are $5/5 = 1$ and $30/30 = 1$. The matrix $Z_{\text{toy}} = [z_1 \ z_2]$ therefore supplies an orthonormal basis of the two-dimensional change space. Every change satisfying mass conservation for the weighted sum can be written as $v = z_1 q_1 + z_2 q_2$.

This change basis differs from the invariant basis. The invariant basis identifies quantities that cannot change. The columns of Z identify directions along which a state can change without changing those quantities. In the full model, five independent invariant rows leave fifteen independent change directions. Let $Z \in \mathbb{R}^{20 \times 15}$ have orthonormal columns spanning $\text{null}(A)$. The fundamental subspace relation gives

$$\text{range}(\nu^\top) = \text{null}(A), \quad AZ = 0, \quad Z^\top Z = I_{15}. \quad (3.29)$$

Every component change consistent with mass conservation can consequently be written as Zq for a unique $q \in \mathbb{R}^{15}$ in the selected orthonormal basis. The coordinates

q describe an admissible direction of material redistribution. They do not identify a unique sequence of the 28 kinetic processes.

For one invariant row, the target for sample i is the influent sum $b_{ih} = \sum_j A_{hj} c_{in,ij}$. A feasible effluent must satisfy $\sum_j A_{hj} c_j = b_{ih}$ for each of the five rows and $c_j \geq 0$ for every component. The vertical bar in set notation means that all conditions to its right must hold. For sample i , define the target $b_i = Ac_{in,i}$ and the feasible set

$$\mathcal{F}_i = \{c \in \mathbb{R}^{20} \mid Ac = b_i, c \geq 0\}. \quad (3.30)$$

The equality describes a 15-dimensional affine space before concentration bounds are applied. An affine space is a linear change space translated by a fixed reference state. Here that reference can be the influent. Non-negativity intersects the affine space with the non-negative orthant, which is the region where every coordinate is non-negative. The intersection is closed because its defining equalities and bounds include their boundary points. It is convex because a weighted average of two feasible states with weights between zero and one retains both their invariant target and non-negative coordinates. A non-negative influent belongs to the set, so the set is nonempty and the influent supplies a feasible anchor.

For the hypothetical three-pool influent $(8, 1, 1)^\top$, the conditions are $c_A + 2c_B + c_C = 11$ and $c_A, c_B, c_C \geq 0$. If only one pool is nonzero, the three boundary states are $(11, 0, 0)^\top$, $(0, 5.5, 0)^\top$, and $(0, 0, 11)^\top$. The feasible region is the triangle between those states in the inventory plane. The state $(6, 1.5, 2)^\top$ lies inside the triangle. A state $(12, -0.5, 0)^\top$ lies in the same inventory plane because $12 - 1 = 11$, but it fails the concentration bound. The equality selects the plane and non-negativity selects the admissible part of it.

Figure 3.2 displays this triangle using c_A and c_B as its axes. The inventory equality supplies the third coordinate through $c_C = 11 - c_A - 2c_B$, so every plotted point represents a complete three-pool state.

The sloping edge has $c_C = 0$. The two coordinate edges have $c_A = 0$ or $c_B = 0$. Inside the triangle, all three components are positive and can redistribute the same inventory in different proportions. The cross-marked point preserves the inventory but has negative pool B . The figure therefore shows how concentration bounds reduce the inventory plane to a particular feasible region without selecting a unique kinetic state.

A two-component hypothetical system with sum fixed by mass conservation $c_1 + c_2 = 10$ illustrates the geometry. The equality permits states such as $(-1, 11)$, $(4, 6)$, and $(12, -2)$. Non-negativity retains only the line segment between $(0, 10)$ and $(10, 0)$. Clipping $(-1, 11)$ to $(0, 11)$ sets its negative coordinate to zero but changes the sum to 11. Enforcing the equality alone and enforcing non-negativity alone therefore solve different problems. Simultaneous enforcement requires the intersection.

The geometry also explains why a feasible prediction need not be kinetically correct. Several non-negative states can share all five invariant values. A state with the wrong

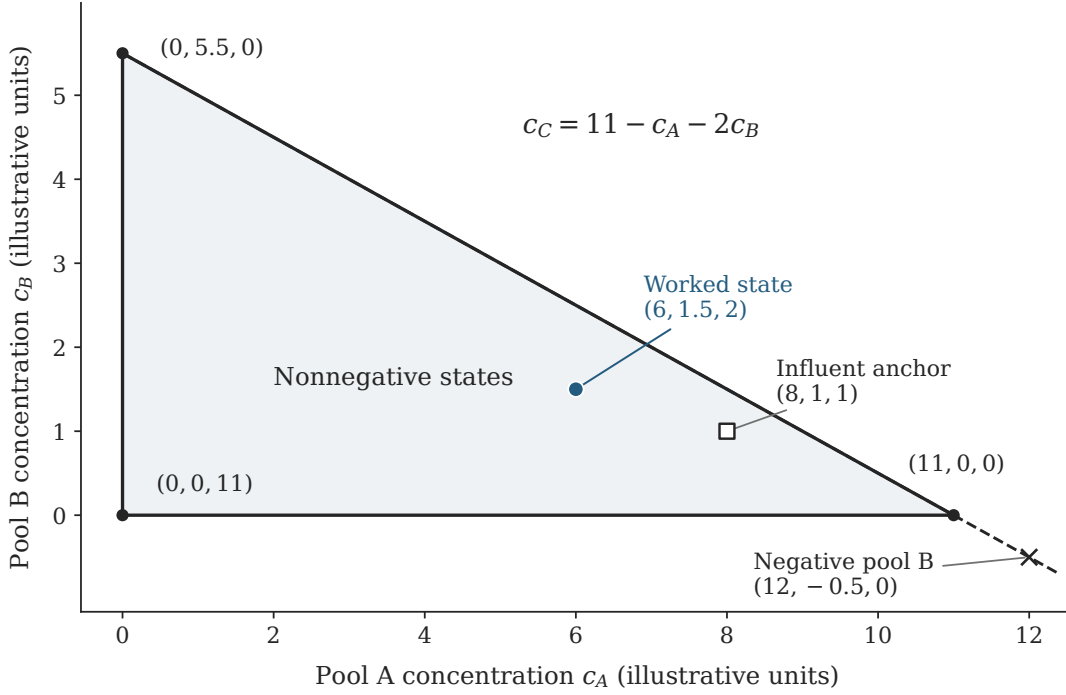


Figure 3.2: Hypothetical non-negative states for the weighted inventory $c_A + 2c_B + c_C = 11$. The shaded triangle includes its boundary. Full state labels retain the third component even though only two coordinates are plotted.

distribution between ammonium and nitrate may satisfy the invariant equality. A state with an implausible biomass concentration may do the same. The feasible set establishes the declared material accounting and sign restrictions. It does not establish that the 28 process rates balance hydraulics and aeration at that state. This distinction is central to the constraint methods of Kircher and Votsmeier (2025) and to the broader physical-consistency perspective of Valente et al. (2025).

3.7 Reported water quality quantities and hidden component errors

Reported water quality quantities are calculated after a component state has been predicted. Each quantity is first a scalar weighted sum of the components. The coeffi-

cients in Table 3.1 give

$$\begin{aligned}
\text{COD} &= S_F + S_A + S_I + X_I + X_S + X_H + X_{PAO} \\
&\quad + X_{PHA} + X_{AOB} + X_{NOB}, \\
\text{TN} &= S_{NH4} + S_{NO2} + S_{NO3} + 0.01S_I + 0.02X_I \\
&\quad + 0.07(X_H + X_{PAO} + X_{AOB} + X_{NOB}), \\
\text{TP} &= S_{PO4} + 0.01X_I + 0.02(X_H + X_{PAO} + X_{AOB} + X_{NOB}) \\
&\quad + X_{PP} + X_{MeP}/4.87, \\
\text{TSS} &= 0.75(X_I + X_S) + 0.90(X_H + X_{PAO} + X_{AOB} + X_{NOB}) \\
&\quad + 3.23X_{PP} + 0.60X_{PHA} + X_{MeP} + X_{MeOH}.
\end{aligned}$$

For example, the factor 0.07 has units of g nitrogen per g biomass COD. Multiplying it by X_H on the biomass COD concentration basis produces a nitrogen concentration. The factor 3.23 has units of g suspended solids per g stored phosphorus. The sum for each reported quantity therefore combines contributions already converted to the same output basis.

A hypothetical reporting state containing only $X_H = 100 \text{ g COD m}^{-3}$ and $X_{PP} = 2 \text{ g P m}^{-3}$ gives

$$\begin{aligned}
\text{COD} &= 100, & \text{TN} &= 0.07(100) = 7, \\
\text{TP} &= 0.02(100) + 2 = 4, & \text{TSS} &= 0.90(100) + 3.23(2) = 96.46.
\end{aligned}$$

Each numerical output uses its own reported constituent unit. This sparse example teaches the composition calculation and is not asserted to be a reactor steady state. The matrix map records these same four sums by placing the COD coefficients in its first row, the TN coefficients in its second row, and the remaining two coefficient groups in their corresponding rows. With the coefficients in Table 3.1,

$$y = I_{comp}c, \quad y = (\text{COD}, \text{TN}, \text{TP}, \text{TSS})^\top, \quad I_{comp} \in \mathbb{R}^{4 \times 20}. \quad (3.31)$$

The four rows convert the component vector to g COD m^{-3} , g N m^{-3} , g P m^{-3} , and g TSS m^{-3} , respectively. All coefficients are non-negative. A non-negative component state therefore gives non-negative reported quantities.

The nitrogen map excludes dissolved dinitrogen. This reporting convention makes TN different from an inventory used for mass conservation that includes nitrogen gas. In a hypothetical redistribution of 3 g N m^{-3} from nitrate to dissolved dinitrogen, the reported TN decreases by 3 g N m^{-3} while the complete nitrogen inventory can remain unchanged. Conversely, a phosphorus inventory that includes every modeled phosphorus pool can coincide with a reporting row. The relationship between a reported total and an invariant must be checked from their coefficients rather than inferred from the word “total.”

The same coefficient table explains constituent conversion. A hypothetical X_H concentration of 100 g COD m^{-3} contributes 100 g COD m^{-3} to COD, 7 g N m^{-3} to TN, 2 g P m^{-3} to TP, and 90 g TSS m^{-3} to TSS under the adopted map. A hypothetical X_{PP} concentration of 2 g P m^{-3} contributes 2 g P m^{-3} to TP and $6.46 \text{ g TSS m}^{-3}$ to TSS. These examples illustrate fixed accounting factors rather than evidence about a treatment outcome.

Aggregation can hide prediction errors. Equal and opposite errors in ammonium and nitrate cancel in reported TN because both have unit nitrogen weights. A negative ammonium value can even be hidden by a sufficiently large nitrate prediction. Likewise, different biomass and substrate errors can compensate in COD while producing distinct errors in TSS because their solids factors differ. Four accurate reported quantities therefore cannot establish an accurate or physically admissible 20-component state.

The reporting map and the invariant operator serve different purposes. The map provides engineering summaries. The invariant operator tests reaction-compatible material accounting at the declared boundary. Neither can recover all of the component information lost by aggregation. Component prediction, composite interpretation, mass conservation, and non-negativity therefore remain separate objects of assessment.

3.8 Mixing and spatial transport as model boundaries

The preceding reactor balance assumes a single mixed concentration for each component in a reactor. This assumption removes spatial coordinates from that reactor state. It is a modeling decision rather than a claim that every treatment unit is uniform. Biofilms, trickling filters, and aerated vessels with incomplete mixing can have concentration gradients and transport limitations (Sadino-Riquelme et al., 2023; Wik, 2003; T. C. Zhang & Bishop, 1994). These effects provide useful context for interpreting the scope of a mixed-reactor surrogate.

A small spatial control volume illustrates the additional accounting. For one component, let V be its fixed volume and $\bar{c}$ its average concentration. Let $F_{\text{adv,in}}$ and $F_{\text{adv,out}}$ be advective mass flow rates. Let $F_{\text{diff,in}}$ and $F_{\text{diff,out}}$ be diffusive mass flow rates. The finite-volume balance is

$$V \frac{d\bar{c}}{dt} = F_{\text{adv,in}} - F_{\text{adv,out}} + F_{\text{diff,in}} - F_{\text{diff,out}} + Vr. \quad (3.32)$$

The symbol r is the net reaction production rate per unit volume. It can be negative when the component is consumed. Every term on the right has units of component mass per time. For a hypothetical volume of 2 m^3 , advective inflow and outflow of 5 and 3 g d^{-1} , diffusive inflow and outflow of 2 and 0 g d^{-1} , and total reaction consumption of 3 g d^{-1} , the accumulation is $5 - 3 + 2 - 0 - 3 = 1 \text{ g d}^{-1}$. Dividing by the volume gives

$d\bar{c}/dt = 0.5 \text{ g m}^{-3} \text{ d}^{-1}$. This calculation shows why the reaction term alone cannot determine a local concentration change.

In a one-dimensional domain with coordinate z , the advective flux is vc and the diffusive flux is $-D_z \partial c / \partial z$. Here v is the velocity and D_z is a dispersion or diffusion coefficient. Dividing the balance of a slice by its volume and taking the limit as its thickness decreases gives

$$\frac{\partial c_j}{\partial t} = -\frac{\partial(vc_j)}{\partial z} + \frac{\partial}{\partial z} \left(D_z \frac{\partial c_j}{\partial z} \right) + \sum_{p=1}^{n_p} \nu_{pj} \rho_p + s_j. \quad (3.33)$$

The term s_j represents a specified external source per unit volume. The first two terms express spatial transport. The summation is the same component-by-component reaction accounting used in the mixed reactor. Boundary conditions specify how material enters and leaves the spatial domain. An invariant weighted sum cancels compatible reaction terms, but the transport and external-source terms still remain in its balance.

This spatial formulation explains how the theoretical ideas can extend to other treatment models. It is not an additional spatial reactor in the present study. Chapter 6 uses mixed reactor stages and a layered clarifier. The stages represent sequential treatment conditions, while the clarifier layers represent solids transport and inventory. Their distinct boundaries must be respected when mass conservation is imposed.

3.9 A common basis for assessing surrogate responses

The component formulation links mechanistic modeling, statistical prediction, physical enforcement, interpretation, and operating choice. The mechanistic model supplies the reference treatment state. The statistical predictor approximates that state from the influent and controls. Projection returns a state within the specified physical set. Interpretation follows the fitted response, and optimization uses that response to select operating settings. Substrate, nutrient, biomass, and solids quantities provide the engineering basis for examining each stage.

Paired raw and projected assessments use the same fitted prediction to identify the effect of enforcement. A comparison between separately trained predictors also includes differences in fitting and hyperparameter selection. The structured regression adds intermediate coupled and affine states that can be examined before final projection. These stages connect the observed output to the part of the calculation that supplied it.

For a connected plant, assessment follows the mixer, reactor stages, clarifier outlets, and solids inventory. Location-specific errors describe where joint projection changes the response. Plant aggregates summarize the resulting prediction across the treatment

system. The quantities used in the intended operating decision guide this assessment (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019).

Table 3.2 records the evidence associated with these assessment questions. Input transformations, learned scales, regularization, and hyperparameter selection use development information. Reserved responses provide the final prediction assessment. This data separation applies to the full procedure, including its learned model and subsequent constrained calculation (Kaufman et al., 2012).

Table 3.2: Assessment questions and the evidence used to connect surrogate responses to engineering decisions.

Assessment question	Evidence	Engineering quantity
Predictive agreement	Held-out component and composite errors with defined scales	Substrate, nutrients, biomass, and solids concentrations
Physical admissibility	Equality residuals, non-negativity checks, and inventory bounds	Conserved inventories and feasible component states
Sample efficiency	Common learning curves and variation across partitions	Error at the available simulation budget
Inspectable structure	Named coefficient blocks, physical units, and complete-response sensitivities	Dependence on influent and operating inputs
Operating quality	Independent mechanistic evaluation at selected controls	Treatment outcomes, engineering requirements, and objective values
Computational effort	Development, search, prediction, projection, and verification times	Time required to deliver an evaluated decision

The numerical checks use stated tolerances and residual definitions. Equalities define the mathematical constraints, while residuals describe their satisfaction by the returned numerical state. Reporting these quantities in original units or a defined scaled form makes the physical calculation available for examination (Boyd & Vandenberghe, 2004; Buzzi-Ferraris & Manenti, 2013). Chapters 4–6 apply this common assessment basis to their reactor, structured-regression, and connected-plant formulations.

Chapter 4

Learning the Reactor Response and Enforcing Physical Admissibility

4.1 Rationale

A surrogate learns the relationship between a reactor input and its steady-state effluent. Its statistical task is to reproduce component concentrations from examples. Its physical task is to return a non-negative component vector with the invariant values appropriate to the supplied influent. These tasks are related because the mechanistic targets satisfy the reactor equations. They remain distinct because an ordinary regression loss does not explicitly restrict every prediction to the sample-specific feasible set (Hansen et al., 2024; Kashinath et al., 2021).

The comparison includes 13 statistical surrogates. The set spans boosting, randomized ensembles, kernel regression, neighborhood regression, latent-variable regression, sparse multiresponse regression, and neural networks. This breadth addresses the model-selection concern raised by Bhosekar and Ierapetritou (2018). The surveys of Borisov et al. (2024) and Schwartz-Ziv and Armon (2022) further explain why a neural architecture alone should not determine the comparator set for a tabular prediction problem. The purpose of the common protocol is to compare explicitly specified predictors for the same physical state.

4.2 Theory

4.2.1 Reactor inputs and raw predictions

For sample i , the two operating values are entered first. They are followed by the twenty influent concentrations in the established component order. Written as a list, the input begins with $(\text{HRT}_i, u_{\text{air},i}, S_{O,\text{in},i}, S_{F,\text{in},i})$ and ends with the remaining influent coordinates through $X_{\text{MeOH},\text{in},i}$. The component list and its units are fixed before any learner is fitted. Transposing the two input blocks into rows permits their horizontal

concatenation. The final transpose restores the column convention. The complete input for sample i is

$$x_i = (u_i^\top, c_{in,i}^\top)^\top \in \mathbb{R}^{22}, \quad u_i = (\text{HRT}_i, u_{air,i})^\top. \quad (4.1)$$

A fitted surrogate is a function that takes these twenty-two numbers and returns twenty component values. The subscript m identifies which fitted method is used. For component j , its prediction can be written as $c_{raw,ij}^{(m)} = f_{m,j}(x_{i1}, \dots, x_{i22})$. The superscript (m) identifies the model and is not an exponent. Collecting its twenty scalar output functions, a fitted surrogate f_m produces the raw component prediction

$$c_{raw,i}^{(m)} = f_m(x_i) \in \mathbb{R}^{20}. \quad (4.2)$$

The raw prediction is assessed in its returned physical units. A post prediction projection then produces a second prediction for the same sample. Pairing the two states permits the effect of physical enforcement to be assessed separately from the choice of learner, data partition, and preprocessing.

4.2.2 Statistical response models

Boosting and randomized tree ensembles

XGBoost builds an additive sequence of regression trees. A regression tree applies a sequence of input comparisons and assigns the value associated with the reached terminal region. For one output, an additive tree model has the scalar form

$$f_j(x) = b_j + \beta_1 h_{1j}(x) + \beta_2 h_{2j}(x) + \dots + \beta_T h_{Tj}(x).$$

Here h_{tj} is the response from tree t , β_t determines its contribution, and b_j is a baseline. The summation notation $b_j + \sum_{t=1}^T \beta_t h_{tj}(x)$ records the same additions. A hypothetical baseline of 10 and two tree contributions of -2 and 1 produce 9. Each new tree addresses the error left by the current ensemble while regularization discourages an unnecessarily complex fit. Its split structure represents changes in response across input regions, and repeated additions permit interactions and nonlinear response shapes. The computational design and regularized objective of Chen and Guestrin (2016) motivate its inclusion as a flexible tabular baseline.

LightGBM also uses gradient-boosted trees, but grows trees through leaf-wise expansion. This strategy can allocate more detail to regions where an additional split offers a larger loss reduction. The efficiency-oriented design of Ke et al. (2017) makes the method relevant when each candidate fit must be repeated over several validation partitions. Its learning rate, leaf count, and minimum leaf support jointly control how closely it follows the training response.

CatBoost uses symmetric boosted trees and a Bayesian bootstrap in the specified regression setting. The work of Prokhorenkova et al. (2018) addresses bias associated

with categorical-feature handling and boosting. The reactor benchmark evaluates this learner on continuous inputs. Each level of a symmetric tree uses the same splitting condition across nodes at that level. This structural restriction gives a different approximation class from the leaf-wise trees of LightGBM.

Adaptive boosting (AdaBoost) regression combines shallow trees through adaptive emphasis on observations that are difficult to predict. The base trees have depth three. The estimator count, learning rate, and loss choice control how strongly the ensemble responds to poorly fitted examples. Drucker (1997) develops the regression extension of boosting that supports this comparator. These four boosting methods fit one regressor for each effluent component. All 20 regressors within a method use one shared selected hyperparameter vector.

Random Forest averages randomized regression trees. For component j , the average of T tree responses is $f_j(x) = [h_{1j}(x) + \dots + h_{Tj}(x)]/T$. Hypothetical tree values of 2, 3, and 4 give a mean of 3. Variation in sampled observations and candidate features reduces dependence among trees, so the average is less sensitive to a particular tree fit (Breiman, 2001). Extra Trees adds randomization of split thresholds (Geurts et al., 2006). Both methods fit the 20 outputs jointly in native multiresponse ensembles. Joint output handling lets the same tree partition support several component predictions. The projection subsequently enforces their stoichiometric relationships.

Tree averaging can preserve non-negativity when all leaf responses are non-negative. Mass conservation also requires the balance target of the new influent. For a hypothetical two-component system, two training responses with totals of 10 and 20 have an equally weighted average with total 15. If a new influent requires total 12, the average is positive but inconsistent with mass conservation for that sample. This distinction is caused by the sample-specific target b_i , rather than by whether the predictor averages positive observations.

Kernel, neighborhood, and latent-variable regression

Support vector regression (SVR) fits a function with an error-insensitive region around each training response. For scalar error e and positive insensitive width ϵ , its error contribution is zero when $|e| \leq \epsilon$ and $|e| - \epsilon$ otherwise. This can be recorded as $\max(0, |e| - \epsilon)$. With a hypothetical width of 0.2, an error of 0.1 contributes zero and an error of -0.5 contributes 0.3. Predictions within that region incur no corresponding error penalty. A linear kernel gives a globally linear input mapping. A radial kernel allows a nonlinear mapping through similarity to training inputs. Its similarity has the form

$$K(z, z') = \exp \left[-\gamma \sum_{\ell=1}^{22} (z_{\ell} - z'_{\ell})^2 \right].$$

Here z and z' contain the standardized input coordinates used by this learner. The positive coefficient γ controls how quickly similarity decreases with squared distance.

Identical inputs give similarity one. Greater distance gives a smaller value. The penalty parameter balances fit and regularity, while the insensitive width sets the tolerated response discrepancy. Smola and Schölkopf (2004) explain these mechanisms and the importance of their parameterization. The benchmark fits one regressor per component with one shared setting vector.

The k -nearest-neighbor (k -NN) predictor finds training observations close to a new input and combines their 20-component responses. For one component and a selected neighborhood $\mathcal{N}(x)$, weighted averaging means

$$f_j(x) = \frac{\sum_{i \in \mathcal{N}(x)} w_i c_{out,ij}}{\sum_{i \in \mathcal{N}(x)} w_i}.$$

The neighborhood size controls the degree of smoothing. Uniform weights give every selected neighbor equal influence, whereas distance weights give closer observations greater influence. Hypothetical neighbor values 2 and 6 give 4 under equal weights. Weights 3 and 1 instead give $[3(2) + 1(6)]/(3 + 1) = 3$. The local behavior analyzed by Mack (1981) explains why input scaling and distance choice matter. An unscaled alkalinity coordinate spanning hundreds of numerical units could dominate a much smaller aeration coordinate even though both influence the reactor. Standardization avoids using native numerical range as an implicit measure of scientific importance.

Partial least squares (PLS) regression represents the inputs and joint response through latent directions chosen to capture their covariance. One latent score is a weighted input sum $t_h = w_{1h}z_1 + \dots + w_{22h}z_{22}$, where z contains standardized inputs. An output is then a baseline plus a weighted sum of the retained scores. Substituting each score's input sum shows why ordinary partial least squares remains a linear input mapping. It can summarize strongly related input variables with a relatively small number of latent components. These weighted statistical combinations differ from the physical effluent components predicted by the model. The chemometric formulation described by Wold et al. (2001) provides an inspectable linear baseline. Its wastewater use is illustrated by Woo et al. (2009), who employs a kernel extension for process-variable estimation, and by C. Yang et al. (2021), who combines partial least squares with another predictor for adaptive effluent-quality prediction. The benchmark retains ordinary partial least squares with one to six latent components while predicting all 20 physical effluent components.

Shared sparse coefficient models

Multi-task Lasso fits one coefficient matrix for all component outputs. A predicted output is a baseline plus a sum of input-coefficient products. For two hypothetical inputs and two outputs, the coefficient rows can be $(3, 4)$ and $(0, 0)$. The first input contributes to both outputs and the second contributes to neither. The Euclidean length of the first row is $\sqrt{3^2 + 4^2} = 5$, and the length of the second row is zero.

Adding row lengths gives 5. This operation penalizes an input's complete coefficient group rather than adding the absolute value of every coefficient separately. If row $B_{j,\cdot}$ contains the coefficients of input j across those outputs, the row-group penalty is

$$\|B\|_{2,1} = \sum_j \|B_{j,\cdot}\|_2. \quad (4.3)$$

This penalty can set an entire input row to zero. It therefore promotes a shared selection of inputs rather than 20 unrelated selections. The feature-sharing principle of Argyriou et al. (2008), the multi-task Lasso algorithm of Liu et al. (2009), and the grouped-variable framework of Yuan and Lin (2006) provide the relevant foundations.

Multi-task Elastic Net combines the row-group penalty with a squared Frobenius penalty. The squared Frobenius norm is the sum of the squares of all matrix entries. For the hypothetical coefficient rows just given, it is $3^2 + 4^2 + 0^2 + 0^2 = 25$. With penalty strength $\lambda > 0$ and mixing ratio $r \in (0, 1)$, the combined penalty has the form

$$\lambda r \sum_j \sqrt{\sum_{f=1}^{20} B_{jf}^2} + \frac{\lambda(1-r)}{2} \sum_j \sum_{f=1}^{20} B_{jf}^2.$$

For the two-output illustration only, taking $\lambda = 0.2$ and $r = 0.5$ gives $0.2(0.5)(5) + 0.2(0.5)(25)/2 = 1.75$. The row-group part encourages selection and the squared part shrinks the retained coefficients. This extends the elastic-net principle of Zou and Hastie (2005) to a joint response structure, with multiresponse computation described by Simon et al. (2013). Both methods use standardized inputs and targets. Their global coefficient matrices can be inspected to identify fitted associations. A nonzero row identifies an input that contributes to the fitted prediction.

These two predictors and partial least squares are grouped as structurally inspectable surrogates. The grouping identifies their globally available coefficient structures. Physical projection enforces the influent-specific balances and non-negativity of their returned predictions.

Neural predictors

The multilayer perceptron (MLP) uses three fully connected hidden layers with 128, 64, and 32 units and one joint 20-output layer. One unit first forms a weighted sum, such as $b + w_1 z_1 + w_2 z_2$, and then applies an activation. A rectified linear activation returns the larger of zero and that sum. For hypothetical values $b = 0.2$, $w_1 = 0.5$, $w_2 = -1$, $z_1 = 2$, and $z_2 = 0.3$, the sum is $0.2 + 0.5(2) - 0.3 = 0.9$ and the activated value is 0.9. If the sum were -0.9 , the activated value would be zero. Each hidden layer applies such operations to the values supplied by the preceding layer. Backpropagation updates the weights according to the prediction loss (Rumelhart et al., 1986). The approximation result of Hornik et al. (1989) motivates the expressive capacity of feedforward networks. The topology considerations discussed by Curteanu and Cartwright (2011) motivate the stated architecture and training controls.

TabNet applies sequential attentive feature selection. At a decision step, a feature mask multiplies each input coordinate by its selected attention weight. If two hypothetical standardized inputs are $(2, 1)$ and their mask weights are $(0.8, 0.2)$, the masked inputs are $(1.6, 0.2)$. A different step can use different weights and therefore examine a different input emphasis. The selected features then contribute to a joint output layer. Arik and Pfister (2021) develop this approach for tabular learning. The attention masks show how the model weights its inputs during prediction. TabNet remains a neural comparator even though its internal masks can be inspected.

An epoch is one pass through the training data during neural fitting. Patience is the allowed number of consecutive epochs without the specified improvement before stopping. These fitting quantities describe the use of simulated reactor cases.

Both neural predictors start from random initialization and use only the simulated training partition available to that fit. The multilayer perceptron uses all available training rows and stops on training-loss convergence or at 250 epochs. Validation-based early stopping is disabled. Its training-loss patience is 15 epochs. TabNet's epoch count is selected in the inner search and ranges from 20 to 50. Its patience is zero, so a refit runs the selected epoch count without reserving an additional response-based stopping partition.

4.2.3 Simultaneous physical requirements

Now let the first hypothetical concentration be 11 and the second be -1 . Their sum is $11 + (-1) = 10$, so the inventory equality is satisfied. The second component fails non-negativity. Replacing it by zero changes the sum to $11 + 0 = 11$. To return to the non-negative part of the line defined by mass conservation, both coordinates can be changed. The state with first concentration 10 and second concentration zero has sum ten and passes both checks.

The individual adjustments make the distinction explicit,

$$\begin{aligned}\text{raw state} &= (11, -1)^\top, \\ \text{clipped state} &= (11 + 0, -1 + 1)^\top = (11, 0)^\top, \\ \text{jointly feasible state} &= (11 - 1, -1 + 1)^\top = (10, 0)^\top.\end{aligned}$$

Clipping changes only the second coordinate. The joint projection also compensates in the first coordinate. An equality projection alone can retain a negative component, while clipping alone can violate an equality. The final assessment therefore checks both properties on the same returned state (Haddad et al., 2010; Kircher & Votsmeier, 2025).

Figure 4.1 places the three states on the line defined by mass conservation and its non-negative segment. Their positions show which requirement each state satisfies.

The raw point is on the extended line defined by mass conservation but below the concentration boundary. The clipped point reaches a non-negative second coordinate

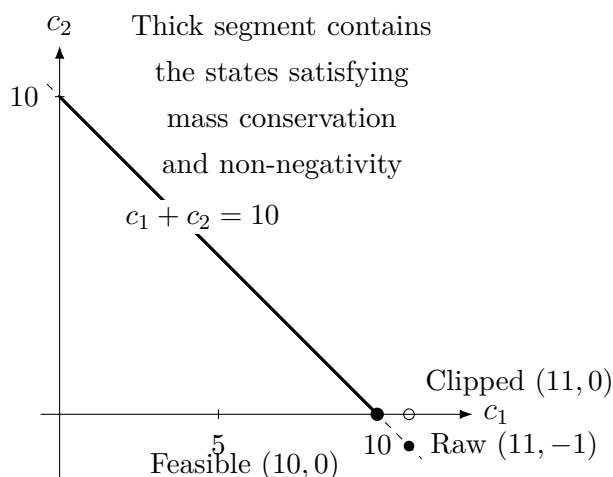


Figure 4.1: Hypothetical two-component geometry showing why clipping a negative concentration can violate a sum fixed by mass conservation.

while leaving that line. The jointly projected point lies at the end of the thick feasible segment. The geometry shows the two properties checked on each returned state.

Separate marginal summaries can also be misleading. If one prediction satisfies only mass conservation and another satisfies only non-negativity, neither is jointly feasible. For a hypothetical collection of two such states, each marginal compliance frequency is one half, while the joint compliance frequency is zero. The correct denominator is the number of states checked, and both conditions are tested within each state before counting joint compliance.

4.2.4 The Kircher–Votsmeier projection for mass conservation and non-negativity

A positive provisional target and relative-error geometry

Kircher and Votsmeier (2025) develop a projection that combines mass conservation with non-negativity for mechanistic-kinetics surrogates. Its transfer to the reactor prediction problem uses the invariant operator A , the basis Z , and the feasible influent anchor. The projection depends on the completed component prediction rather than on predictor internals. It can therefore be applied to every comparator without changing its training objective or architecture. Its inputs are the full component vector, the corresponding non-negative influent, and the physical operators.

The first step compares each raw component with its positive floor. If a raw component is above the floor, its provisional value is the raw value. Otherwise, its provisional value is the floor. For component j , the operation is $c_{+,ij}^{(m)} = \max(c_{raw,ij}^{(m)}, \varepsilon_{c,j})$. A hypothetical raw pair $(-0.2, 3)$ becomes $(10^{-8}, 3)$ when both native-unit floors have value 10^{-8} . The corresponding inverse values are 10^8 and $1/3$. The floor gives a positive denominator for every provisional concentration. The subsequent projection enforces

mass conservation.

A diagonal weighting matrix places each inverse concentration on its diagonal and zero elsewhere. For the hypothetical positive pair $(1, 2)^\top$, its action on a discrepancy column h is

$$W = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}, \quad Wh = \begin{bmatrix} h_1 \\ h_2/2 \end{bmatrix}.$$

No component is mixed with another in this operation. Each component discrepancy is divided by its own provisional concentration, so both weighted coordinates are relative discrepancies. The first step forms a strictly positive provisional target,

$$c_{+,i}^{(m)} = \max(c_{raw,i}^{(m)}, \epsilon_c), \quad W_i^{(m)} = \text{Diag}(\mathbf{1} \oslash c_{+,i}^{(m)}). \quad (4.4)$$

The maximum and division are componentwise. Each entry of ϵ_c is 10^{-8} in its component's native unit. The provisional target supplies both a reference state and finite positive weights. The subsequent mass conservation and non-negativity stages determine the final physical prediction.

For a component change h_j , the weighted discrepancy is $h_j/c_{+,ij}^{(m)}$. The geometry therefore penalizes relative movement. A hypothetical movement of 1 g m^{-3} corresponds to a relative discrepancy of 0.01 at a component value of 100 g m^{-3} and 10 at a component value of 0.1 g m^{-3} . Small coordinates receive strong protection. Large coordinates can absorb larger absolute changes at the same relative cost. Relative-error projection and variance-standardized accuracy therefore assess redistribution using different scales.

Selecting a candidate that satisfies mass conservation

Changes consistent with mass conservation can first be written one component at a time. If Z has fifteen change columns, coordinate j of a change is $Z_{j1}q_1 + \cdots + Z_{j,15}q_{15}$. Adding that change to $c_{in,ij}$ gives the candidate component. Since each column of Z has zero invariant product, every such combination preserves the influent target.

A complete two-component hypothetical calculation shows how these coordinates are chosen. Both components use the same abstract concentration basis. Let the required sum be four and the feasible anchor be $(2, 2)^\top$. The raw pair is $(1, 2)^\top$, which sums to three. The provisional pair is unchanged because both values exceed the floor. A normalized invariant row and a normalized change column are

$$A_{\text{ex}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad Z_{\text{ex}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Their product is $(1 - 1)/2 = 0$. A scalar change amount t gives the candidate $(2 + t, 2 - t)^\top$. In the normalized basis its coordinate is $q = \sqrt{2}t$, because $Z_{\text{ex}}q = (t, -t)^\top$. The sum remains four for every t . All states consistent with mass conservation for sample i have the form

$$c = c_{in,i} + Zq, \quad q \in \mathbb{R}^{15}. \quad (4.5)$$

For the hypothetical pair, the provisional target change from the anchor is $(-1, 0)^\top$. The difference between an allowed change and this target change is $(t+1, -t)^\top$. Applying the inverse provisional concentrations gives $(t+1, -t/2)^\top$. Its squared Euclidean norm is

$$E(t) = (t+1)^2 + \frac{t^2}{4}.$$

At $t = 0$, this value is one. Choosing a negative t moves the first component toward its raw value while requiring an increase in the second to preserve the sum. A derivative describes the change in the objective for a small change in t . The derivative of $(t+1)^2$ is $2(t+1)$ and that of $t^2/4$ is $t/2$. Adding them gives the rate of change of the complete objective. The best compromise can be found by differentiating,

$$\frac{dE}{dt} = 2(t+1) + \frac{t}{2} = \frac{5}{2}t + 2.$$

Setting the derivative to zero gives $t = -4/5$. The second derivative is $5/2 > 0$, so this point is the unique minimum. The resulting candidate is $(6/5, 14/5)^\top = (1.2, 2.8)^\top$. Its relative discrepancies from the provisional target are 0.2 and 0.4, giving squared objective $0.2^2 + 0.4^2 = 0.20$. A candidate $(1.5, 2.5)^\top$ instead gives $0.5^2 + 0.25^2 = 0.3125$. The weighting favors the first candidate because the smaller provisional component receives stronger protection against absolute movement.

Figure 4.2 shows this calculation as a geometric comparison. Curves around the raw pair connect states with the same projection objective. The line defined by mass conservation restricts the candidate to states whose two components sum to four.

The relative-weighted ellipse is taller than it is wide because movement in the second component is divided by two. A larger absolute change in that component can therefore have the same weighted cost as a smaller change in the first. The circular geometry assigns equal cost to equal absolute changes. Both chosen points satisfy the sum fixed by mass conservation, but the distance definition determines where each geometry touches that line. The feasible anchor locates the reference from which changes satisfying mass conservation are constructed and need not be the nearest candidate.

For the full problem, let $d_j = c_{+,ij}^{(m)} - c_{in,ij}$ denote the provisional change and let $w_j = 1/c_{+,ij}^{(m)}$. The scalar objective adds all twenty squared relative discrepancies,

$$E(q) = \sum_{j=1}^{20} w_j^2 \left(\sum_{k=1}^{15} Z_{jk} q_k - d_j \right)^2.$$

The inner sum builds one component of the change satisfying mass conservation. Subtracting d_j gives its discrepancy from the target change. Multiplication by w_j^2 scales its square. The outer sum combines these component contributions. The notation argmin identifies the coordinate vector at which this objective is smallest, rather than the value of the objective itself. The projection selects the change satisfying mass conservation closest to the provisional target change in the relative-error geometry,

$$q_i^{*,(m)} = \operatorname{argmin}_{q \in \mathbb{R}^{15}} \left\| W_i^{(m)} [Zq - (c_{+,i}^{(m)} - c_{in,i})] \right\|_2^2. \quad (4.6)$$

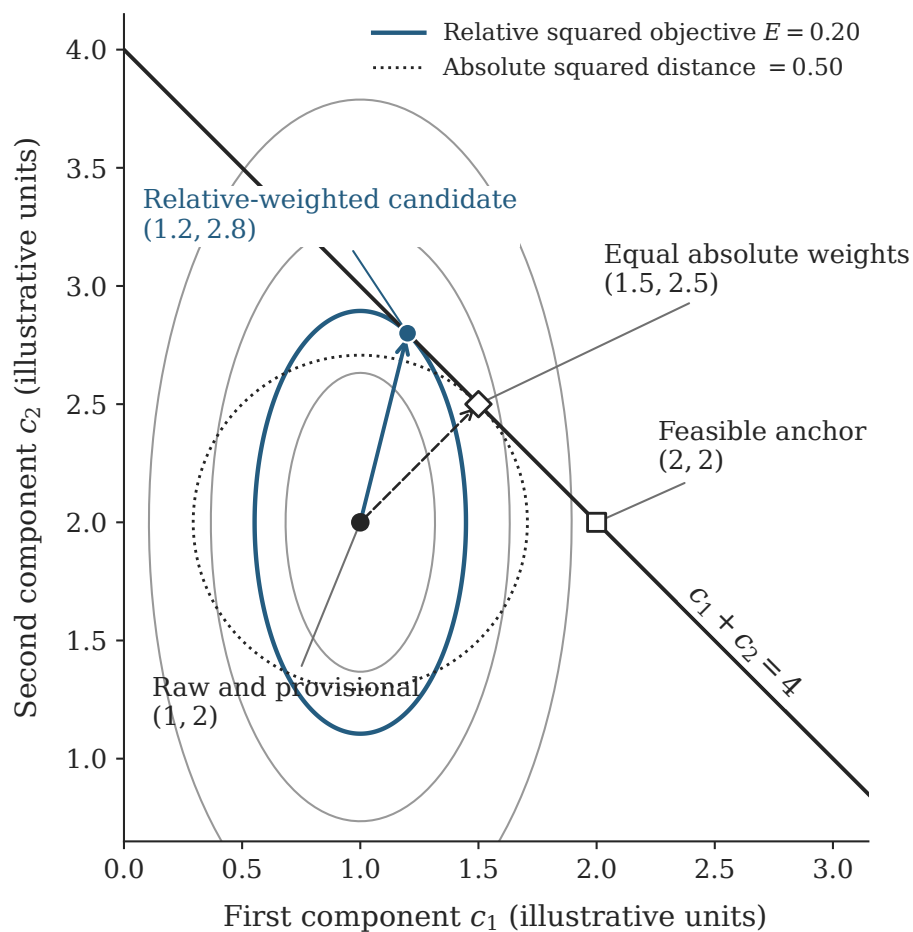


Figure 4.2: Hypothetical projection for mass conservation with relative and equal absolute weights. The highlighted ellipse first touches the line defined by mass conservation at the relative-weighted candidate. The dotted circle touches it at the equal-absolute-weight candidate.

The squared objective is a quadratic function of q . To minimize it, differentiate with respect to each coordinate q_ℓ . A partial derivative, written with ∂ , varies this one coordinate while holding the others fixed. Differentiating the square and then its inner sum gives

$$\frac{\partial E}{\partial q_\ell} = 2 \sum_{j=1}^{20} w_j^2 Z_{j\ell} \left(\sum_{k=1}^{15} Z_{jk} q_k - d_j \right).$$

Set this derivative to zero, divide by two, and move the target-change contribution to the right-hand side. For every ℓ , the resulting equation is

$$\sum_{k=1}^{15} \left(\sum_{j=1}^{20} w_j^2 Z_{j\ell} Z_{jk} \right) q_k = \sum_{j=1}^{20} w_j^2 Z_{j\ell} d_j.$$

The coefficient of q_k is a weighted dot product of columns ℓ and k of Z . Because $W_i^{(m)}$ is diagonal, its square places w_j^2 on the diagonal. Collecting these fifteen equations gives a 15×15 coefficient matrix and a fifteen-coordinate right-hand side. Differentiation gives

$$Z^\top (W_i^{(m)})^2 Z q_i^{*,(m)} = Z^\top (W_i^{(m)})^2 (c_{+,i}^{(m)} - c_{in,i}). \quad (4.7)$$

For the hypothetical pair, the compact coefficient calculation gives $Z_{\text{ex}}^\top W^2 Z_{\text{ex}} = (1 + 1/4)/2 = 5/8$. Its right-hand side is $-1/\sqrt{2}$. Dividing gives $q^* = -8/(5\sqrt{2}) = -4\sqrt{2}/5$, which equals $\sqrt{2}t$ for the scalar minimum already obtained. The scalar and matrix calculations therefore return the same candidate. For any nonzero vector v , the full problem's associated quadratic form equals $\|W_i^{(m)} Z v\|_2^2 > 0$. If Zv were zero, multiplying by $Z^\top$ would give $v = Z^\top Z v = 0$, contradicting the choice of v . Positive diagonal weights cannot turn a nonzero column into zero. The squared norm is consequently strictly positive. A positive-definite matrix has a strictly positive such product in every nonzero direction. The matrix is therefore positive definite and the minimizer is unique.

The selected coordinates build the component change through the fifteen sums $\sum_k Z_{jk} q_k^*$. Adding each sum to its matching influent coordinate constructs the state. In the hypothetical pair, this operation is $(2, 2)^\top + (-0.8, 0.8)^\top = (1.2, 2.8)^\top$. Both entries are non-negative and their invariant value is $(1.2 + 2.8)/\sqrt{2} = 4/\sqrt{2}$, equal to the anchor value. This example needs no projection interpolation for non-negativity. The candidate satisfying mass conservation is

$$c_{cand,i}^{(m)} = c_{in,i} + Z q_i^{*,(m)}. \quad (4.8)$$

Since $AZ = 0$, it satisfies $A c_{cand,i}^{(m)} = A c_{in,i} = b_i$ in exact arithmetic. If the provisional target is already consistent with mass conservation, its change lies in $\text{null}(A)$. The least-squares residual can then be zero and the candidate reproduces that target to numerical tolerance.

Equation (4.7) provides the derivation rather than the preferred numerical solution. The weighted design $W_i^{(m)} Z$ is solved through an orthogonal-triangular factorization.

Singular-value decomposition with relative cutoff 10^{-12} is the fallback. No explicit inverse of the normal-equation matrix is formed. Near-zero provisional targets can produce very unequal weights, so solving the original weighted system avoids the extra conditioning penalty associated with forming a normal-equation inverse.

The largest feasible interpolation toward the candidate

The candidate is also checked for non-negativity. A different hypothetical three-component example isolates the boundary problem. Its quantity used for mass conservation is the unweighted sum, with feasible influent $(8, 1, 1)^\top$ and candidate satisfying mass conservation $(4, -0.5, 6.5)^\top$. Both sums are ten. Subtracting the influent from the candidate gives the direction $(-4, -1.5, 5.5)^\top$. A fraction α of that direction gives individual concentrations

$$c_1(\alpha) = 8 - 4\alpha, \quad c_2(\alpha) = 1 - 1.5\alpha, \quad c_3(\alpha) = 1 + 5.5\alpha.$$

The first component requires $8 - 4\alpha \geq 0$, so $\alpha \leq 2$. The second requires $1 - 1.5\alpha \geq 0$, so $\alpha \leq 2/3$. The third increases and gives no upper bound for $\alpha \geq 0$. Restricting α to $[0, 1]$ makes $2/3$ the first boundary point. Define

$$\delta_i^{(m)} = c_{cand,i}^{(m)} - c_{in,i} = Zq_i^{*,(m)}. \quad (4.9)$$

The segment $c_{in,i} + \alpha\delta_i^{(m)}$, with $0 \leq \alpha \leq 1$, satisfies mass conservation because $A\delta_i^{(m)} = 0$. For a general decreasing coordinate, start with $c_{in,ij} + \alpha\delta_{ij}^{(m)} \geq 0$. Moving the negative change to the other side gives $c_{in,ij} \geq \alpha(-\delta_{ij}^{(m)})$. The factor $-\delta_{ij}^{(m)}$ is positive, so dividing by it preserves the inequality direction. For every decreasing coordinate, non-negativity requires

$$\alpha \leq \frac{c_{in,ij}}{-\delta_{ij}^{(m)}} \quad \text{when } \delta_{ij}^{(m)} < 0. \quad (4.10)$$

The minimum of these bounds is the first point at which the segment reaches the non-negative-orthant boundary.

If every candidate coordinate is non-negative, the entire direction is allowed and $\alpha = 1$. If at least one candidate coordinate is negative, the smallest decreasing-coordinate ratio supplies a bound below one. Taking that bound reaches the concentration boundary. Multiplication by $1 - \eta$ retreats slightly from the boundary to limit a sign change caused by rounding. With numerical safety margin $\eta = 10^{-12}$, the accepted interpolation factor and final state are

$$\alpha_i^{(m)} = \begin{cases} 1, & c_{cand,i}^{(m)} \geq 0, \\ (1 - \eta) \min_{j \mid \delta_{ij}^{(m)} < 0} \frac{c_{in,ij}}{-\delta_{ij}^{(m)}}, & \text{otherwise,} \end{cases} \quad (4.11)$$

$$\tilde{c}_i^{(m)} = c_{in,i} + \alpha_i^{(m)} \delta_i^{(m)}.$$

If the candidate is negative somewhere, at least one bound lies below one. Non-negativity of the influent makes every bound non-negative. The factor is restricted

to $[0, 1]$ against roundoff. Decreasing coordinates satisfy $\tilde{c}_{ij}^{(m)} \geq \eta c_{in,ij} \geq 0$, and increasing coordinates remain non-negative. For invariant row h , the final weighted component sum is $\sum_j A_{hj} c_{in,ij} + \alpha \sum_j A_{hj} \delta_{ij}^{(m)}$. The second sum is zero and the first is b_{ih} . Mass conservation follows from

$$A\tilde{c}_i^{(m)} = A c_{in,i} + \alpha_i^{(m)} A Z q_i^{*,(m)} = b_i. \quad (4.12)$$

The unrestricted boundary factor is the largest feasible interpolation factor. The safety margin makes the returned factor slightly smaller when projection is necessary.

A hypothetical three-component system with sum fixed by mass conservation 10 illustrates this safeguard. Let the feasible influent be $(8, 1, 1)^\top$ and a candidate satisfying mass conservation be $(4, -0.5, 6.5)^\top$. The direction is $(-4, -1.5, 5.5)^\top$. Its decreasing-coordinate bounds are 2 and $2/3$, so the largest feasible factor is $2/3$. Ignoring the numerical safety margin only for this illustration gives $(16/3, 0, 14/3)^\top$. The sum remains 10 and every component is non-negative. The first coordinate changes even though the second coordinate determines the boundary. One scalar shortens the complete direction satisfying mass conservation.

If an influent coordinate is zero and the candidate direction decreases that coordinate, its bound is zero. The final state then returns to the influent anchor. This returns a feasible state. Likewise, a component that is absent from the invariant rows can still move during interpolation because the complete direction is shortened. Recording the factor and activation frequency is therefore necessary for interpreting the projection.

Numerical acceptance and returned predictions

Numerical acceptance tests the returned state rather than assuming that the exact algebra eliminates rounding error. For row h , form the residual $r_h = \sum_j A_{hj} \tilde{c}_{ij}^{(m)} - b_{ih}$. Square its five values, add them, and take the square root to obtain the mass conservation norm. Separately compare each of the twenty concentrations with the negative of its own tolerance. A hypothetical concentration of -5×10^{-11} passes a 10^{-10} sign tolerance, whereas -2×10^{-10} fails it. These examples illustrate the numerical sign check. A returned projection is accepted only when

$$\|A\tilde{c}_i^{(m)} - b_i\|_2 \leq \tau_{mc}, \quad \tilde{c}_{ij}^{(m)} \geq -\tau_{nn,j} \quad \text{for every } j, \quad (4.13)$$

with $\tau_{mc} = 10^{-10}$ in the fixed invariant-coordinate convention and $\tau_{nn,j} = 10^{-10}$ in the native unit of component j . These checks distinguish the exact mathematical construction from its finite-precision realization. A finite raw vector is required.

The final state satisfies the declared invariants and sign restrictions. The least-squares stage minimizes weighted distance over the affine space, and the interpolation stage selects a feasible point on the influent-to-candidate line segment. The prediction error uses training standard deviations, while the projection weights use the positive

provisional concentrations. These definitions give the two calculations distinct objectives.

Figure 4.3 brings the projection steps and their checks into one prediction procedure. The non-negativity decision concerns the candidate satisfying mass conservation. The final numerical audit applies to the state returned by either branch.

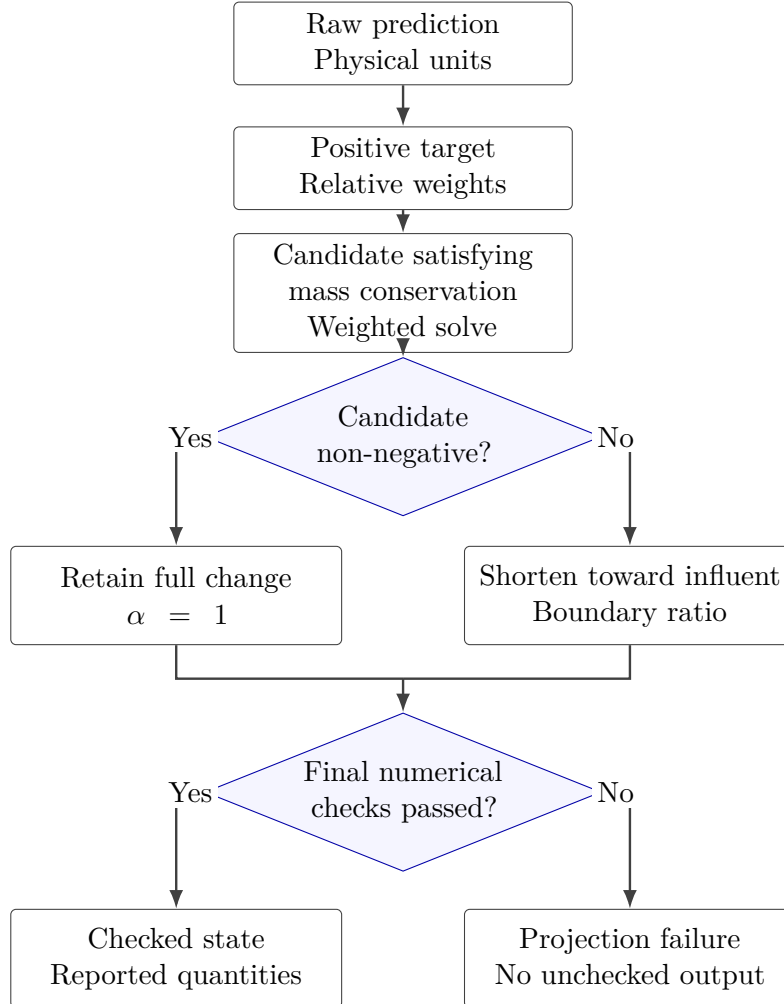


Figure 4.3: Post prediction mass conservation and non-negativity procedure. Positive-target construction, projection for mass conservation, and final-state acceptance have separate roles. The diagram shows the sequence used for each prediction.

The provisional positive target supplies a reference and weights for the mass conservation solve. A non-negative candidate can retain the full change satisfying mass conservation, while a candidate crossing a concentration boundary requires shortening toward the feasible influent. Both branches reach the same final checks. This arrangement applies the same numerical acceptance criteria to each returned state.

Each raw or projected composite is obtained by multiplying the relevant component values by the coefficients already defined in Table 3.1 and adding them. The coefficients do not change between the two prediction types. A paired comparison therefore changes only the component state supplied to the four scalar sums. Raw and

projected composites are calculated with the same map,

$$y_{raw,i}^{(m)} = I_{comp}c_{raw,i}^{(m)}, \quad \tilde{y}_i^{(m)} = I_{comp}\tilde{c}_i^{(m)}. \quad (4.14)$$

Feasibility is checked on the complete component state before this aggregation.

4.3 Methodology

4.3.1 Constructing a mechanistic learning domain

The single-reactor dataset is designed to contain 10,000 accepted steady states. Each candidate varies all 22 input coordinates. A randomized, scrambled, strength-one Latin hypercube places one sample in each stratum of every marginal input distribution. A stratum is one interval in a partition of an input range. A marginal distribution describes one coordinate, such as influent ammonium, considered separately from the other inputs. The construction uses seed 42 and no subsequent design optimization. Linear scaling maps the unit-hypercube samples to the bounds in Table 4.1. Latin hypercube sampling distributes coverage across marginal ranges without requiring a full Cartesian grid (McKay et al., 1979).

The standalone domain varies the operating and influent inputs within the declared simulation bounds. Its alkalinity interval uses the stated native coordinate. The coupled plant uses its own operating and influent bounds.

For each candidate input, nonlinear least squares solves Equation (3.14) with numerical state bounds $[10^{-8}, 1500]$ in each component's native unit. The cost-change, step-size, and gradient convergence tolerances are 10^{-9} . Each solve allows at most 1200 function evaluations. The generation procedure has an overall attempt cap of 25 times the target retained count. A state is retained only if the solver converges and the maximum absolute component-balance residual is no greater than 10^{-9} on the fixed native component-unit-per-day coordinates. The residual criterion uses the native component-unit-per-day coordinates.

Two physically informed starting states are tried before dynamic relaxation. The first blends the sampled influent with a preceding accepted effluent when one is available. It lowers readily consumed substrate coordinates, retains biomass pools, and estimates dissolved oxygen from dilution and transfer. The second applies stronger substrate depletion and biomass enrichment. Starting vectors are restricted to $[10^{-6}, 1500]$. If both steady solves fail, a stiff backward differentiation formula integrates the dynamic balance for 20 days and supplies a new initial state. The state bounds and starting-state restrictions apply to concentrations. The 28 kinetic process rates are evaluated without an artificial ceiling or clipping.

Solver acceptance, invariant residual, and minimum concentration are recorded separately. A converged bounded least-squares calculation can terminate without meeting

Table 4.1: Design bounds for the standalone reactor learning domain. The intervals define the sampled input ranges.

Input	Meaning	Interval	Unit
HRT	Hydraulic operating input	[6, 36]	h
u_{air}	Dimensionless aeration input	[0.5, 2.5]	1
S_O	Influent dissolved oxygen	[0, 0.5]	g O ₂ m ⁻³
S_F	Influent fermentable substrate	[20, 180]	g COD m ⁻³
S_A	Influent acetate	[5, 80]	g COD m ⁻³
S_{NH_4}	Influent ammonium nitrogen	[12, 55]	g N m ⁻³
S_{NO_2}	Influent nitrite nitrogen	[0, 3]	g N m ⁻³
S_{NO_3}	Influent nitrate nitrogen	[0, 8]	g N m ⁻³
S_{N_2}	Influent dissolved dinitrogen	[0, 2]	g N m ⁻³
S_{PO_4}	Influent soluble phosphate	[2, 18]	g P m ⁻³
S_I	Influent soluble inert organic matter	[10, 90]	g COD m ⁻³
S_{ALK}	Influent alkalinity	[80, 260]	mol m ⁻³
X_I	Influent particulate inert matter	[20, 120]	g COD m ⁻³
X_S	Influent slowly biodegradable substrate	[60, 280]	g COD m ⁻³
X_H	Influent heterotrophic biomass	[15, 100]	g COD m ⁻³
X_{PAO}	Influent phosphate-accumulating biomass	[5, 60]	g COD m ⁻³
X_{PP}	Influent stored polyphosphate	[2, 20]	g P m ⁻³
X_{PHA}	Influent stored polyhydroxyalkanoates	[1, 30]	g COD m ⁻³
X_{AOB}	Influent ammonia-oxidizing biomass	[0.5, 8]	g COD m ⁻³
X_{NOB}	Influent nitrite-oxidizing biomass	[0.5, 8]	g COD m ⁻³
X_{MeP}	Influent precipitated metal phosphate	[0, 12]	g TSS m ⁻³
X_{MeOH}	Influent metal hydroxide	[0, 12]	g TSS m ⁻³

the required balance residual. A positive state can also fail mass conservation. Retaining these separate diagnostics prevents one successful numerical check from standing in for all others. The surrogate inputs contain the sampled operating and influent coordinates. Initialization choices and solver history are not predictive features.

4.3.2 Preprocessing and validation without response leakage

A fold is one reserved group of cases in cross-validation. Successive fits change which group is withheld. Nested cross-validation separates an inner comparison used to select hyperparameters from an outer comparison used to assess the selected model. Here each case contains influent and operating inputs paired with the mechanistic effluent state.

The evaluation uses nested subsets at

$$N \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (4.15)$$

A seed-42 permutation determines the nested row order and persistent master fold identifiers before subsets are formed. Every smaller subset is contained in each larger subset. A row retains its outer-fold role at every scale. Five outer folds train on $0.8N$ observations and score the remaining $0.2N$. Thus a design size of 500 gives 400 training rows and 100 scored rows per outer iteration. A design size of 10,000 gives 8000 training rows and 2000 scored rows. The five iterations change which fold is withheld, so each row supplies a scored prediction from a fit that excludes it. The same rows and fold identifiers are used by every surrogate and both prediction types.

Hyperparameters are selected using four inner folds within the largest outer-training partition. The scored outer fold is absent from the search. Within an 8000-row outer-training pool, an inner iteration fits on 6000 rows and scores 2000 different rows from that pool. Those inner-validation rows are not the 2000 rows reserved for outer assessment. The inner score selects the setting, and the outer score assesses a model fitted with that selected setting. For each outer fold, its selected setting is retained over the smaller nested subsets. Learning curves therefore examine the response to sample size under a fixed fold-specific setting. The bias–variance discussion of Belkin et al. (2019) cautions against assuming that error must vary monotonically with model capacity or available data. The learning-curve review of Viering and Loog (2023) supports treating the complete curve as an object of assessment.

Every transformation is fitted within the applicable training partition. During inner selection, that partition is the inner-training split. During outer scoring, it is the outer-training subset at the current size. During extrapolation assessment, it is the full in-domain training pool. A training coordinate has one mean and one standard deviation for that fit. If its M training values are $v_1, \dots, v_M$, these quantities are

calculated in three steps,

$$\mu = \frac{v_1 + \cdots + v_M}{M}, \quad \sigma^2 = \frac{(v_1 - \mu)^2 + \cdots + (v_M - \mu)^2}{M}, \quad \sigma = \sqrt{\sigma^2}.$$

The second quantity is the population variance. The population convention divides by the training count M . A z-score subtracts the mean and divides by the standard deviation, giving $z = (v - \mu)/\sigma$. Subtracting the mean centers the coordinate and division puts its numerical variation on a unit-standard-deviation scale. The transformation requires $\sigma > 0$.

For hypothetical training values 2, 2, 6, and 6, the mean is 4. Their centered values are -2 , -2 , 2, and 2, so the variance is $(4 + 4 + 4 + 4)/4 = 4$ and the standard deviation is 2. A validation value of 6 therefore becomes $(6 - 4)/2 = 1$. The validation value is not added to the training values before calculating the mean or variance. If this coordinate is an output, a predicted z-score of 0.5 returns to physical units as $4 + 2(0.5) = 5$.

The mean and population standard deviation of a training coordinate define its z-score transformation. Validation inputs are transformed with those training quantities, and target-scaled predictions are restored to physical units before projection or physical diagnostics. Every target has its own inverse transformation $\hat{c}_j = \mu_j + \sigma_j \hat{z}_j$. A single common inverse scale would be incorrect for components with different units or variability.

Table 4.2: Output handling and preprocessing for the statistical comparators. Scaling uses only the training partition applicable to each fit.

Surrogates	Output handling	Scaling
XGBoost, LightGBM, CatBoost	Separate component regressors	None
AdaBoost	Separate component regressors	Targets only
Random Forest	Joint tree response	None
Extra Trees	Joint tree response	Targets only
Support vector regression	Separate component regressors	Inputs and targets
k -nearest neighbors	Joint neighborhood response	Inputs and targets
Partial least squares, Multi-task Lasso, Multi-task Elastic Net	Joint coefficient structure	Inputs and targets
Multilayer perceptron, TabNet	Joint neural output	Inputs and targets

Tree partitions depend on coordinate ordering, so a positive affine input rescaling preserves candidate threshold partitions. Distance-based, covariance-based, and regu-

larized models require more deliberate scaling because relative numerical magnitudes affect their objectives. Separate-output tree regressors do not combine different component units within one joint fitting loss. The target-only scaling of Extra Trees and AdaBoost is retained as a specified benchmark choice. Independent of those training choices, every scored component error is standardized with the designated training reference. Training target scaling and evaluation normalization are distinct operations.

No missing-value imputation, response-based outlier removal, or target filtering is applied. No validation or extrapolation response is used to estimate a scaling parameter. A hypothetical decision to standardize a validation target with the standard deviation of its own fold would change the scoring reference using data outside training. Retaining a training reference instead asks how large an error is relative to the variability the learner was allowed to observe.

4.3.3 Hyperparameter selection

Each surrogate uses a seeded tree-structured Parzen search with 100 trials per study and no pruning or timeout. One trial proposes a complete parameter setting, fits it within the prescribed inner-training partitions, and calculates its validation objective. The objective is an error value to be minimized. It differs from a parameter value such as tree depth or penalty strength. The search mechanism is supported by the optimization framework of Akiba et al. (2019). The objective is the mean squared standardized component error over the inner validation folds. Five studies select the settings for the five largest-scale outer-training partitions. A sixth, independent four-fold study selects the setting for a final full-domain fit. The design specifies $6(100) = 600$ trials per surrogate and $13(600) = 7800$ across all 13 surrogates.

The search ranges, fixed controls, and 100-trial allocation define a common fitting protocol for the thirteen predictors.

4.3.4 Evaluation of error, feasibility, and projection size

Component errors on a training-defined scale

Let $\mathcal{T}$ denote the applicable training reference and let $\sigma_j^{(\mathcal{T})}$ be the population standard deviation of target component j in that reference. For one assessed concentration, subtract the reference effluent value from the predicted value to obtain its signed physical error. Divide this error by the training standard deviation of the same component. For example, a hypothetical prediction of 3 g N m^{-3} against a reference of 2 g N m^{-3} has signed error 1 g N m^{-3} . A training standard deviation of 2 g N m^{-3} gives standardized error 0.5. The constituent units cancel in the division. For prediction type t ,

Table 4.3: Search domains and fixed controls for tree ensemble surrogates. Integer limits are inclusive. “Log” denotes a log-uniform search.

Surrogate	Search domain and fixed controls
XGBoost	80–320 trees. Learning rate 0.01–0.2 (log). Depth 3–8. Minimum child weight 0.5–8 (log). Row and feature fractions 0.6–1. Absolute-value penalty 10^{-6} –1 (log). Squared penalty 10^{-3} –10 (log). Histogram or approximate trees. Squared-error objective.
LightGBM	80–320 trees. Learning rate 0.01–0.2 (log). 15–63 leaves. Depth unbounded or in $\{4, 6, 8, 10\}$. Minimum child rows 5–40. Row and feature fractions 0.6–1 with sub-sampling frequency 1. Absolute-value penalty 10^{-6} –1 (log). Squared penalty 10^{-6} –10 (log). Regression objective.
CatBoost	80–320 iterations. Learning rate 0.01–0.2 (log). Depth 4–8. Squared penalty 0.01–20 (log). Bagging temperature 0–5. Random strength 10^{-3} –5 (log). Root mean squared error loss and Bayesian bootstrap.
AdaBoost	50–250 depth-three trees. Learning rate 0.01–1 (log). Linear, squared, or exponential regression loss.
Random Forest	100–400 trees. Depth unbounded or in $\{6, 10, 14, 18\}$. Minimum split size 2–10. Minimum leaf size 1–4. Feature fraction 0.5–1. Bootstrap enabled or disabled.
Extra Trees	100–400 trees. Depth unbounded or in $\{6, 10, 14, 18\}$. Minimum split size 2–10. Minimum leaf size 1–4. Feature fraction 0.5–1. Bootstrap enabled or disabled.

Table 4.4: Search domains and fixed controls for kernel, neighbor, multiresponse, and neural surrogates. Integer limits are inclusive. “Log” denotes a log-uniform search.

Surrogate	Search domain and fixed controls
Support vector regression	Penalty 0.1–10 (log). Error-insensitive width 0.01–0.5 (log). Linear or radial kernel. Radial coefficient set from standardized input variance or inverse input dimension. Tolerance 10^{-3} – 10^{-2} (log). Shrinking enabled. At most 20,000 iterations. Kernel cache of 256 megabytes per output regressor.
k -nearest neighbors	1–15 neighbors. Uniform or distance weights. Automatic, ball-tree, k -dimensional tree, or brute-force search. Leaf size 15–60. Minkowski, Euclidean, or Manhattan distance. Minkowski order in $\{1, 2, 3\}$.
Partial least squares	1–6 latent components. 200–1000 maximum iterations. Tolerance 10^{-8} – 10^{-4} (log). Internal scaling disabled because external training-partition scaling is applied.
Multi-task Elastic Net	Penalty strength 10^{-6} –10 (log). Row-group mixing ratio 0.05–0.95. Cyclic coordinate descent. At most 20,000 iterations and tolerance 10^{-6} .
Multi-task Lasso	Penalty strength 10^{-6} –10 (log). Cyclic coordinate descent. At most 20,000 iterations and tolerance 10^{-6} .
Multilayer perceptron	Hidden widths (128, 64, 32). Rectified linear, hyperbolic tangent, or logistic activation. Adaptive moment optimization. Squared-weight penalty 10^{-6} – 10^{-2} (log). Batch size in $\{64, 128, 256\}$. Learning rate 5×10^{-4} – 10^{-2} (log). Training-loss tolerance 10^{-4} – 10^{-3} (log). Shuffling enabled. Training-loss patience 15 epochs. At most 250 epochs.
TabNet	Equal decision and attention widths in $\{8, 16, 24\}$. Three or four attention steps. Relaxation coefficient 1–2. Sparsity coefficient 10^{-6} – 10^{-2} (log). Adaptive moment learning rate 0.003–0.03 (log). Sparsemax or entmax-1.5 masks. 20–50 selected epochs. Batch size in $\{512, 1024\}$. Virtual batch size 128. Patience zero.

which is either raw or projected, define

$$\xi_{ij}^{(m,t)} = \frac{c_{ij}^{(m,t)} - c_{out,ij}}{\sigma_j^{(\mathcal{T})}}. \quad (4.16)$$

The normalized mean squared error (nMSE), normalized root mean squared error (nRMSE), and normalized mean absolute error (nMAE) combine these standardized errors. A two-observation, two-component hypothetical example shows their order of calculation. Suppose the training standard deviations are 2 and 4. Let the reference and predicted values be

Observation	Reference		Prediction	
	Component 1	Component 2	Component 1	Component 2
1	2	4	3	2
2	4	8	2	12

Values within each component use that component's own physical basis. The four physical errors are 1, -2 , -2 , and 4. Dividing each by its corresponding training standard deviation gives

$$\xi = \begin{bmatrix} 1/2 & -2/4 \\ -2/2 & 4/4 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5 \\ -1 & 1 \end{bmatrix}.$$

There are four scored component-observation pairs. Their squared errors sum to $0.25 + 0.25 + 1 + 1 = 2.5$. Their mean squared standardized error is $2.5/4 = 0.625$, and its square root is approximately 0.7906. Their absolute errors sum to $0.5 + 0.5 + 1 + 1 = 3$, so the mean absolute standardized error is $3/4 = 0.75$. The square is taken before averaging for nMSE. The square root is taken after that average for nRMSE. Absolute values are taken before averaging for nMAE.

For the twenty-component assessment, the first summation accumulates errors across observations and the second accumulates them across components. Division by $20n$ gives every scored pair equal weight after standardization. The compact definitions are

$$\text{nMSE}_{m,t} = \frac{1}{20n} \sum_{i=1}^n \sum_{j=1}^{20} (\xi_{ij}^{(m,t)})^2, \quad (4.17)$$

$$\text{nRMSE}_{m,t} = \sqrt{\text{nMSE}_{m,t}}, \quad (4.18)$$

$$\text{nMAE}_{m,t} = \frac{1}{20n} \sum_{i=1}^n \sum_{j=1}^{20} |\xi_{ij}^{(m,t)}|. \quad (4.19)$$

The component state is the primary scoring space because it is the prediction target and the space in which constraints are imposed. The root mean squared score emphasizes larger standardized errors. The absolute score gives a complementary average deviation. The normalized measures require a positive training standard deviation for every component.

A hypothetical error of 1 g m^{-3} has standardized magnitude 0.1 for a component with training standard deviation 10 g m^{-3} . The same physical error has magnitude 10 for a component with training standard deviation 0.1 g m^{-3} . The score gives equal numerical weight to standardized component variation.

The coefficient of determination compares squared prediction error with the error of predicting one constant value for every observation. That constant is the true mean within the scored partition. For component 1 of the hypothetical example, the true mean is $(2 + 4)/2 = 3$. Predicting 3 for both observations would give squared errors $(2 - 3)^2 + (4 - 3)^2 = 2$. The actual predictions give squared errors $(3 - 2)^2 + (2 - 4)^2 = 5$. Dividing 5 by 2 and subtracting from one gives $R^2 = 1 - 5/2 = -1.5$. For component 2, the true mean is 6. The corresponding baseline square sum is $(4 - 6)^2 + (8 - 6)^2 = 8$, and the prediction square sum is $(2 - 4)^2 + (12 - 8)^2 = 20$. Its score is also $1 - 20/8 = -1.5$. The negative scores mean that the predictions are worse than their scored-partition constant baselines.

The mean used in this comparison is $\bar{c}_{out,j} = (c_{out,1j} + \dots + c_{out,nj})/n$. It belongs to scoring and is not used to fit the model. The denominator of R^2 therefore differs from the training-defined scale of nMSE. For a scored partition with true mean $\bar{c}_{out,j}$, the component coefficient of determination and its equal-weight average are

$$R_{j,m,t}^2 = 1 - \frac{\sum_i (c_{ij}^{(m,t)} - c_{out,ij})^2}{\sum_i (c_{out,ij} - \bar{c}_{out,j})^2}, \quad \bar{R}_{m,t}^2 = \frac{1}{20} \sum_{j=1}^{20} R_{j,m,t}^2. \quad (4.20)$$

The macro-average requires a positive scored-partition variance for each component. The coefficient can be negative when the prediction is worse than the scored-partition mean. It complements the standardized error but uses a different response-variability reference.

The macro-average adds the twenty component R^2 values and divides by twenty. Likewise, pooled nRMSE is the square root of the mean of component squared standardized errors. It is generally different from the arithmetic mean of component root errors. Hypothetical component mean squared standardized errors of 1 and 9 give pooled root error $\sqrt{(1 + 9)/2} = \sqrt{5}$, whereas the mean of their separate roots is $(1 + 3)/2 = 2$. This distinction prevents differently constructed aggregate scores from being treated as interchangeable.

Physical-unit errors are retained for individual components. Secondary composite error is calculated separately for COD, TN, TP, and TSS. For a composite, calculate one signed error per observation, square the errors, average them over observations, and take the square root. Dividing that physical-unit root error by the training standard deviation of the composite gives its normalized root error. A hypothetical root error of 5 g COD m^{-3} divided by a training standard deviation of 10 g COD m^{-3} gives 0.5. The normalized root mean squared error of composite q is its physical-unit root mean squared error divided by $\sigma_q^{(T)}$, the training standard deviation of that composite. This composite score uses the variability of one reported quantity. Composite standard

deviations are calculated after applying the common reporting map to the training targets.

Violation frequencies and their magnitudes

The mass conservation residual begins with five scalar errors. For invariant row h , its error is $r_h = \sum_j A_{hj}c_j - b_{ih}$. Its Euclidean magnitude is $\sqrt{r_1^2 + \dots + r_5^2}$. A hypothetical two-row residual of $(3, 4)^\top$ has magnitude five in its fixed invariant-coordinate convention. This example explains the norm.

The negative magnitude uses a different operation. First divide each concentration by its training standard deviation. Then retain a negative coordinate if one is present and replace every non-negative coordinate by zero. Finally add the absolute values of the retained negatives. Hypothetical standardized values $(-2, 1.5, -2)$ become $(-2, 0, -2)$ and have absolute-sum magnitude four. A positive component cannot cancel a negative component in this diagnostic. The mass conservation residual and standardized total negative magnitude of a state c are

$$v_{mc,i}(c) = \|Ac - b_i\|_2, \quad v_{nn,i}^{\text{std}}(c) = \left\| \min(c \oslash \sigma^{(\mathcal{T})}, 0) \right\|_1. \quad (4.21)$$

The minimum is componentwise. The mass conservation residual uses the same full-precision orthonormal basis for every sample. Its numerical value uses the adopted invariant-coordinate convention. The negative magnitude uses training standard deviations to compare different component bases without summing incompatible physical units.

A frequency counts observations satisfying a stated failure condition and divides that count by the assessed observation count. The indicator $\mathbb{K}[\cdot]$ returns one when its bracketed condition is true and zero otherwise. For a hypothetical four-observation assessment, suppose the mass conservation failure indicators are $(1, 0, 1, 0)$ and the non-negativity failure indicators are $(0, 1, 1, 0)$. The two frequencies are $(1+0+1+0)/4 = 0.5$ and $(0+1+1+0)/4 = 0.5$. An observation with several negative components is still one failing observation in this row-level count. The two row-level violation frequencies are

$$\pi_{mc} = \frac{1}{n} \sum_i \mathbb{K}[v_{mc,i}(c_i) > \tau_{mc}], \quad \pi_{nn} = \frac{1}{n} \sum_i \mathbb{K}[\min_j(c_{ij} + \tau_{nn,j}) < 0]. \quad (4.22)$$

A state is counted as non-negative only when every component passes its tolerance. Component-specific violation frequencies identify which coordinates cross the boundary. In the four-observation example, only the fourth observation passes both conditions, so its joint feasible fraction is one quarter. The third fails both conditions and appears in both marginal failure counts. Adding the two failure frequencies would therefore count it twice. A joint indicator applies both pass conditions to the same observation before averaging. The joint feasible fraction is

$$\pi_{\text{feasible}} = \frac{1}{n} \sum_i \mathbb{K}[v_{mc,i}(c_i) \leq \tau_{mc} \text{ and } \min_j(c_{ij} + \tau_{nn,j}) \geq 0]. \quad (4.23)$$

Mass conservation and non-negativity frequencies are not added to calculate this fraction because the same state can violate both requirements.

Mean, median, and maximum violation magnitudes are retained along with the frequencies. A hypothetical set of many concentrations just below zero can have a high violation frequency but small total negative magnitude. Another set with one strongly negative state can have low frequency and a large maximum magnitude. The two summaries answer different questions. Physical-unit negative magnitudes are also retained separately for each component.

Paired accuracy effects and displacement

The paired accuracy effect subtracts the raw score from the projected score on the same assessment observations. If hypothetical raw and projected nRMSE values are 0.40 and 0.35, the difference is $0.35 - 0.40 = -0.05$. If the projected value were 0.45, the difference would be $+0.05$. This sign convention expresses whether error decreases or increases. Component movement and mass conservation residuals are recorded separately. For each outer fold, the effect on the primary error score is

$$\Delta \text{nRMSE}_{m,N,g} = \text{nRMSE}_{m,\text{projected},N,g} - \text{nRMSE}_{m,\text{raw},N,g}. \quad (4.24)$$

A negative difference indicates lower error after projection, and a positive difference indicates higher error. Differences are calculated within paired folds before summary. Component and composite differences are retained because a single aggregate score can conceal opposite changes among outputs.

Displacement measures the difference between the projected prediction and its own raw prediction. Calculate one change per component, divide it by that component's training standard deviation, square it, sum the squares, and take the square root. For the hypothetical weighted-projection pair already calculated, the changes are $1.2 - 1 = 0.2$ and $2.8 - 2 = 0.8$. If its illustrative training standard deviations are 0.1 and 0.4, the standardized changes are 2 and 2. Their Euclidean displacement is $\sqrt{2^2 + 2^2} = 2\sqrt{2}$. The example deliberately uses a training scale different from the provisional concentration scale, as the two operations serve different purposes. The sample-level standardized displacement is

$$d_{c,i}^{\text{std},(m)} = \left\| (\tilde{c}_i^{(m)} - c_{\text{raw},i}^{(m)}) \oslash \boldsymbol{\sigma}^{(\mathcal{T})} \right\|_2. \quad (4.25)$$

This norm measures how far the projection moves the state relative to training variability. It measures movement rather than accuracy. A large displacement can reduce violations of the specified physical constraints while moving away from the true kinetic response. Mean, median, and maximum displacement, componentwise physical-unit movement, the fraction requiring backtracking, and the distribution of α are therefore assessed together.

Learning curves retain the full error trajectory over Equation (4.15). An area summary connects the errors at neighboring sample sizes by straight segments. The area under one segment is its width multiplied by the average of its two endpoint heights. For hypothetical error heights 0.8 and 0.6 at design sizes 500 and 1450, the segment area is $(1450 - 500)(0.8 + 0.6)/2 = 665$. If the next hypothetical height is 0.4 at size 2400, the next area is $(2400 - 1450)(0.6 + 0.4)/2 = 475$. The total area over those two segments is 1140. Dividing by the size interval $2400 - 500 = 1900$ gives an average height of 0.6.

The full design has ten adjacent intervals. Calculate the trapezoidal area of each interval, add those areas, and divide by $10000 - 500 = 9500$. Division by the total interval converts the area into an average error height. As a descriptive summary, the normalized area under an error curve is calculated by trapezoidal integration,

$$\mathcal{A}_{m,t} = \frac{1}{N_{11} - N_1} \sum_{\ell=1}^{10} \frac{E_{m,t}(N_\ell) + E_{m,t}(N_{\ell+1})}{2} (N_{\ell+1} - N_\ell), \quad (4.26)$$

where $E_{m,t}(N)$ is the mean held-out error at size N . This quantity has the same scale as the selected error measure. It summarizes behavior across the declared size interval alongside the individual size scores. Training and validation error at the largest size provide a separate descriptive gap. The complete learning curve shows how error responds to sample size (Viering & Loog, 2023).

Each in-domain metric is summarized as the arithmetic mean and sample standard deviation of five outer-fold values. If these values are $v_1, \dots, v_5$, first calculate $\bar{v} = (v_1 + \dots + v_5)/5$. Then calculate $s = \sqrt{[(v_1 - \bar{v})^2 + \dots + (v_5 - \bar{v})^2]/4}$. The denominator is four because this summary uses the sample standard deviation of five fold values. It differs from the population standard deviation used to standardize a training coordinate. The standard deviation describes partition-to-partition variation.

4.3.5 Testing inputs beyond the learning domain

Extrapolation assessment uses separately simulated inputs outside the training hyperrectangle. The final setting for each surrogate is selected with four folds on all 10,000 in-domain states. Its preprocessing and predictor are then refitted on that complete pool. No extrapolation response enters selection, fitting, scaling, or calibration. This separation addresses the domain-shift concern discussed in scientific machine learning by Karniadakis et al. (2021) and Shen et al. (2023).

Four inputs can exceed their upper learning bounds. Exceedance is measured relative to the width of the original input interval. For HRT, the width is $36 - 6 = 30$ h. A hypothetical input of 39 h exceeds the upper bound by 3 h, giving exceedance $3/30 = 0.10$. Dividing by the original width allows a comparable severity description for inputs with different units. For input ℓ with bounds $[L_\ell, U_\ell]$, define its normalized

exceedance as

$$e_{il} = \frac{x_{il} - U_\ell}{U_\ell - L_\ell}. \quad (4.27)$$

The mild design uses $e_{il} \in [0.10, 0.25]$ and the severe design uses $e_{il} \in [0.35, 0.60]$. Reversing the exceedance calculation gives $x_{il} = U_\ell + e_{il}(U_\ell - L_\ell)$. For HRT, the mild endpoints are $36 + 0.10(30) = 39$ h and $36 + 0.25(30) = 43.5$ h. The severe endpoints are $36 + 0.35(30) = 46.5$ h and $36 + 0.60(30) = 54$ h. The generator targets a mixture of 70% one-input excursions and 30% two-input excursions. All remaining inputs remain within the original bounds. Every candidate therefore lies outside the training hyperrectangle and consequently outside the training-point convex hull. The hyperrectangle is the set satisfying every original coordinate interval. Every weighted average of training points remains inside those intervals, so a point violating even one interval is also outside their convex hull.

Table 4.5: Fixed exterior intervals for the two extrapolation designs. The intervals define the exterior input conditions.

Input	Learning interval	Mild interval	Severe interval	Unit
HRT	[6, 36]	[39, 43.5]	[46.5, 54]	h
u_{air}	[0.5, 2.5]	[2.7, 3.0]	[3.2, 3.7]	1
S_{NH4}	[12, 55]	[59.3, 65.8]	[70.1, 80.8]	g N m ⁻³
X_S	[60, 280]	[302, 335]	[357, 412]	g COD m ⁻³

Each severity design targets 600 accepted steady states and retains the same solver, initialization sequence, and balance-residual criterion. Candidate, accepted, failed, valid-but-unselected, and retained counts are distinguished. The realized one-input and two-input proportions are also distinguished from their design targets. The exterior assessment measures prediction error across the four selected inputs.

Extrapolation errors are standardized with population component standard deviations from all 10,000 in-domain targets. Composite normalization uses the corresponding full-domain composite standard deviations. The nMSE, nRMSE, nMAE, mass conservation and non-negativity diagnostics, displacement, and backtracking diagnostics are calculated on identical raw and projected cases. The coefficient of determination is secondary because the deliberately shifted response distribution changes its denominator.

Each exterior metric summarizes 600 cases from one full-domain refit. Satisfaction of the projected constraints is interpreted separately from the predictive accuracy of extrapolation. A state can remain non-negative and consistent with mass conservation while its estimated nitrogen distribution or biomass abundance is inaccurate.

4.3.6 Computational cost assessment

Computation is partitioned into hyperparameter search, preprocessing and fitting, raw inference, projection alone, and complete projected inference. Search time is reported once for each distinct study because a largest-scale outer setting is reused at all eleven sizes. Summing that same search duration over the nested size rows would count the selection cost repeatedly. Fitting time excludes the search and reflects the preparation required for the specified training partition.

Inference uses a deterministic batch of 512 rows constructed by cycling through the relevant validation inputs. The batch size stays fixed even when fewer than 512 unique validation observations are available. Cycling reuses input rows to make a fixed-sized timing workload. Prediction errors retain their original assessment counts. Five warm-up calls precede 30 timed repetitions for each path. Sorting the thirty batch durations puts them in increasing order. Their median is the average of the fifteenth and sixteenth values. The median batch latency is divided by 512 and expressed as milliseconds per sample. For a hypothetical median batch duration of 102.4 ms, the reported per-sample quantity would be $102.4/512 = 0.2$ ms. For in-domain assessment, the five fold medians are summarized by their arithmetic mean and sample standard deviation. Exterior timing uses the same batching and repetitions for one final fit.

Projection timing includes positive-target construction, relative weighting, the weighted solve, mass conservation and non-negativity checks, and any projection interpolation for non-negativity. Complete projected inference includes both raw prediction and projection. All physical operators, returned component states, preprocessing, and metrics use double precision. TabNet’s internal tensors use single precision before its predictions return to the common physical-coordinate calculations. Pseudorandom generation and stochastic fitting use streams derived from seed 42. Deterministic coordinate-descent settings are retained for the multi-task models.

The timing measures describe the declared batch protocol. Complete prediction time includes the physical projection that accompanies each statistical prediction.

4.4 Results

4.4.1 Mechanistic reference-state checks

The learning domain contained 10,000 accepted steady states. Each exterior design contained 600 accepted states. Every attempted candidate passed the specified steady-state balance check, and every retained concentration was strictly positive. The realized exterior designs contained 70% one-input excursions and 30% two-input excursions. Table 4.6 reports the invariant residual separately from solver acceptance.

The largest invariant residual in the severe design was 2.91×10^{-10} , compared with the 10^{-10} tolerance imposed on projected predictions. The mechanistic acceptance rule

Table 4.6: Accepted mechanistic reference states and their maximum invariant residuals. The residual uses the fixed invariant-coordinate convention. Solver acceptance uses the component-balance residual.

Assessment domain	Attempted	Accepted	Maximum $\ A_{c_{out}} - A_{c_{in}}\ _2$
Learning domain	10,000	10,000	4.90×10^{-11}
Mild exterior inputs	600	600	5.67×10^{-11}
Severe exterior inputs	600	600	2.91×10^{-10}

used the maximum component-balance residual of 10^{-9} in the native component-unit-per-day coordinates. These two residuals have different definitions. The full-precision physical operators satisfied $\max_{h,p} |(A\nu^\top)_{hp}| = 4.25 \times 10^{-16}$. The small invariant residuals were consistent with numerical rather than structural departures from the specified balances.

4.4.2 Prediction within the learning domain

At $N = 10,000$, the multilayer perceptron had the lowest observed raw nRMSE of 0.150 ± 0.012 . CatBoost followed at 0.230 ± 0.006 , then LightGBM, XGBoost, and TabNet. Table 4.7 gives the paired results for all thirteen predictors. The comparisons use the same five outer folds.

Table 4.7: Component prediction error at $N = 10,000$. Raw and projected nRMSE values are five-fold means with sample standard deviations. The paired difference is projected minus raw and is calculated before rounding.

Surrogate	Raw nRMSE	Projected nRMSE	Difference
XGBoost	0.254 ± 0.006	0.255 ± 0.006	$+0.0006$
LightGBM	0.249 ± 0.008	0.250 ± 0.007	$+0.0007$
CatBoost	0.230 ± 0.006	0.229 ± 0.006	-0.0013
AdaBoost	0.525 ± 0.004	0.581 ± 0.028	$+0.0561$
Random Forest	0.718 ± 0.005	0.813 ± 0.020	$+0.0947$
Extra Trees	0.563 ± 0.005	0.608 ± 0.025	$+0.0445$
Support vector regression	0.364 ± 0.011	0.356 ± 0.011	-0.0082
k -nearest neighbors	0.607 ± 0.007	0.621 ± 0.021	$+0.0142$
Partial least squares	0.721 ± 0.006	0.745 ± 0.031	$+0.0239$
Multi-task Elastic Net	0.536 ± 0.008	0.516 ± 0.008	-0.0200
Multi-task Lasso	0.536 ± 0.008	0.516 ± 0.008	-0.0200
Multilayer perceptron	0.150 ± 0.012	0.149 ± 0.012	-0.0016
TabNet	0.272 ± 0.027	0.275 ± 0.024	$+0.0032$

Projection reduced the mean nRMSE for five predictors and increased it for eight.

The largest reductions were approximately 0.020 for each multi-task linear model. Random Forest had the largest increase of 0.095. The multilayer perceptron retained the lowest projected error of 0.149 ± 0.012 . Its macro-average R^2 increased from 0.977 to 0.978. Random Forest’s macro-average R^2 decreased from 0.484 to 0.332. However, its nMAE decreased from 0.497 to 0.467. Its squared-error and absolute-error scores therefore changed in opposite directions.

Figure 4.4 shows the three component-level measures together. The relative positions of raw and projected bars differ across measures. This is most apparent for the randomized tree ensembles and partial least squares.

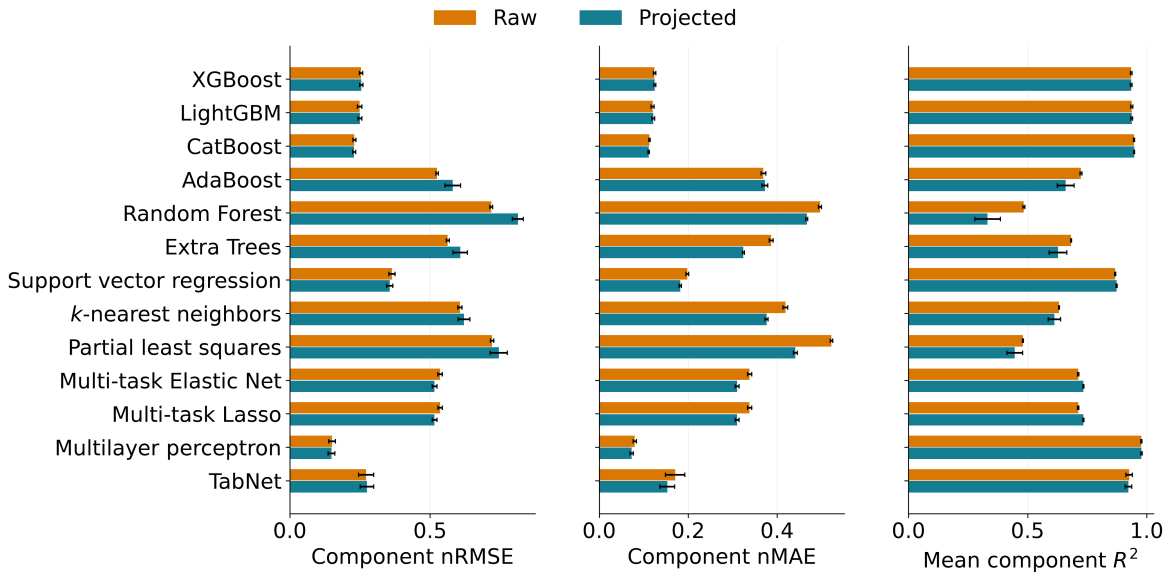


Figure 4.4: Raw and projected component accuracy at $N = 10,000$. Panels show nRMSE, nMAE, and the mean component R^2 . Error bars show one sample standard deviation across five outer folds.

4.4.3 Response to the amount of training data

Every predictor had lower raw nRMSE at $N = 10,000$ than at $N = 500$. Figure 4.5 shows the complete trajectories. XGBoost and LightGBM had the lowest observed mean raw errors at $N = 500$, with values of 0.435 and 0.439. The multilayer perceptron improved from 0.524 to 0.150. TabNet improved from 0.938 to 0.150 and showed the largest endpoint reduction among the predictors.

The lowest-error predictor changed with the data budget. The boosted trees performed well at the smallest design size. The multilayer perceptron attained the lowest error at the largest size. TabNet still improved from 0.298 at $N = 9050$ to 0.272 at $N = 10,000$. Its final reduction was approximately 0.0266. Its error continued to decrease at the largest evaluated sizes.

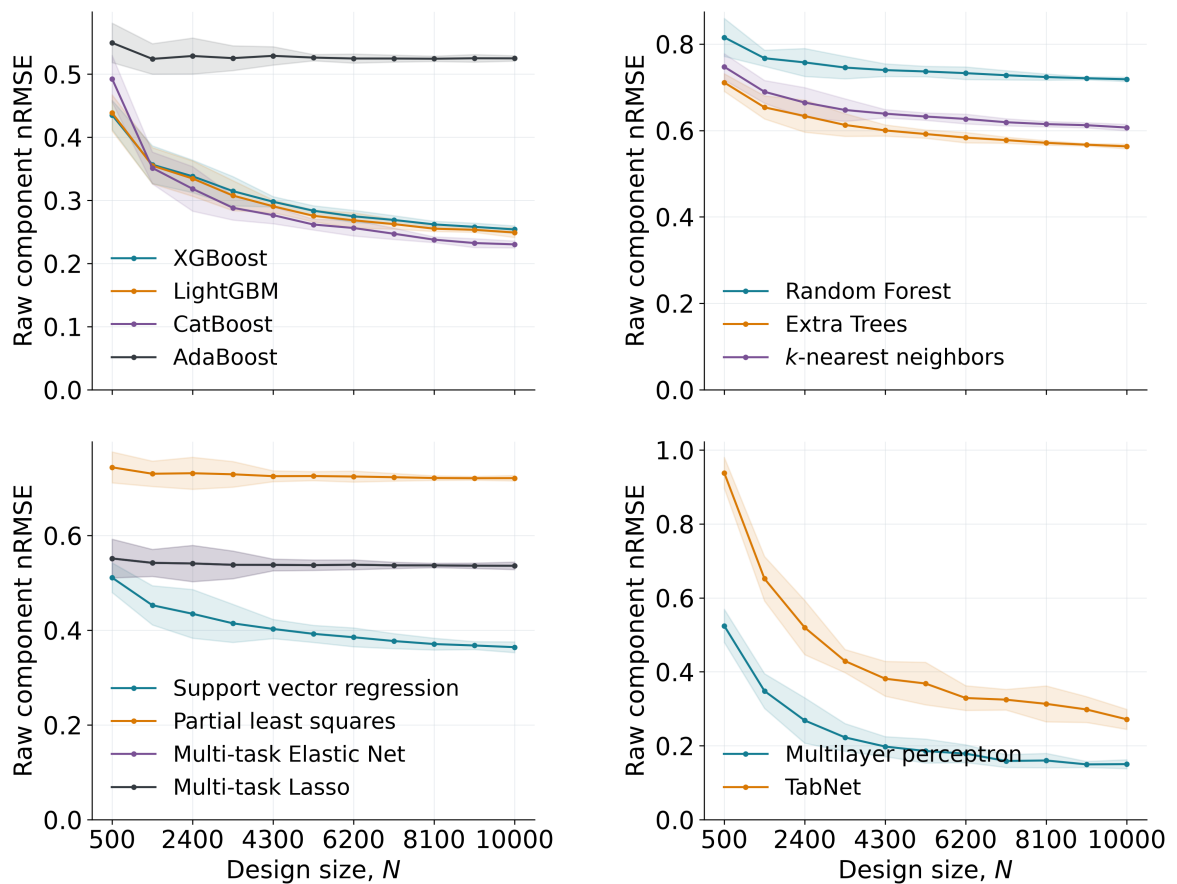


Figure 4.5: Raw component prediction error across the eleven nested design sizes. Separate panels group the predictors by their response representation. Lines show the five-fold mean nRMSE. Shaded regions show one sample standard deviation across folds.

The effect of projection also varied with design size. Both multi-task linear models benefited at all eleven sizes. Support vector regression benefited at every size from $N = 5250$ onward. The projection effect for the multilayer perceptron and TabNet changed sign across the size sequence. These trajectories were obtained with the prescribed fold-specific settings selected at the largest size.

Figure 4.6 separates this adjustment effect from the raw learning curves. A negative cell indicates lower error after projection. The persistent reductions for the multi-task models contrast with the mixed effects for the more flexible predictors.

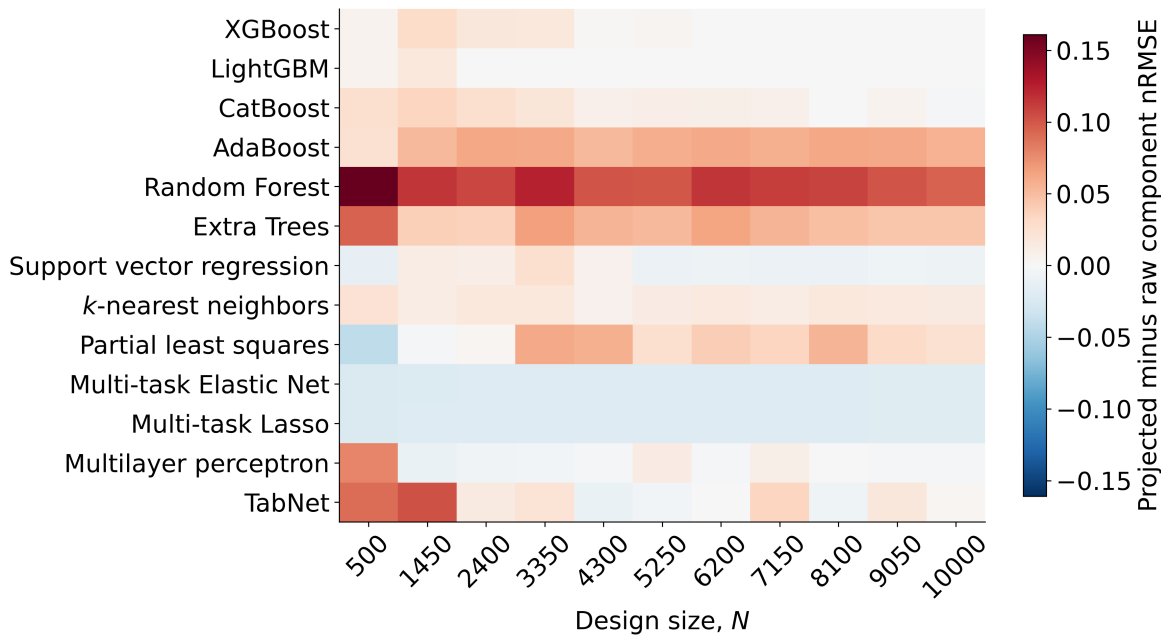


Figure 4.6: Projected minus raw mean component nRMSE for every predictor and design size. Negative values indicate improvement. Both prediction types use the same assessment folds and training-defined component scales.

4.4.4 Physical enforcement and component redistribution

Every raw assessment state failed the declared mass conservation tolerance at every design size. At $N = 10,000$, AdaBoost, Random Forest, Extra Trees, and the nearest-neighbor predictor nevertheless returned only non-negative raw concentrations. The other predictors had row-level non-negativity violation rates from 39.0% to 76.8%. Table 4.8 reports their rates and the size of the subsequent adjustment.

Projection produced no mass conservation or non-negativity failures in the 750,750 prediction instances evaluated across all thirteen predictors and eleven design sizes. These prediction instances include repeated assessment across nested sizes. The maximum projected invariant residual was 2.06×10^{-13} . Interpolation for non-negativity was activated in 2489 instances, or 0.332%. Random Forest, partial least squares, and

Table 4.8: Physical diagnostics at $N = 10,000$. Entries are five-fold means. Every raw mass conservation violation rate was 100%. Every projected mass conservation and non-negativity violation rate was zero.

Surrogate	Raw negative states (%)	Interpolation activated (%)	Mean d_c^{std}
XGBoost	39.72	0.00	0.300
LightGBM	38.95	0.00	0.301
CatBoost	59.06	0.00	0.242
AdaBoost	0.00	0.30	0.901
Random Forest	0.00	1.42	2.002
Extra Trees	0.00	0.88	1.528
Support vector regres- sion	76.78	0.01	0.363
k -nearest neighbors	0.00	0.25	1.192
Partial least squares	54.31	0.93	1.808
Multi-task Elastic Net	71.06	0.00	0.482
Multi-task Lasso	71.05	0.00	0.482
Multilayer perceptron	58.58	0.00	0.189
TabNet	60.48	0.04	0.545

Extra Trees accounted for 904, 574, and 566 activations, respectively. Together they accounted for 82.1% of the activations.

Figure 4.7 compares negative raw states with adjustment size and interpolation use. These are separate properties. Random Forest had no negative raw states but required the largest mean displacement and the most frequent interpolation at the largest design size.

At the largest design size, projection reduced component nRMSE in 188 of the 260 predictor–component comparisons. Both multi-task linear models improved eighteen of twenty components. The multilayer perceptron improved seventeen. CatBoost, partial least squares, and support vector regression each improved sixteen. In contrast, the heterotrophic biomass component X_H had higher error for every predictor. Random Forest’s fermentable-substrate component S_F increased from nRMSE 0.887 to 1.993. Its 142 interpolated predictions accounted for that component’s error increase, while the other 9858 predictions retained essentially the same squared error.

Figure 4.8 locates the component-level changes. The larger positive changes in a few components explain why an aggregate squared-error measure can increase even when most component scores improve.

The composite results also differed by quantity. Every predictor had lower TN and TP normalized root error after projection. Every predictor had higher COD and TSS normalized root error. The reported composites therefore responded differently to the

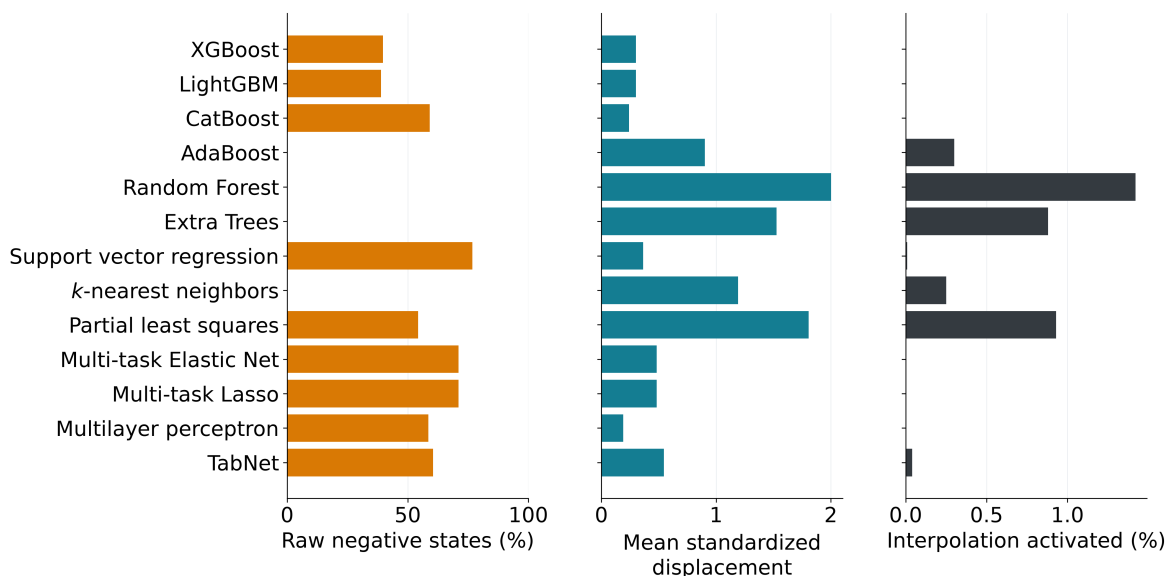


Figure 4.7: Physical diagnostics at $N = 10,000$. Panels show the fraction of raw states with negative components, mean standardized displacement, and interpolation activation. Values are five-fold means. All raw states failed the mass conservation tolerance, while every projected state passed both imposed physical checks.

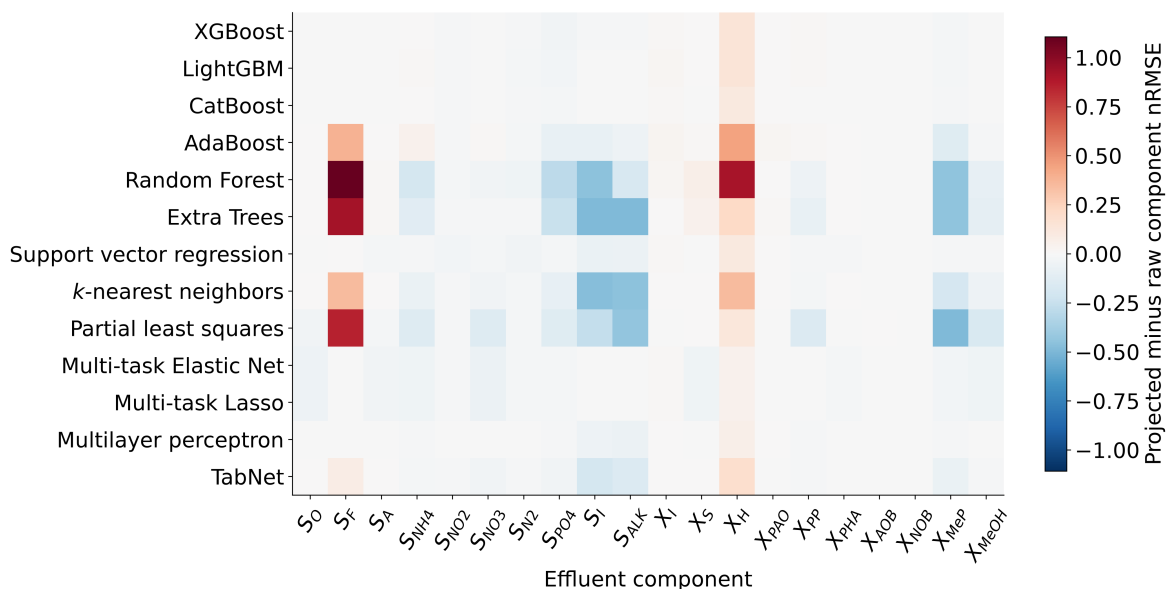


Figure 4.8: Projected minus raw component nRMSE at $N = 10,000$. Each cell is the difference between five-fold mean scores for one predictor and one component. Negative values indicate improvement. The same color limits apply to all cells.

same component adjustment.

Figure 4.9 presents the corresponding root errors in physical units. TN and TP improved while COD and TSS worsened for every predictor. The component projection thus changed the accuracy of different reporting quantities in different directions.

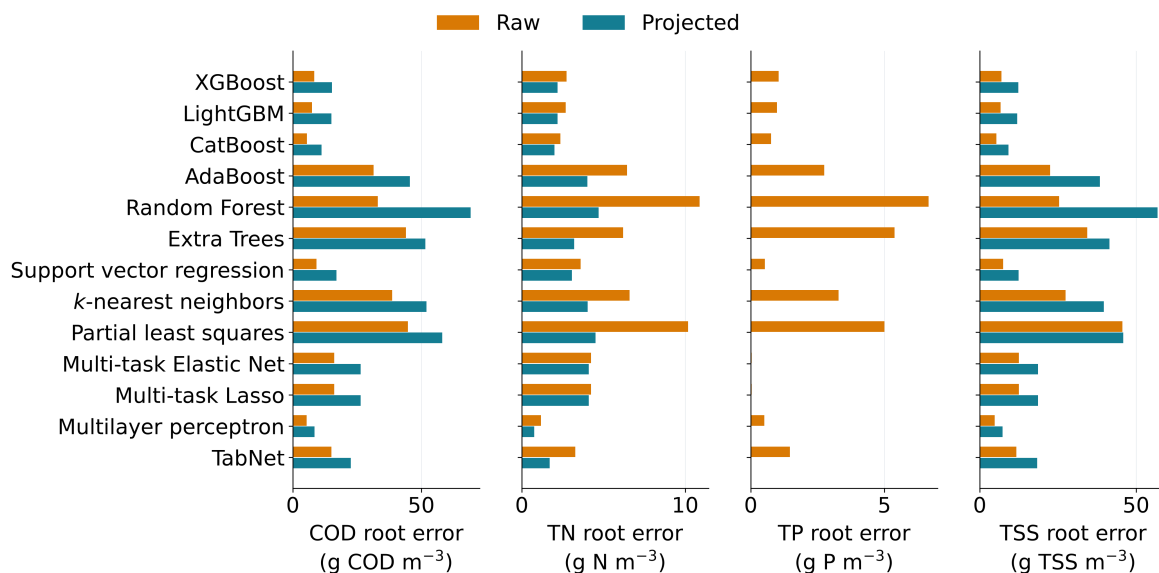


Figure 4.9: Raw and projected root mean squared errors for COD, TN, TP, and TSS at $N = 10,000$. Bars show means of five fold-specific root errors. Each panel uses the physical unit of its own reported quantity. These errors are distinct from the standardized component nRMSE.

4.4.5 Prediction beyond the learning domain

Both raw and projected error increased from mild to severe exterior inputs for every predictor. The multilayer perceptron had the lowest error in both designs. Its projected nRMSE was 0.251 for mild inputs and 0.521 for severe inputs. TabNet followed at 0.333 and 0.547. Table 4.9 reports the full comparison under the common full-domain normalization.

Projection reduced aggregate nRMSE in eleven of thirteen mild comparisons and nine of thirteen severe comparisons. AdaBoost and Random Forest worsened in both designs. Extra Trees and the nearest-neighbor predictor also worsened in the severe design. Component error decreased in 192 of 260 mild comparisons and 193 of 260 severe comparisons. Every predictor improved a majority of its component scores in each exterior design.

Figure 4.10 uses a common horizontal scale for the two severity levels. The larger severe-domain errors remain visible even for the most accurate predictors. Both prediction types had larger errors under severe exterior inputs.

Table 4.9: Component nRMSE beyond the learning domain. Each entry summarizes 600 accepted mechanistic states from one full-domain refit.

Surrogate	Mild exterior inputs		Severe exterior inputs	
	Raw	Projected	Raw	Projected
XGBoost	0.424	0.420	0.861	0.853
LightGBM	0.419	0.413	0.848	0.842
CatBoost	0.393	0.387	0.836	0.828
AdaBoost	0.692	0.711	1.036	1.139
Random Forest	0.908	0.996	1.242	1.489
Extra Trees	0.717	0.679	1.084	1.222
Support vector regression	0.555	0.538	0.989	0.948
k -nearest neighbors	0.760	0.732	1.113	1.147
Partial least squares	0.908	0.841	1.267	1.190
Multi-task Elastic Net	0.750	0.688	1.146	1.044
Multi-task Lasso	0.751	0.688	1.146	1.044
Multilayer perceptron	0.255	0.251	0.531	0.521
TabNet	0.336	0.333	0.563	0.547

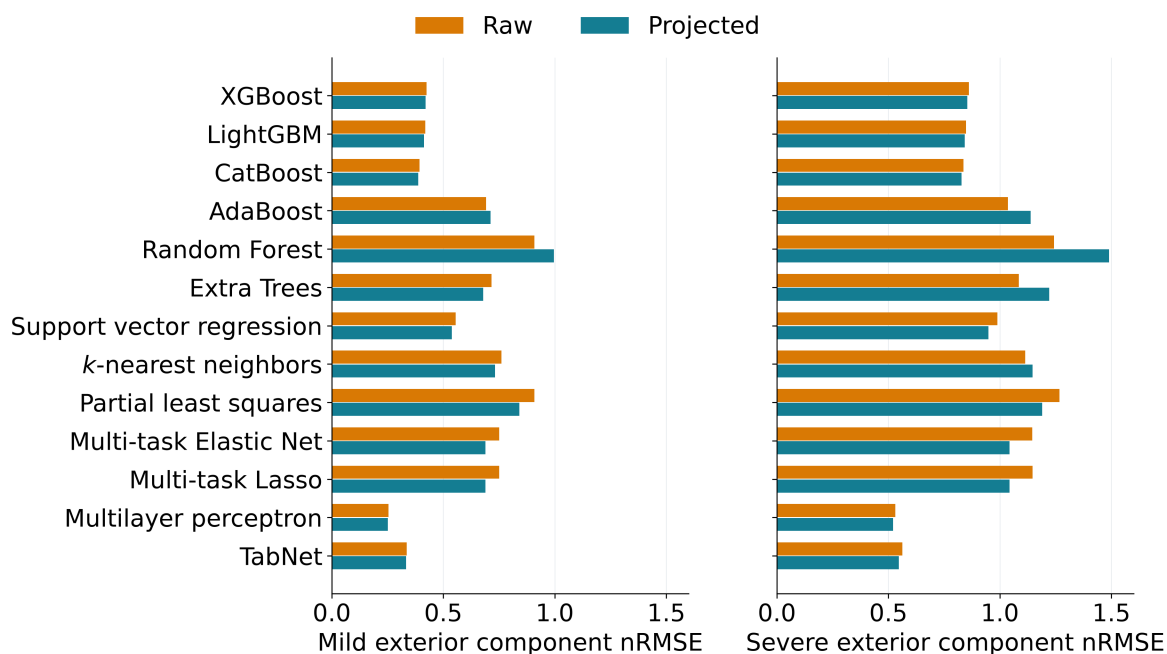


Figure 4.10: Raw and projected component nRMSE for mild and severe exterior inputs. Each severity includes 600 independently simulated reference states. Both panels use full-domain training scales and a common error-axis range.

All 15,600 exterior projected prediction instances passed the imposed physical checks. Their maximum invariant residual was 1.72×10^{-13} . Interpolation was activated in 68 instances, or 0.436%. Combining the interior and exterior assessments gave 766,350 projected prediction instances with no failures of the declared mass conservation and non-negativity checks. The increase in severe-domain error occurred despite this complete constraint compliance.

4.4.6 Fitting and prediction costs

Table 4.10 separates model preparation from repeated prediction. At the largest design size, mean fitting time ranged from approximately 0.024 s for Multi-task Elastic Net to 56.02 s for support vector regression. Mean search time ranged from 14.21 s for Multi-task Lasso to 11,082.77 s for support vector regression. These preparation costs were incurred before repeated use of the fitted predictor.

Figure 4.11 shows these differences on logarithmic axes. Projection dominated the inference cost of the fastest linear predictors. The two panels report preparation time and repeated prediction time separately.

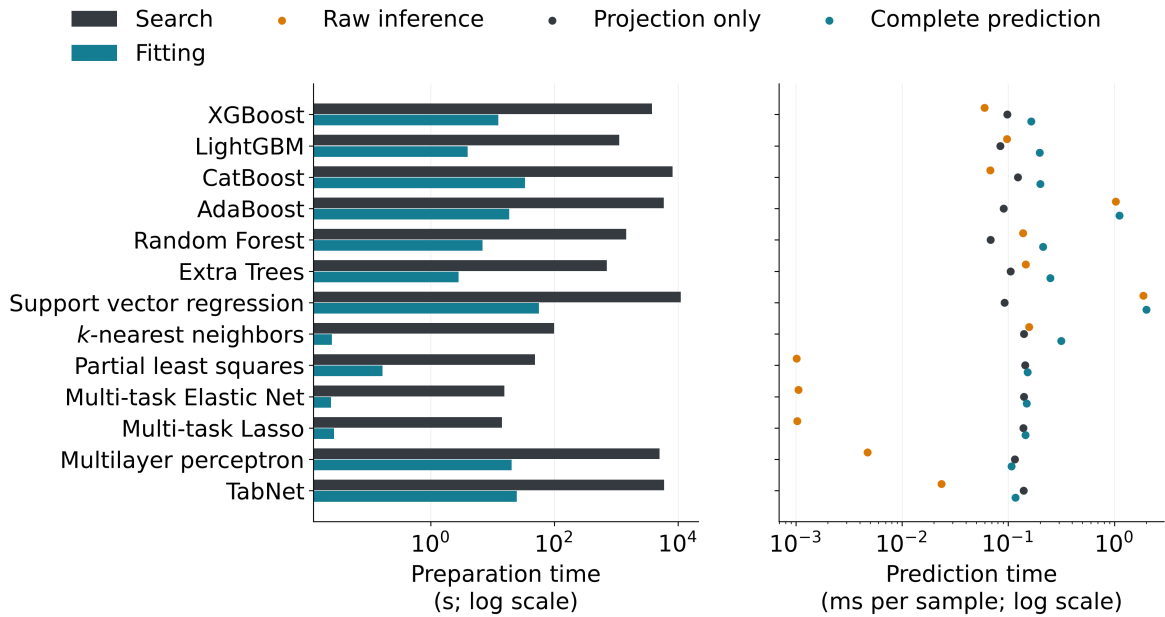


Figure 4.11: Model preparation and repeated prediction costs at $N = 10,000$. The left panel separates search and fitting in seconds. The right panel separates raw inference, projection alone, and complete prediction in milliseconds per sample under the fixed batch protocol. Both panels show five-fold means on logarithmic axes. Complete prediction was timed separately.

The multilayer perceptron had the lowest observed complete batch-normalized prediction cost of 0.1071 ms per sample. Support vector regression had the largest at 2.0008 ms. Projection alone ranged from 0.0681 to 0.1442 ms per sample. It dominated

Table 4.10: Five-fold mean computational costs at $N = 10,000$. Search and fitting are in seconds. Prediction costs are in milliseconds per sample under the fixed 512-row batch protocol. Complete prediction includes raw inference and projection.

Surrogate	Search	Fitting	Raw	Projection	Complete
XGBoost	3810.95	12.46	0.0597	0.0976	0.1649
LightGBM	1126.04	3.96	0.0971	0.0840	0.1974
CatBoost	8165.42	33.30	0.0677	0.1235	0.1997
AdaBoost	5894.14	18.55	1.0310	0.0901	1.1101
Random Forest	1452.33	6.89	0.1376	0.0681	0.2117
Extra Trees	705.21	2.83	0.1461	0.1053	0.2484
Support vector regression	11082.77	56.02	1.8736	0.0922	2.0008
k -nearest neighbors	99.34	0.03	0.1574	0.1401	0.3148
Partial least squares	48.75	0.17	0.0010	0.1442	0.1515
Multi-task Elastic Net	15.59	0.02	0.0011	0.1406	0.1493
Multi-task Lasso	14.21	0.03	0.0010	0.1387	0.1447
Multilayer perceptron	5016.24	20.46	0.0047	0.1151	0.1071
TabNet	5971.04	24.73	0.0235	0.1398	0.1170

the cost of the fastest raw linear and neural predictions. The complete multilayer perceptron time of 0.1071 ms was smaller than the separately measured projection-only time of 0.1151 ms. Each path has its own repeated batch measurements, and the complete prediction cost uses the directly measured complete path.

4.5 Discussion

4.5.1 Mass conservation and prediction error

Consider a hypothetical two-component prediction problem in which both components use one material basis. The first reference concentration is 6 and the second is 4 in arbitrary concentration units. Their sum is $6 + 4 = 10$. The raw prediction gives 8 for the first component and 3 for the second. Its sum is $8 + 3 = 11$, so its inventory residual is $11 - 10 = 1$. Both coordinates are non-negative, but the state is inconsistent with mass conservation under the specified equality.

Suppose a hypothetical projection for mass conservation lowers the first prediction by one unit and leaves the second unchanged. The projected sum is $7 + 3 = 10$. The component errors can be calculated directly,

$$\text{raw errors} = (8 - 6, 3 - 4) = (2, -1),$$

$$\text{projected errors} = (7 - 6, 3 - 4) = (1, -1),$$

$$\text{raw squared distance} = 2^2 + (-1)^2 = 5,$$

$$\text{projected squared distance} = 1^2 + (-1)^2 = 2.$$

The corresponding Euclidean distances are $\sqrt{5} \approx 2.236$ and $\sqrt{2} \approx 1.414$. Their values follow from comparison with the reference state. The mass conservation test instead uses the predicted sums. In compact component notation, the raw state is $(8, 3)^\top$, the reference state is $(6, 4)^\top$, and the chosen projected state is $(7, 3)^\top$. This calculation assesses feasibility and error on the same hypothetical prediction.

4.5.2 Statistical accuracy and physical admissibility

The paired assessment gave a direct counterexample to using low error as a physical acceptance rule. The multilayer perceptron had the lowest raw component error at the largest size, yet every raw state failed the mass conservation tolerance and 58.58% contained a negative component beyond tolerance. The positive outputs of the averaging-based predictors also failed mass conservation. Their responses combined cases with different influent-specific balance targets. Positivity of the contributing states did not preserve the target of a new case.

This result supports the distinction between learning a response and enforcing a constraint discussed by Kashinath et al. (2021) and Hansen et al. (2024). The projection of Kircher and Votsmeier (2025) addressed the second task across all thirteen predictor types. Its observed constraint compliance did not depend on the predictor having the lowest statistical error. A final physical acceptance check is therefore justified when a subsequent calculation requires the full non-negative component state and the adopted invariant balances.

Violation frequency and magnitude also require separate interpretation. A state was counted as failing when any one of twenty components crossed the tolerance. For CatBoost, raw non-negativity incidence increased from 12.8% at $N = 500$ to 59.1% at $N = 10,000$, even as aggregate error decreased. Its overall fit improved over the same interval. The negative predictions were concentrated in small substrate and nitrogen pools. A more accurate aggregate response can still contain small departures at a zero boundary.

4.5.3 Why enforcement can increase prediction error

The projection minimized relative movement from a positive provisional target. The accuracy score measured error relative to training variability. These were different objectives. The method also shortened one complete direction when a candidate crossed a concentration boundary. Neither operation was designed to minimize the assessed nRMSE.

The S_F result illustrates the effect of shared interpolation. Its column in the invariant operator was numerically zero. The mass conservation solve consequently left it essentially unchanged before interpolation. When another component required a shorter direction, S_F was also moved toward its much larger influent value. A safeguard

needed by one component could therefore worsen an otherwise acceptable prediction of another. Random Forest’s 142 affected full-size states provide a concrete example of this mechanism.

The X_H losses had a different explanation. Its relatively large concentrations made an absolute adjustment less costly under inverse-concentration weighting. Such an adjustment could satisfy the invariants while moving away from the mechanistic biomass target. The opposing changes in TN, TP, COD, and TSS further showed that mass conservation did not determine every reported composite. Near-alignment of the TP reporting vector with the phosphorus invariant explained its very small projected error. COD and TSS remained sensitive to the redistribution of organic and particulate components.

Both multi-task linear models benefited consistently. The largest aggregate losses occurred in predictors with larger displacements and more interpolation activations. Paired error, displacement, and activation summaries identified this trade-off more clearly than a feasible fraction alone.

4.5.4 Data support, extrapolation, and practical use

The strong small-data performance of boosted trees is consistent with the tabular comparisons discussed by Borisov et al. (2024) and Shwartz-Ziv and Armon (2022). The multilayer perceptron had the lowest error at the largest data budget. TabNet also continued to improve as more mechanistic states became available. Learning-curve interpretation requires the full trajectory and its validation design (Viering & Loog, 2023).

Exterior inputs increased error even when every projected state remained feasible. The changed error under exterior inputs is consistent with the role of input-domain coverage discussed by Shen et al. (2023) and Valente et al. (2025). The projection maintained the imposed component balances and non-negativity throughout both exterior designs.

The timing results supported rapid repeated evaluation after preparation. They also showed that a very fast raw predictor could spend most of its complete prediction time on physical enforcement. A deployment choice must therefore compare complete prediction cost rather than raw inference alone.

The evidence establishes reliable enforcement of the specified mass conservation and non-negativity conditions for the evaluated simulated steady states. It also establishes that the statistical cost of enforcement depends on the predictor, component, and assessment domain.

Chapter 5

Interpretable Regression with Checked Physical Predictions

5.1 Rationale

A component prediction is useful for engineering analysis when its dependence on the inputs can be examined and its material accounting can be checked. These requirements address different questions. A transparent regression equation explains how the predicted state depends on operating conditions and influent loads. A mass conservation test determines whether that state preserves the inventories represented by the reaction network. A non-negativity test determines whether each predicted concentration is physically meaningful. An explicit equation can fail either physical test. A physically projected prediction can also remain difficult to interpret if the underlying regression is opaque.

Invariant-constrained second-order regression (ICSOR) combines an explicit polynomial driver, learned dependence among effluent components, and checked prediction. A polynomial driver is the fitted expression containing inputs, squares, and pairwise products before component coupling is applied. In this chapter, deployment means evaluating the fitted model at a new influent and operating setting and returning a physically checked component state. The driver retains separate terms for operating conditions, influent composition, curvature, and interactions. The coupling relation expresses how the driver contributes to the predicted component state. Deployment then checks that state and applies a constrained projection when necessary. Together, these stages provide an inspectable regression and a checked effluent state.

The preference for an inspectable regression follows the argument of Rudin (2019) that transparent model structures can support decisions that require scrutiny. It also follows the broader development of models that combine learned relations with physical structure discussed by Shen et al. (2023). The fitted relations are available for examination, comparison with process knowledge, and testing across operating conditions.

Activated sludge nutrient removal provides a strong reason to retain interactions. Carbon availability influences denitrification and the activity of organisms that store phosphorus. Oxygen availability influences both carbon oxidation and nitrification. Wentzel et al. (1992) describe these connected pathways, and Guerrero et al. (2011) examine the role of carbon source in the competition between phosphorus-storing organisms and denitrifiers. A regression that represents an aeration effect independently of loading can overlook such dependencies. An interaction term can describe an association between an operating action and a loading condition while keeping that association visible.

The prediction target is the effluent component vector $c_{out} \in \mathbb{R}^F$. The adopted process basis has $F = 20$ components and $P = 28$ reactions. The operating vector $u \in \mathbb{R}^{M_{op}}$ has $M_{op} = 2$ entries for HRT and aeration level. The full influent vector is $c_{in} \in \mathbb{R}^F$. Component concentrations retain their physical units. The four reported composites are obtained from a fixed map $I_{comp} \in \mathbb{R}^{4 \times F}$. That map is applied after component prediction.

5.2 Theory

5.2.1 Mass conservation in the adopted state space

The link between a component balance and a regression constraint can first be seen in a hypothetical conversion of one component into another. Suppose that one unit of component 1 is converted into one unit of component 2. The two components are expressed in a common inventory unit for this illustration. Their changes satisfy

$$c_{out,1} - c_{in,1} = -\xi, \quad c_{out,2} - c_{in,2} = \xi. \quad (5.1)$$

Adding these equations eliminates the unknown conversion amount. The relation available to the regression is $c_{out,1} + c_{out,2} = c_{in,1} + c_{in,2}$. With an illustrative influent $(6, 4)^\top$ and conversion amount 2, the effluent is $(4, 6)^\top$. Both totals are 10. Many other non-negative pairs also have that total. The balance defines a set of regression outputs with the same total inventory.

The same calculation can be written using the small matrices

$$\nu = \begin{bmatrix} -1 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad A\nu^\top = 1(-1) + 1(1) = 0. \quad (5.2)$$

These hypothetical matrices use reduced dimensions to make each operation visible. The adopted reactor uses the stated 20-component basis, reaction matrix, and operating domain.

The stoichiometric matrix $\nu \in \mathbb{R}^{P \times F}$ records reactions as rows and components as columns. Let $V_0 \in \mathbb{R}^{F \times K}$ span its right null space, where $K = 5$. The invariant operator is

$$A = V_0^\top \in \mathbb{R}^{K \times F}, \quad \nu V_0 = 0, \quad A\nu^\top = 0. \quad (5.3)$$

The columns of V_0 are independent, so A has full row rank. With an orthonormal null-space basis, $AA^\top = I_K$. A different full-rank basis describes the same mass conservation conditions, although its row scaling changes the numerical size of a soft invariant penalty.

For the control volume and boundary accounting represented by the adopted process model, a reaction-induced change has the form

$$c_{out} - c_{in} = \nu^\top \xi, \quad \xi \in \mathbb{R}^P. \quad (5.4)$$

The reaction-progress vector ξ expresses the net contribution of the reactions in concentration-equivalent units. Multiplication by A removes that unknown contribution and gives

$$A(c_{out} - c_{in}) = 0. \quad (5.5)$$

This construction uses the mass conservation logic of the photochemical surrogate framework of Sturm and Wexler (2020). Sturm and Wexler (2022) further show how a stoichiometric transformation can connect learned reaction information to state changes satisfying mass conservation. The same linear algebra applies here, but the component combinations used for mass conservation are those of the activated sludge reaction network.

Equation (5.5) uses the modeled boundary in Equation (5.4). External gas transfer, chemical dosing, solids withdrawal, and other boundary fluxes enter the balance where they occur. With an explicit boundary contribution s , the relation is $A(c_{out} - c_{in} - s) = 0$.

For a new influent state, the component set used by checked deployment is

$$\mathcal{F}(c_{in}) = \{c \in \mathbb{R}^F \mid Ac = Ac_{in}, c \geq \mathbf{0}\}. \quad (5.6)$$

When $c_{in} \geq \mathbf{0}$, this set contains c_{in} and is therefore nonempty. Its members satisfy the modeled linear invariants and component non-negativity within the declared boundary.

The reported composites provide another reason to work in component space. Distinct component states can have the same COD, TN, TP, and TSS values. A negative component can also be hidden by positive contributions to an aggregate. When I_{comp} is entrywise non-negative, a non-negative component state produces non-negative composites. The reverse implication does not generally hold. Mass conservation and non-negativity are therefore tested before the component state is collapsed into the four reported quantities.

5.2.2 A partitioned second-order driver

Separating operation from loading

Operating conditions and influent composition play different engineering roles. HRT and aeration define aspects of the reactor environment. Influent components define

the material presented to that environment. This distinction is consistent with the operating-sensitivity analysis of de Araujo et al. (2013) and the separation of operational and influent uncertainty examined by Machado et al. (2014).

Start with two hypothetical operating entries u_1, u_2 and two influent entries c_1, c_2 . A second-order model uses the individual entries, their squares, and their pairwise products. The operating products are $u_1^2, u_1u_2, u_2u_1, u_2^2$. The feature vector keeps both orders as an accounting convention, with $u_1u_2 = u_2u_1$.

For two column vectors, the Kronecker operation multiplies the entire second vector by the first entry of the first vector, then by its second entry. Thus

$$\begin{aligned} u \otimes u &= \begin{bmatrix} u_1u_1 \\ u_1u_2 \\ u_2u_1 \\ u_2u_2 \end{bmatrix}, \quad c \otimes c = \begin{bmatrix} c_1c_1 \\ c_1c_2 \\ c_2c_1 \\ c_2c_2 \end{bmatrix}, \\ u \otimes c &= \begin{bmatrix} u_1c_1 \\ u_1c_2 \\ u_2c_1 \\ u_2c_2 \end{bmatrix}. \end{aligned} \tag{5.7}$$

For the illustrative values $u = (2, 3)^\top$ and $c = (4, 5)^\top$, these three blocks are respectively $(4, 6, 6, 9)^\top$, $(16, 20, 20, 25)^\top$, and $(8, 10, 12, 15)^\top$. Including the baseline 1 and the four first-order entries gives 17 ordered features in this reduced example. The same construction applies to the larger adopted input vector.

The feature vector retains both roles through the ordered expansion

$$\phi(u, c_{in}) = \begin{bmatrix} 1 \\ u \\ c_{in} \\ u \otimes u \\ c_{in} \otimes c_{in} \\ u \otimes c_{in} \end{bmatrix} \in \mathbb{R}^D, \quad D = 1 + M_{op} + F + M_{op}^2 + F^2 + M_{op}F. \tag{5.8}$$

The symbol $\otimes$ denotes the Kronecker product. The adopted dimensions give $D = 467$. The ordered expansion includes both u_iu_j and u_ju_i , and both $c_{in,i}c_{in,j}$ and $c_{in,j}c_{in,i}$. These pairs are equal as numerical features. Their interpretation requires the symmetry convention developed below.

Multiplication by regression coefficients can also be read one entry at a time. For an illustrative first driver, take the baseline coefficient as 1, operating coefficients as $(2, -1)$, and influent coefficients as $(0.5, 0)$. Let its operating quadratic row be $(0.1, 0.1, 0.1, 0)$, influent quadratic row be $(0, 0, 0, -0.05)$, and operating-load row be

$(0, 0, 0.25, 0)$. Applying these rows to the numerical feature blocks gives

$$\begin{aligned}
 r_1 = 1 &+ \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\
 &+ \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \\ 6 \\ 9 \end{bmatrix} \\
 &+ \begin{bmatrix} 0 & 0 & 0 & -0.05 \end{bmatrix} \begin{bmatrix} 16 \\ 20 \\ 20 \\ 25 \end{bmatrix} \\
 &+ \begin{bmatrix} 0 & 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 8 \\ 10 \\ 12 \\ 15 \end{bmatrix} \\
 &= 1 + 1 + 2 + 1.6 - 1.25 + 3 = 7.35.
 \end{aligned} \tag{5.9}$$

Each row-vector multiplication is a sum of coefficient times feature. It is the same operation as an ordinary linear regression, but the supplied features include nonlinear products. A second hypothetical driver $r_2 = -u_1 + u_2 + 0.2c_2$ would equal 2 at the same inputs. The coupled relation converts the two driver values into predicted concentrations.

Before grouping the coefficients into matrices, the first driver in this illustration can be written as

$$r_1 = 1 + 2u_1 - u_2 + 0.5c_1 + 0.1u_1^2 + 0.2u_1u_2 - 0.05c_2^2 + 0.25u_2c_1. \tag{5.10}$$

The coefficient 0.2 of u_1u_2 is the sum of the two ordered coefficients 0.1. In a general row, the scalar calculation is $r_f = \sum_{\ell=1}^D B_{f\ell}\phi_\ell$. Carrying out this sum for all components gives the driver $r \in \mathbb{R}^F$ through

$$r(u, c_{in}) = B\phi(u, c_{in}), \quad B \in \mathbb{R}^{F \times D}. \tag{5.11}$$

The driver is an intermediate regression quantity supplied to the coupled prediction relation. Its coefficients have no sign restriction, so the regression can represent both increasing and decreasing effects. Non-negativity is imposed on component states where it has a physical meaning. The driver block form is

$$\begin{aligned}
 r(u, c_{in}) = &b + W_u u + W_{in} c_{in} + \Theta_{uu}(u \otimes u) \\
 &+ \Theta_{cc}(c_{in} \otimes c_{in}) + \Theta_{uc}(u \otimes c_{in}).
 \end{aligned} \tag{5.12}$$

Here $b \in \mathbb{R}^F$ is a baseline term. The matrices $W_u \in \mathbb{R}^{F \times M_{op}}$ and $W_{in} \in \mathbb{R}^{F \times F}$ contain first-order operating and influent terms. The matrices $\Theta_{uu} \in \mathbb{R}^{F \times M_{op}^2}$ and

$\Theta_{cc} \in \mathbb{R}^{F \times F^2}$ contain curvature within each input group. The matrix $\Theta_{uc} \in \mathbb{R}^{F \times M_{op} F}$ contains operating-load interactions.

Illustrative operating and loading effects

Consider an illustrative driver for one component with aeration level a and influent ammonium concentration n . A term of the form

$$r_f = \beta_0 + \beta_a a + \beta_n n + \beta_{aa} a^2 + \beta_{an} an \quad (5.13)$$

has an aeration derivative $\partial r_f / \partial a = \beta_a + 2\beta_{aa} a + \beta_{an} n$. The effect described by this derivative changes with both aeration level and ammonium loading. The derivative of this hypothetical polynomial combines the linear aeration term, curvature, and the aeration-ammonium interaction.

The same reasoning applies to simultaneous influent loads. A carbon-ammonium interaction allows the modeled association with carbon availability to depend on nitrogen loading. A phosphorus-storage interaction allows the modeled association with dissolved phosphate to depend on a particulate storage pool. These terms provide a transparent approximation to a nonlinear response involving the microbial pathways described by Henze et al. (1999).

The second-order library represents smooth local curvature and pairwise input interactions. Explicit nonlinear libraries make candidate response structures available for inspection, as illustrated by Brunton et al. (2016).

Symmetry and the meaning of cross terms

For a two-entry vector, direct multiplication gives

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = q_{11}x_1^2 + (q_{12} + q_{21})x_1x_2 + q_{22}x_2^2. \quad (5.14)$$

Only the sum of the two off-diagonal coefficients enters this polynomial. Replacing them by their average leaves that sum unchanged. For example,

$$Q = \begin{bmatrix} 0.1 & 0.3 \\ -0.1 & 0 \end{bmatrix}, \quad \frac{Q + Q^\top}{2} = \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0 \end{bmatrix}. \quad (5.15)$$

At $x = (2, 3)^\top$, the first matrix gives $0.4 + 1.8 - 0.6 = 1.6$. The symmetric matrix gives $0.4 + 0.6 + 0.6 = 1.6$. The asymmetric coefficients cannot be distinguished from these predictions. Their squared sums are different. The original matrix has squared sum $0.01 + 0.09 + 0.01 = 0.11$, while its symmetric part has squared sum $0.01 + 0.01 + 0.01 = 0.03$. The removed part has squared sum 0.08 and contributes no prediction. This is why a positive coefficient penalty favors the symmetric coefficient matrix.

For component f , reshape the corresponding rows of Θ_{uu} and Θ_{cc} into square matrices $Q_{uu}^{(f)}$ and $Q_{cc}^{(f)}$. A quadratic term can then be written as $x^\top Qx$. For any real square matrix,

$$x^\top Qx = x^\top \text{sym}(Q)x, \quad \text{sym}(Q) = \frac{Q + Q^\top}{2}. \quad (5.16)$$

The skew-symmetric part contributes nothing to the prediction. It can nevertheless increase the coefficient norm because

$$\|Q\|_F^2 = \|\text{sym}(Q)\|_F^2 + \|\text{skew}(Q)\|_F^2. \quad (5.17)$$

Replacing each square slice by its symmetric part preserves every fitted value and cannot increase a positive ridge penalty. The projection is applied after each driver update. With a strictly positive driver ridge weight, the exact minimum-norm solution has no skew-symmetric part.

Symmetry also determines how a coefficient display is read. If $q_{ij} = q_{ji}$, the two displayed off-diagonal cells together contribute $2q_{ij}x_ix_j$. A displayed cell is therefore half the coefficient of the corresponding unordered cross term. For an illustrative matrix with $q_{12} = q_{21} = 0.3$, the polynomial contribution is $0.6x_1x_2$. Reading either cell as the entire interaction coefficient would halve its contribution.

Each component has $M_{op}(M_{op} + 1)/2 = 3$ distinct operating quadratic terms and $F(F + 1)/2 = 210$ distinct influent quadratic terms. Together with the baseline, linear terms, and operating-load interactions, the library has 276 distinct polynomial terms per component. These counts define the structural dimensions of the polynomial library.

5.2.3 Learned coupling among effluent components

Consider two hypothetical component equations in which each channel has a coupling coefficient 0.2 to the other channel. With driver values 4 and 2, their equations are

$$c_1 = 4 + 0.2c_2, \quad c_2 = 2 + 0.2c_1. \quad (5.18)$$

Substituting the second equation into the first gives $c_1 = 4 + 0.2(2 + 0.2c_1)$. Moving the remaining c_1 term to the left gives $0.96c_1 = 4.4$. Therefore $c_1 = 55/12$ and $c_2 = 35/12$. The solution is approximately $(4.5833, 2.9167)^\top$. These hypothetical values demonstrate the arithmetic of a simultaneous regression relation.

Moving the coupled terms to the left makes the same calculation a matrix solve,

$$\Gamma = \begin{bmatrix} 0 & 0.2 \\ 0.2 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & -0.2 \\ -0.2 & 1 \end{bmatrix}, \quad R \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}. \quad (5.19)$$

The determinant of R is $1 - 0.2^2 = 0.96$. Its inverse is

$$R^{-1} = \frac{1}{0.96} \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix}. \quad (5.20)$$

Multiplying this inverse by the driver gives the same two scalar solutions. The inverse also explains why a change in one driver can change both predicted components.

The regression includes a matrix $\Gamma \in \mathbb{R}^{F \times F}$ that describes fitted dependence among the effluent components. Define

$$R = I_F - \Gamma, \quad Rc = r(u, c_{in}). \quad (5.21)$$

Equivalently, the equation for component f is $c_f = r_f + \sum_{j \neq f} \Gamma_{fj} c_j$. This form separates the input-dependent driver from dependence among the predicted component channels. The diagonal of Γ is fixed to zero to avoid a redundant self-coupling coefficient.

The sign of Γ_{fj} describes reinforcement or compensation within the fitted simultaneous relation. Activated sludge reactions involve several components at once, and the stoichiometric matrix specifies those transformations. The coupling matrix is learned from state patterns. Its entries describe fitted dependence associated with correlated transformations, loading patterns, and the allocation of effects between the driver and coupling blocks.

When R is nonsingular, the raw component prediction is $R^{-1}r$. Multiplication by R^{-1} redistributes or amplifies the response represented by the driver. For interpretation, the driver, coupling matrix, and complete raw mapping are examined together.

The adopted coupling constraints require

$$\Gamma_{ff} = 0, \quad |\Gamma_{fj}| \leq \gamma_{abs} \quad (f \neq j), \quad \kappa_2(I_F - \Gamma) \leq \kappa_{max}. \quad (5.22)$$

The condition number is the ratio of the largest to smallest singular value of the coupled-system matrix. A nearly singular matrix can amplify small driver perturbations and numerical errors. The condition check controls the complete matrix separately from the bounds on individual entries.

The small symmetric matrix in Equation (5.19) illustrates this measure without a large singular-value calculation. In the direction $(1, 1)^\top$, multiplication by R scales the vector by 0.8. In the direction $(1, -1)^\top$, it scales the vector by 1.2. These orthogonal directions give singular values 0.8 and 1.2, so the condition number is $1.2/0.8 = 1.5$.

If the two illustrative couplings were 0.99, the same directions would have scales 0.01 and 1.99. The condition number would be 199. A driver error of magnitude ϵ in each component along the first direction would produce a state error of 100ϵ in each component. This hypothetical choice illustrates amplification at a coupling magnitude of 0.99, above the adopted bound.

Even a small entry bound can permit singularity in a larger matrix. In a hypothetical four-component matrix with every off-diagonal entry equal to $1/3$, every row sum of Γ is one. Hence $\Gamma \mathbf{1} = \mathbf{1}$ and $(I - \Gamma)\mathbf{1} = 0$. Each entry is smaller than the adopted bound 0.368, but the coupled-system matrix is singular. This example explains why entry bounds and the condition-number test perform separate tasks.

Shrinking a candidate coupling matrix toward zero can provide a safe candidate because $I_F - 0 = I_F$. A zero coupling matrix gives a condition number of one. A

prescribed $\kappa_{max} > 1$ therefore permits this fallback. The nonlinear condition-number requirement defines a generally nonconvex admissible coupling set. It is evaluated after the convex box-constrained estimation problem proposes a coupling update.

5.3 Methodology

5.3.1 Fitting in prediction space

The coupled training objective

The fitting problem adjusts coefficients to approximate the mechanistic effluent concentrations. It also uses auxiliary fitted states, which are concentration variables optimized for the training cases. These variables help express fitting constraints and penalties. They differ from the component predictions calculated for a new influent after fitting.

The squared error of one prediction is the square of its difference from the reference. For several components and observations, fitting adds these squared differences. A compact matrix norm represents this sum. For any matrix M with entries M_{ij} ,

$$\|M\|_F^2 = \sum_i \sum_j M_{ij}^2. \quad (5.23)$$

For example, a residual matrix $\begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$ has squared norm $1^2 + (-2)^2 + 3^2 + 0^2 = 14$.

The norm is $\sqrt{14}$, while a least-squares objective uses 14. Squaring prevents positive and negative errors from canceling when they are added.

Let i identify an observation, f a component, k an invariant, and ℓ a feature. The entry $\hat{c}_{if}$ is the fitted concentration, while $\phi_{i\ell}$ is the corresponding feature entry. Before grouping the five training contributions into norms, their scalar form is

$$\begin{aligned} J = & \sum_{i=1}^N \sum_{f=1}^F (c_{out,if} - \hat{c}_{if})^2 \\ & + \lambda_{inv} \sum_{i=1}^N \sum_{k=1}^K \left[\sum_{f=1}^F A_{kf} (\hat{c}_{if} - c_{in,if}) \right]^2 \\ & + \lambda_{sys} \sum_{i=1}^N \sum_{f=1}^F \left[\hat{c}_{if} - \sum_{j=1}^F \Gamma_{fj} \hat{c}_{ij} - \sum_{\ell=1}^D B_{f\ell} \phi_{i\ell} \right]^2 \\ & + \lambda_B \sum_{f=1}^F \sum_{\ell=1}^D B_{f\ell}^2 + \lambda_\Gamma \sum_{f=1}^F \sum_{j=1}^F \Gamma_{fj}^2. \end{aligned} \quad (5.24)$$

The second contribution first computes an inventory residual by adding component changes with their invariant weights. It then squares that residual. The third contribution compares the fitted component state, after coupling, with the driver generated

by the inputs. The last two sums penalize coefficient magnitude rather than prediction residuals.

The dimensions of the matrices show how these sums are assembled. The entry in row i , column k of $(\hat{C} - C_{in})A^\top$ is the inventory residual in the second sum. The entry in row i , column f of $\hat{C}R^\top$ is $\hat{c}_{if} - \sum_j \Gamma_{fj}\hat{c}_{ij}$. The transpose appears because observations are stored as rows, while the single-observation equation $Rc = r$ uses column vectors.

For N training observations, $\Phi \in \mathbb{R}^{N \times D}$ contains feature vectors as rows. The matrices C_{in} and C_{out} contain influent and reference effluent states in the same row convention. An auxiliary fitted-state matrix $\hat{C} \in \mathbb{R}_+^{N \times F}$ separates the predicted training states from the regression coefficients. The training criterion is

$$\begin{aligned} J(B, \Gamma, \hat{C}) = & \|C_{out} - \hat{C}\|_F^2 + \lambda_{inv} \|(\hat{C} - C_{in})A^\top\|_F^2 \\ & + \lambda_{sys} \|\hat{C}(I_F - \Gamma)^\top - \Phi B^\top\|_F^2 \\ & + \lambda_B \|B\|_F^2 + \lambda_\Gamma \|\Gamma\|_F^2. \end{aligned} \quad (5.25)$$

The estimation problem minimizes this criterion subject to non-negative fitted training states, zero coupling diagonal, coupling box bounds, and the conditioning safeguard. All penalty weights are non-negative. The adopted ridge weights and coupled-system weight are strictly positive.

The first term measures component prediction error. The second term discourages disagreement with the modeled invariants. The third term encourages agreement between the fitted states and the coupled regression equation. The final two terms limit coefficient magnitude. The non-negative fitted training state balances prediction error, invariant mismatch, and coupled-system mismatch through these weighted terms.

An illustrative one-observation calculation makes the five contributions concrete. Take $c_{out} = (3, 1)^\top$, $c_{in} = (2, 2)^\top$, $\hat{c} = (2.5, 1.2)^\top$, and $A = [1 \ 1]$. Use the hypothetical coupling matrix in Equation (5.19). For a single baseline feature, take $B = (2, 1)^\top$, so the driver is $(2, 1)^\top$. The five unweighted contributions are

$$\begin{aligned} \text{prediction error} &= (3 - 2.5)^2 + (1 - 1.2)^2 = 0.29, \\ \text{inventory error} &= (2.5 + 1.2 - 2 - 2)^2 = 0.09, \\ \text{system error} &= (2.5 - 0.2(1.2) - 2)^2 \\ &\quad + (1.2 - 0.2(2.5) - 1)^2 = 0.1576, \\ \text{driver squared sum} &= 2^2 + 1^2 = 5, \\ \text{coupling squared sum} &= 0.2^2 + 0.2^2 = 0.08. \end{aligned} \quad (5.26)$$

With illustrative weights $\lambda_{inv} = 2$, $\lambda_{sys} = 3$, $\lambda_B = 0.1$, and $\lambda_\Gamma = 0.5$, the objective is $0.29 + 2(0.09) + 3(0.1576) + 0.1(5) + 0.5(0.08) = 1.4828$. These hypothetical weights illustrate the relative cost assigned to each discrepancy. Table 5.1 gives the adopted numerical fitting weights.

The auxiliary state is a variable in the fitting problem. The raw prediction for a new input is obtained by solving the learned coupled relation. Residual penalties guide this fit using the general penalty principle of physics-informed training developed by Raissi et al. (2019). Checked deployment then enforces mass conservation and non-negativity on the returned state.

The objective is expressed in physical component coordinates. Components have different units and numerical scales, so the magnitudes of their residuals and coefficients reflect those conventions. The four penalty weights also depend on the fixed feature and invariant scaling. Consistent rescaling transforms the features, coefficients, invariant operator, reporting map, and penalty weights together. Pinheiro et al. (2025) discuss the influence of feature scaling on learning. For physical interpretation, unit consistency is as necessary as numerical conditioning.

Deterministic block updates

The term $\hat{C}\Gamma^\top$ makes the joint objective nonconvex. Holding two variable blocks fixed gives a simpler problem in the remaining block. The fitting procedure updates the driver, coupling matrix, and fitted states in that order within each cycle.

This distinction can be checked with the reduced residual $(1 - \gamma z)^2$, where γ is a coupling and z a fitted component. If z is fixed, this is a quadratic function of γ . If γ is fixed, it is a quadratic function of z . It is not jointly convex. At $(\gamma, z) = (0.2, 5)$ and $(0.1, 10)$, the residual is zero. At the midpoint $(0.15, 7.5)$, it is $(1 - 1.125)^2 = 1/64$. A convex function could not exceed the average of the two zero endpoint values at their midpoint. The product between a fitted state and a coupling coefficient accounts for the nonconvex joint objective.

For fixed Γ and $\hat{C}$, the driver update is a ridge-regularized linear least-squares problem. Define $T = \hat{C}R^\top$. For one output row, write its coefficient column as β and its target column from T as t . The part of the objective that depends on this row is

$$\lambda_{sys} \sum_{i=1}^N \left(t_i - \sum_{\ell=1}^D \phi_{i\ell} \beta_\ell \right)^2 + \lambda_B \sum_{\ell=1}^D \beta_\ell^2. \quad (5.27)$$

Differentiate with respect to one coefficient β_h . The derivative of the squared residual contributes its residual times the feature ϕ_{ih} , while the ridge term contributes $2\lambda_B \beta_h$. Setting the derivative to zero gives

$$\lambda_{sys} \sum_{i=1}^N \phi_{ih} \left(\sum_{\ell=1}^D \phi_{i\ell} \beta_\ell - t_i \right) + \lambda_B \beta_h = 0. \quad (5.28)$$

Writing this equation for every coefficient gives

$$(\lambda_{sys} \Phi^\top \Phi + \lambda_B I_D) \beta = \lambda_{sys} \Phi^\top t. \quad (5.29)$$

The matrix $\Phi^\top \Phi$ contains sums of products of feature columns. The vector $\Phi^\top t$ contains sums of feature times target. Adding the positive ridge weight to the diagonal prevents a zero-curvature direction when feature columns are duplicated.

For a hypothetical calculation with one coefficient and no intercept, let $\Phi = (1, 2)^\top$, $t = (2, 3)^\top$, and both weights equal one. The objective becomes

$$(2 - \beta)^2 + (3 - 2\beta)^2 + \beta^2 = 13 - 16\beta + 6\beta^2. \quad (5.30)$$

Its derivative is $-16 + 12\beta$, so the minimizer is $\beta = 4/3$. The matrix calculation gives $\Phi^\top \Phi = 1^2 + 2^2 = 5$, $\Phi^\top t = 1(2) + 2(3) = 8$, and $(5 + 1)\beta = 8$. Both calculations agree. Without the penalty, the coefficient would be $8/5$. The reduction illustrates ridge shrinkage for one coefficient. The adopted driver still includes its baseline and all stated features.

Assembling the normal equations for all output rows gives the exact solution

$$B^\top = \left(\lambda_{sys} \Phi^\top \Phi + \lambda_B I_D \right)^{-1} \lambda_{sys} \Phi^\top \hat{C} R^\top. \quad (5.31)$$

The positive driver ridge weight makes the matrix in this expression positive definite, even when ordered quadratic columns are duplicated. In computation, the corresponding linear system is solved. An explicit matrix inverse is unnecessary. The square quadratic slices are then symmetrized without changing their predictions.

For fixed B and $\hat{C}$, define $E = \hat{C} - \Phi B^\top$.

The coupling calculation can be read one output row at a time. For row f , the residual at observation i is $E_{if} - \sum_{j \neq f} \hat{c}_{ij} \Gamma_{fj}$. Its minimization uses the other fitted component columns as predictors of E_{if} , with a ridge penalty and box bounds. In a hypothetical row with one permitted coupling γ , two predictor values 1 and 2, and residual targets 0.4 and 0.8, equal unit weights give

$$(0.4 - \gamma)^2 + (0.8 - 2\gamma)^2 + \gamma^2 = 0.8 - 4\gamma + 6\gamma^2. \quad (5.32)$$

The derivative $-4 + 12\gamma$ vanishes at $\gamma = 1/3$. This value lies below the adopted absolute entry bound 0.368. If the illustrative box bound were instead 0.25, the convex scalar problem would attain its minimum at 0.25. The conditioning test is applied to the complete coupling matrix after these row-wise calculations.

Collecting these scalar residuals across all output rows gives the coupling proposal

$$\min_{\Gamma} \quad \lambda_{sys} \|E - \hat{C} \Gamma^\top\|_F^2 + \lambda_\Gamma \|\Gamma\|_F^2 \quad (5.33)$$

subject to its zero diagonal and box bounds. This is a convex quadratic program. Its candidate is subsequently screened using Equation (5.22). A candidate that fails the condition check is shrunk toward zero or replaced by an admissible fallback. The screened proposal is evaluated against the previous admissible coupling matrix using the complete objective.

For fixed B and Γ , the fitted-state update separates over the training rows. For sample i , write $\hat{c}_i$ as a column vector.

Expand the three residual sums for one observation before collecting their coefficients. The data term contributes $\hat{c}^\top \hat{c} - 2c_{out}^\top \hat{c}$. The invariant term contributes

$\lambda_{inv}\hat{c}^\top A^\top A\hat{c} - 2\lambda_{inv}c_{in}^\top A^\top A\hat{c}$. The system term contributes $\lambda_{sys}\hat{c}^\top R^\top R\hat{c} - 2\lambda_{sys}(B\phi)^\top R\hat{c}$. Terms independent of $\hat{c}$ do not affect its minimizer. Collecting its quadratic terms gives H_C , and collecting its linear terms gives h_i .

Write the expanded quadratic as $\hat{c}_i^\top H_C \hat{c}_i - 2h_i^\top \hat{c}_i$. The entry $(H_C)_{fg}$ multiplies the ordered product $\hat{c}_{if}\hat{c}_{ig}$, and $(h_i)_f$ supplies its linear term. For distinct components, the two mirrored entries contribute together, following the quadratic symmetry convention. Let δ_{fg} equal one when $f = g$ and zero otherwise. Reading the coefficients entry by entry gives

$$\begin{aligned} (H_C)_{fg} &= \delta_{fg} + \lambda_{inv} \sum_{k=1}^K A_{kf} A_{kg} + \lambda_{sys} \sum_{j=1}^F R_{jf} R_{jg}, \\ (h_i)_f &= c_{out,if} + \lambda_{inv} \sum_{k=1}^K A_{kf} \sum_{g=1}^F A_{kg} c_{in,ig} \\ &\quad + \lambda_{sys} \sum_{j=1}^F R_{jf} \sum_{\ell=1}^D B_{j\ell} \phi_{i\ell}. \end{aligned} \tag{5.34}$$

For example, the invariant contribution to $(H_C)_{12}$ adds the product of the first and second component weights over the invariant rows. The system contribution adds the product of columns 1 and 2 of R . These products explain why correcting one fitted component can change the preferred value of another.

Completing the square explains what the non-negative solve is adjusting. With $c_0 = H_C^{-1}h_i$,

$$\hat{c}^\top H_C \hat{c} - 2h_i^\top \hat{c} = (\hat{c} - c_0)^\top H_C (\hat{c} - c_0) - h_i^\top H_C^{-1} h_i. \tag{5.35}$$

The unrestricted minimizer is c_0 . When it contains a negative entry, the constrained problem seeks the closest non-negative state in the quadratic geometry defined by H_C . Off-diagonal entries tilt that geometry, so ordinary coordinate clipping need not give the closest state.

For example, take hypothetical $H_C = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ and $h_i = (-1, 2)^\top$. These coefficients can arise from unit data, invariant, and system weights with $A = [1 \ 1]$ and $R = I$. A signed driver can make the first entry of h_i negative. The unrestricted solution is $c_0 = (-5/8, 7/8)^\top$. Clipping gives $(0, 7/8)^\top$. Instead, set the constrained first component to zero and minimize the remaining expression $3c_2^2 - 4c_2$. Its derivative $6c_2 - 4$ gives $c_2 = 2/3$.

At $(0, 2/3)^\top$, the derivative in the first coordinate is $2(c_2 + 1) = 10/3$, which is positive. Increasing that zero component would increase the objective, so the boundary value is appropriate. The objective is $-4/3$ at this state and $-77/64$ at the clipped state. The constrained solution has the smaller objective. The negative objective values are possible because the constants removed during expansion are absent. The complete residual-based training criterion remains non-negative.

The general coefficient arrays derived above are assembled in the compact form

$$\begin{aligned} H_C &= I_F + \lambda_{inv} A^\top A + \lambda_{sys} R^\top R, \\ h_i &= c_{out,i} + \lambda_{inv} A^\top A c_{in,i} + \lambda_{sys} R^\top B \phi_i. \end{aligned} \quad (5.36)$$

Each row minimizes $\hat{c}_i^\top H_C \hat{c}_i - 2h_i^\top \hat{c}_i$ over $\hat{c}_i \geq \mathbf{0}$. The identity term makes H_C positive definite, so the row minimizer is unique. Clipping the unconstrained solution componentwise generally does not solve this problem because the off-diagonal terms in H_C connect component adjustments.

The quadratic-program method of Stellato et al. (2018) provides an algorithmic foundation for the constrained coupling and fitted-state blocks. These blocks use their quadratic structure directly. A generic stochastic first-order method could also optimize a related criterion, but it would require separate mechanisms to control non-negativity, coupling bounds, and conditioning. The block formulation keeps each requirement attached to an explicit subproblem.

Descent, stopping, and numerical safeguards

Exact minimization over a block cannot increase the objective when its admissible set contains the previous block value. Since $J \geq 0$, an objective sequence that is monotone non-increasing converges in value. The convergence analysis of Razaviyayn et al. (2013) relates blockwise minimization to stationary solutions under its stated conditions.

The conditioning screen requires particular care. Shrinking a coupling proposal can increase the training criterion relative to the previous admissible coupling matrix. The procedure therefore evaluates the objective after screening and retains the previous admissible matrix when a screened candidate cannot provide an acceptable non-increasing update. Solver tolerances also require an explicit allowance for numerical error. Monotone descent is assessed for the accepted objective-decreasing updates after screening.

Each restart begins with an admissible coupling matrix and non-negative fitted states. A ridge solve supplies the initial driver. Full cycles continue until an iteration limit is reached or the slope of the trailing objective history is sufficiently small. A running minimum records the best admissible state reached. When several restarts are used, the admissible state with the smallest recorded objective is retained. A nearly flat objective triggers the prescribed stopping rule.

Figure 5.1 separates the three fitting updates from the checks that make an update acceptable. The scalar calculations in Equations (5.30) and (5.32) illustrate the first and third computational boxes in the left column. The non-negative-state example explains the return path in the right column.

The cycle returns to the driver because each block changes the conditions faced by the others. The conditioning screen acts on the coupling proposal before the fitted-

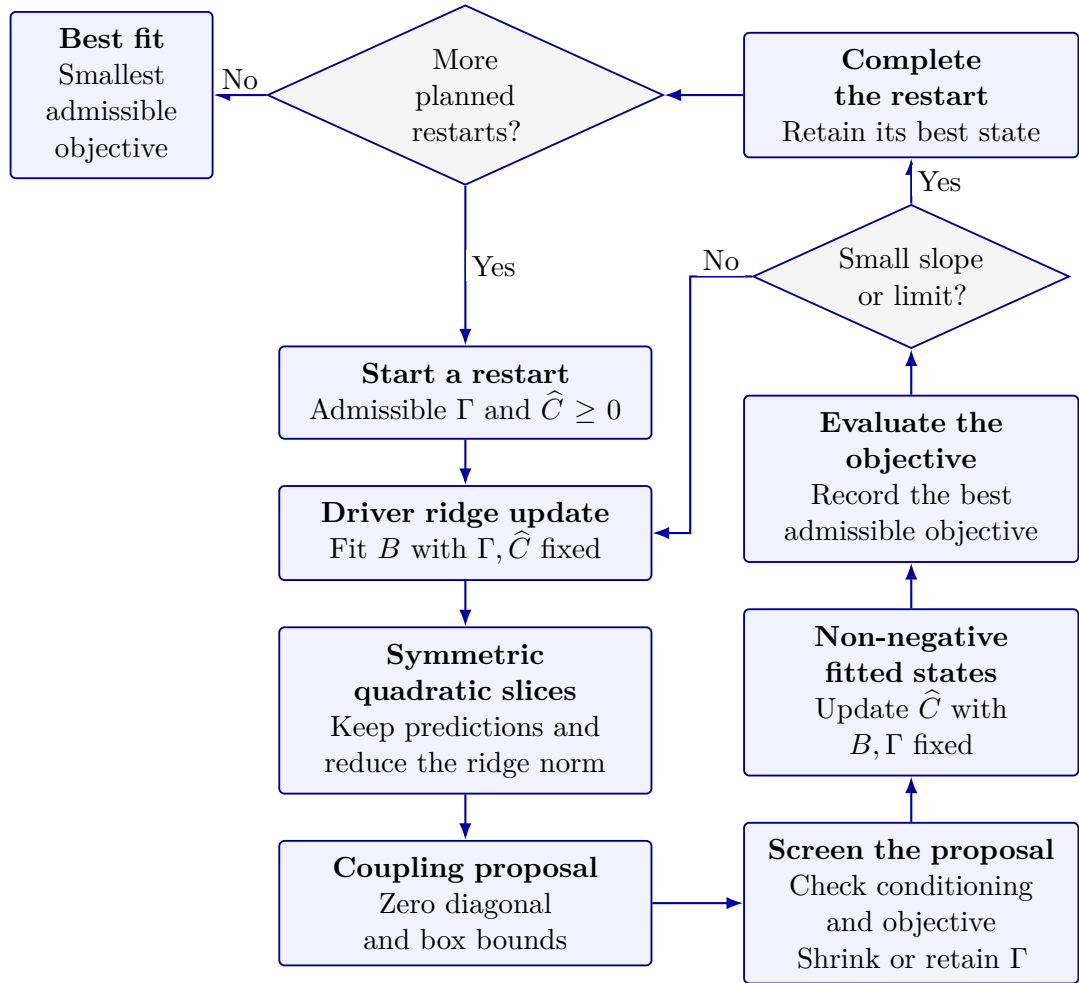


Figure 5.1: Conceptual fitting cycle with explicit symmetry, conditioning, descent, and non-negativity checks. The prescribed restart count can be one.

state update. The full objective is then recorded after the cycle. The restart decision concerns the prescribed search procedure, while the final selection compares admissible objective values. These steps define the accepted sequence of updates for the nonconvex joint fit.

Table 5.1: Specified fitting settings for the structured regression.

Setting	Value
Baseline feature	Included
Invariant penalty λ_{inv}	8.68×10^{-3}
Coupled-system penalty λ_{sys}	1.27×10^{-3}
Driver ridge penalty λ_B	6.09×10^{-5}
Coupling ridge penalty λ_Γ	6.55×10^{-4}
Off-diagonal bound γ_{abs}	0.368
Maximum full fitting cycles	60
Restarts	1
Trailing objective window	12 cycles
Objective slope tolerance	2.85×10^{-7}

Table 5.1 specifies numerical fitting choices. The condition threshold κ_{max} remains an additional prescribed numerical safeguard and is checked independently of the coefficient bound. Hyperparameter assessment uses validation observations within the training pool. Held-out test observations do not determine penalty weights or stopping choices. Fitting settings remain fixed within a sample-size comparison so that changes in the learning curve reflect changing sample availability under a common model specification.

5.3.2 Checked deployment

The raw coupled state

For a new pair $(u_*, c_{in,*})$, the fitted parameters give

$$\hat{R} = I_F - \hat{\Gamma}, \quad d_* = \hat{B}\phi(u_*, c_{in,*}), \quad \hat{R}c_{raw} = d_*. \quad (5.37)$$

The notation $c_{raw} = \hat{R}^{-1}d_*$ describes the mapping mathematically. A linear-system solve evaluates it numerically. The accepted conditioning bound limits the sensitivity of that solve. The raw state is returned only when it satisfies both the invariant and non-negativity checks.

For a vector, the Euclidean norm is the square root of the sum of squared entries. Thus an illustrative invariant-residual vector $(3, 4)^\top$ has norm $\sqrt{3^2 + 4^2} = 5$. The absolute norm adds the absolute entries instead. For an illustrative concentration

vector $(-0.3, 0.1, -0.4)^\top$, retaining only negative entries gives $(-0.3, 0, -0.4)^\top$, whose absolute norm is 0.7. The first calculation measures the size of a mass conservation discrepancy. The second measures total negative concentration in the stated numerical coordinates.

For any candidate c , define the diagnostic residuals

$$v_{mc}(c) = \|A(c - c_{in,*})\|_2, \quad v_{nn}(c) = \|\min(c, \mathbf{0})\|_1. \quad (5.38)$$

The minimum is taken componentwise. The adopted diagnostic tolerance is 10^{-10} . Algebraic equality and non-negativity define the mathematical requirement. Residuals below the numerical tolerance define its computational assessment. Invariant residual magnitudes use the stated scaling of A .

Affine projection for mass conservation

The nearest state satisfying mass conservation can first be derived for the hypothetical two-component inventory. Write the raw prediction as $(p_1, p_2)^\top$ and the required total as T . The two unknown projected concentrations minimize $(c_1 - p_1)^2 + (c_2 - p_2)^2$ while satisfying $c_1 + c_2 = T$. Introduce a multiplier η for that equality and write

$$\mathcal{L} = \frac{1}{2}(c_1 - p_1)^2 + \frac{1}{2}(c_2 - p_2)^2 + \eta(c_1 + c_2 - T). \quad (5.39)$$

The factor one-half simplifies differentiation without changing the minimizer. Differentiating with respect to the two concentrations and the multiplier gives

$$c_1 - p_1 + \eta = 0, \quad c_2 - p_2 + \eta = 0, \quad c_1 + c_2 = T. \quad (5.40)$$

The first two equations give $c_1 = p_1 - \eta$ and $c_2 = p_2 - \eta$. Substituting them into the inventory equality gives $p_1 + p_2 - 2\eta = T$. Thus

$$\begin{aligned} \eta &= \frac{p_1 + p_2 - T}{2}, \\ c_1 &= p_1 - \frac{p_1 + p_2 - T}{2}, \\ c_2 &= p_2 - \frac{p_1 + p_2 - T}{2}. \end{aligned} \quad (5.41)$$

The unweighted distance shares the inventory discrepancy equally between these two coordinates. With $p_1 = 12$, $p_2 = 1$, and $T = 10$, the discrepancy is 3 and the multiplier is 1.5. The projected state is $(10.5, -0.5)^\top$. The equality is satisfied even though the second component is negative. This example identifies the inequality that is absent from an equality-only projection.

When the raw state fails its checks, the first projection solves

$$c_{aff} = \arg \min_{c \in \mathbb{R}^F} \|c - c_{raw}\|_2^2 \quad \text{subject to} \quad Ac = Ac_{in,*}. \quad (5.42)$$

Introduce a Lagrange multiplier $\eta \in \mathbb{R}^K$. Stationarity gives $c - c_{raw} + A^\top \eta = 0$. Substitution into the equality gives $AA^\top \eta = A(c_{raw} - c_{in,*})$. The full row rank of A therefore gives the unique solution

$$c_{aff} = c_{raw} - A^\top (AA^\top)^{-1} A(c_{raw} - c_{in,*}). \quad (5.43)$$

The fixed matrix $A^\top (AA^\top)^{-1} A$ can be prepared once. The affine projection is then a matrix-vector operation for each new prediction.

Each matrix step follows the same scalar derivation. The multiplier column contains one multiplier per independent inventory. The vector $A^\top \eta$ converts these inventory projections into component adjustments. Solving $AA^\top \eta = A(c_{raw} - c_{in,*})$ determines the adjustments needed to remove the inventory residual. For the two-component illustration,

$$AA^\top = 2, \quad A^\top (AA^\top)^{-1} A = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}. \quad (5.44)$$

Taking the illustrative influent as $(6, 4)^\top$, the raw-minus-influent difference is $(6, -3)^\top$. Multiplying it by this projection matrix gives $(1.5, 1.5)^\top$, which is subtracted from the raw state. This recovers the scalar calculation without guessing how much either component should change.

This projection is an orthogonal projection onto the affine set defined by mass conservation in the chosen numerical component coordinates. It removes the part of the prediction error normal to that set and retains the part tangent to it. The projection distance uses the stated component unit convention. Sturm and Silva (2025) develop related projections for mass conservation and explain why species-dependent weights matter when concentration scales differ.

The affine state is checked for both mass conservation residuals and non-negativity before deployment. The equality follows analytically from Equation (5.43), but checking its residual catches numerical or operator inconsistency. An affine state that satisfies mass conservation can still contain a negative concentration. That possibility requires a projection with inequality constraints.

Joint mass conservation and non-negativity

The absolute-distance projection can also be written before introducing compact vector inequalities. For the state that satisfies mass conservation but violates non-negativity $(10.5, -0.5)^\top$, require $c_1 + c_2 = 10$ and $c_1, c_2 \geq 0$. Introduce non-negative adjustment bounds ρ_1, ρ_2 . The constraints

$$\begin{aligned} \rho_1 &\geq c_1 - 10.5, & \rho_1 &\geq 10.5 - c_1, \\ \rho_2 &\geq c_2 + 0.5, & \rho_2 &\geq -0.5 - c_2 \end{aligned} \quad (5.45)$$

make ρ_1 at least the absolute first adjustment and ρ_2 at least the absolute second adjustment. Minimizing $w_1 \rho_1 + w_2 \rho_2$ with positive weights makes each bound equal

to the corresponding absolute adjustment. No absolute-value function remains in the optimization constraints.

Set $c_2 = t$ to impose the inventory equality, so $c_1 = 10 - t$ and $0 \leq t \leq 10$. Both absolute adjustments then equal $t + 0.5$. The objective is $(w_1 + w_2)(t + 0.5)$, which increases with t . Its minimum is at $t = 0$, giving $(10, 0)^\top$. For equal unit weights, its cost is 1. At $t = 1$, the state would be $(9, 1)^\top$, with cost 3. This direct comparison explains the optimal projection in the reduced problem.

When the affine state fails the non-negativity check, the final projection solves

$$\begin{aligned} \hat{c} = \arg \min_{c, \rho \in \mathbb{R}^F} \quad & w_c^\top \rho \\ \text{subject to} \quad & Ac = Ac_{in,*}, \\ & c \geq \mathbf{0}, \\ & -\rho \leq c - c_{aff} \leq \rho, \\ & \rho \geq \mathbf{0}. \end{aligned} \tag{5.46}$$

For strictly positive fixed weights w_c , minimization makes $\rho_f = |c_f - c_{aff,f}|$ at an optimum. The objective is therefore the weighted absolute adjustment $\sum_f w_{c,f} |c_f - c_{aff,f}|$. The program minimizes adjustment from the affine state while retaining the fitted regression coefficients.

The program is linear because the absolute values are represented by auxiliary variables and linear inequalities. The revised-simplex method described by Huangfu and Hall (2018) provides an algorithmic foundation for solving this type of projection. The mathematical feasible set is nonempty whenever the supplied influent state is non-negative under the adopted boundary relation. The positive weights make the adjustment objective bounded below and discourage unconstrained movement of any component. A minimizer exists, although several minimizing states can exist when the absolute-distance objective has a flat face.

Large weights make changes to a component more costly. Small weights permit larger changes when those changes help satisfy the specified constraints. Weights are fixed before deployment using the intended component scaling and scientific priorities. A sensitivity analysis of the weights can examine how much the returned state depends on that modeling choice. Equal numerical weights also embody a choice because component units and typical magnitudes differ.

Three hypothetical components show how weights can change an admissible allocation. Let the inventory row be $[1 \ 1 \ 1]$ and the affine reference satisfying mass conservation be $(-1, 4, 7)^\top$, whose total is 10. Restoring its first component to zero requires removal of one unit from another component. Two feasible candidates are $(0, 3, 7)^\top$ and $(0, 4, 6)^\top$. For weights $(1, 10, 1)^\top$, their costs are respectively $1(1) + 10(1) = 11$ and $1(1) + 1(1) = 2$. The second component is more costly to change, so removing material from the third is preferable. With weights $(1, 1, 10)^\top$, the two costs become 2 and 11.

The preferred allocation reverses. Both allocations preserve the same inventory, while the weights determine the preferred adjustment.

Kircher and Votsmeier (2025) show why a projection for mass conservation needs simultaneous non-negativity enforcement in reaction-kinetics surrogates. Their scaled-space treatment protects low-concentration species during projection. Equation (5.46) addresses the same joint requirement through a weighted absolute-distance problem for activated sludge components. It imposes the adopted linear invariants and non-negativity on the returned state.

Illustrative projection and the failure of clipping

Consider an illustrative two-component system with $A = [1 \ 1]$, a total influent inventory of 10, and a raw prediction $c_{raw} = (12, 1)^\top$. The raw total is 13. The affine formula subtracts 1.5 from each component and gives $c_{aff} = (10.5, -0.5)^\top$. The total is now 10, but the second component is negative.

Clipping that negative value to zero produces $(10.5, 0)^\top$, which violates the total inventory. With equal projection weights, Equation (5.46) instead returns $(10, 0)^\top$. Both constraints hold. These hypothetical values illustrate the projection geometry.

The deployment order uses this geometry directly. A feasible raw state needs no adjustment. An infeasible raw state is first projected onto the mass conservation set. A feasible affine state can then be returned. The linear program is required only when the affine state still fails non-negativity. Every returned candidate is checked under the declared numerical tolerances. An unsuccessful numerical solve is recorded as a prediction failure.

Figure 5.2 makes the branch conditions explicit. Both physical checks appear at the raw and affine stages. The last projection branch also requires successful optimization and an independent admissibility check.

The hypothetical state $(12, 1)^\top$ follows the two No branches because its total is incorrect and its affine projection contains a negative component. The weighted projection supplies $(10, 0)^\top$, which reaches the return branch only after the stated checks pass. A state that satisfies both conditions at an earlier branch reaches the same reporting step. The different paths therefore change the projection effort while retaining the same acceptance requirements.

The component state returned by this rule is the deployed output. Its reported quantities are

$$\hat{y}_{\text{comp}} = I_{\text{comp}} \hat{c}. \quad (5.47)$$

The deployment constraints enforce admissibility, and the stage-wise assessment measures how often each projection is needed.

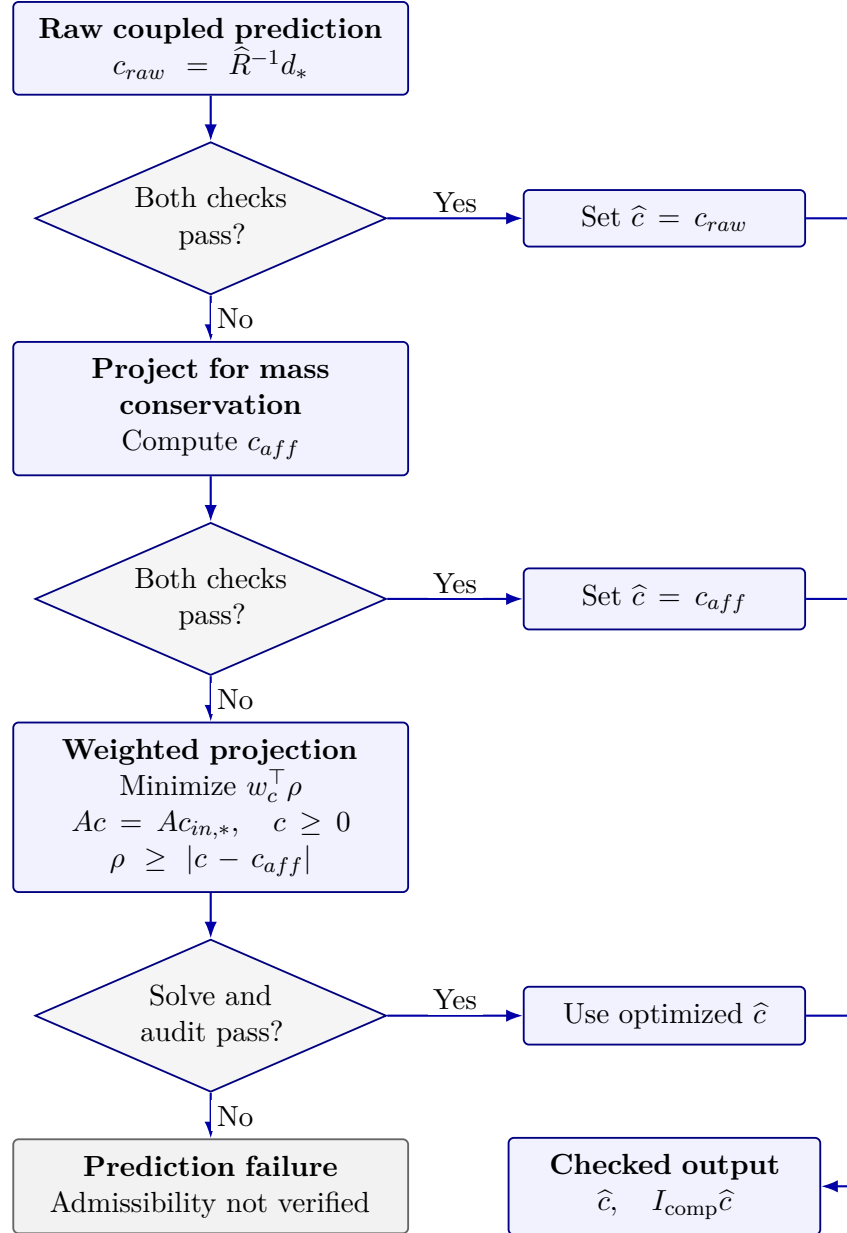


Figure 5.2: Checked deployment of one predicted component state. Accepted branches merge before the common reporting map is applied. A failed final audit produces a reported prediction failure.

5.3.3 Interpreting the complete prediction map

Driver coefficients and back projection

The transformation from driver coefficients to output coefficients can be computed with small matrices. Use the hypothetical coupling matrix in Equation (5.19), a reduced feature vector $(1, a)^\top$, and driver coefficients

$$B = \begin{bmatrix} 4 & 1 \\ 2 & 0 \end{bmatrix}, \quad r = \begin{bmatrix} 4 + a \\ 2 \end{bmatrix}. \quad (5.48)$$

The first coefficient column describes the baseline, and the second describes dependence on the hypothetical scalar input a . Multiplication through the coupling inverse gives

$$\begin{aligned} R^{-1}B &= \frac{1}{0.96} \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 2 & 0 \end{bmatrix} \\ &= \frac{1}{0.96} \begin{bmatrix} 4.4 & 1 \\ 2.8 & 0.2 \end{bmatrix} = \begin{bmatrix} 55/12 & 25/24 \\ 35/12 & 5/24 \end{bmatrix}. \end{aligned} \quad (5.49)$$

The second component now depends on a , even though its driver coefficient for a was zero. This dependence is introduced by the simultaneous coupling solve.

For a hypothetical reported quantity $y = c_1 + 2c_2$, its reporting row is $[1 \ 2]$. Its transformed coefficient row is

$$[1 \ 2] R^{-1}B = [125/12 \ 35/24]. \quad (5.50)$$

At $a = 2$, the driver is $(6, 2)^\top$, the raw component state is $(20/3, 10/3)^\top$, and the reported quantity is $40/3$. The transformed row gives the same value through $125/12 + (35/24)2 = 40/3$. The hypothetical reporting row provides a reduced arithmetic example.

The rows of $\hat{B}$ refer to component drivers. Coupling and the reporting map convert these rows into COD, TN, TP, and TSS coefficients. The raw component coefficient matrix is

$$B_{\text{raw}} = \hat{R}^{-1}\hat{B}, \quad c_{\text{raw}} = B_{\text{raw}}\phi. \quad (5.51)$$

The corresponding map in reported-composite space is

$$B_{\text{raw,comp}} = I_{\text{comp}}\hat{R}^{-1}\hat{B}, \quad y_{\text{raw,comp}} = B_{\text{raw,comp}}\phi. \quad (5.52)$$

This transformation carries the driver coefficients through the learned coupling and then through the reporting operator. Inspection of a COD quadratic block uses its row in $B_{\text{raw,comp}}$, rather than an arbitrary row of $\hat{B}$.

Figure 5.3 shows the complete raw prediction chain using the named feature groups. It keeps the component driver, simultaneous coupling solve, and reporting operation visible as separate mathematical steps.

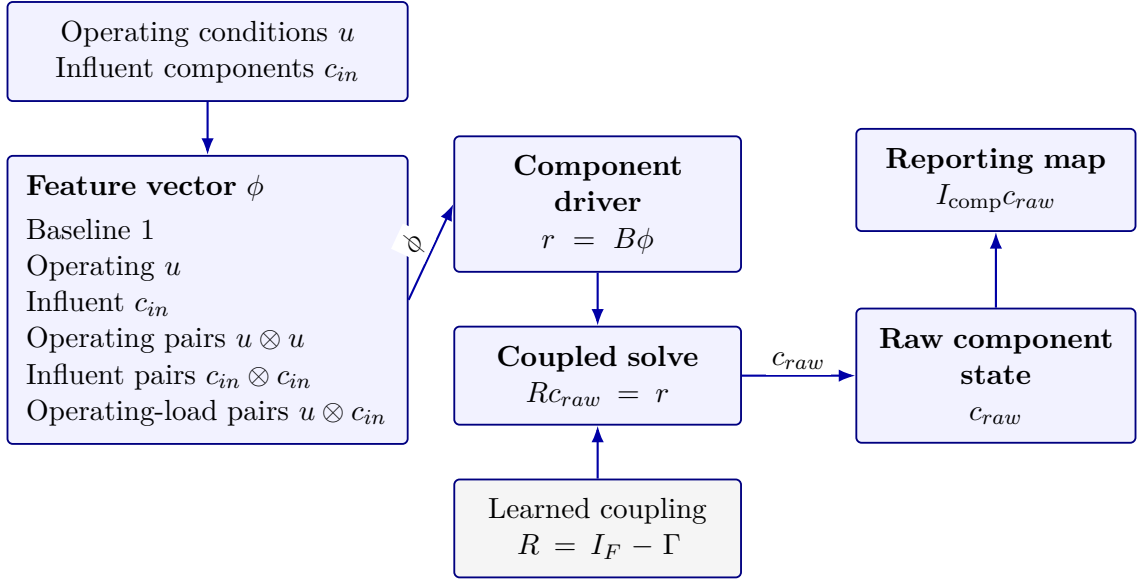


Figure 5.3: Named feature blocks and the transformation from driver to raw component and reported-composite predictions. Composing the three operations gives the raw reported coefficient map $I_{\text{comp}}R^{-1}B$.

In the two-feature example, the second driver has no direct coefficient for the scalar input, but the coupling path gives its raw component a nonzero input coefficient. The reporting path combines both component coefficients. Reading the paths in this order explains the back projection in Equations (5.49) and (5.50). Checked deployment acts on the raw component state before the final reported quantities are calculated.

The named blocks can be displayed as coefficient maps. Operating terms are grouped separately from influent terms. Symmetric quadratic maps display both ordered cells while reporting unique cross terms consistently. Operating-load maps show which operating variables interact with which influent components. The coupling matrix is displayed separately because it has a different interpretation from the input-driver terms.

A coefficient-display threshold is used only to simplify reporting after fitting. For each reported-composite raw-regression map, the threshold is 10^{-4} times its largest absolute coefficient. For the coupling matrix, the threshold is 10^{-4} times its largest absolute off-diagonal entry. The fitted mapping is retained in full, and the threshold determines which entries appear in the coefficient display. Symmetric quadratic summaries count unique upper-triangular terms. A small coefficient can still have a large contribution when multiplied by a large feature. Interpretation therefore examines coefficient magnitude, input scale, and the contribution of the corresponding term over the declared domain.

Analytical jacobians of the raw regression

Differentiation of a scalar polynomial shows why a coefficient and a marginal response differ. For the illustrative driver in Equation (5.10),

$$\begin{aligned}\frac{\partial r_1}{\partial u_1} &= 2 + 0.2u_1 + 0.2u_2, \\ \frac{\partial r_1}{\partial u_2} &= -1 + 0.2u_1 + 0.25c_1, \\ \frac{\partial r_1}{\partial c_1} &= 0.5 + 0.25u_2, \quad \frac{\partial r_1}{\partial c_2} = -0.1c_2.\end{aligned}\tag{5.53}$$

At $u = (2, 3)^\top$ and $c = (4, 5)^\top$, these derivatives are 3, 0.4, 1.25, and -0.5 . In particular, the first derivative is 3 although the linear coefficient of u_1 is 2. The operating quadratic term and interaction with u_2 supply the remaining contribution.

The derivative of the second illustrative driver $r_2 = -u_1 + u_2 + 0.2c_2$ with respect to u_1 is -1 . Differentiate the coupled equations with the coupling held fixed. Their derivative equations are $\partial c_1/\partial u_1 - 0.2\partial c_2/\partial u_1 = 3$ and $-0.2\partial c_1/\partial u_1 + \partial c_2/\partial u_1 = -1$. Thus the same inverse used for prediction gives

$$\begin{bmatrix} \partial c_1/\partial u_1 \\ \partial c_2/\partial u_1 \end{bmatrix} = \frac{1}{0.96} \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 35/12 \\ -5/12 \end{bmatrix}.\tag{5.54}$$

For the hypothetical reporting row $[1 \ 2]$, the reported derivative is $35/12 + 2(-5/12) = 25/12$. Differentiating the driver, solving through the coupling, and applying the reporting weights are three separate operations.

Putting several derivatives in one matrix gives a Jacobian. Its rows identify the output being differentiated, and its columns identify the input being varied. At the illustrative inputs, the operating-driver Jacobian and its coupled transformation are

$$J_{r,u} = \begin{bmatrix} 3 & 0.4 \\ -1 & 1 \end{bmatrix}, \quad R^{-1}J_{r,u} = \begin{bmatrix} 35/12 & 5/8 \\ -5/12 & 9/8 \end{bmatrix}.\tag{5.55}$$

The first column is the derivative calculation above. The second follows by solving the same coupled system for the derivative column $(0.4, 1)^\top$. A Jacobian therefore collects several scalar sensitivity calculations rather than introducing a different kind of response.

A coefficient describes a term in the polynomial. A Jacobian describes the change in the whole mapping at an operating point. For component f , let $H_{uc}^{(f)} \in \mathbb{R}^{M_{op} \times F}$ reshape its interaction coefficients so that the cross term is $u^\top H_{uc}^{(f)} c_{in}$. The driver can then be expressed as

$$\begin{aligned}r_f &= b_f + (W_u)_{f,:}u + (W_{in})_{f,:}c_{in} + u^\top Q_{uu}^{(f)}u \\ &\quad + c_{in}^\top Q_{cc}^{(f)}c_{in} + u^\top H_{uc}^{(f)}c_{in}.\end{aligned}\tag{5.56}$$

With symmetric square slices, its gradients are

$$\begin{aligned}\nabla_u r_f &= (W_u)_{f,:}^\top + 2Q_{uu}^{(f)}u + H_{uc}^{(f)}c_{in}, \\ \nabla_{c_{in}} r_f &= (W_{in})_{f,:}^\top + 2Q_{cc}^{(f)}c_{in} + (H_{uc}^{(f)})^\top u.\end{aligned}\tag{5.57}$$

The quadratic gradients in Equation (5.57) follow by differentiating each scalar product. For $x^\top Qx = \sum_i \sum_j q_{ij}x_i x_j$, differentiation with respect to x_h collects the terms with $i = h$ and those with $j = h$. Their sum is $\sum_j (q_{hj} + q_{jh})x_j$. It becomes the h th entry of $2Qx$ when Q is symmetric. For $u^\top Hc = \sum_i \sum_j h_{ij}u_i c_j$, differentiation with respect to u_i gives $\sum_j h_{ij}c_j$, while differentiation with respect to c_j gives $\sum_i h_{ij}u_i$. These are the entries of Hc and $H^\top u$.

Stacking the transposed gradients gives $J_{r,u} \in \mathbb{R}^{F \times M_{op}}$ and $J_{r,c} \in \mathbb{R}^{F \times F}$. The raw-state and raw-composite Jacobians follow through the same back projection as the coefficients,

$$\begin{aligned}J_{raw,u} &= \hat{R}^{-1} J_{r,u}, & J_{raw,c} &= \hat{R}^{-1} J_{r,c}, \\ J_{raw,comp,u} &= I_{comp} J_{raw,u}, & J_{raw,comp,c} &= I_{comp} J_{raw,c}.\end{aligned}\tag{5.58}$$

These derivatives describe the complete fitted raw regression at the specified inputs, including curvature and interactions. The projection-stage derivatives are calculated below.

An illustrative aeration sensitivity can be inspected at low and high ammonium loading while holding the remaining inputs fixed. A difference in the two derivatives identifies a loading-dependent association within the fitted model. Oxygen-transfer limitations and heterotrophic competition provide process context for assessing such a pattern, as discussed by Stenstrom and Song (1991). Comparison with that process context provides a plausibility check on the fitted response pattern.

Sensitivity after the affine projection

Return to the two-feature example in Equation (5.49). Its raw derivative with respect to a is $(25/24, 5/24)^\top$, whose total is $5/4$. This raw response changes the total component inventory. At fixed influent inventory, an affine projection must remove that total change. For the two-component invariant,

$$\begin{aligned}P_A &= \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \\ \frac{\partial c_{aff}}{\partial a} &= P_A \begin{bmatrix} 25/24 \\ 5/24 \end{bmatrix} = \begin{bmatrix} 5/12 \\ -5/12 \end{bmatrix}.\end{aligned}\tag{5.59}$$

The two entries now sum to zero. For the reporting row $[1 \ 2]$, the derivative is $5/12 + 2(-5/12) = -5/12$. The corresponding raw derivative was $35/24$. The projection changes both its magnitude and sign because it enforces a different material-accounting condition on the complete returned state.

The influent reference also changes when an influent component is perturbed. In this reduced example, the driver contains no influent term. Increasing either influent component by δ increases the required total by δ . The unweighted affine projection then adds $\delta/2$ to each predicted component. Its influent derivative is the matrix $\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, even though the raw influent derivative is zero. This contribution comes from the right-hand side of the mass conservation equality.

Define the complementary projection matrices

$$Q_A = A^\top (AA^\top)^{-1} A, \quad P_A = I_F - Q_A. \quad (5.60)$$

Then $c_{aff} = P_A c_{raw} + Q_A c_{in}$. With the fitted coefficients and invariant operator held fixed, its Jacobians are

$$J_{aff,u} = P_A J_{raw,u}, \quad J_{aff,c} = P_A J_{raw,c} + Q_A. \quad (5.61)$$

The second expression includes the change in the mass conservation reference when the influent changes. The term Q_A accounts for the changing influent reference.

Multiplication by A gives $AJ_{aff,u} = 0$ and $AJ_{aff,c} = A$. An operating perturbation changes the affine prediction only along directions that preserve its fixed influent inventories. An influent perturbation changes those inventories consistently with the influent reference. These relations provide analytical checks on the sensitivity calculation.

When the affine branch is used over a neighborhood of inputs, its coefficients and Jacobians describe that branch. The complete deployment rule includes transitions between raw, affine, and linear-program branches, each with its corresponding sensitivity.

Active constraints in the final projection

An explicit boundary in the same hypothetical model shows the change in sensitivity caused by non-negativity. Keep the total influent inventory equal to 10. The affine components obtained from Equation (5.49) are

$$c_{aff,1} = \frac{35}{6} + \frac{5a}{12}, \quad c_{aff,2} = \frac{25}{6} - \frac{5a}{12}. \quad (5.62)$$

Their sum is 10. Near $a = 10$, the first component is positive, while the second reaches zero at $a = 10$. Immediately below this boundary the affine state is feasible and its derivative is $(5/12, -5/12)^\top$. Above the boundary the affine second component is negative. The positive-weight absolute projection returns $(10, 0)^\top$, so its derivative with respect to a is $(0, 0)^\top$. The final projected state therefore has a different derivative on each side of the boundary. The boundary is characterized by its two one-sided derivatives.

In the projected branch, the binding equalities are $c_1 + c_2 = 10$ and $c_2 = 0$. Differentiating them gives $\partial c_1 / \partial a + \partial c_2 / \partial a = 0$ and $\partial c_2 / \partial a = 0$. These two equations

directly give the zero derivative. The general active-set calculation uses the same idea with all binding inventory and adjustment constraints.

The linear program can make a component concentration equal to zero or bind one of the absolute-adjustment inequalities. These binding constraints define an active set. If the optimal solution is unique and its nondegenerate optimal basis remains unchanged over a local neighborhood, the solution varies affinely with the relevant right-hand sides in that neighborhood. A branch-specific sensitivity can then be derived from that fixed basis.

At an active-set transition, the derivative can change abruptly. When several optimal projected states exist, the returned solution can also depend on the rule used to select among them. Local deployment sensitivity is assessed by differentiating a verified fixed basis where appropriate or by perturbing admissible inputs and resolving the entire deployment problem. The perturbation size and any branch changes are recorded as part of that sensitivity assessment.

This distinction prevents three different quantities from being conflated. A driver coefficient belongs to $\hat{B}$. A raw-state sensitivity passes through $\hat{R}^{-1}$. A deployed-state sensitivity also depends on the mass conservation reference and active projection constraints. Each quantity answers a different question about the model.

Following sensitivity through projection

Interpretation follows the complete prediction map. Coupling transforms the driver into component predictions. The reporting map combines components, and active output constraints change their local derivatives. Physical-unit interpretation identifies a driver coefficient, a raw prediction derivative, or a derivative of the final projected response (Brunton et al., 2016; Shen et al., 2023).

A second hypothetical calculation isolates the effect of projection using different raw equations. Let a dimensionless operating input be a , and let two hypothetical raw concentration equations be

$$c_{raw,1}(a) = 4 + a, \quad c_{raw,2}(a) = 3 + 2a.$$

The raw first-component slope is one and the second-component slope is two. At $a = 2$, the concentrations are 6 and 7, with a total of 13. Suppose the required total is ten and the projection minimizes the sum of squared adjustments with equal weights. If the first adjustment is d_1 and the second is d_2 , mass conservation requires $d_1 + d_2 = -3$. Substitute $d_2 = -3 - d_1$ into the squared-adjustment objective and differentiate,

$$\begin{aligned} d_1^2 + d_2^2 &= d_1^2 + (-3 - d_1)^2, \\ \frac{d}{dd_1} [d_1^2 + (-3 - d_1)^2] &= 4d_1 + 6 = 0. \end{aligned}$$

Both adjustments are -1.5 , giving a state satisfying mass conservation with concentrations 4.5 and 5.5. For a general a , the excess total is $(4 + a) + (3 + 2a) - 10 = 3a - 3$.

Equal-weight projection subtracts one half of that excess from each coordinate,

$$c_{aff,1}(a) = 4 + a - \frac{3a - 3}{2} = 5.5 - 0.5a,$$

$$c_{aff,2}(a) = 3 + 2a - \frac{3a - 3}{2} = 4.5 + 0.5a.$$

The projected slopes are -0.5 and 0.5 . The first changes sign even though its raw slope is positive. The sum of the projected slopes is zero, consistent with a total that does not depend on a .

Non-negativity introduces another stage. The affine first component reaches zero at $a = 11$. For $a > 11$, the affine state lies outside the segment satisfying mass conservation and non-negativity. In this equal-weight two-component example, the nearest jointly feasible state is then $(0, 10)^\top$. Both local output slopes are zero while that boundary remains active. The slope changes at the transition point. This calculation illustrates why a raw coefficient, an affine sensitivity, and a final constrained sensitivity have distinct meanings.

Figure 5.4 plots the hypothetical first-component equations and the slopes calculated from them. The left panel shows the concentration returned at each stage. The right panel shows the corresponding local rate of change with respect to the operating input.

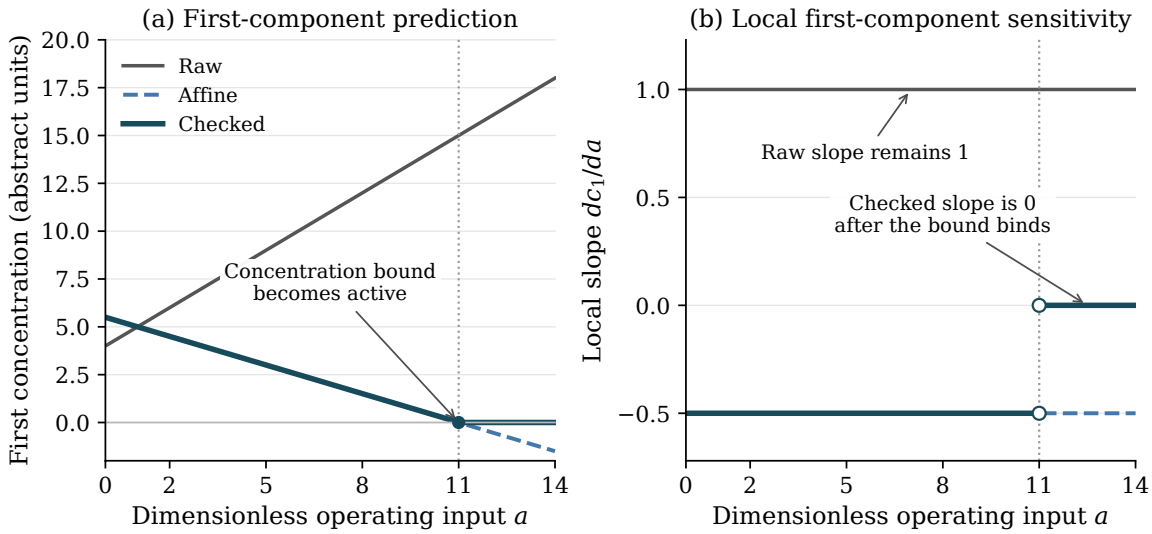


Figure 5.4: Hypothetical first-component response and sensitivity before and after mass conservation and non-negativity enforcement. Open circles mark the two one-sided checked slopes at the active-boundary transition.

Before the boundary becomes active, the affine and checked curves coincide. Their negative first-component slope differs from the positive raw slope because mass conservation redistributes the operating response between the two components. After the boundary becomes active, the checked concentration stays at zero and its local slope is zero. The transition is characterized by the two one-sided checked slopes.

5.3.4 Assessment design for the structured regression

Common inputs and comparator models

The structured-regression assessment uses a design collection of 10,000 accepted steady-state reactor cases. Every method receives the same 22 physical inputs and predicts the same 20 effluent components. Its reported composites are obtained from the common composition matrix. The input domain, reaction model, simulation bounds, and acceptance criterion remain those of the declared reactor benchmark. The comparison therefore changes the predictor while holding the modeled process and prediction task fixed.

The nine unconstrained comparators are XGBoost, LightGBM, CatBoost, AdaBoost, random forest, support vector regression, k -nearest neighbors, partial least squares, and a fully connected multilayer perceptron. These methods represent boosted and randomized tree ensembles, kernel regression, local neighbor averaging, latent linear regression, and nonlinear neural regression. Their foundations include the ensemble construction of Breiman (2001), the boosting formulations of Drucker (1997), Chen and Guestrin (2016), Ke et al. (2017), and Prokhorenkova et al. (2018), and the latent regression of Wold et al. (2001). The comparators have no invariant penalty, auxiliary non-negative fitted-state matrix, or learned component-coupling equation during fitting. The latent components used by partial least squares are statistical input combinations. The shared 20-component prediction target remains the physical effluent state for every comparator.

The neural comparator has hidden layers with 128, 64, and 32 units and logistic activation. Dense connections map the 22 inputs through these layers to 20 component outputs. Its capacity is evaluated under the same samples and reported-composite map as the other predictors. A transparent polynomial and a neural network can therefore differ in their internal structure without changing the supervised target.

Training partitions and hyperparameter selection

The primary assessment uses a row-wise 80% training and 20% held-out split with random seed 42. The specified collection gives a training pool of 8000 rows and a held-out set of 2000 rows. Hyperparameter selection uses only the training pool. Half of that pool forms a 4000-row tuning subset. A further 80% and 20% split gives 3200 tuning-training rows and 800 tuning-validation rows. The tuning design uses one internal validation split.

Each predictor receives 100 hyperparameter trials under a seeded tree-structured Parzen estimator with seed 42. No trial pruning or wall-clock timeout is imposed. Bergstra et al. (2011) develop this search approach, while Akiba et al. (2019) provide the associated hyperparameter optimization framework. The selection criterion is mean squared error in the shared physical component space on the internal validation set.

Selected settings are held fixed for the primary assessment and the repeated sample-size design. The final primary fit uses the full 8000-row training pool.

For the unconstrained comparators, input and output standardization parameters are estimated from the relevant training partition. Predictions are transformed back to physical component coordinates before any invariant check, reporting map, or error assessment. ICSOR remains in physical coordinates so that its driver, invariant operator, and deployed constraints use the same component convention. Internal-validation transformations are estimated from tuning-training rows only. The training partitions determine the means, standard deviations, and hyperparameters. Projection weights are fixed before held-out assessment.

Table 5.2 gives the specified numerical configurations. These values define fitted model settings rather than measured predictive performance. The internal search compares candidate configurations according to the stated validation objective.

Table 5.2: Specified configurations of the nine unconstrained comparators.

Comparator		Configuration
XGBoost		309 trees, learning rate 0.125, maximum depth 3, minimum child weight 6.84, row fraction 0.915, feature fraction 0.828, absolute penalty 1.18×10^{-3} , and squared penalty 0.382.
LightGBM		302 trees, learning rate 0.0809, 16 leaves, maximum depth 8, minimum leaf count 14, row fraction 0.845, feature fraction 0.869, absolute penalty 3.17×10^{-3} , and squared penalty 3.36×10^{-4} .
CatBoost		236 boosting iterations, learning rate 0.172, depth 4, leaf squared penalty 0.0266, bagging temperature 0.0277, and random strength 0.0657.
AdaBoost		236 estimators, learning rate 0.999, and squared regression loss.
Random forest		390 trees, maximum depth 14, minimum split count 3, minimum leaf count 1, feature fraction 0.500, and no bootstrap resampling.
Support vector regression		Polynomial kernel of degree 4, regularization parameter 0.118, insensitive-error width 0.0458, constant offset 0.889, and solver tolerance 8.96×10^{-4} . The kernel scale is the reciprocal of the product of input dimension and training input variance.
k -nearest neighbors		15 neighbors, inverse-distance weighting, Manhattan distance, and a k -dimensional search tree with leaf size 40.
Partial least squares	least	6 latent components, maximum 887 fitting iterations, and tolerance 9.91×10^{-5} .
Multilayer perceptron	per-	Hidden widths (128, 64, 32), logistic activation, adaptive moment estimation, squared penalty 6.33×10^{-4} , batch size 64, initial learning rate 1.08×10^{-3} , maximum 500 iterations, and tolerance 1.63×10^{-5} . Training rows are not shuffled, and early stopping is disabled.

Scoring states and physical diagnostics

The scored ICSOR output is the final checked component state. For comparator accuracy assessment, the raw component prediction receives the same affine projection for mass conservation in Equation (5.43). The composition map is then applied to the projected state. This comparison describes the accuracy of a checked structured prediction and comparator predictions after affine projection. Comparator accuracy scoring uses the affine state, while ICSOR accuracy scoring uses the final state from checked deployment.

The numerical assessment used an operator A_{num} obtained by rounding the null-space basis, setting negligible coefficients to zero, and dividing each row by its first nonzero coefficient. The fitting, affine projection, and constrained deployment calculations used this operator in place of the ideal operator A . The reported violation rates use these adopted numerical balances and the stated operator scaling.

Physical diagnostics use the direct raw states of the comparators. For ICSOR, diagnostics distinguish the raw coupled state, affine state, and final deployed state. This stage separation identifies the component state assessed by each diagnostic. The empirical balance diagnostic is $v_{mc}^{num}(c) = \|A_{num}(c - c_{in})\|_2$. The non-negativity diagnostic remains v_{nn} in Equation (5.38). At each stage, the violation frequency for diagnostic v is

$$f_v = \frac{100}{n} \sum_{i=1}^n \mathbf{1}\{v(c_i) > 10^{-10}\}, \quad (5.63)$$

where n is the number of assessed observations and v is either v_{mc}^{num} or v_{nn} . Frequencies, residual magnitudes, adjustment distances, and deployment-branch use describe different aspects of the same prediction procedure. Predictive error is reported alongside these physical diagnostics.

Composite error measures

The differences among the five measures can first be computed for two hypothetical observations of one reported quantity. Take reference values (2, 4) and predictions (3, 2). Their residuals are (1, -2). Squaring them gives (1, 4), while taking absolute values gives (1, 2). The mean squared error is $(1+4)/2 = 2.5$, its square root is approximately 1.5811, and the mean absolute error is $(1+2)/2 = 1.5$. These measures describe different summaries of the same two errors.

The absolute relative errors are $1/2$ and $2/4$, so their mean is 0.5 . The reference mean is 3. Its total squared variation is $(2-3)^2 + (4-3)^2 = 2$. The prediction residual squared sum is 5. Therefore the coefficient of determination is $1 - 5/2 = -1.5$. Its negative value says that the hypothetical predictions have greater squared error than assigning the reference mean to both observations. These values illustrate the definitions on a hypothetical pair of observations.

For reported quantity $q \in \{1, 2, 3, 4\}$, define the prediction residual $e_{iq} = \hat{y}_{iq} - y_{iq}$. The five measures are mean squared error, root mean squared error, mean absolute error, mean absolute relative error, and the coefficient of determination. They are computed per reported quantity and summarized across the four quantities. The squared and absolute measures are

$$E_{2,q} = \frac{1}{n} \sum_i e_{iq}^2, \quad E_{root,q} = \sqrt{E_{2,q}}, \quad E_{1,q} = \frac{1}{n} \sum_i |e_{iq}|. \quad (5.64)$$

For strictly nonzero reference values, the relative measure is

$$E_{rel,q} = \frac{1}{n} \sum_i \frac{|e_{iq}|}{|y_{iq}|}. \quad (5.65)$$

It is reported as a ratio. Multiplication by 100 converts it to a percentage. A zero reference value makes this expression undefined, and a very small value can dominate it. Any zero-reference convention is declared explicitly before comparing predictors. Targets with undefined relative error are identified under that convention.

The per-target coefficient of determination is

$$R_q^2 = 1 - \frac{\sum_i e_{iq}^2}{\sum_i (y_{iq} - \bar{y}_q)^2}. \quad (5.66)$$

A target with zero reference variance makes this expression undefined. A negative value is possible when the predictor has a greater squared error than the reference-mean predictor. This coefficient measures prediction error relative to the target variance.

The aggregate squared error pools the four composite residuals. Its root is

$$E_{root} = \left(\frac{1}{4n} \sum_{q=1}^4 \sum_{i=1}^n e_{iq}^2 \right)^{1/2}. \quad (5.67)$$

Aggregate absolute and relative errors use the corresponding mean over observations and composites. Aggregate R^2 uses the mean of the four target-specific coefficients. The composite coordinates retain their own physical units. Their pooled error is a numerical benchmark summary under that coordinate convention. Per-target scores remain necessary to show which reported quantities contribute to an aggregate comparison.

Pooling before taking the square root also matters. If four hypothetical target mean squared errors are 1, 9, 1, and 9, the pooled mean squared error is 5 and its root is $\sqrt{5}$, approximately 2.2361. The arithmetic mean of the four target root errors is instead $(1 + 3 + 1 + 3)/4 = 2$. These are different summaries. Equation (5.67) specifies the first calculation.

The primary ordering measure is aggregate root mean squared error. The other measures provide complementary assessments. Squared error emphasizes larger residuals, absolute error gives average deviation, relative error emphasizes deviations in relation to the reference magnitude, and R^2 compares the squared residual with reference variation. Predictor orderings are compared separately under each measure.

Repeated sample-size assessment

The sample-size design uses

$$n \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (5.68)$$

At each size, ten subsamples are drawn without replacement from the design collection. Deterministic seeds are derived from seed 42. Each run uses a row-wise 80% training and 20% held-out split. All ten predictors receive the same sampled rows and partitions. The design specifies 110 fits per predictor across the size schedule. The assessment uses repeated random splitting.

The fixed hyperparameters isolate sensitivity to available observations under one model specification. The mean and spread of each held-out score are summarized across the ten runs. Learning curves describe predictive performance across the entire sample-size schedule. Geman et al. (1992) explain the relation between flexibility and the bias-variance problem, and Viering and Loog (2023) review the varied shapes of learning curves.

The learning-curve area can first be calculated over a reduced hypothetical schedule. Suppose an error measure has values 6, 4, and 3 at sample sizes 100, 200, and 400. Between the first two sizes, a trapezoid has width 100 and mean height $(6 + 4)/2 = 5$, giving area 500. Between the next two, its width is 200 and mean height $(4 + 3)/2 = 3.5$, giving area 700. The total area is 1200. Dividing by the total width $400 - 100 = 300$ gives a normalized area of 4. A larger interval contributes more to this average than a smaller interval, even when each has one pair of endpoints. This hypothetical schedule illustrates the area calculation for unequal sample-size intervals.

For successive retained sizes $n_1, \dots, n_S$, write the mean error of predictor m under measure j as $\overline{M}_{m,j}(n_s)$. The general numerical calculation is

$$\frac{1}{n_{max} - n_{min}} \sum_{s=1}^{S-1} (n_{s+1} - n_s) \frac{\overline{M}_{m,j}(n_s) + \overline{M}_{m,j}(n_{s+1})}{2}. \quad (5.69)$$

Let $\overline{M}_{m,j}(n)$ denote the mean held-out measure j for predictor m at sample size n . The normalized area under its learning curve is

$$L_{m,j} = \frac{1}{n_{max} - n_{min}} \int_{n_{min}}^{n_{max}} \overline{M}_{m,j}(n) \, dn. \quad (5.70)$$

The trapezoidal rule evaluates this integral over the stated sample sizes. Normalization by the sample-size interval gives an average height of the learning curve and preserves the units of measure j . The normalization denominator is the sample-size interval. Cawley (2011) and Guyon et al. (2011) provide related trajectory summaries in active learning. The present design varies the number of randomly sampled observations.

A rank discards the numerical distance between scores and keeps their order. For four hypothetical predictor errors 3.0, 3.0, 4.5, and 6.0, the dense ranks are 1, 1, 2,

and 3. The tied first two predictors receive the same rank. There is no skipped rank after the tie. The difference between the third and fourth errors does not influence the difference between their ranks. If one predictor receives ranks 1, 2, and 2 in three illustrative comparisons, its mean rank is $(1 + 2 + 2)/3 = 5/3$. The actual assessment averages over the stated combinations of target, measure, and size.

For each measure j , reported quantity q , and retained size n_s , predictors receive a dense rank $r_{m,j,q}(n_s)$ from their mean held-out score. Lower squared, root, absolute, and relative errors receive better ranks. Larger R^2 values receive better ranks. Equal scores receive equal ranks, and the next distinct score receives the next consecutive rank. With $S = 11$ sizes and five measures, the summaries are

$$\bar{r}_{m,j} = \frac{1}{4S} \sum_{q=1}^4 \sum_{s=1}^S r_{m,j,q}(n_s), \quad \bar{r}_m = \frac{1}{5} \sum_{j=1}^5 \bar{r}_{m,j}. \quad (5.71)$$

These summaries describe relative ordering across the stated comparisons. Demšar (2006) develops broader rank-based model comparisons, while Shevchenko et al. (2024) emphasize variability in benchmarking. Mean ranks and absolute errors are reported together to describe both ordering and error magnitude.

At the largest sample size, the generalization gap is

$$\Delta E_{root,m} = E_{root,m}^{held-out} - E_{root,m}^{training}. \quad (5.72)$$

A positive gap indicates greater held-out error. A small gap can accompany either accurate prediction or high-error underfitting, so it is interpreted with the corresponding absolute scores. Training time is recorded for every repeated fit and summarized as a distribution across runs. Deployment effort is assessed separately through prediction time, projection-branch use, and constrained-solve effort. This separation makes the cost of learning distinct from the cost of returning a checked state.

For a hypothetical predictor with training error 0.5 and held-out error 0.8, the gap is 0.3. Another with errors 5.0 and 5.1 has a gap of only 0.1. The second has the smaller gap but much larger absolute errors. This arithmetic shows how absolute error and the training-to-held-out gap describe different aspects of prediction.

5.3.5 Interpretability assessment

The interpretability assessment examines the six driver blocks, the coupling matrix, and the corresponding component and composite raw-regression maps. It compares fitted response patterns with the declared process domain. The reported coefficient summaries describe the response structure of one fitted model. Brunton et al. (2016) connect interpretable nonlinear terms to the problem of recovering structure from data, while Machado et al. (2014) show how operational and influent uncertainty complicate parameter identification in activated sludge models.

The driver and coupling parameters can trade off in the relation $Rc = B\phi$. Different joint stationary solutions can predict similar component states with different coefficients. Symmetry removes the redundancy of ordered self-quadratic pairs. Coefficient signs and magnitudes are interpreted together with the complete fitted response.

Prediction error and physical diagnostics are assessed separately for the raw regression, the affine state, and the final deployed state. The comparison distinguishes what the regression learns from what the projection enforces. The magnitude of the projection is also examined because a feasible deployed state can lie far from its raw prediction. Computational assessment records fitting effort and deployment effort separately. These measures describe the cost of fitting the structured regression and enforcing its physical conditions.

The assessment evaluates steady-state component prediction in the stated input domain and control volume. The raw regression supplies an inspectable approximation, and checked deployment enforces the adopted numerical invariants and non-negativity on the returned state.

5.4 Results

5.4.1 Accuracy on the fixed held-out partition

Table 5.3 reports the four-composite scores on the 2000 held-out cases. ICSOR used its final checked state. The other predictors used their affine-aligned states for accuracy scoring. The multilayer perceptron had the lowest pooled root mean squared error of 4.38. LightGBM followed at 5.30. Support vector regression and ICSOR were closely spaced at 5.97 and 5.98. These scores pool the native composite coordinates under Equation (5.67). The pooled scores retain the native composite-coordinate convention.

ICSOR's pooled root error was 36.5% higher than the multilayer perceptron's and 12.8% higher than LightGBM's. It remained below XGBoost and CatBoost on this measure. The mean absolute relative error of 0.0256 corresponded to 2.56%. The fixed-split comparison quantified the accuracy trade-off associated with the structured prediction procedure.

Figure 5.5 compares root error, absolute error, and mean composite R^2 . ICSOR ranked fourth by pooled root error on this partition.

Table 5.4 separates the composite contributions. ICSOR's COD root error of 7.77 was 0.18 above LightGBM's and below that of the other non-neural predictors. Its TN and TSS errors were 3.55 and 8.36. The multilayer perceptron had the lowest COD, TN, and TSS errors. Every method had a TP error of 0.0361 after the specified alignment and reporting operations. TP therefore did not distinguish the predictors in this assessment.

Figure 5.6 shows why the aggregate comparison needs separate target scores. COD

Table 5.3: Fixed-split held-out composite performance. ICSOR is scored after checked deployment. Comparators are scored after affine alignment without a non-negativity constraint. Relative absolute error is a ratio rather than a percentage.

Predictor	E_{root}	E_1	Mean R^2	E_{rel}
Multilayer perceptron	4.38	2.55	0.9915	0.0123
LightGBM	5.30	3.21	0.9806	0.0203
Support vector regression	5.97	3.49	0.9713	0.0244
ICSOR	5.98	3.70	0.9700	0.0256
XGBoost	6.12	3.81	0.9733	0.0244
CatBoost	6.45	4.07	0.9727	0.0250
AdaBoost	21.27	13.15	0.8643	0.0653
k -nearest neighbors	23.61	13.87	0.8535	0.0557
Partial least squares	28.24	17.05	0.7808	0.0681
Random Forest	29.00	16.93	0.7934	0.0632

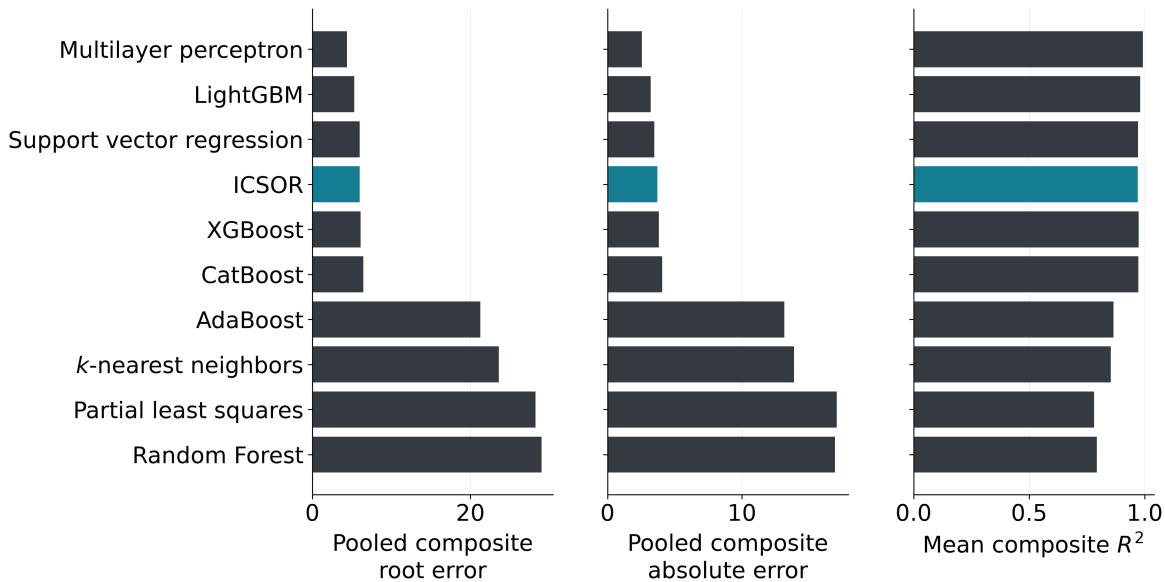


Figure 5.5: Fixed-split composite accuracy on 2000 held-out cases. Teal bars identify ICSOR after checked deployment. Gray bars identify comparators after affine alignment. Pooled root and absolute errors retain the native composite-coordinate convention. The right panel averages four composite-specific R^2 values.

Table 5.4: Fixed-split root mean squared error for each composite. COD, TN, TP, and TSS use g COD m⁻³, g N m⁻³, g P m⁻³, and g TSS m⁻³, respectively.

Predictor	COD	TN	TP	TSS
Multilayer perceptron	6.48	1.58	0.0361	5.66
LightGBM	7.59	2.77	0.0361	6.85
Support vector regression	8.87	3.50	0.0361	7.20
ICSOR	7.77	3.55	0.0361	8.36
XGBoost	8.78	3.27	0.0361	7.88
CatBoost	9.13	3.23	0.0361	8.53
AdaBoost	38.30	5.58	0.0361	17.68
<i>k</i> -nearest neighbors	38.24	4.23	0.0361	27.37
Partial least squares	43.22	5.09	0.0361	36.00
Random Forest	45.87	4.00	0.0361	35.26

and TSS contributed much larger numerical errors than TP. The separate target panels retain the constituent units used for each reported quantity.

5.4.2 Sample efficiency and generalization

ICSOR had the lowest mean held-out root error at the smallest design size of 500 cases. Its value was 10.79, compared with 11.26 for XGBoost and 11.45 for LightGBM. The multilayer perceptron ranked fifth at 13.26. At the largest design size, the repeated-split means were 4.62 for the multilayer perceptron, 5.27 for LightGBM, 5.84 for support vector regression, and 5.94 for ICSOR. The ordering therefore changed as more training cases became available.

ICSOR had the lowest reported normalized root-error curve area of 6.33. LightGBM was close at 6.35 and had the best mean dense rank of 2.2. The multilayer perceptron had the second-best mean rank of 2.5. ICSOR and XGBoost each had a reported mean rank of 4.3. ICSOR led the curve-area comparison, while LightGBM led the mean-rank comparison.

ICSOR's held-out minus training root-error gap was 0.22. This was smaller than the gaps for the multilayer perceptron, LightGBM, XGBoost, and support vector regression. Partial least squares also had a relatively small gap of 0.38, but its held-out root error was 27.96. ICSOR combined a small gap with substantially lower absolute error than partial least squares.

Figure 5.7 presents the three trajectory summaries separately. ICSOR had the lowest root-error curve area, whereas LightGBM had the best mean rank. The two summaries answer different comparative questions.

Figure 5.8 compares the smallest and largest design sizes for four leading predictors. ICSOR led this group at 500 cases, while the multilayer perceptron led at 10,000 cases.

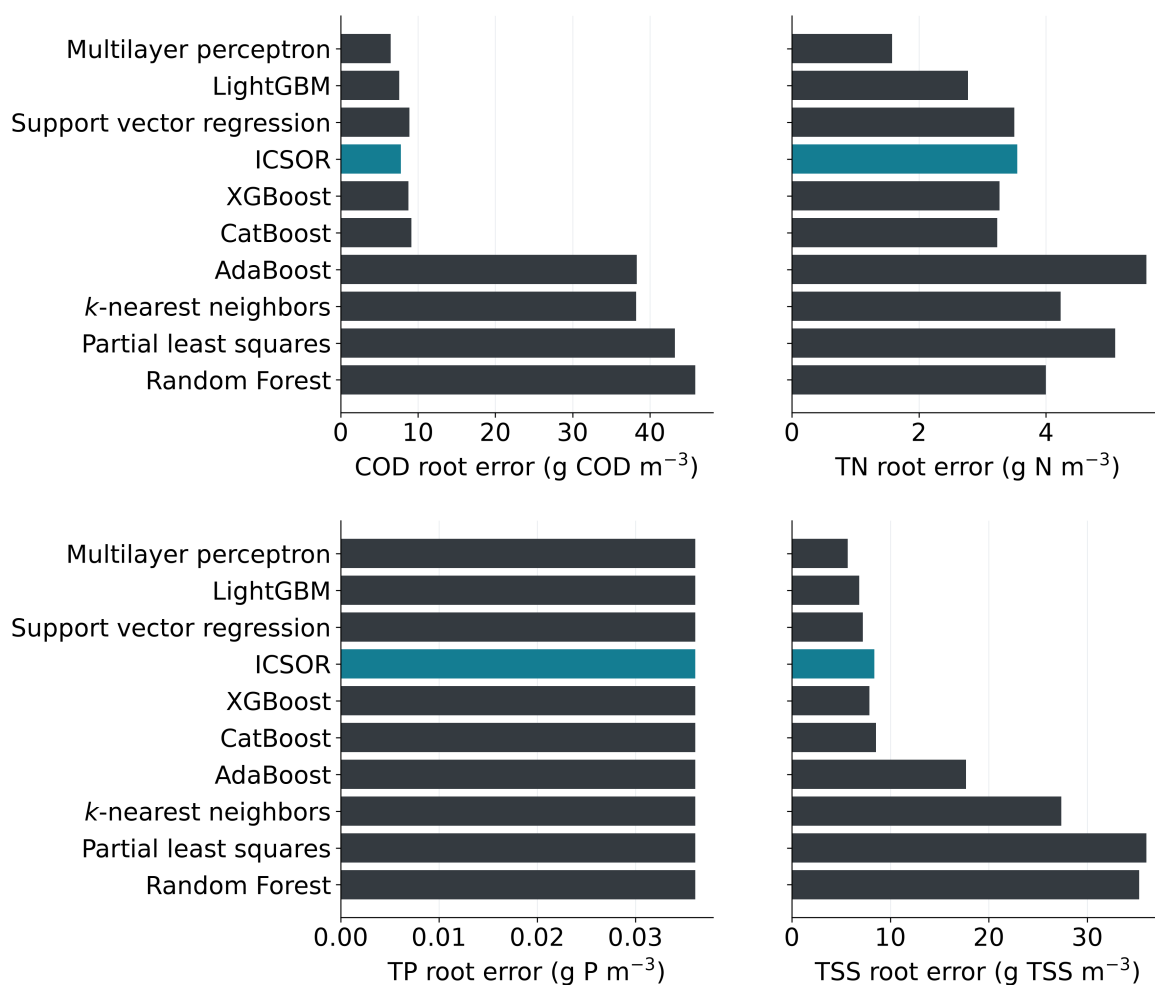


Figure 5.6: Fixed-split root mean squared error for COD, TN, TP, and TSS. Teal identifies ICSOR and gray identifies the comparators. Each panel has its own constituent unit and horizontal scale. All methods have the same reported TP error under the specified alignment and reporting procedure.

Table 5.5: Repeated-split composite assessment over eleven design sizes and ten runs per size. Final error and the held-out minus training gap refer to $n = 10,000$. The learning-curve area is normalized by the design-size interval. Lower error and curve-area values indicate better performance. The gap measures the separation between training and held-out error.

Predictor	Final E_{root}	Normalized curve area	Mean rank	ΔE_{root}
Multilayer perceptron	4.62	6.75	2.5	0.65
LightGBM	5.27	6.35	2.2	1.99
Support vector regression	5.84	6.90	4.0	1.19
ICSOR	5.94	6.33	4.3	0.22
XGBoost	6.10	7.01	4.3	1.44
CatBoost	6.26	7.60	5.0	0.69
AdaBoost	21.42	20.29	8.2	0.23
k -nearest neighbors	23.76	25.77	7.0	23.66
Partial least squares	27.96	30.17	9.4	0.38
Random Forest	28.78	30.31	8.2	18.71

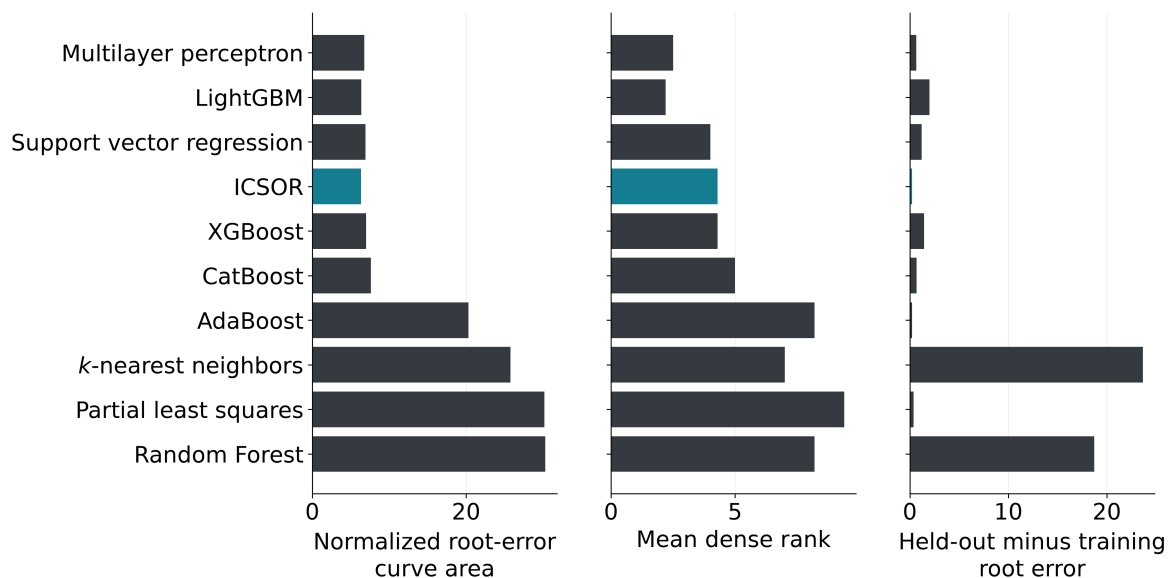


Figure 5.7: Normalized root-error curve area, mean dense rank, and held-out minus training root-error gap. Curve areas and ranks summarize eleven design sizes with ten repeated splits per size. The gap refers to the largest design size. Teal identifies ICSOR. The gap is reported alongside the held-out error.

The two endpoints show the change in the leading predictor between the smallest and largest designs.

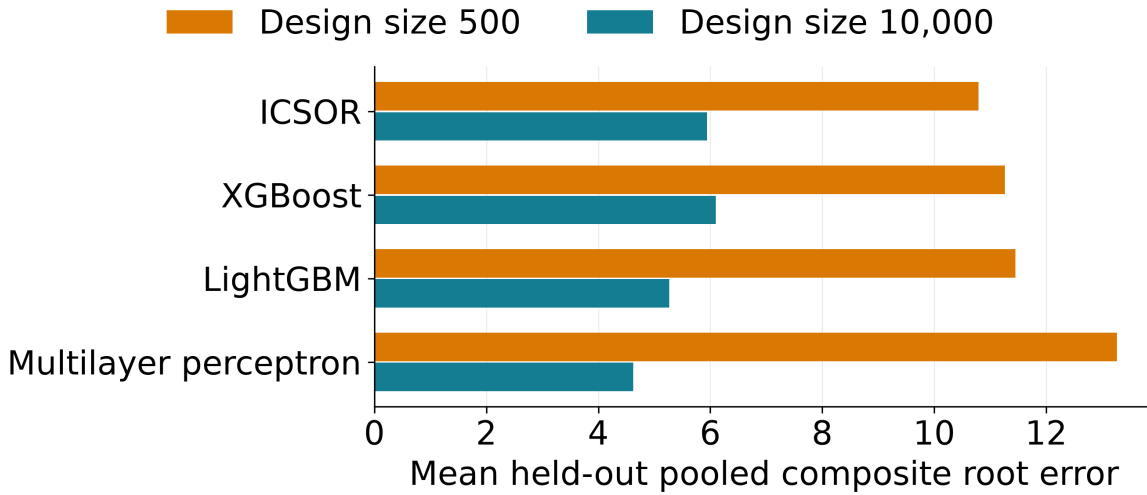


Figure 5.8: Mean held-out pooled composite root error at design sizes 500 and 10,000 for ICSOR, XGBoost, LightGBM, and the multilayer perceptron. Values summarize ten repeated splits. The panels show repeated-split endpoint means at the two specified design sizes.

5.4.3 Physical checks and deployment branches

Table 5.6 distinguishes the final ICSOR output from the raw outputs of the comparators. The final structured predictions had no violations of the adopted numerical balances or non-negativity tolerance. Every comparator raw state violated the adopted balance tolerance. AdaBoost, Random Forest, and the nearest-neighbor model nevertheless had no negative raw states. The other raw non-negativity violation rates ranged from 46.00% to 70.10%.

No raw coupled ICSOR prediction passed both physical checks. All failed the balance tolerance, although 21.7% were non-negative. Affine projection restored the adopted balances for every case and also returned a non-negative state for 436 cases, or 21.8%. Those cases were returned after the affine stage. The remaining 1564 cases, or 78.2%, required the weighted linear projection. The solve used a mean of 23.5 iterations and a maximum of 40. Every final state passed both checks.

Figure 5.9 keeps the comparator diagnostics and ICSOR’s deployment stages separate. The rise from zero raw balance compliance to complete final compliance occurred through projection. The branch counts show that the constrained solve was used for most deployed predictions.

These stage counts identify the source of the final compliance. The affine stage restored the adopted balances, and the constrained projection supplied non-negativity for

Table 5.6: Component-state violation frequencies on the 2000 held-out cases. ICSOR uses its final checked state. Comparators use their raw state, not the affine-aligned state used for accuracy scoring. Balance violations refer to A_{num} and the declared 10^{-10} tolerance.

Predictor	Balance violations (%)	Negative states (%)
Multilayer perceptron	100.0	58.20
LightGBM	100.0	46.00
Support vector regression	100.0	68.65
ICSOR	0.0	0.00
XGBoost	100.0	64.95
CatBoost	100.0	70.10
AdaBoost	100.0	0.00
k -nearest neighbors	100.0	0.00
Partial least squares	100.0	48.70
Random Forest	100.0	0.00

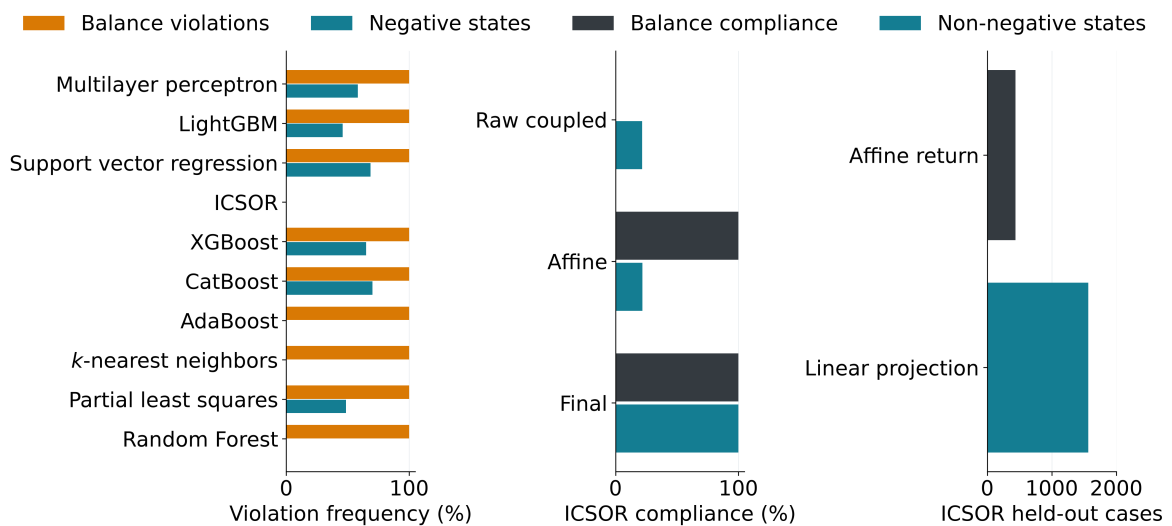


Figure 5.9: Physical diagnostics and ICSOR deployment on 2000 held-out cases. The left panel compares final ICSOR states with raw comparator states. The middle panel separates raw coupled, affine, and final ICSOR compliance. The right panel gives the numbers returned after affine projection and after constrained linear projection. Balance checks use the adopted numerical operator A_{num} .

the remaining cases. The complete deployment procedure returned zero final violation rates.

5.4.4 Computational effort

At the smallest design size, mean ICSOR fitting time was 21.5 s. LightGBM, XGBoost, and the multilayer perceptron required 4.72, 3.16, and 4.99 s, respectively. At the largest size, ICSOR required 94.4 s. The faster tree ensembles required 4.13–8.13 s, while partial least squares and the nearest-neighbor model required 0.228 and 2.70 s. ICSOR was nevertheless faster to fit than AdaBoost at 117.7 s, support vector regression at 136.8 s, and the multilayer perceptron at 161.6 s.

The recursive constrained fitting procedure was more costly than the fastest comparators and faster than three comparators at the largest design size. Fitting time was assessed separately from deployment effort. The 78.2% linear-projection activation rate showed that most fixed-split predictions required an optimization step after regression evaluation. The reported iteration counts described this deployment workload.

Figure 5.10 separates the two design sizes on logarithmic time axes. The smaller-size panel compares four predictors, and the larger-size panel compares ICSOR with five alternatives. The comparison reports the time required for the repeated model fits.

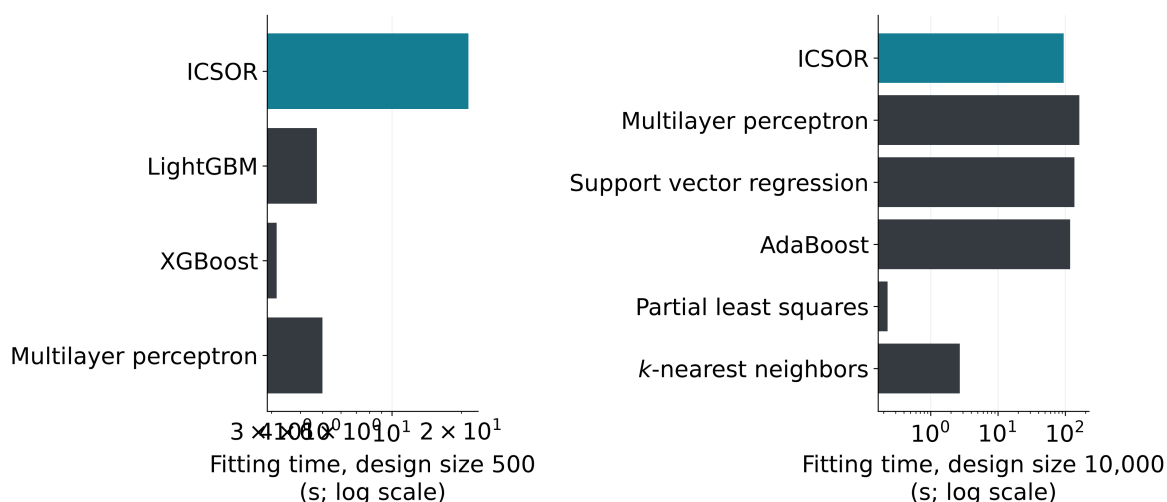


Figure 5.10: Mean fitting times for the specified predictors at design sizes 500 and 10,000. Teal identifies ICSOR. Each panel uses a logarithmic seconds axis. The values concern repeated model fits and exclude deployment calculations.

5.4.5 Inspectable coefficient structure

Table 5.7 reports the COD raw-composite coefficient counts and the shared coupling entries. The polynomial summary counted 276 unique candidates. Adding the 380 off-diagonal coupling candidates gave 656 entries in this display. The display combined

the COD polynomial row with the shared coupling entries. The display retained 469 entries, or 71.5%, under the specified post-fitting thresholds.

Table 5.7: Coefficients exceeding the reporting thresholds in one fitted model. Polynomial blocks describe the COD row of $B_{\text{raw,comp}}$. The coupling matrix is shared across component outputs and uses its own threshold. Symmetric quadratic blocks count unique upper-triangular terms.

Coefficient block	Candidates	Retained	Retained (%)
Baseline b	1	1	100.0
Operating terms W_u	2	2	100.0
Influent terms W_{in}	20	19	95.0
Operating curvature Θ_{uu}	3	3	100.0
Operating-load interactions Θ_{uc}	40	29	72.5
Influent curvature Θ_{cc}	210	44	21.0
Shared off-diagonal coupling Γ	380	371	97.6
Combined display	656	469	71.5

The influent curvature block retained 44 of 210 unique terms. In contrast, 371 of 380 coupling entries exceeded their reporting threshold. The fitted structure therefore combined selective influent curvature with dense component coupling. Selectivity was concentrated in the influent-curvature block, while the coupling matrix was dense.

The largest absolute COD influent-curvature entry was the dissolved-oxygen self term of -5.8097 . Each mirrored dissolved-oxygen and dinitrogen cell had value -1.7411 . Their combined unordered interaction coefficient was -3.4822 . Other prominent terms linked dissolved oxygen with metal phosphate, ammonia-oxidizing biomass, soluble phosphate, nitrite-oxidizing biomass, and nitrite. These values described the raw composite map after coupling and reporting.

Figure 5.11 displays the retained counts and selected oxygen-related curvature terms. Its interaction values combine the two mirrored cells. The comparison describes the fitted polynomial under its native input units.

5.5 Discussion

5.5.1 A structured approximation under limited data

ICSOR's lowest small-design error and lowest reported root-error curve area supported a data-efficiency benefit in the evaluated domain. Its explicit second-order terms defined a structured response approximation. The invariant-aware fitting objective and learned component coupling supplied additional structure. The combined design performed well when fewer simulated states were available. This interpretation

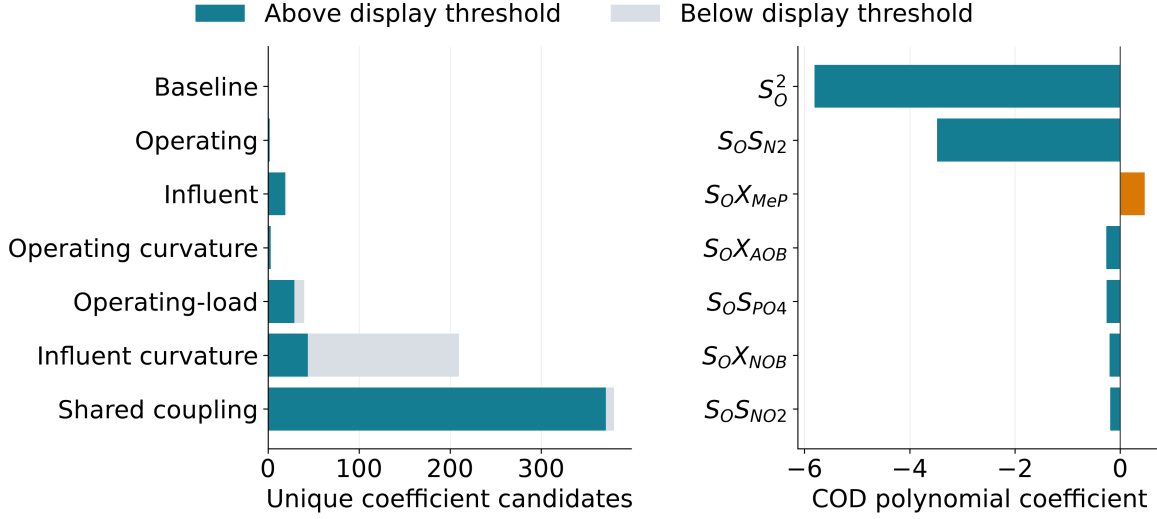


Figure 5.11: Coefficient structure of one fitted ICSOR model. The left panel separates entries above and below the display thresholds for the COD polynomial blocks and shared coupling. The right panel shows selected COD influent-curvature coefficients with mirrored interaction entries combined. Coefficient units depend on the corresponding input terms. The displayed values describe the raw composite mapping.

is consistent with the role of model structure in hybrid learning discussed by Shen et al. (2023) and with the changing model orderings described by Viering and Loog (2023).

The leading predictor depended on the assessment measure. LightGBM had the best mean rank across targets, metrics, and sizes. The multilayer perceptron had the lowest full-size error. ICSOR’s curve area was 6.33, close to LightGBM’s 6.35. Selection based on full-size composite error favored the multilayer perceptron under this protocol. An inspectable equation, data efficiency, and checked final states provided additional engineering criteria for selecting ICSOR.

The training-to-held-out gap and absolute error described complementary properties. Partial least squares combined a small gap with high prediction error. ICSOR combined a gap of 0.22 with substantially lower held-out error, while the leading flexible predictors achieved still lower absolute error. Reporting both measures clarified the relation between generalization and accuracy.

5.5.2 Fitting guidance and final-state enforcement

The physical assessment showed how the complete procedure supplied its returned state. The soft invariant penalty guided the fitted training states, and the learned coupled equation generated the raw prediction for each new input. Every fixed-split raw prediction required affine projection. The affine state passed both checks in 21.8% of cases, and the weighted linear projection handled the remaining 78.2%.

The complete deployment procedure returned zero final balance and non-negativity

violations. Stage-wise diagnostics identified the affine and constrained projections as the source of this compliance. This separation between constrained fitting and output enforcement follows the approaches examined by Hansen et al. (2024) and Kircher and Votsmeier (2025).

The numerical assessment used the rounded and normalized operator A_{num} with a tolerance of 10^{-10} . Accuracy scoring used checked ICSOR states and affine-aligned comparator states. Physical comparator diagnostics used their raw states. These specified procedures produced the 36.5% root-error difference between ICSOR and the multilayer perceptron. The shared alignment and reporting convention also produced the common TP error of 0.0361.

5.5.3 What the coefficient structure explains

The named coefficient blocks exposed operating dependence, loading dependence, curvature, and component coupling for direct inspection. This form of interpretability follows the preference for transparent models discussed by Rudin (2019). Coupling and the reporting map translated the component-driver coefficients into composite response equations.

The retained counts showed where the fitted representation was selective and where it remained dense. Most unique influent-curvature terms were below the display threshold, while almost all coupling entries exceeded their separate threshold. The display retained 469 of 656 entries. The complete fitted mapping retained its coefficients, with reporting thresholds controlling the detail visible in the display.

The prominent oxygen-related curvature terms were consistent with oxygen-sensitive reaction pathways in the adopted process model. Stenstrom and Song (1991) provide process context for oxygen dependence in nitrification, while Wentzel et al. (1992) describe interactions among nutrient-removal pathways. The dissolved-oxygen self coefficient and the oxygen-dinitrogen interaction therefore identified fitted response patterns that could be compared with the reaction model. Their contributions were interpreted in the native input units and through the complete coupled mapping.

The polynomial supplied analytical raw sensitivities, while the affine projection modified them to preserve the influent inventories. Equations (5.58) and (5.61) described these two stages. The frequent constrained branch made active-set sensitivity relevant to the final returned response. Figure 5.4 demonstrated this change through an explicit example in which the first-component slope changed from positive to negative and then became zero at an active non-negativity boundary.

5.5.4 Engineering implications

The structured model combined competitive composite prediction, favorable low-data behavior, and final states that passed the imposed numerical checks. Its

second-order terms and coupling map made the fitted response available for inspection. Checked deployment then enforced the adopted material accounting and component non-negativity before reporting COD, TN, TP, and TSS.

These properties involved distinct computational costs. ICSOR required more fitting time than the fastest tree and linear comparators. At 10,000 cases, it fitted faster than AdaBoost, support vector regression, and the multilayer perceptron. The constrained deployment solve was used for most fixed-split predictions. Separate fitting and deployment measurements therefore described the effort required to construct the approximation and return a checked state for engineering analysis.

Chapter 6

Connected plant surrogates and operating optimization

6.1 Rationale

Mixed liquor is the wastewater and suspended biomass within the biological reactors. Clarifier overflow is the upper outlet carrying treated water. Clarifier underflow is the lower outlet carrying concentrated sludge. Their concentrations and flow rates determine the material discharged, recycled, and wasted.

An activated sludge plant repeatedly moves material between biological reactors and a secondary clarifier. Settled biomass returns to the biological treatment train. Mixed liquor may return to an upstream stage to support nitrogen removal. A separate waste stream removes accumulated solids. Aeration, recycle, and wasting therefore affect more than one treatment unit. The account of biological treatment given by Henze et al. (2008) explains why these decisions need a common plant balance. The clarification and thickening model of Härtel and Pöpel (1992) adds the storage and separation processes that determine which solids return to the reactors.

The operating question is how to choose these settings while balancing treated-water quality and resource use. de Araujo et al. (2013) formulate activated sludge operation as a constrained economic problem and examine the sensitivity of the resulting decisions. Padrón-Páez et al. (2020) examine wastewater treatment design through multiple objectives. Aboagye et al. (2021) organize treatment-network design and sustainability assessment within mathematical programming. These studies make a common point. A preferred decision depends on the objective, the feasible operating region, and the quantities used to measure treatment and resource use.

A process surrogate can reduce the cost of evaluating the same plant at many candidate settings. McBride and Sundmacher (2019) describe surrogate construction from experiments and process simulations. Bhosekar and Ierapetritou (2018) connect surrogate prediction to feasibility analysis and optimization. Applications include the

Kriging treatment model of He et al. (2023), the machine learning process model within predictive control developed by Bernardelli et al. (2020), and the unit surrogates used for wastewater resource-recovery optimization by Durkin et al. (2024). Each application links prediction to a decision problem. Its objective and constraints determine which prediction errors matter most.

A physically admissible prediction for one reactor does not by itself establish a consistent plant prediction. The predicted clarifier underflow may disagree with the predicted return-sludge contribution at the mixer. Separate predictions may also assign different amounts of an inventory used for mass conservation to connected streams. ICSOR supplies a useful starting point because it separates second-order statistical prediction from enforcement of declared physical relations (Selerio, 2026). Extending this principle to a plant requires a joint response and a joint projection. The projection must account for the mixer, all reactors, both clarifier outlets, and the solids stored in the clarifier.

Three questions remain distinct. A network prediction can satisfy the imposed balances. It can also approximate a mechanistic reference accurately. Finally, it can lead an optimizer to a useful operating decision. The surrogate comparisons studied by Pedrozo et al. (2025) make optimization performance a separate endpoint from predictive accuracy. The gray-box methods of Hameed et al. (2026) also retain checks against the original model when managing approximation error. The connected formulation developed here therefore includes independent mechanistic evaluation of selected controls. The connected surrogate is termed Extended ICSOR. The direct route solves a smooth nonlinear programming (NLP) problem and is termed Smooth NLP.

6.2 Theory

6.2.1 Hydraulic connections and reactor inventories

Flow through the treatment train

Let Q_0 denote the fresh-influent flow in $\text{m}^3 \text{d}^{-1}$ and let $c_{\text{in}} \in \mathbb{R}_+^F$ denote its component concentrations. The plant has n_r completely mixed reactors and an L -layer clarifier. The reactor concentrations are $c_1, \dots, c_{n_r}$. The mixer concentration is m . Clarifier overflow and underflow concentrations are c_E and c_U . The overflow is the final plant outlet, so $c_{\text{out}} = c_E$.

The dimensionless ratios r_I , r_R , and w describe mixed-liquor recycle, return activated sludge, and waste activated sludge flow relative to Q_0 . Mixed-liquor recycle leaves the final reactor before the clarifier. Return and waste sludge share the clarifier underflow concentration. Each ratio is an actual flow divided by Q_0 . Denote the train, clarifier-feed, underflow, and overflow flows by Q_P, Q_C, Q_U, Q_E . The mixer, reactor

splitter, underflow splitter, and external water balance respectively give

$$Q_P = Q_0 + r_I Q_0 + r_R Q_0, \quad (6.1)$$

$$Q_C = Q_P - r_I Q_0, \quad (6.2)$$

$$Q_U = r_R Q_0 + w Q_0, \quad (6.3)$$

$$Q_E = Q_0 - w Q_0. \quad (6.4)$$

The second equation removes the mixed-liquor recycle before clarification. The third includes both underflow destinations. Dividing each hydraulic balance by Q_0 gives

$$q_P = 1 + r_I + r_R, \quad q_C = 1 + r_R, \quad q_U = r_R + w, \quad q_E = 1 - w. \quad (6.5)$$

Here q_P is the reactor-train flow ratio and q_C is the clarifier-feed flow ratio. The identity $q_C = q_E + q_U$ confirms the clarifier water balance. The adopted domain requires $r_I \geq 0$, $r_R \geq 0$, $0 \leq w < 1$, and $q_U > 0$. The outlet concentrations are then defined by positive outlet flows.

Figure 6.1 places these flow balances on the physical connections of the five-reactor plant. The two recycle routes return material to the mixer, while treated water and waste sludge cross the external boundary.

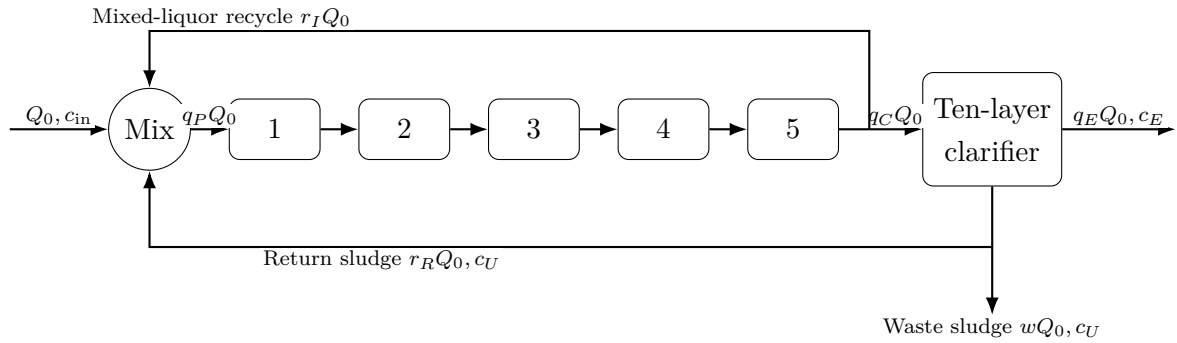


Figure 6.1: Connected plant configuration used for the operating study. Reactors 1 and 2 are unaerated. Reactors 3–5 have independent aeration controls.

The diagram shows why the reactor-train and clarifier-feed flows differ. Mixed-liquor recycle leaves before the clarifier, while return sludge leaves the clarifier underflow. Their concentrations consequently come from different locations. The mixer calculation uses each stream's own flow and concentration rather than assigning one shared recycle concentration.

The mixer balance is a component balance rather than a balance of a single water-quality total. For component f , its incoming and outgoing mass rates satisfy

$$Q_P m_f = Q_0 c_{in,f} + r_I Q_0 c_{n_r,f} + r_R Q_0 c_{U,f}. \quad (6.6)$$

Each product has units of g d^{-1} when the component basis is g m^{-3} . Divide by Q_0 and then by q_P to obtain

$$m_f = \frac{c_{in,f} + r_I c_{n_r,f} + r_R c_{U,f}}{1 + r_I + r_R}. \quad (6.7)$$

For two components, the same calculation is performed independently in each row,

$$\begin{bmatrix} q_P & 0 \\ 0 & q_P \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} = \begin{bmatrix} c_{in,1} + r_I c_{nr,1} + r_R c_{U,1} \\ c_{in,2} + r_I c_{nr,2} + r_R c_{U,2} \end{bmatrix}. \quad (6.8)$$

The complete concentration vector therefore satisfies

$$q_P m = c_{in} + r_I c_{nr} + r_R c_U. \quad (6.9)$$

A high mixer concentration can therefore result from internal circulation. It does not imply that an external pollutant load has been created. The component-resolved framework of X. Zhang et al. (2026) treats these recirculation and discharge streams explicitly.

A hypothetical mixing calculation makes the units and recycle effect explicit. Let $Q_0 = 10,000 \text{ m}^3 \text{ d}^{-1}$, $r_I = 2$, $r_R = 0.5$, and $w = 0.02$. The four physical flows are 35,000, 15,000, 5,200, and 9,800 $\text{m}^3 \text{ d}^{-1}$ in the order Q_P, Q_C, Q_U, Q_E . For one particulate component, specify fresh, final-reactor, and underflow concentrations of 200, 1,000, and 2,000 g m^{-3} . The mixer calculation is

$$q_P m_f = 200 + 2(1,000) + 0.5(2,000) = 3,200, \quad (6.10)$$

$$m_f = 3,200/3.5 = 914.29 \text{ g m}^{-3}. \quad (6.11)$$

The physical incoming mass rates are 2,000, 20,000, and 10,000 kg d^{-1} . Their sum is the 32,000 kg d^{-1} leaving the mixer. The mixer concentration exceeds the fresh concentration because the two return streams contain concentrated material. These hypothetical streams illustrate the flow-weighted mixer calculation.

The nominal stage HRT is measured on the fresh-flow basis. A volume divided by a flow gives time. Thus V_i/Q_0 has units of days, and multiplying by 24 converts it to hours. The actual flow through the reactor is $Q_P = Q_0 q_P$, so its actual residence time is V_i/Q_P . The inverse of that actual time is the dilution rate. With reactor volume V_i ,

$$h_i = \frac{24V_i}{Q_0}, \quad H = \sum_{i=1}^{n_r} h_i, \quad d_i = \frac{Q_0 q_P}{V_i} = \frac{24q_P}{h_i}. \quad (6.12)$$

The factors h_i and H are in hours. The dilution rate d_i is in d^{-1} . The actual residence time during one passage through stage i is h_i/q_P . For a hypothetical plant with $H = 24$ h, $r_I = 2$, and $r_R = 0.5$, the reactor-train flow ratio is 3.5. Its train residence time per passage is about 6.86 h. Equal tanks then have a residence time per passage of about 1.37 h. These illustrative numbers explain why nominal HRT and recycle flow must be interpreted together.

Fixed stage fractions can be represented by $h_i = f_i^V H$, where $f_i^V > 0$ and $\sum_i f_i^V = 1$. The case study uses equal fractions. Varying H at fixed Q_0 varies biological reactor volume. The resulting problem combines capacity selection and operating decisions.

For an existing plant with installed tanks, the volumes would be fixed and H would be removed from the free-control vector.

Let ϑ collect only settings that vary freely. A constant setting remains a parameter. Treating it as a varying regression coordinate would create a zero-variance feature. A carbon or chemical addition may also enter the control vector. An addition represented only by a source term has negligible liquid carrier flow. An appreciable carrier flow requires a new hydraulic stream and revised downstream dilution rates.

Biological conversion and non-negativity

The reaction network uses the component and process model of Chapter 3. Its stoichiometric matrix is $\nu \in \mathbb{R}^{P \times F}$ and its process-rate vector in stage i is $\rho_i \in \mathbb{R}_+^P$. For component f , the dynamic component equation is

$$V_i \frac{dc_{i,f}}{dt} = Q_P c_{i-1,f} - Q_P c_{i,f} + V_i \sum_{p=1}^P \nu_{pf} \rho_{i,p} + V_i \omega_i (e_{S_O})_f + V_i \xi_{i,f}. \quad (6.13)$$

The reaction sum combines the change caused by each process. A negative ν_{pf} consumes component f , while a positive entry produces it. The oxygen selector is one for the oxygen component and zero elsewhere. At steady state, the accumulation term is zero. Dividing the remaining equation by V_i gives

$$0 = d_i (c_{i-1,f} - c_{i,f}) + \sum_{p=1}^P \nu_{pf} \rho_{i,p} + \omega_i (e_{S_O})_f + \xi_{i,f}. \quad (6.14)$$

Putting all F component equations into one vector gives the steady reactor equation, with $c_0 = m$,

$$0 = d_i (c_{i-1} - c_i) + \nu^\top \rho_i (c_i, \vartheta) + k_L a_i(a_i) (S_O^* - c_{i,S_O}) e_{S_O} + \xi_i(\vartheta). \quad (6.15)$$

The vector e_{S_O} selects dissolved oxygen. The parameter S_O^* is its saturation concentration. The vector ξ_i represents declared stage additions. Each term has units of component concentration per day. The treatment-model framework of Henze et al. (2000) separates hydraulic transport, stoichiometric conversion, and external sources in this way.

At a zero component concentration, the combined reaction and external-source terms must not direct that component below zero. With non-negative inflow and a locally Lipschitz balance function defined on the boundary, this condition preserves the non-negative state space of the dynamic model (Haddad et al., 2010). Numerical acceptance checks concentrations and reaction rates independently of the solver's stopping message.

The same distinction matters for a surrogate. A fitted quadratic equation can predict a negative concentration even when every training state is non-negative. Polynomial approximation imposes no sign restriction. The work of Kircher and Votsmeier (2025) addresses the simultaneous requirements of mass conservation and non-negativity in kinetic surrogates. Both requirements need explicit treatment.

Stage invariants and the external plant balance

Consider a hypothetical reaction that transfers material from component 1 to component 2 without changing their combined amount. Its stoichiometric and inventory rows are

$$\nu = \begin{bmatrix} -1 & 1 \end{bmatrix}, \quad A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}. \quad (6.16)$$

The factor $1/\sqrt{2}$ gives the inventory row unit length. Matrix multiplication shows the cancellation directly,

$$A\nu^\top = \frac{1}{\sqrt{2}} [1(-1) + 1(1)] = 0. \quad (6.17)$$

At an illustrative dilution rate of 2 d^{-1} , take upstream concentrations $(12, 3)$ and downstream concentrations $(8, 7) \text{ g m}^{-3}$, with a reaction rate of $8 \text{ g m}^{-3} \text{ d}^{-1}$. The component equations are

$$0 = 2(12 - 8) - 8, \quad (6.18)$$

$$0 = 2(3 - 7) + 8. \quad (6.19)$$

Adding them removes the unknown reaction rate. Both upstream and downstream totals are 15 g m^{-3} . The normalized invariant is that total divided by $\sqrt{2}$. The full reactor model applies the same cancellation to more components and more processes.

Let $A \in \mathbb{R}^{K \times F}$ contain independent inventory rows such that

$$A\nu^\top = 0, \quad Ae_{S_O} = 0, \quad \text{rank}(A) = K. \quad (6.20)$$

The rows are normalized to unit Euclidean length. Unit row normalization does not make them orthogonal. Physically named inventories and an orthonormal basis can span the same subspace defined by mass conservation while using different coordinates. The stoichiometric fixed-layer formulation of Sturm and Wexler (2022) illustrates the value of separating quantities used for mass conservation from the rates that redistribute them.

For inventory row k , multiply component equation f by A_{kf} and sum over f . The reaction contribution becomes

$$\sum_{f=1}^F A_{kf} \sum_{p=1}^P \nu_{pf} \rho_{i,p} = \sum_{p=1}^P \left(\sum_{f=1}^F A_{kf} \nu_{pf} \right) \rho_{i,p} = 0. \quad (6.21)$$

The inner sum is entry (k, p) of $A\nu^\top$. Oxygen transfer also disappears because $Ae_{S_O} = 0$. The remaining scalar equation is

$$d_i \left(\sum_f A_{kf} c_{i-1,f} - \sum_f A_{kf} c_{i,f} \right) + \sum_f A_{kf} \xi_{i,f} = 0. \quad (6.22)$$

Moving the transport difference to the other side and dividing by the positive d_i gives the remaining stage relation,

$$A(c_i - c_{i-1}) = d_i^{-1} A\xi_i(\vartheta). \quad (6.23)$$

If there is no stage addition, $Ac_i = Ac_{i-1}$. For example, nitrification can transfer nitrogen from ammonium to oxidized forms while denitrification transfers it to dissolved nitrogen gas. These changes do not remove nitrogen from the model's complete nitrogen inventory. A reporting quantity that excludes dissolved nitrogen gas can decrease while the inventory used for mass conservation remains unchanged. Nutrient-form interpretation in U.S. Environmental Protection Agency (2009) supports the need to inspect individual species as well as reported totals.

In the two-component example, adding component 1 at $2 \text{ g m}^{-3} \text{ d}^{-1}$ changes the downstream concentrations to (9, 7) when the same reaction rate is used. The two balances are $2(12 - 9) - 8 + 2 = 0$ and $2(3 - 7) + 8 = 0$. Their total changes by $2/2 = 1 \text{ g m}^{-3}$. Thus the invariant transition must include a dose whenever that dose carries the quantity used for mass conservation. A zero invariant difference would incorrectly remove the added material.

Internal recycle streams also cancel from an external invariant balance. First add the stage transitions. Every intermediate Ac_i appears once with a plus sign and once with a minus sign. This gives

$$A(c_{n_r} - m) = \sum_i \frac{V_i}{Q_0 q_P} A\xi_i, \quad (6.24)$$

$$q_P Ac_{n_r} - q_P Am = D_{\text{add}}, \quad D_{\text{add}} = \sum_i \frac{V_i}{Q_0} A\xi_i. \quad (6.25)$$

Next multiply the mixer balance by A and substitute it for $q_P Am$,

$$q_P Ac_{n_r} = Ac_{\text{in}} + r_I Ac_{n_r} + r_R Ac_U + D_{\text{add}}, \quad (6.26)$$

$$(q_P - r_I) Ac_{n_r} = Ac_{\text{in}} + r_R Ac_U + D_{\text{add}}. \quad (6.27)$$

Since $q_P - r_I = q_C$, the left-hand side is the clarifier-feed inventory flow. The clarifier balance gives a second expression for that same quantity,

$$q_C Ac_{n_r} = q_E Ac_E + (r_R + w) Ac_U. \quad (6.28)$$

Equating the expressions and subtracting $r_R Ac_U$ cancels the return-sludge flow. The mixed-liquor recycle already canceled in $q_P - r_I$. The remaining external relation is

$$Ac_{\text{in}} + \sum_{i=1}^{n_r} \frac{V_i}{Q_0} A\xi_i = q_E Ac_E + w Ac_U. \quad (6.29)$$

Only fresh feed, stage additions, effluent, and waste sludge remain. The recycle ratios still influence the state and treatment processes. Their cancellation means that they are internal transfers rather than external sources or sinks.

For ASM2d-TSN, $F = 20$, $P = 28$, and $K = 5$. The five inventories concern soluble inert material, total nitrogen including dissolved nitrogen gas, total phosphorus, an alkalinity-related combination, and a metal-related combination. The last two

unnormalized combinations are

$$I_{\text{alk}} = S_{\text{ALK}} - \frac{S_{\text{NH}_4}}{14} + \frac{S_{\text{NO}_2}}{14} + \frac{S_{\text{NO}_3}}{14} - \frac{S_{\text{PO}_4}}{31}, \quad (6.30)$$

$$I_{\text{metal}} = 3.45X_{\text{MeP}} + 4.87X_{\text{MeOH}}. \quad (6.31)$$

The nitrogen and phosphorus rows use the constituent weights in Table 3.1. The nitrogen inventory also gives dissolved nitrogen gas unit weight. The stoichiometric rank is 15 under the stated parameterization. The inventory identities are audited rather than inferred from a reporting label such as TN.

6.2.2 Clarification and solids storage

A nonreactive separator with two outlet flows

The clarifier uses a nonreactive settling model. Nonreactive means that the adopted clarifier equations transport and separate components without biological or chemical conversion within the unit. Soluble components follow the liquid. Particulate components can settle and concentrate in the underflow. Takács et al. (1991) provide the settling and thickening formulation used for the aggregate solids profile. Jeppsson and Diehl (1996) explain why component propagation through a clarifier requires attention beyond a single aggregate solids concentration.

Selection means taking designated entries of a vector without changing them. For the illustrative order $(S_A, S_{\text{NH}_4}, X_H, X_I)$, the two selector matrices are

$$E_S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad E_X = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (6.32)$$

Thus $E_S c = (S_A, S_{\text{NH}_4})^\top$ and $E_X c = (X_H, X_I)^\top$. Transposing a selector puts selected entries back into their original positions and fills the other positions with zeros. In this example,

$$E_S^\top E_S = \text{diag}(1, 1, 0, 0), \quad E_X^\top E_X = \text{diag}(0, 0, 1, 1). \quad (6.33)$$

Their sum is the identity matrix because every component belongs to exactly one group. Their cross-product is zero because the groups do not overlap. For the complete ordered model, $E_S = [I_{10} \ 0]$ and $E_X = [0 \ I_{10}]$. In general, let $E_S \in \mathbb{R}^{F_S \times F}$ and $E_X \in \mathbb{R}^{F_X \times F}$ select soluble and particulate components. They satisfy

$$E_S^\top E_S + E_X^\top E_X = I_F, \quad E_S E_X^\top = 0. \quad (6.34)$$

The column t_X maps component concentrations to TSS. Its entries are non-negative and its soluble entries are zero.

For the four-component order above, $t_X = (0, 0, 0.90, 0.75)^\top$. If $X_H = 100$ and $X_I = 40$ g COD m⁻³, the TSS calculation is $0.90(100) + 0.75(40) = 120$ g TSS m⁻³. The two factors convert component COD bases to suspended-solids mass. Adding the

native numerical concentrations without these conversion factors would give a different quantity.

For one component, mass conservation across the clarifier gives

$$Q_C c_{nr,f} = Q_E c_{E,f} + Q_U c_{U,f}. \quad (6.35)$$

The statistical response uses component flows divided by Q_0 ,

$$g_E = q_E c_E, \quad g_U = q_U c_U. \quad (6.36)$$

Their units are the same as concentration because the flow ratios are dimensionless. The actual component discharge rates are $Q_0 g_E$ and $Q_0 g_U$. Writing balances in these coordinates avoids dividing by a small underflow ratio inside the equality system. The component balance and soluble pass-through are

$$g_E + g_U = q_C c_{nr}, \quad E_S g_U = q_U E_S c_{nr}. \quad (6.37)$$

The two relations imply $E_S g_E = q_E E_S c_{nr}$. Settling therefore cannot remove a soluble nutrient directly in this model.

For a soluble component f , pass-through fixes $c_{U,f} = c_{nr,f}$. Substitute $g_{U,f} = q_U c_{nr,f}$ into the total balance to obtain

$$g_{E,f} = (q_C - q_U) c_{nr,f} = q_E c_{nr,f}, \quad (6.38)$$

$$c_{E,f} = g_{E,f} / q_E = c_{nr,f}. \quad (6.39)$$

For a hypothetical feed concentration of 20 g m^{-3} and flow ratios $q_C = 1.5$, $q_U = 0.52$, and $q_E = 0.98$, the feed, underflow, and overflow component flows are 30, 10.4, and 19.6 g m^{-3} on the fresh-flow basis. Both outlet concentrations remain 20 g m^{-3} .

The particulate densification requirement is

$$E_X g_U \geq q_U E_X c_{nr}. \quad (6.40)$$

After division by q_U , each underflow particulate concentration is at least its feed concentration. Non-negative overflow and the total component balance also prevent the underflow from recovering more than the incoming particulate flow.

For an illustrative particulate component, take $q_C = 1.5$, $q_U = 0.52$, and feed concentration 100 g m^{-3} . Densification requires $g_U \geq 52 \text{ g m}^{-3}$ on the fresh-flow basis. Mass conservation and non-negative overflow require $g_U \leq 150 \text{ g m}^{-3}$. Thus the underflow recovery fraction lies between $0.52/1.5$ and one. A soluble component with the same feed concentration instead has $g_U = 52$ and unchanged soluble outlet concentrations. These hypothetical examples describe the soluble and particulate constraints.

An admissible illustrative choice within that particulate interval is $g_U = 140$. Mass conservation then gives $g_E = 150 - 140 = 10$. Dividing by the respective flow ratios gives $c_U = 140/0.52 = 269.23$ and $c_E = 10/0.98 = 10.20 \text{ g m}^{-3}$. The underflow has

recovered $140/150 = 93.33\%$ of the incoming component flow. The concentrations alone do not give that recovery fraction because the outlet water flows differ.

For any particulate component with positive feed concentration, the same reasoning gives

$$\frac{q_U}{q_C} \leq \frac{g_{U,f}}{q_C c_{n_r,f}} \leq 1. \quad (6.41)$$

Multiplication by the non-negative TSS weights preserves inequalities. Hence $X_U \geq X_N$. The solids balance $q_C X_N = q_E X_E + q_U X_U$ then gives

$$q_E X_E \leq (q_C - q_U) X_N = q_E X_N. \quad (6.42)$$

Division by $q_E > 0$ establishes the other endpoint ordering.

Applying t_X gives feed and outlet solids concentrations

$$X_N = t_X^\top c_{n_r}, \quad X_E = \frac{t_X^\top g_E}{q_E}, \quad X_U = \frac{t_X^\top g_U}{q_U}. \quad (6.43)$$

The declared constraints imply $X_E \leq X_N \leq X_U$. They bound separation but do not determine its magnitude. The nonlinear settling calculation and an empirical endpoint model supply additional information through different routes.

The layer-resolved mechanistic model

Clarifier layers are numbered from the surface to the hopper. Their TSS concentrations are $s_1, \dots, s_L$, with $s_1 = X_E$ and $s_L = X_U$. The exact settling velocity at concentration s is

$$v(s, X_N) = \max \left\{ 0, \min \left[v_{\max}, v_0 \left(e^{-r_h(s - f_{\text{ns}} X_N)} - e^{-r_p(s - f_{\text{ns}} X_N)} \right) \right] \right\}. \quad (6.44)$$

The gravitational solids flux is $sv(s, X_N)$. The nonsettleable fraction f_{ns} makes the settling offset depend on the feed concentration. At an interface, a receiver concentration above X_t activates a limitation by the smaller adjacent gravitational flux. Otherwise, the upper-layer gravitational flux passes to the receiver. Hydraulic transport uses the overflow velocity above the feed layer and the underflow velocity below it.

Each finite-volume layer equation is its incoming solids flux minus its outgoing solids flux, multiplied by $A_{\text{cl}}/V_{\text{cl},\ell}$. The feed layer also receives $Q_0 q_C X_N / V_{\text{cl},\ell_F}$. Internal interface fluxes cancel when these equations are summed. At steady state, the resulting total solids balance agrees with Equation (6.37) after application of t_X .

A three-layer illustration shows that cancellation without hiding it inside a summation. Place the feed in layer 2. Let B_{12} be the net solids rate from layer 2 toward layer 1 and B_{23} the net rate from layer 2 toward layer 3. These signed rates combine hydraulic and gravitational transport in g d^{-1} . The layer mass equations are

$$V_1 \dot{s}_1 = B_{12} - Q_E s_1, \quad (6.45)$$

$$V_2 \dot{s}_2 = Q_C X_N - B_{12} - B_{23}, \quad (6.46)$$

$$V_3 \dot{s}_3 = B_{23} - Q_U s_3. \quad (6.47)$$

Adding them removes both interface rates. At steady state, the remaining equation is $Q_C X_N = Q_E X_E + Q_U X_U$. This cancellation holds even though the interface rates depend nonlinearly on the neighboring concentrations. Mass conservation of the whole clarifier therefore does not determine its internal fluxes or profile.

The mechanistic state retains one aggregate solids concentration per layer. At a steady state, particulate outlet components are reconstructed with the final-reactor particulate composition,

$$E_X c_E = \frac{s_1}{X_N} E_X c_{n_r}, \quad E_X c_U = \frac{s_L}{X_N} E_X c_{n_r}. \quad (6.48)$$

Soluble outlet components equal their feed concentrations. The steady component mapping is defined for $X_N > 0$ and uses the final-reactor particulate composition (Jeppsson & Diehl, 1996).

An exact interval for aggregate inventory under the endpoint envelope

Inventory is concentration multiplied by volume, summed over the vessel. For an illustrative four-layer clarifier, it is

$$M = V_1 X_E + V_2 s_2 + V_3 s_3 + V_4 X_U. \quad (6.49)$$

For example, four 100-m³ layers with concentrations (10, 200, 600, 1,000) g m⁻³ store $100(10 + 200 + 600 + 1,000) = 181,000$ g, or 181 kg. The general clarifier's stored solids mass is

$$M_{cl} = \sum_{\ell=1}^L V_{cl,\ell} s_\ell, \quad V_{cl} = \sum_{\ell=1}^L V_{cl,\ell}. \quad (6.50)$$

An outlet concentration cannot determine this mass. Two clarifiers can have the same endpoints and different internal storage. Whole-plant solids retention time therefore requires the inventory as well as the two outlet solids flows. The benchmark formulation of Alex et al. (2008) includes reactor and settler inventories in its sludge-age calculation.

The mechanistic acceptance envelope requires

$$X_E \leq s_\ell \leq X_U, \quad \ell = 2, \dots, L-1. \quad (6.51)$$

This is an endpoint envelope. It does not require monotonic concentration from one layer to the next. With positive layer volumes, minimizing the inventory assigns every internal layer X_E . Maximizing it assigns every internal layer X_U .

For four layers, the envelope bounds each of the two unknown terms separately,

$$V_2 X_E + V_3 X_E \leq V_2 s_2 + V_3 s_3 \quad (6.52)$$

$$\leq V_2 X_U + V_3 X_U. \quad (6.53)$$

Adding the known endpoint terms gives

$$(V_1 + V_2 + V_3) X_E + V_4 X_U \leq M, \quad (6.54)$$

$$M \leq V_1 X_E + (V_2 + V_3 + V_4) X_U. \quad (6.55)$$

For the hypothetical four equal layers above, these values are 103,000 and 301,000 g. The 181,000-g inventory lies inside the interval. If both internal concentrations were instead 400 g m^{-3} , the mass would still be 181,000 g. Thus matching endpoints and total inventory does not identify the internal distribution.

Figure 6.2 shows the same four-layer arithmetic as concentration bars. Changing the two internal layers changes stored mass while the surface and hopper concentrations remain fixed. The left and right panels attain the lower and upper interval endpoints derived from the envelope.

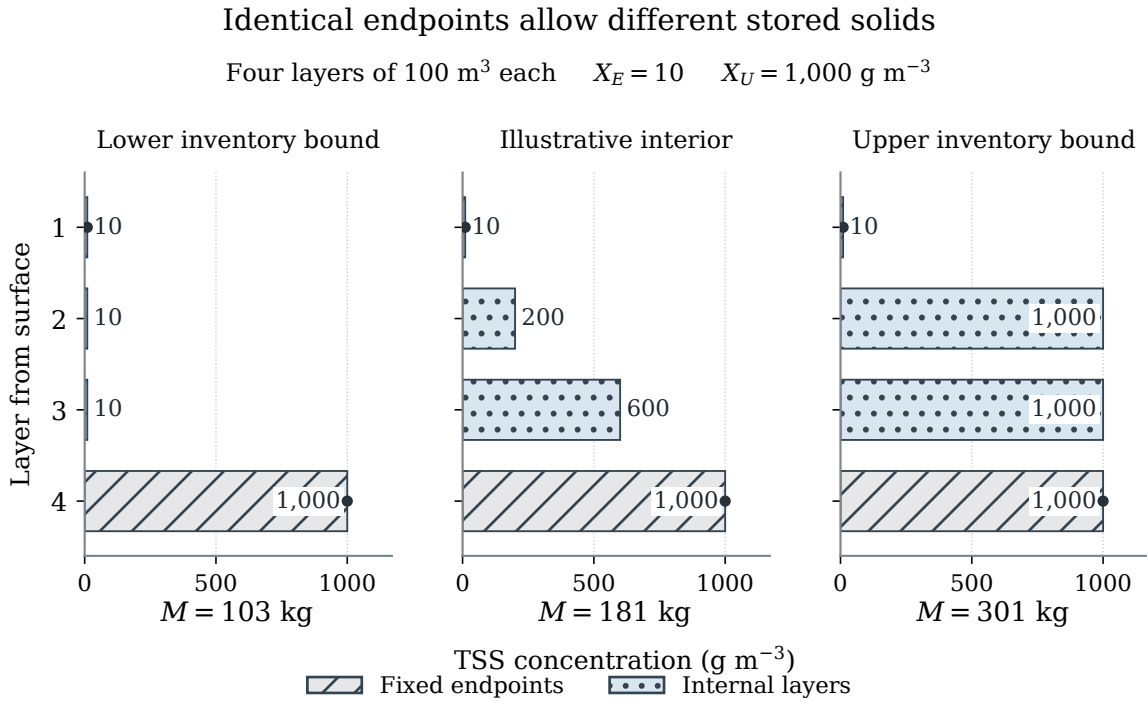


Figure 6.2: Hypothetical four-layer profiles with identical overflow and underflow concentrations. Each layer has volume 100 m^3 . The lower-bound, illustrative interior, and upper-bound profiles store 103, 181, and 301 kg of solids. Hatching identifies the fixed endpoints, while dotted fills identify internal layers. The diagram represents the adopted endpoint envelope.

The internal layers in the middle panel need not resemble those of another profile with the same inventory. Their total mass is the retained scalar information. The nonlinear layer equations supply the additional information needed to determine a mechanistic profile.

With L layers, the identical argument replaces $V_2 + V_3$ by $\sum_{\ell=2}^{L-1} V_{\text{cl},\ell}$. That internal-volume sum is $V_{\text{cl}} - V_{\text{cl},1} - V_{\text{cl},L}$. Collecting coefficients of each endpoint gives the general

interval,

$$M_{\text{cl}}^- = (V_{\text{cl}} - V_{\text{cl},L})X_E + V_{\text{cl},L}X_U, \quad (6.56)$$

$$M_{\text{cl}}^+ = V_{\text{cl},1}X_E + (V_{\text{cl}} - V_{\text{cl},1})X_U, \quad (6.57)$$

$$M_{\text{cl}}^- \leq M_{\text{cl}} \leq M_{\text{cl}}^+. \quad (6.58)$$

These bounds are necessary and sufficient for a profile with the prescribed endpoints and envelope to have the stated inventory. To see sufficiency, give every internal layer a common concentration s_* between X_E and X_U . The inventory is then

$$V_{\text{cl},1}X_E + V_{\text{cl},L}X_U + (V_{\text{cl}} - V_{\text{cl},1} - V_{\text{cl},L})s_*. \quad (6.59)$$

This expression moves continuously through the full interval as s_* varies. For equal-volume layers, it also gives a simple constructive profile for any admitted inventory.

As an illustrative calculation, take ten layers of 600 m^3 , $X_E = 10 \text{ g m}^{-3}$, and $X_U = 5,000 \text{ g m}^{-3}$. The allowable inventory interval is 3,054,000–27,006,000 g. Each inventory within this interval has an envelope-compatible profile. Joint projection enforces the interval, and layer-resolved mechanistic evaluation checks the flux equations.

For these ten equal layers, the calculation can be written out as

$$M_{\text{cl}} = 600(X_E + s_2 + s_3 + \cdots + s_9 + X_U), \quad (6.60)$$

$$M_{\text{cl}}^- = 600(9X_E + X_U), \quad (6.61)$$

$$M_{\text{cl}}^+ = 600(X_E + 9X_U). \quad (6.62)$$

The lower endpoint has nine layers at X_E and one at X_U . The upper endpoint has one layer at X_E and nine at X_U . This explains both coefficients in the compact interval rather than treating them as fitted constants.

6.2.3 Statistical prediction of the connected response

A response that preserves network and storage information

For a teaching example with one reactor and two components, denote a soluble component by S and a particulate component by X . The stored response is the explicit nine-entry column

$$\chi_{\text{toy}} = (m_S, m_X, c_{1,S}, c_{1,X}, g_{E,S}, g_{E,X}, g_{U,S}, g_{U,X}, M_{\text{cl}})^\top. \quad (6.63)$$

Adding a second reactor inserts $c_{2,S}$ and $c_{2,X}$ before the outlet coordinates and increases the length to eleven. Every reactor contributes F entries. The mixer and two outlets each contribute another F , while the clarifier inventory contributes one. The connected surrogate therefore predicts

$$\chi = \text{col}(m, c_1, \dots, c_{n_r}, g_E, g_U, M_{\text{cl}}) \in \mathbb{R}^{n_\chi}, \quad n_\chi = (n_r + 3)F + 1. \quad (6.64)$$

Here `col` stacks its arguments into one column. The response preserves concentrations needed for stage inventories, outlet flows needed for material balances, and clarifier storage needed for solids retention time. It omits the internal layer profile. The full mechanistic state is

$$y = \text{col}(c_1, \dots, c_{n_r}, s_1, \dots, s_L) \in \mathbb{R}_+^{n_y}, \quad n_y = n_r F + L. \quad (6.65)$$

The mapping from y to χ reconstructs the mixer, both outlets, and inventory deterministically. This mapping lets two different response dimensions support the same engineering calculations.

For the five-reactor case, the response count is $20 + 5(20) + 20 + 20 + 1 = 161$. The full-state count is $5(20) + 10 = 110$. Table 6.1 makes the response ordering explicit. A position identifies a coordinate and does not imply that neighboring entries have the same physical unit.

Table 6.1: Blocks in the 161-coordinate connected response.

Block	Positions	Coordinates
Mixer	1–20	$m_1, \dots, m_{20}$
Reactor 1	21–40	$c_{1,1}, \dots, c_{1,20}$
Reactor 2	41–60	$c_{2,1}, \dots, c_{2,20}$
Reactor 3	61–80	$c_{3,1}, \dots, c_{3,20}$
Reactor 4	81–100	$c_{4,1}, \dots, c_{4,20}$
Reactor 5	101–120	$c_{5,1}, \dots, c_{5,20}$
Overflow flows	121–140	$g_{E,1}, \dots, g_{E,20}$
Underflow flows	141–160	$g_{U,1}, \dots, g_{U,20}$
Clarifier inventory	161	M_{cl}

Complete second-order features and ridge estimation

The statistical input includes every free control and every fresh-influent component. Standardization first subtracts a fitting-set mean and then divides by a fitting-set standard deviation. For one operating coordinate and one influent coordinate,

$$z_1 = \frac{\vartheta_1 - \bar{\vartheta}_1}{s_{\vartheta_1}}, \quad z_2 = \frac{c_{\text{in},1} - \bar{c}_{\text{in},1}}{s_{\text{in},1}}. \quad (6.66)$$

These are dimensionless coordinates. For hypothetical aeration 0.8, fitting mean 0.5, and fitting scale 0.1, the standardized value is 3. For hypothetical ammonium concentration 40 g m^{-3} with mean 30 and scale 10 g m^{-3} , the value is 1. The different native units no longer determine the regression penalty. For fitting-set centers and positive diagonal scales, define

$$z = \text{col} \left[D_{\vartheta, \text{fit}}^{-1}(\vartheta - \bar{\vartheta}), D_{\text{in}, \text{fit}}^{-1}(c_{\text{in}} - \bar{c}_{\text{in}}) \right] \in \mathbb{R}^{d_z}, \quad d_z = d_{\vartheta} + F. \quad (6.67)$$

For two standardized inputs, their symmetric product matrix is

$$zz^\top = \begin{bmatrix} z_1^2 & z_1 z_2 \\ z_2 z_1 & z_2^2 \end{bmatrix}. \quad (6.68)$$

Because multiplication commutes, $z_1 z_2$ and $z_2 z_1$ are the same feature. One triangular copy therefore gives the unique quadratic entries $(z_1^2, z_1 z_2, z_2^2)$. Including the linear entries and intercept produces six terms,

$$(1, z_1, z_2, z_1^2, z_1 z_2, z_2^2)^\top. \quad (6.69)$$

Three inputs would give an intercept, three linear terms, three squares, and three distinct cross-products, for ten features. In general there are d_z squares and $d_z(d_z - 1)/2$ cross-products. Their sum is $d_z(d_z + 1)/2$. The complete quadratic feature family is

$$r(z) = \text{col} \left[z, \text{vech}(zz^\top) \right], \quad (6.70)$$

$$\phi(\vartheta, c_{\text{in}}) = \text{col} \left[1, D_{r, \text{fit}}^{-1}(r(z) - \bar{r}) \right], \quad (6.71)$$

$$n_\phi = 1 + d_z + \frac{d_z(d_z + 1)}{2}. \quad (6.72)$$

The operator vech retains one triangular copy of the symmetric matrix. It includes every square and every pairwise product once. A product between ammonium loading and aeration allows their joint effect to differ from the sum of their separate effects. A square of the recycle coordinate permits local curvature. These terms approximate the input-response relation.

Quadratic features undergo their own fitting-set centering and scaling after they are constructed. For the hypothetical standardized inputs $(3, 1)$, the unscaled nonconstant terms are $(3, 1, 9, 3, 1)$. If the first squared feature has fitting mean 1 and standard deviation 2, its final value is $(9 - 1)/2 = 4$. If the cross-product has fitting mean 0.2 and standard deviation 0.5, its final value is $(3 - 0.2)/0.5 = 5.6$. This second standardization prevents squares with large variance from receiving a weaker effective ridge penalty merely because they are numerically larger.

A monomial is a single polynomial term, such as z_1^2 or $z_1 z_2$. The feature set containing each distinct term once spans the same quadratic functions as an ordered feature set that retains both $z_1 z_2$ and $z_2 z_1$. Their two coefficients enter prediction only through their sum. For an effective cross-product coefficient b , an ordered feature set can assign $b/2$ to each duplicate. The squared coefficient penalty is then $b^2/2$, compared with b^2 for one unique coefficient. Thus removing duplicates preserves the polynomial family but changes the regularization convention unless the penalty is adjusted. The connected fit selects its penalty for the stated feature set containing each distinct term once. For 27 inputs, the count is $1 + 27 + 378 = 406$.

For response coordinate k , write its standardized target as $\tilde{\chi}_{ik} = (\chi_{ik} - \bar{\chi}_k)/s_{\chi, k}$.

Its fitted value is $C_{k0} + \sum_{j=1}^{n_\phi-1} C_{kj}\phi_{ij}$. The scalar ridge objective for all responses is

$$\frac{1}{nn_\chi} \sum_{i=1}^n \sum_{k=1}^{n_\chi} \left(\tilde{\chi}_{ik} - \sum_{j=0}^{n_\phi-1} C_{kj}\phi_{ij} \right)^2 + \frac{\gamma}{n_\chi} \sum_{k=1}^{n_\chi} \sum_{j=1}^{n_\phi-1} C_{kj}^2. \quad (6.73)$$

The index $j = 0$ is the intercept, with feature value one. It appears in prediction and does not appear in the penalty sum. Dividing a physical prediction error by $s_{\chi,k}$ is equivalent to weighting its squared error by $1/s_{\chi,k}^2$. Response scaling therefore gives each coordinate comparable importance in the joint statistical score.

For example, an illustrative concentration error of 10 g m^{-3} with a response scale of 20 g m^{-3} contributes $0.5^2 = 0.25$ before averaging. An inventory error of $10,000 \text{ g}$ with a scale of $200,000 \text{ g}$ contributes $0.05^2 = 0.0025$. Comparing the unscaled numbers would give the opposite impression. The scales define what counts as a large error.

Let $\Phi_{\mathcal{I}}$ contain the feature rows for fitting indices $\mathcal{I}$ and let $\tilde{Y}_{\mathcal{I}}$ contain standardized joint responses. With $R_\phi = \text{diag}(0, 1, \dots, 1)$, the coefficient matrix solves

$$\hat{C}_\gamma = \arg \min_C \left\{ \frac{\|\tilde{Y}_{\mathcal{I}} - \Phi_{\mathcal{I}} C^\top\|_F^2}{|\mathcal{I}|n_\chi} + \frac{\gamma}{n_\chi} \|C R_\phi\|_F^2 \right\}. \quad (6.74)$$

The intercept is unpenalized. A positive γ regularizes correlated quadratic features. Hastie et al. (2009) explain this stabilization and the corresponding trade-off between model flexibility and coefficient shrinkage. The raw prediction is

$$\chi_{\text{raw}} = \bar{\chi} + D_\chi \hat{C}_\gamma \phi(\vartheta, c_{\text{in}}). \quad (6.75)$$

The response scale D_χ is fixed from fitting data. It places the inventory, component concentrations, and outlet flows on comparable statistical scales. Joint projection then applies the physical constraints to the raw prediction.

Differentiating the scalar ridge objective for one response column with coefficient vector β gives its normal equations,

$$(\Phi^\top \Phi + n\gamma R_\phi) \hat{\beta} = \Phi^\top \tilde{\chi}. \quad (6.76)$$

The term $n\gamma R_\phi$ adds positive curvature to the nonintercept coefficient directions. In a hypothetical one-feature teaching problem, take three feature and target values both equal to $(-\sqrt{3/2}, 0, \sqrt{3/2})$. Their means are zero and their population standard deviations are one. Symmetry gives a zero intercept. With slope b , the objective becomes $(1-b)^2 + \gamma b^2$. Differentiation gives $2(b-1) + 2\gamma b = 0$, so $\hat{b} = 1/(1+\gamma)$. For $\gamma = 1$, the fitted slope is 0.5 rather than the unpenalized value of one. This hypothetical calculation shows coefficient shrinkage.

Finally, the physical prediction is recovered coordinate by coordinate as $\chi_{\text{raw},k} = \bar{\chi}_k + s_{\chi,k} \sum_j \hat{C}_{kj} \phi_j$. A hypothetical mean of 100 g m^{-3} , scale of 20 g m^{-3} , and standardized prediction of 0.5 give 110 g m^{-3} . This return to physical units is required before imposing component balances.

The quadratic feature family is retained from ICSOR (Selerio, 2026). Its coefficients describe the raw statistical relation. If projection is active, a raw coefficient alone does not give the derivative of the final projected response. That derivative also depends on the constraints, their control dependence, and the active inequalities.

A separate positive model for overflow solids

Low overflow TSS calls for a different error scale from large reactor solids concentrations. An absolute error of 1 g m^{-3} is 1% of 100 g m^{-3} and 100% of 1 g m^{-3} . A joint standardized response score may not adequately describe this distinction at the discharge point. The relative-error argument of Tofallis (2015) motivates a separate logarithmic model.

For positive mechanistic targets, define

$$\ell_E = \log(X_E/X_\star), \quad X_\star = 1 \text{ mg L}^{-1}. \quad (6.77)$$

The ratio inside the logarithm is dimensionless. The scalar ridge fit uses the same complete quadratic features,

$$\hat{\beta}_{E,\gamma_E} = \arg \min_{\beta} \left\{ \frac{\|\ell_{E,\mathcal{I}} - \Phi_{\mathcal{I}}\beta\|_2^2}{|\mathcal{I}|} + \gamma_E \|R_{\phi}\beta\|_2^2 \right\}. \quad (6.78)$$

Its concentration-scale prediction is

$$\widehat{X}_{E,\log} = X_\star \exp(\hat{\beta}_{E,\gamma_E}^\top \phi) > 0. \quad (6.79)$$

A twofold concentration error has the same logarithmic discrepancy at low and high concentrations. Exponentiation gives a positive statistical estimate that enters the joint mass conservation projection.

For hypothetical targets of 2 and 8 mg L^{-1} , the dimensionless logarithms are $\log 2$ and $\log 8$. If a single fitted log value represents both equally, squared log error is minimized at

$$\frac{\log 2 + \log 8}{2} = \log 4. \quad (6.80)$$

Back-transformation gives $1 \times \exp(\log 4) = 4 \text{ mg L}^{-1}$. The arithmetic mean is 5 mg L^{-1} , which explains why exponentiation is not an unbiased concentration-scale mean adjustment. A fitted log value of -2 instead gives 0.1353 mg L^{-1} . It remains positive even though it is below the reference concentration $X_\star$. The reference establishes units and does not impose a lower bound.

Under squared logarithmic error, direct exponentiation represents a conditional geometric center rather than an unbiased concentration-scale mean. No concentration-scale bias adjustment is applied. The separate model also avoids imposing a constant solids-removal fraction across all influents and controls. Such a fraction would disregard variation in feed composition, hydraulics, recycle, and biological state. Its fitted

endpoint supplies the empirical closure introduced in Chapter 2. The physical balances leave the division of solids between the two outlets undetermined. The fitted overflow solids concentration supplies the additional relation used in joint projection. It remains a statistical estimate rather than a mass conservation law.

6.2.4 Joint projection onto the declared plant constraints

Equalities and inequalities at fixed controls

The projection adjusts every connected response block together. The required matrix rows can first be assembled for a hypothetical one-reactor, two-component plant. Use the nine-coordinate order already given for χ_{toy} . Let $r_I = 1$, $r_R = 0.5$, and $w = 0.1$, so $q_P = 2.5$, $q_C = 1.5$, $q_U = 0.6$, and $q_E = 0.9$. These hypothetical values illustrate assembly of the constraint rows. The soluble component is S and the particulate component X has unit TSS weight. Suppose their sum is fixed by mass conservation in the reactor and the fresh concentrations are $(30, 20) \text{ g m}^{-3}$.

The two mixer rows, one inventory row, two clarifier rows, soluble pass-through, and fitted endpoint are then

$$2.5m_S - c_{1,S} - \frac{5}{6}g_{U,S} = 30, \quad (6.81)$$

$$2.5m_X - c_{1,X} - \frac{5}{6}g_{U,X} = 20, \quad (6.82)$$

$$c_{1,S} + c_{1,X} - m_S - m_X = 0, \quad (6.83)$$

$$g_{E,S} + g_{U,S} - 1.5c_{1,S} = 0, \quad (6.84)$$

$$g_{E,X} + g_{U,X} - 1.5c_{1,X} = 0, \quad (6.85)$$

$$g_{U,S} - 0.6c_{1,S} = 0, \quad (6.86)$$

$$g_{E,X} = 0.9(10) = 9. \quad (6.87)$$

The last row illustrates a fitted overflow concentration of 10 g m^{-3} . The inventory row is written without its harmless unit-length factor to keep the numbers clear. Multiplying a zero equality by a positive factor does not change its solutions. In the same coordinate order, the explicit equality matrix is

$$\begin{bmatrix} 2.5 & 0 & -1 & 0 & 0 & 0 & -5/6 & 0 & 0 \\ 0 & 2.5 & 0 & -1 & 0 & 0 & 0 & -5/6 & 0 \\ -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1.5 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1.5 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & -0.6 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \chi_{\text{toy}} = \begin{bmatrix} 30 \\ 20 \\ 0 \\ 0 \\ 0 \\ 0 \\ 9 \end{bmatrix}. \quad (6.88)$$

Every row expresses one previously derived relation. A zero entry means that the coordinate is absent from that relation. The inventory coordinate has a zero column because these equalities do not determine stored solids.

The illustrative response

$$\chi_{\text{toy}} = (30, 50, 30, 50, 27, 9, 18, 66, 18,000)^\top \quad (6.89)$$

satisfies all seven rows. For example, its particulate mixer row is $2.5(50) - 50 - (5/6)(66) = 20$. Its clarifier particulate row is $9 + 66 = 1.5(50)$. Underflow concentration is $66/0.6 = 110 \text{ g m}^{-3}$. These arithmetic checks verify the linear constraints.

The full physical equality operator $\mathcal{H}(\vartheta)\chi = b(\vartheta, c_{\text{in}})$ collects

$$\text{col} \begin{bmatrix} q_P m - r_I c_{n_r} - (r_R/q_U)g_U - c_{\text{in}} \\ \{A(c_i - c_{i-1}) - d_i^{-1}A\xi_i\}_{i=1}^{n_r} \\ g_E + g_U - q_C c_{n_r} \\ E_S g_U - q_U E_S c_{n_r} \end{bmatrix} = 0. \quad (6.90)$$

The empirical overflow relation is appended as

$$t_X^\top g_E = q_E \widehat{X}_{E,\log}(\vartheta, c_{\text{in}}). \quad (6.91)$$

The complete operator is denoted $\mathcal{H}_{\log}$ with right-hand side $b_{\log}$. It has $2F + n_r K + F_S + 1$ rows. The last row fixes total overflow TSS. It does not prescribe its particulate composition.

The remaining teaching-plant requirements are also individual matrix rows. With three 100-m³ layers, $V_{\text{cl}} = 300 \text{ m}^3$ and $V_1 = V_3 = 100 \text{ m}^3$. Densification is $0.6c_{1,X} - g_{U,X} \leq 0$. The inventory inequalities follow by substituting $X_E = g_{E,X}/0.9$ and $X_U = g_{U,X}/0.6$ into the endpoint bounds and multiplying by $0.9(0.6) = 0.54$,

$$120g_{E,X} + 90g_{U,X} - 0.54M_{\text{cl}} \leq 0, \quad (6.92)$$

$$-60g_{E,X} - 180g_{U,X} + 0.54M_{\text{cl}} \leq 0. \quad (6.93)$$

The lower row has coefficient $120 = 0.6(300 - 100)$ and coefficient $90 = 0.9(100)$. These are hydraulic and volume factors. They are not regression coefficients. At the illustrative response above, densification gives $30 - 66 = -36$, while both inventory rows give $-2,700$. The corresponding inventory interval is 13,000–23,000 g, which contains 18,000 g.

The three inequality rows can also be displayed in the response order,

$$\begin{bmatrix} 0 & 0 & 0 & 0.6 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 120 & 0 & 90 & -0.54 \\ 0 & 0 & 0 & 0 & 0 & -60 & 0 & -180 & 0.54 \end{bmatrix} \chi_{\text{toy}} \leq 0. \quad (6.94)$$

Non-negativity would add one row $-\chi_{\text{toy},k} \leq 0$ for each of the nine coordinates. The inequality matrix has the same column order as the equality matrix. Changing that order in one operator would constrain the wrong physical quantities.

For any plant size, collecting the same row patterns gives the physical inequality operator,

$$\mathcal{G}(\vartheta)\chi = \text{col} \begin{bmatrix} q_U E_X c_{n_r} - E_X g_U \\ q_U (V_{cl} - V_{cl,L}) t_X^\top g_E + q_E V_{cl,L} t_X^\top g_U - q_E q_U M_{cl} \\ q_E q_U M_{cl} - q_U V_{cl,1} t_X^\top g_E - q_E (V_{cl} - V_{cl,1}) t_X^\top g_U \end{bmatrix} \leq 0. \quad (6.95)$$

The first block enforces particulate densification. The last two rows are the inventory bounds multiplied by the positive product $q_E q_U$. This multiplication removes divisions without changing feasibility. Non-negativity requires $\chi \geq 0$ separately.

At fixed (ϑ, c_{in}) , every constraint is affine in χ . This remains true for the overflow closure because its fitted right-hand side is then a known scalar. The endpoint-envelope inequalities also remain affine even though their coefficients depend on the controls. The distinction between fixed-control linearity and control dependence is central to the numerical formulation.

The scaled nearest-response problem

Projection size is first defined for an individual response coordinate. If the raw value is $\chi_{\text{raw},k}$ and its positive scale is s_k , then

$$u_k = \frac{\chi_k - \chi_{\text{raw},k}}{s_k}. \quad (6.96)$$

The scalar projection objective is $\frac{1}{2} \sum_{k=1}^{n_\chi} u_k^2$. Thus a change of two scale units in one component costs $\frac{1}{2}(2)^2 = 2$, while a change of half a scale unit costs $\frac{1}{2}(0.5)^2 = 0.125$. With $D_\chi = \text{diag}(s_1, \dots, s_{n_\chi})$, the response relation becomes $\chi = \chi_{\text{raw}} + D_\chi u$.

For a mixer component row, the substitution is explicit,

$$q_P(m_{\text{raw},f} + s_{m,f} u_{m,f}) - r_I(c_{\text{raw},n_r,f} + s_{n_r,f} u_{n_r,f}) \quad (6.97)$$

$$- \frac{r_R}{q_U}(g_{\text{raw},U,f} + s_{U,f} u_{U,f}) = c_{\text{in},f}. \quad (6.98)$$

The scales multiply unknown projections, while all raw values are known at the current inputs. Moving the raw terms to the right gives one linear equation in u . The other balances are transformed in the same way. Let D_b and D_g be positive diagonal row scales. The projected standardized displacement solves

$$\hat{u} = \arg \min_{u \in \mathbb{R}^{n_\chi}} \frac{1}{2} u^\top u, \quad (6.99)$$

$$\text{subject to } D_b^{-1} \{ \mathcal{H}_{\log}(\vartheta)(\chi_{\text{raw}} + D_\chi u) - b_{\log}(\vartheta, c_{\text{in}}) \} = 0,$$

$$\chi_{\text{raw}} + D_\chi u \geq 0,$$

$$D_g^{-1} \mathcal{G}(\vartheta)(\chi_{\text{raw}} + D_\chi u) \leq 0,$$

$$\hat{\chi} = \chi_{\text{raw}} + D_\chi \hat{u}. \quad (6.100)$$

The identity Hessian defines a strictly convex quadratic program. In physical coordinates, its metric is $D_x^{-\top} D_x^{-1}$. Response scaling determines which physical changes count as large projections. Positive constraint-row scaling changes numerical conditioning while leaving the feasible set unchanged. Sturm and Silva (2025) motivate species-sensitive weighting for projection. Boyd and Vandenberghe (2004) supply the convex optimization principles used for feasibility, uniqueness, and optimality.

The physically constrained surrogate combines the original ICSOR separation of prediction and projection with a connected plant response (Selerio, 2026). The manifold projection developed by Valente et al. (2025) provides related output-projection context. Analytic neural-output constraints in Beucler et al. (2021), flux learning for mass conservation in Sturm and Wexler (2020), and constrained training in Ruckstuhl et al. (2021) provide alternative locations for physical enforcement. Li et al. (2024) include governing-equation residuals in physics-informed unit-operation models, including an activated sludge reactor. A residual penalty makes disagreement costly during fitting. It differs from an equality that each admitted prediction must satisfy. The present plant balances and inventory interval follow from the declared process model.

For a nonempty feasible set, closedness, coercivity, and strict convexity give a unique projected response. Feasibility is checked jointly for the fitted endpoint, mass conservation, non-negativity, and densification. For instance, demanding overflow solids above every feed solids concentration allowed by the other balances contradicts $X_E \leq X_N$. The empirical closure is therefore capable of making an otherwise feasible physical polyhedron empty. An infeasible fitted endpoint is recorded as a failed projection.

Figure 6.3 distinguishes the three inputs to the joint projection. The raw response supplies the starting point for projection. Derived physical relations define balance and envelope requirements. The separate logarithmic prediction supplies a fitted endpoint equality. Its empirical origin remains relevant even though it enters the same equality system as the physical balances.

The projected response minimizes standardized adjustment within the stated response scales and feasible set. The physical rows and fitted endpoint jointly define that set.

An independently auditable optimality certificate

Consider a separate hypothetical two-component projection. The raw prediction is $(-2, 9)$, its coordinate scales are $(1, 2)$, and the required total fixed by mass conservation is 10. The physical-coordinate problem is

$$\min_{c_1, c_2} \frac{1}{2}(c_1 + 2)^2 + \frac{1}{8}(c_2 - 9)^2, \quad c_1 + c_2 = 10, \quad c_1 \geq 0, \quad c_2 \geq 0. \quad (6.101)$$

The second coefficient is $1/(2 \times 2^2)$. A change of one physical unit in component 2 costs less than the same change in component 1 because its scale is larger.

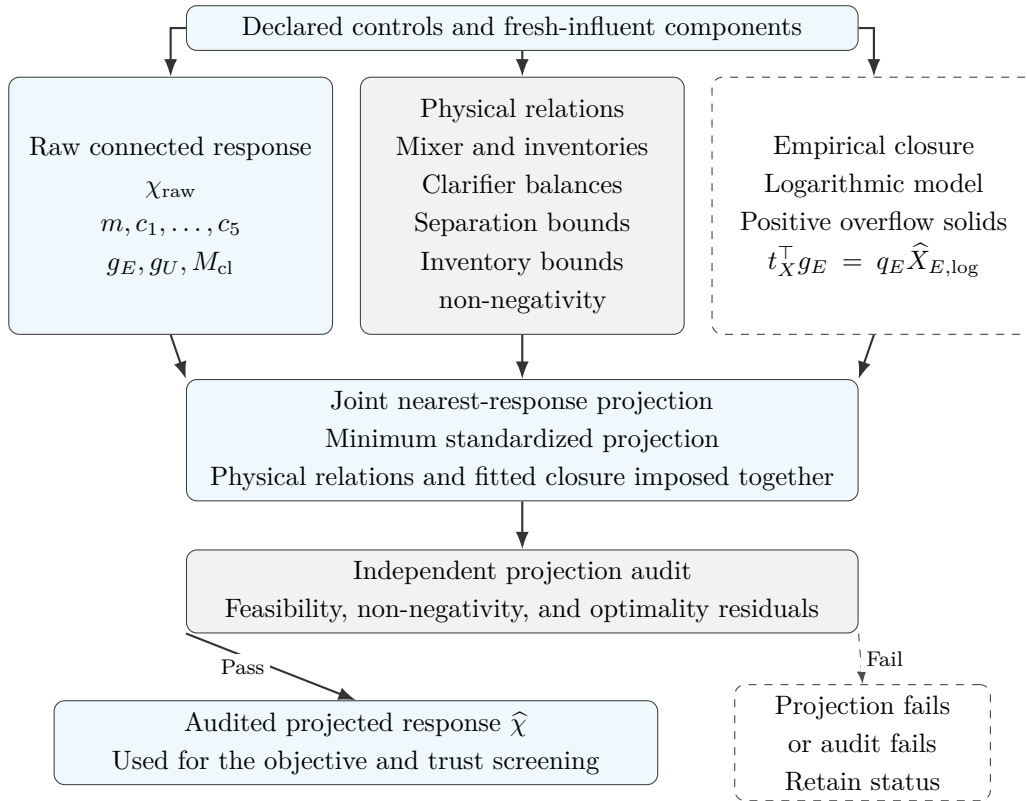


Figure 6.3: Conceptual information flow in the connected projection. The dashed closure block denotes an empirically fitted relation rather than a physical identity. Projection corrects all response blocks jointly. The audit checks the imposed relations at their numerical tolerances.

First omit non-negativity and introduce an equality multiplier λ . Stationarity and the total fixed by mass conservation give

$$c_1 + 2 + \lambda = 0, \quad (6.102)$$

$$\frac{1}{4}(c_2 - 9) + \lambda = 0, \quad (6.103)$$

$$c_1 + c_2 = 10. \quad (6.104)$$

The first two equations imply $c_1 = -2 - \lambda$ and $c_2 = 9 - 4\lambda$. Their sum gives $7 - 5\lambda = 10$, so $\lambda = -0.6$ and $(c_1, c_2) = (-1.4, 11.4)$. Mass conservation alone leaves a negative component.

Now impose the active boundary $c_1 = 0$, which forces $c_2 = 10$. Use multiplier $\mu_1 \geq 0$ for $-c_1 \leq 0$ and $\mu_2 \geq 0$ for $-c_2 \leq 0$. The Lagrangian is the objective plus $\lambda(c_1 + c_2 - 10) - \mu_1 c_1 - \mu_2 c_2$. Since component 2 is positive, its inequality is inactive and $\mu_2 = 0$. Its stationarity equation gives $\lambda = -0.25$. Component 1 then gives $2 - 0.25 - \mu_1 = 0$, so $\mu_1 = 1.75$. Both inequality multipliers are non-negative, and both products $\mu_j c_j$ vanish. These checks certify the boundary solution rather than merely declaring it plausible.

In standardized coordinates, $u = (2, 0.5)^\top$. The same problem has

$$E = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad e = 3, \quad G_u = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad g = \begin{bmatrix} -2 \\ 9 \end{bmatrix}. \quad (6.105)$$

The equality is $u_1 + 2u_2 = 3$. The inequality rows are the two physical non-negativity requirements after substitution. This example explains the standardized form before its use for the larger plant.

At fixed inputs, write the standardized problem as

$$\min_u \frac{1}{2} u^\top u, \quad Eu = e, \quad G_u u \leq g. \quad (6.106)$$

Let $\lambda^=$ be unrestricted equality multipliers and $\lambda^\leq \geq 0$ inequality multipliers. The optimality conditions are primal feasibility, dual feasibility, stationarity, and complementarity,

$$u + E^\top \lambda^= + G_u^\top \lambda^\leq = 0, \quad (6.107)$$

$$\lambda_j^\leq (g_j - (G_u u)_j) = 0 \quad \text{for every } j. \quad (6.108)$$

To derive the dual value, collect the linear terms in the Lagrangian. Define $v = E^\top \lambda^= + G_u^\top \lambda^\leq$. Completing the square gives

$$\frac{1}{2} u^\top u + v^\top u - e^\top \lambda^= - g^\top \lambda^\leq = \frac{1}{2} \|u + v\|_2^2 - \frac{1}{2} \|v\|_2^2 \quad (6.109)$$

$$- e^\top \lambda^= - g^\top \lambda^\leq. \quad (6.110)$$

The first term has minimum zero at $u = -v$. Minimizing over u therefore gives the dual value,

$$d(\lambda^=, \lambda^\leq) = -\frac{1}{2} \|E^\top \lambda^= + G_u^\top \lambda^\leq\|_2^2 - e^\top \lambda^= - g^\top \lambda^\leq. \quad (6.111)$$

For primal- and dual-feasible quantities, the gap is

$$\Gamma = \frac{1}{2}\|u + E^\top \lambda^\leftarrow + G_u^\top \lambda^\leq\|_2^2 + (\lambda^\leq)^\top (g - G_u u) \geq 0. \quad (6.112)$$

It vanishes exactly when stationarity and complementarity hold. This gives a mathematical certificate of the unique projection (Boyd & Vandenberghe, 2004). Numerical acceptance checks the individual residuals independently. A small reported gap does not replace a non-negativity or equality check.

The hypothetical two-component example permits every certificate number to be checked. With $\lambda^\leftarrow = -0.25$ and $\lambda^\leq = (1.75, 0)^\top$,

$$E^\top \lambda^\leftarrow + G_u^\top \lambda^\leq = (-2, -0.5)^\top = -u, \quad (6.113)$$

$$g - G_u u = (0, 10)^\top, \quad (6.114)$$

$$\frac{1}{2}u^\top u = \frac{1}{2}(4 + 0.25) = 2.125, \quad (6.115)$$

$$d = -2.125 - 3(-0.25) - [-2(1.75)] = 2.125. \quad (6.116)$$

The primal and dual values agree. Stationarity is zero coordinate by coordinate, and the complementarity sum is $1.75(0) + 0(10) = 0$. The gap is consequently zero. A multiplier with the wrong sign would fail dual feasibility even if some other residual happened to be small.

Every distinct trial receives a projection solve from an independent initialization. An operator-splitting quadratic solver of the type studied by Stellato et al. (2020) supplies a numerical candidate. Multipliers are reconstructed independently through bounded-variable least squares for the audit. Numerical retries change the solver settings while preserving the mathematical problem.

Error bounds in the projection metric

Let $\mathcal{C}$ be the fixed-input feasible set and use the norm $\|v\|_{D_\chi^{-1}} = \|D_\chi^{-1}v\|_2$. This norm has the scalar meaning

$$\|v\|_{D_\chi^{-1}}^2 = \sum_{k=1}^{n_\chi} \frac{v_k^2}{s_k^2}. \quad (6.117)$$

To see the error argument, temporarily work in standardized coordinates. Write $a_k = \chi_{\text{raw},k}/s_k$, $q_k = \hat{\chi}_k/s_k$, and $t_k = \chi_{\text{ref},k}/s_k$. The transformed feasible set is convex because the scales define an invertible linear change of coordinates.

For any feasible comparison point z , the segment $q + \tau(z - q)$ remains feasible for $0 \leq \tau \leq 1$. The projection objective cannot decrease when leaving its minimizer along that segment. Its derivative at zero must therefore satisfy

$$\sum_k (q_k - a_k)(z_k - q_k) \geq 0. \quad (6.118)$$

When the reference itself is feasible, take $z = t$. Reversing both signs gives $\sum_k (a_k - q_k)(q_k - t_k) \geq 0$. The componentwise identity

$$(a_k - t_k)^2 = (a_k - q_k)^2 + (q_k - t_k)^2 \quad (6.119)$$

$$+ 2(a_k - q_k)(q_k - t_k) \quad (6.120)$$

then implies, after summing over coordinates,

$$\sum_k (q_k - t_k)^2 \leq \sum_k (a_k - t_k)^2 - \sum_k (a_k - q_k)^2. \quad (6.121)$$

The last sum is exactly the squared standardized projection. Returning to the physical-coordinate notation, if a mechanistic reference χ_{ref} belongs exactly to $\mathcal{C}$, the projection variational inequality gives

$$\|\hat{\chi} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 - \|\hat{u}\|_2^2. \quad (6.122)$$

Thus the projection cannot increase aggregate squared error in this particular metric to an exactly feasible target. Component errors, physical-unit scores, and operating objectives are assessed separately.

In the hypothetical weighted two-component problem, take the feasible reference $(3, 7)$. Raw squared error is $(-2 - 3)^2 + (9 - 7)^2/4 = 26$. Projected squared error is $(0 - 3)^2 + (10 - 7)^2/4 = 11.25$. Projection size squared is $2^2 + 0.5^2 = 4.25$. The bound reads $11.25 \leq 26 - 4.25 = 21.75$. The second component's absolute error increases from 2 to 3 even though the aggregate weighted error decreases. The weighted aggregate and individual component errors therefore describe different effects.

The fitted overflow equality usually differs from the mechanistic endpoint. A physically consistent mechanistic target can therefore lie outside $\mathcal{C}$. Define

$$\delta_{\text{feas}} = \|P_{\mathcal{C}}(\chi_{\text{ref}}) - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}. \quad (6.123)$$

Let p be the closest feasible point to t in standardized coordinates. The previous segment argument now applies to $z = p$. Split the remaining inner product into a feasible-point part and a reference-distance part,

$$\sum_k (a_k - q_k)(q_k - t_k) = \sum_k (a_k - q_k)(q_k - p_k) \quad (6.124)$$

$$+ \sum_k (a_k - q_k)(p_k - t_k). \quad (6.125)$$

The first term is non-negative. The magnitude of the second is at most

$$\left[\sum_k (a_k - q_k)^2 \right]^{1/2} \left[\sum_k (p_k - t_k)^2 \right]^{1/2} = r_{\text{corr}} \delta_{\text{feas}}. \quad (6.126)$$

This is the scalar Cauchy–Schwarz inequality. It bounds how far an infeasible reference can weaken the projection argument. Let $r_{\text{corr}} = \|\chi_{\text{raw}} - P_{\mathcal{C}}(\chi_{\text{raw}})\|_{D_{\chi}^{-1}}$. Apply the projection variational inequality with the feasible comparison point $P_{\mathcal{C}}(\chi_{\text{ref}})$. Expanding

the squared error and bounding the remaining inner product gives

$$\|P_C(\chi_{\text{raw}}) - \chi_{\text{ref}}\|_{D_X^{-1}}^2 \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_X^{-1}}^2 - r_{\text{corr}}^2 + 2r_{\text{corr}}\delta_{\text{feas}} \quad (6.127)$$

$$\leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_X^{-1}}^2 + \delta_{\text{feas}}^2. \quad (6.128)$$

Taking square roots therefore gives the more easily interpreted bound

$$\|P_C(\chi_{\text{raw}}) - \chi_{\text{ref}}\|_{D_X^{-1}} \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_X^{-1}} + \delta_{\text{feas}}. \quad (6.129)$$

The distance includes any finite numerical infeasibility of the reference. The reference distance is evaluated against both the physical identities and empirical equality. The weighted-projection discussion in Sturm and Silva (2025) likewise separates mass conservation from predictive improvement. The original-model evaluation checks biological kinetics, nonlinear settling, and local dynamic stability.

For a hypothetical illustration of the empirical-closure issue, let the total fixed by mass conservation still be 10 but also require the first component to equal a fitted value of 4. The feasible set then contains only (4, 6). Suppose the mechanistic reference and raw prediction are both (2, 8) with scales (1, 2). They satisfy the physical total and non-negativity but do not satisfy the fitted equality. Raw error is zero. Projection introduces weighted error $\sqrt{(4-2)^2 + (6-8)^2/4} = \sqrt{5}$. The reference distance to the set is also $\sqrt{5}$, so the general bound is met exactly. Physical identities alone would have given the wrong premise for the stronger guarantee.

Control-dependent sensitivities

With fixed constraint matrices and an affine parameter-dependent right-hand side, a parametric quadratic projection can have a piecewise affine solution (Tøndel et al., 2003). Here the projection matrices and endpoint also vary with the controls. Hydraulic coefficients, inverse flow ratios, the input features, and the exponential endpoint all depend on ϑ . The projected plant response is generally piecewise smooth where regularity holds. Active-set transitions define boundaries between its smooth regions.

The differentiation can first be seen in a scalar hypothetical projection,

$$\min_u \frac{1}{2}u^2, \quad b(\theta)u = e(\theta), \quad (6.130)$$

where θ is one control and $b(\theta) \neq 0$. Stationarity and feasibility are $u + b\lambda = 0$ and $bu = e$. Written together, they form the two-by-two system

$$\begin{bmatrix} 1 & b \\ b & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ e \end{bmatrix}. \quad (6.131)$$

Differentiate the two scalar equations with respect to θ . The product rule gives $u' + b'\lambda + b\lambda' = 0$ and $b'u + bu' = e'$. Moving known products to the right leaves

$$\begin{bmatrix} 1 & b \\ b & 0 \end{bmatrix} \begin{bmatrix} u' \\ \lambda' \end{bmatrix} = \begin{bmatrix} -b'\lambda \\ e' - b'u \end{bmatrix}. \quad (6.132)$$

The matrix is unchanged. Its right-hand side now contains derivatives of both coefficients and targets.

For $b = 1 + \theta$ and $e = \theta$, the solution is $u = \theta/(1 + \theta)$ and $\lambda = -\theta/(1 + \theta)^2$. At the hypothetical control $\theta = 1$, the values are $u = 0.5$ and $\lambda = -0.25$. The derivative system becomes

$$\begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} u' \\ \lambda' \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.5 \end{bmatrix}. \quad (6.133)$$

Its second row gives $u' = 0.25$ and its first gives $\lambda' = 0$. Direct differentiation of $u = \theta/(1 + \theta)$ gives the same $u' = 1/(1 + \theta)^2$. Ignoring b' would instead differentiate a problem whose constraint coefficient did not vary.

For a stable active set, stack equality rows and active inequality rows into $B(\vartheta)$ and their right-hand sides into $b_a(\vartheta)$. The stationary system is

$$\begin{bmatrix} I & B^\top \\ B & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda_a \end{bmatrix} = \begin{bmatrix} 0 \\ b_a \end{bmatrix}. \quad (6.134)$$

The componentwise product rule is now applied to $u + B^\top \lambda_a = 0$ and $Bu = b_a$,

$$\partial_j u + B^\top \partial_j \lambda_a = -(\partial_j B)^\top \lambda_a, \quad (6.135)$$

$$B \partial_j u = \partial_j b_a - (\partial_j B) u. \quad (6.136)$$

For a control ϑ_j , putting these derivative equations into one block system gives

$$\begin{bmatrix} I & B^\top \\ B & 0 \end{bmatrix} \begin{bmatrix} \partial_j u \\ \partial_j \lambda_a \end{bmatrix} = \begin{bmatrix} -(\partial_j B)^\top \lambda_a \\ \partial_j b_a - (\partial_j B) u \end{bmatrix}. \quad (6.137)$$

Since D_χ is fixed after fitting, the physical response derivative is $\partial_j \hat{\chi} = \partial_j \chi_{\text{raw}} + D_\chi \partial_j \hat{u}$. For the endpoint, first differentiate the scalar fitted logarithm, which gives $\partial_j (\hat{\beta}^\top \phi) = \hat{\beta}^\top \partial_j \phi$. Exponential differentiation then gives $\partial_j \hat{X}_{E,\log} = \hat{X}_{E,\log} \hat{\beta}^\top \partial_j \phi$. Applying the product rule to its flow-weighted value gives the endpoint derivative,

$$\partial_j (q_E \hat{X}_{E,\log}) = (\partial_j q_E) \hat{X}_{E,\log} + q_E \hat{X}_{E,\log} \hat{\beta}_{E,\gamma_E}^\top \partial_j \phi. \quad (6.138)$$

Ignoring this term would differentiate a different optimization problem.

These derivatives are used only when active rows are independent, the stationary matrix is well conditioned, active inequality multipliers are strictly positive, and independent perturbation checks pass. Near degeneracy, different active descriptions can correspond to the same unique response. Uniqueness of the projection therefore does not ensure a reliable classical derivative. A value-only outer method remains available when the derivative chain cannot be validated.

6.2.5 Treatment priorities, resource measures, and operating safeguards

A common engineering objective

The effluent-quality map in Table 3.1 gives

$$z_E = I_{\text{comp}} c_E = \text{col}(\text{COD}, \text{TN}, \text{TP}, \text{TSS}). \quad (6.139)$$

Let the four diagonal entries of D_T be $T_{\text{COD}}, T_{\text{TN}}, T_{\text{TP}}, T_{\text{TSS}}$. Dividing each effluent quantity by its own scale gives a dimensionless quality contribution. The scalar weighted sum is

$$\begin{aligned} j_Q = & \varpi_{Q,\text{COD}} \frac{z_{E,\text{COD}}}{T_{\text{COD}}} + \varpi_{Q,\text{TN}} \frac{z_{E,\text{TN}}}{T_{\text{TN}}} \\ & + \varpi_{Q,\text{TP}} \frac{z_{E,\text{TP}}}{T_{\text{TP}}} + \varpi_{Q,\text{TSS}} \frac{z_{E,\text{TSS}}}{T_{\text{TSS}}}. \end{aligned} \quad (6.140)$$

For positive fixed normalization D_T and quality priorities $\varpi_Q \geq 0$ with $\mathbf{1}^\top \varpi_Q = 1$, this sum is written compactly as

$$j_Q = \varpi_Q^\top D_T^{-1} z_E. \quad (6.141)$$

The four totals describe different treatment concerns. They do not identify which individual nutrient transformations occur. A nearly constant TN across a reactor can accompany conversion of ammonium to nitrite. A decrease in overflow TP can arise from particulate separation rather than an additional clarifier reaction. The nutrient-control guidance of U.S. Environmental Protection Agency (2009) and component-resolved treatment framework of X. Zhang et al. (2026) support inspecting species and solids flows alongside aggregate quality measures.

Normalizing a bounded setting subtracts its lower bound and divides by the full range. This gives zero at the lower bound and one at the upper bound. For example, $H = 18$ h within 6–36 h gives $(18 - 6)/(36 - 6) = 0.4$. Aeration instead sums physical transfer capacities. Each product $V_i k_L a_i$ has units $\text{m}^3 \text{d}^{-1}$, so its fixed reference has the same units. Wasted solids are proportional to $Q_0 w X_U$. Dividing this rate by its fixed reference $Q_0 w^U X_U^{\text{ref}}$ cancels Q_0 . The resource terms are

$$j_H = \frac{H - H^L}{H^U - H^L}, \quad j_A = \frac{\sum_{i \in \mathcal{A}} V_i k_L a_i}{E_A^{\text{ref}}}, \quad (6.142)$$

$$j_I = \frac{r_I - r_I^L}{r_I^U - r_I^L}, \quad j_R = \frac{r_R - r_R^L}{r_R^U - r_R^L}, \quad (6.143)$$

$$j_W = \frac{w X_U}{w^U X_U^{\text{ref}}}. \quad (6.144)$$

The set $\mathcal{A}$ contains stages that can be aerated. The aeration reference $E_A^{\text{ref}} > 0$ is fixed before optimization. It is the maximum transfer-capacity numerator over the declared operating box. A fixed reference prevents an increase in selected volume from being canceled by a changing denominator.

The aeration proxy measures transfer capacity. Actual oxygen transfer also depends on the dissolved-oxygen deficit in Equation (6.15). The energy-management model of Miederer et al. (2025) distinguishes aeration and other energy demands. The treatment model of X. Zhang et al. (2026) makes the oxygen-transfer driving force explicit. The dimensionless terms are engineering measures of capacity and resource use.

Write the overall coefficients in the same order as the six terms. The complete scalar objective is

$$J = \varpi_1 j_Q + \varpi_2 j_H + \varpi_3 j_A + \varpi_4 j_I + \varpi_5 j_R + \varpi_6 j_W. \quad (6.145)$$

Its compact vector form is

$$J(\vartheta, \chi, \varpi) = \varpi^\top \text{col}(j_Q, j_H, j_A, j_I, j_R, j_W), \quad \varpi \geq 0, \quad \mathbf{1}^\top \varpi = 1. \quad (6.146)$$

The overall vector ϖ separates quality from capacity and resource priorities. The vector ϖ_Q separates priorities among the quality measures. These coefficients are choices about trade-offs. They are not percentages of the objective attained at an operating point.

For the equal-volume case,

$$j_A = \frac{H}{36 \text{ h}} \frac{a_3 + a_4 + a_5}{3}. \quad (6.147)$$

This follows from $V_i = Q_0 H / (24 n_r)$ and $k_L a_i = 47 a_i$. For hypothetical settings with all three aeration coordinates equal to one, reducing H from 36 h to 12 h reduces this proxy from one to one-third. The normalized aeration settings have stayed the same, but the total transfer capacity has changed with volume. A comparison based only on the aeration coordinates would miss this effect.

The baseline priority vectors are

$$\varpi = (0.50, 0.15, 0.20, 0.05, 0.05, 0.05)^\top, \quad \varpi_Q = \frac{1}{4} \mathbf{1}_4. \quad (6.148)$$

Quality has the largest individual coefficient. Aeration has the largest resource coefficient. The importance of oxygen supply and solids age to treatment and energy use is discussed by Leu et al. (2012). The exact coefficients remain case-study preferences. Changes in those preferences define separate operating questions and require separately reported decisions.

A hypothetical objective calculation can now be followed through all six terms. Use effluent values (40, 5, 1, 4) mg L⁻¹ and illustrative fixed quality scales (100, 10, 2, 20) mg L⁻¹. With equal quality priorities, $j_Q = (0.4 + 0.5 + 0.5 + 0.2) / 4 = 0.4$. These illustrative scales are not estimated study quantities. Let $H = 18 \text{ h}$, $(a_3, a_4, a_5) = (0.8, 0.6, 0.4)$, $r_I = 1$, $r_R = 0.5$, $w = 0.02$, and $X_U = 6,000 \text{ g m}^{-3}$.

The total biological volume is $Q_0 H / 24 = 7,500 \text{ m}^3$, so each equal tank has volume 1,500 m³. Aeration capacity is $1,500(47)(0.8 + 0.6 + 0.4) = 126,900 \text{ m}^3 \text{ d}^{-1}$. Table 6.2

Table 6.2: Hypothetical calculation of dimensionless objective contributions.

Term	Calculation	Value	Weighted value
j_Q	$(0.4 + 0.5 + 0.5 + 0.2)/4$	0.4	0.200
j_H	$(18 - 6)/(36 - 6)$	0.4	0.060
j_A	$126,900/423,000$	0.3	0.060
j_I	$(1 - 0)/(4 - 0)$	0.25	0.0125
j_R	$(0.5 - 0.25)/(1.25 - 0.25)$	0.25	0.0125
j_W	$(0.02)(6,000)/[(0.05)(15,000)]$	0.16	0.008

gives every normalized term and weighted contribution. Their sum is 0.353. This hypothetical arithmetic illustrates the declared objective.

The overall priority of 0.50 for quality multiplies j_Q after its four internal quality weights have been applied. A quality priority of one-quarter is therefore not an overall objective coefficient of one-quarter. Also, a resource reference need not be a hard maximum. In particular, j_W can exceed one if its numerator exceeds the declared reference. A denominator used for normalization does not itself impose feasibility.

Solids retention time and clarifier loading

Stored solids and lost solids have different units. For two reactors, the stored mass would be $V_1X_1 + V_2X_2 + M_{cl}$ in grams. Its loss rate would be $Q_EX_E + Q_0wX_U$ in grams per day. Dividing mass by mass per day gives days. The same accounting applies to all five reactors.

External solids leave in the overflow and waste-sludge streams,

$$\dot{M}_X = Q_0(q_EX_E + wX_U). \quad (6.149)$$

Return sludge does not count as an external loss. With $X_i = t_X^\top c_i$, the whole-plant solids retention time is

$$\theta_X = \frac{\sum_{i=1}^{n_r} V_i X_i + M_{cl}}{\dot{M}_X}, \quad \dot{M}_X \geq \epsilon_X > 0. \quad (6.150)$$

Its numerator includes the clarifier inventory. Omitting that mass would underestimate solids age for the same external loss rate. A wasting ratio alone also does not determine solids age because wX_U and the stored mass both matter (Alex et al., 2008).

For a hypothetical accounting example, use $Q_0 = 10,000 \text{ m}^3 \text{ d}^{-1}$, $w = 0.02$, $X_E = 10$, and $X_U = 6,000 \text{ g m}^{-3}$. The two external loss contributions are

$$Q_0q_EX_E = 10,000(0.98)(10) = 98,000 \text{ g d}^{-1}, \quad (6.151)$$

$$Q_0wX_U = 10,000(0.02)(6,000) = 1,200,000 \text{ g d}^{-1}. \quad (6.152)$$

Their sum is $1,298,000 \text{ g d}^{-1}$. At a total reactor volume of $7,500 \text{ m}^3$, assigning every reactor $3,000 \text{ g TSS m}^{-3}$ gives $22,500,000 \text{ g}$ in the reactors. If the clarifier inventory is

7,500,000 g, the total inventory is 30,000,000 g. The illustrative solids retention time is $30,000,000/1,298,000 = 23.11$ d. This hypothetical ratio illustrates the inventory and external-loss accounting.

Clarifier surface loading divides the overflow volume rate by area. Clarifier solids loading divides the feed solids mass rate by area. Since one gram is 10^{-3} kilograms, its conversion appears before the area division. The descriptive clarifier loading measures are

$$\ell_{\text{surface}} = \frac{Q_0 q_E}{A_{\text{cl}}}, \quad \ell_{\text{solids}} = \frac{10^{-3} Q_0 q_C X_N}{A_{\text{cl}}}. \quad (6.153)$$

The factor 10^{-3} converts grams to kilograms. The surface quantity is in m d^{-1} and the solids quantity is in $\text{kg m}^{-2} \text{ d}^{-1}$. These quantities and θ_X are descriptive measures in the baseline comparison.

For the same hypothetical flow with $r_R = 0.5$, $q_C = 1.5$, $A_{\text{cl}} = 1,500 \text{ m}^2$, and $X_N = 3,000 \text{ g m}^{-3}$, the surface loading is $9,800/1,500 = 6.53 \text{ m d}^{-1}$. The feed solids rate is $10,000(1.5)(3,000) = 45,000,000 \text{ g d}^{-1}$, or $45,000 \text{ kg d}^{-1}$. Dividing by area gives $30 \text{ kg m}^{-2} \text{ d}^{-1}$. Using underflow instead of clarifier-feed solids in this formula would calculate a different loading quantity.

The common engineering safeguards are

$$X_U \leq X_U^{\text{max}}, \quad X_N \geq X_F^{\text{min}} > 0, \quad \dot{M}_X \geq \epsilon_X > 0. \quad (6.154)$$

The positive feed-solids limit keeps particulate outlet reconstruction defined. The positive external-loss limit keeps solids retention time defined. The underflow bound represents the admitted thickening or solids-handling range. Their numerical values are plant-specific problem inputs selected before search. An underflow limit should be based on accepted sludge-handling conditions. The feed and loss floors should prevent undefined calculations without excluding physically relevant low-solids operation. These engineering inputs are distinct from the fixed $X_U^{\text{ref}} = 15,000 \text{ g m}^{-3}$ used only to normalize the wasted-solids objective.

Trust diagnostics for operating search

The trust diagnostics are numerical checks used to restrict the operating search to specified levels of data support and model consistency. They assess input coverage, projection size, fitted clarifier behavior, and reactor balance residuals. They are safeguards for selecting candidate aeration, recycle, and wasting settings. Their fixed thresholds define the accepted search region.

The trust diagnostics examine model consistency within the region admitted by projection. The projection diagnostic first squares each standardized movement, averages these squares over the response coordinates, and takes the square root. For the hypothetical two-component projection $u = (2, 0.5)$, it is $\sqrt{(4 + 0.25)/2} = 1.458$. Dividing by the number of coordinates makes its interpretation comparable across response dimensions.

Feature leverage uses the feature vector and the regularized fitting geometry. In a hypothetical intercept-plus-one-feature example with three standardized feature values $(-\sqrt{3/2}, 0, \sqrt{3/2})$ and $\gamma = 1$,

$$\Phi^\top \Phi + 3\gamma R_\phi = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}. \quad (6.155)$$

Its inverse is $\text{diag}(1/3, 1/6)$. At query feature $(1, z)$, leverage is $1/3 + z^2/6$. It is $1/3$ at the center, $1/2$ at $z = 1$, and $11/6$ at $z = 3$. The largest development value in this teaching example is $7/12$. These hypothetical numbers illustrate weaker joint feature support away from the observed range.

The split diagnostic compares each particulate underflow flow with the flow expected from a common recovery factor. In a hypothetical two-particulate example with unit TSS weights, let the feed component concentrations be $(40, 60)$, so $X_N = 100$. An underflow component-flow vector $(20, 30)$ has total 50 and gives numerators $100(20) - 50(40) = 0$ and $100(30) - 50(60) = 0$. Changing the outlet allocation to $(25, 25)$ preserves total underflow solids but gives numerators 500 and -500 . With illustrative fixed split scales of 500, the standardized values are 1 and -1 , and their root mean square is one. Total solids consistency therefore need not establish composition consistency.

The reactor diagnostic first evaluates the kinetic balances, then divides each residual by its fixed balance scale, and finally computes their root mean square. In the hypothetical two-component reactor, a projected state $(9, 6)$ preserves the upstream total of $(12, 3)$. With $d = 2$ and the reaction rate held at 8 for this illustration, its two residuals are $2(12 - 9) - 8 = -2$ and $2(3 - 6) + 8 = 2$. A balance scale of 10 for each component gives standardized residuals $(-0.2, 0.2)$ and a diagnostic of 0.2. The invariant is satisfied even though the kinetic equations are not.

These four fixed diagnostics are written compactly as

$$\kappa_{\text{corr}} = \frac{\|D_\chi^{-1}(\hat{\chi} - \chi_{\text{raw}})\|_2}{\sqrt{n_\chi}}, \quad (6.156)$$

$$\kappa_{\text{lev}} = \phi^\top (\Phi^\top \Phi + n_{\text{dev}} \gamma R_\phi)^{-1} \phi, \quad (6.157)$$

$$\kappa_{\text{split}} = \left[\frac{1}{F_X} \sum_{f \in X} \left\{ \frac{X_N g_{U,f} - (t_X^\top g_U) c_{n_r,f}}{d_{\text{split},f}} \right\}^2 \right]^{1/2}, \quad (6.158)$$

$$\kappa_{\text{reac}} = \frac{\|D_{\text{bal}}^{-1} f_{\text{sm,reactor}}(\vartheta, \hat{\chi})\|_2}{\sqrt{n_r \bar{F}}}. \quad (6.159)$$

The first measures standardized projection size. A large value identifies a raw response that requires a large movement to satisfy the imposed constraints. The projection is constrained and reported separately from the engineering objective.

The second diagnostic is regularized feature leverage. It measures input support within the fitted quadratic feature map. Sahigara et al. (2012) compare applicability-domain assessments and motivate checking where a statistical model is being used. A

control vector inside its coordinate bounds may still have weak support in the joint feature space.

The third diagnostic checks the common particulate-recovery assumption in Equation (6.48). If every particulate component has the same recovery, its numerator vanishes. This diagnostic supplements component mass conservation and densification with a check on the reduced mechanistic composition relation.

The fourth diagnostic checks smooth reactor kinetics at the projected concentrations. It includes hydraulic transport, net biological conversion, and oxygen transfer on separate scales. Clarifier-layer residuals are evaluated separately with the layer-resolved mechanistic model. Its smoothing width is 10^{-8} under the definitions given below.

A row scale is based on the sizes of its separate physical terms before cancellation. Suppose a hypothetical row has terms $(100, -100)$ in one development state and $(200, -200)$ in another. Its net balance is zero in both states, but its terms have substantial magnitude. Their mean squared magnitude is $(100^2 + 100^2 + 200^2 + 200^2)/4 = 25,000$, giving a root mean square of 158.11. Scaling only the canceled net balance would lose this information. For a constraint row with p terms b_{ij} over n_{dev} development states, its fixed scale is

$$d_b = \max \left\{ 1, \left[\frac{1}{n_{\text{dev}} p} \sum_{i=1}^{n_{\text{dev}}} \sum_{j=1}^p b_{ij}^2 \right]^{1/2} \right\}. \quad (6.160)$$

The same convention defines inequality and split scales. The overflow row uses the mechanistic overflow flow and the out-of-fold fitted endpoint flow as its two terms. The floor of one is in the corresponding numerical row units. It is not a concentration floor.

Projection, split, and reactor limits use nearest-rank 95th percentiles of grouped out-of-fold development diagnostics. If sorted values are $v_{(1)}, \dots, v_{(n_{\text{dev}})}$, the limit is $v_{(\lceil 0.95 n_{\text{dev}} \rceil)}$. The leverage limit is the maximum development leverage under the final fitted feature map. These quantities are fixed before optimization. They define empirical search thresholds. A diagnostic with zero limit remains constrained to zero and uses one only as a numerical row scale.

The surrogate decision problem is consequently

$$\begin{aligned} \min_{\vartheta} \quad & J(\vartheta, \hat{\chi}(\vartheta, c_{\text{in}}), \varpi), \\ \text{subject to} \quad & \vartheta \in \mathcal{V}, \\ & g_{\text{eng}}(\vartheta, \hat{\chi}) \leq 0, \\ & g_{\text{trust}}(\vartheta, \chi_{\text{raw}}, \hat{\chi}) \leq 0, \\ & \hat{\chi} \text{ satisfies Equation (6.100).} \end{aligned} \quad (6.161)$$

The lower problem is a convex quadratic projection. The upper problem is generally nonconvex. This distinction follows the nested-problem perspective described by Colson et al. (2007). The trust-region analysis of Alexandrov et al. (1998) explains why

reliable approximation management needs more than a small average prediction error. The four fixed diagnostics supply the admission checks for this operating search.

6.2.6 Bounded products and convex relaxation

Flow and concentration products help explain why process optimization remains nonlinear even when a surrogate is linear in its fitted coefficients. A bounded product provides a simple example. Let $x_L \leq x \leq x_U$, $y_L \leq y \leq y_U$, and let z represent the product xy . The four products

$$(x - x_L)(y - y_L), \quad (x_U - x)(y_U - y), \quad (x_U - x)(y - y_L), \quad (x - x_L)(y_U - y)$$

are non-negative. Expanding the first gives $xy - x_Ly - y_Lx + x_Ly_L \geq 0$. Replacing xy by z gives one lower bound on z . Expanding the other three products gives the remaining bounds. Together they form the McCormick relaxation (McCormick, 1976)

$$\begin{aligned} z &\geq x_Ly + y_Lx - x_Ly_L, \\ z &\geq x_Uy + y_Ux - x_Uy_U, \\ z &\leq x_Uy + y_Lx - x_Uy_L, \\ z &\leq x_Ly + y_Ux - x_Ly_U. \end{aligned} \tag{6.162}$$

The four inequalities are linear in the variables x , y , and z . Every exact product within the stated bounds satisfies them. Their intersection nevertheless permits points that do not satisfy $z = xy$.

For a hypothetical pair with $0 \leq x, y \leq 1$, the bounds become $z \geq 0$, $z \geq x + y - 1$, $z \leq x$, and $z \leq y$. At $x = y = 0.5$, they permit $0 \leq z \leq 0.5$. The exact product is only $z = 0.25$. A favorable objective obtained at another permitted value of z is therefore not, by itself, an attainable process response. Tighter variable bounds can improve the relaxation, but they do not generally replace the exact product relation. Process-network optimization uses this distinction when constructing bounds and assessing solutions (Rubio-Castro et al., 2013; L. Yang et al., 2014).

This example separates a convex relaxation from the physical projection used here. The relaxation enlarges a nonlinear feasible set to obtain an optimization bound. The projection selects a response satisfying the declared mass conservation and non-negativity conditions near a predicted response. The present fixed-configuration plant retains the surrogate-assisted and direct mechanistic searches described below. Its plant connections remain fixed throughout these searches.

6.2.7 A direct mechanistic comparator

The full-state constrained optimization problem

The direct route retains the controls and full reduced mechanistic state. It re-constructs $\chi(\vartheta, c_{\text{in}}, y)$ and evaluates the same engineering objective. Its finite-volume

clarifier equations require a positive feed-solids denominator at solver iterates. An auxiliary variable $\widetilde{X}_F \geq X_F^{\min}$ supplies that denominator, with equality $\widetilde{X}_F = t_X^\top c_{n_r}$ imposed at feasibility.

In a hypothetical one-reactor, two-component, three-layer illustration, the full state is $(c_S, c_X, s_1, s_2, s_3)$. It has five entries rather than the nine entries of the reduced statistical response. A simple conversion rate kc_S gives two reactor equations,

$$0 = d(m_S - c_S) - kc_S, \quad (6.163)$$

$$0 = d(m_X - c_X) + kc_S. \quad (6.164)$$

The three layer equations are those obtained from the interface-flow balances above, divided by their own volumes. The direct problem imposes all five equations rather than replacing the two reactor equations by only their sum fixed by mass conservation. Its three-layer envelope contains two scalar inequalities, $s_1 - s_2 \leq 0$ and $s_2 - s_3 \leq 0$. Four layers would require $(s_1 - s_2, s_1 - s_3, s_2 - s_4, s_3 - s_4) \leq 0$. This ordering explains how the envelope is assembled for ten layers.

The auxiliary denominator also has a concrete role. An infeasible trial can have zero feed solids while a nonzero layer concentration is being adjusted. Division by the actual feed value would then be undefined. An independently bounded $\widetilde{X}_F$ keeps the trial calculation defined. The equality $\widetilde{X}_F = X_N$ must subsequently hold, so this device does not allow a zero-feed trial to be accepted as a feasible physical state. It provides a usable calculation during the search, not an alternative physical feed concentration.

Let f_{sm} collect all $n_r F$ reactor equations and all L smoothed layer equations. Let

$$g_{\text{env}}(y) = \text{col}(s_1 \mathbf{1}_{L-2} - s_{2:L-1}, s_{2:L-1} - s_L \mathbf{1}_{L-2}). \quad (6.165)$$

The direct nonlinear program is

$$\begin{aligned} & \min_{\vartheta, y, \widetilde{X}_F} J(\vartheta, \chi(\vartheta, c_{\text{in}}, y, \widetilde{X}_F), \varpi), \\ & \text{subject to } f_{\text{sm}}(\vartheta, c_{\text{in}}, y, \widetilde{X}_F) = 0, \\ & \quad \widetilde{X}_F = t_X^\top c_{n_r}, \quad \widetilde{X}_F \geq X_F^{\min}, \\ & \quad \vartheta \in \mathcal{V}, \quad y \geq 0, \quad g_{\text{env}}(y) \leq 0, \\ & \quad g_{\text{eng,red}}(\vartheta, \chi) \leq 0. \end{aligned} \quad (6.166)$$

The reduced engineering vector removes the feed-solids row already imposed by the auxiliary equality and lower bound. The auxiliary does not alter a feasible physical state. It keeps the outlet mapping defined while a nonlinear solver moves through infeasible trial states.

Smooth switches and continuation

The exact kinetic and settling functions contain minima, maxima, positive parts, possible zero denominators, and a receiver switch. Derivative-based optimization ben-

efits from smooth approximations to these operations. The chemical-process optimization treatment of Buzzi-Ferraris and Manenti (2013) provides the general derivative and conditioning context. The exact identities

$$\max(a, b) = \frac{a + b + |a - b|}{2}, \quad \min(a, b) = \frac{a + b - |a - b|}{2} \quad (6.167)$$

show where the nondifferentiable absolute value enters. Replacing $|r|$ by $\sqrt{r^2 + \delta^2}$ removes its corner for every $\delta > 0$. The positive part uses the same replacement in $(v + |v|)/2$. A denominator regularization instead replaces $1/b$ by $b/(b^2 + \delta^2)$. These are different operations chosen for different numerical difficulties. For positive width $\delta = \epsilon S$, define

$$\max_{\delta}(a, b) = \frac{1}{2} \left[a + b + \sqrt{(a - b)^2 + \delta^2} \right], \quad (6.168)$$

$$\min_{\delta}(a, b) = \frac{1}{2} \left[a + b - \sqrt{(a - b)^2 + \delta^2} \right], \quad (6.169)$$

$$[v]_{+, \delta} = \frac{1}{2} \left[v + \sqrt{v^2 + \delta^2} \right], \quad (6.170)$$

$$\left(\frac{a}{b} \right)_{\delta} = \frac{ab}{b^2 + \delta^2}. \quad (6.171)$$

For negative v , the positive-part expression is evaluated as $\delta^2/[2(\sqrt{v^2 + \delta^2} - v)]$ to avoid subtraction of nearly equal values. These approximations change the equations near a switch. Independent evaluation uses the exact equations at the selected controls.

For hypothetical values $a = 2$, $b = 1$, and $\delta = 0.1$, the smooth maximum is $(3 + \sqrt{1.01})/2 = 2.00249$ and the smooth minimum is 0.99751. The exact values are 2 and 1. For $v = -0.2$, the smooth positive part is $(-0.2 + \sqrt{0.05})/2 = 0.01180$, while the exact positive part is zero. At $b = 0$, the regularized quotient $ab/(b^2 + \delta^2)$ is defined and equals zero. At $a = 1$ and $b = 0.1$, it is 5 for $\delta = 0.1$, compared with the exact quotient 10. These differences illustrate why smaller widths and an independent exact-model evaluation are needed.

The stable positive-part expression follows by rationalization,

$$\frac{v + \sqrt{v^2 + \delta^2}}{2} = \frac{(v + \sqrt{v^2 + \delta^2})(\sqrt{v^2 + \delta^2} - v)}{2(\sqrt{v^2 + \delta^2} - v)} \quad (6.172)$$

$$= \frac{\delta^2}{2(\sqrt{v^2 + \delta^2} - v)}. \quad (6.173)$$

The numerator becomes $v^2 + \delta^2 - v^2 = \delta^2$. This form avoids canceling a negative v against a nearly equal square root.

For the receiver switch, first measure where the receiver concentration lies across its transition band. Subtracting the lower edge $X_t - h$ and dividing by the band width $2h$ maps that band to $[0, 1]$. For receiver concentration s , the smooth switch is zero below $X_t - h$, one above $X_t + h$, and

$$6\zeta^5 - 15\zeta^4 + 10\zeta^3, \quad \zeta = \frac{s - X_t + h}{2h} \quad (6.174)$$

within the transition band. This polynomial matches the values and first two derivatives at the band endpoints. It gives smooth interpolation between the two gravitational-flux rules.

For the illustrative band $X_t = 3,000$ and $h = 10 \text{ g m}^{-3}$, concentrations 2,990, 3,000, and 3,010 correspond to $\zeta = 0, 0.5, 1$. The switch values are 0, 0.5, and 1. At 2,995, $\zeta = 0.25$ and the switch is 0.10352. At 3,005, it is 0.89648. If the unrestricted gravitational flux is J_1 and the receiver-limited flux is J_2 , interpolation gives $(1 - \sigma)J_1 + \sigma J_2$ for switch value σ . The switch therefore changes a flux rule gradually rather than only replacing a reported concentration.

Each argument scale S is its development-set root mean square with a floor of one. Reactor arguments are pooled across stages. They include total oxidized nitrogen, fermentable substrate plus acetate, the hydrolysis denominator, the polyphosphate and stored-carbon denominators, and remaining storage capacity. Clarifier arguments include $s - f_{\text{ns}}X_N$ and gravitational solids flux. The velocity scale is 250 m d^{-1} . These scales stay fixed throughout continuation.

The direct route reduces smoothing through three stages,

$$(\epsilon, h) = (10^{-6}, 10), (10^{-7}, 3), (10^{-8}, 1), \quad (6.175)$$

where h is in g TSS m^{-3} . Each stage initializes from the preceding stage. Continuation helps a solver approach sharper switches while preserving a usable starting state. The final smooth candidate still requires an exact-model evaluation.

Jacobian structure and globalization

The Jacobian entry J_{jk} is the partial derivative of residual f_j with respect to state coordinate y_k . It describes how one balance changes when one coordinate changes while the others are held fixed. For a hypothetical two-reactor soluble conversion with rate $kc_{i,S}$, soluble pass-through in the recycle gives

$$m_S = \frac{c_{\text{in},S} + (r_I + r_R)c_{2,S}}{q_P}. \quad (6.176)$$

The two soluble residuals are $f_{1,S} = d_1(m_S - c_{1,S}) - kc_{1,S}$ and $f_{2,S} = d_2(c_{1,S} - c_{2,S}) - kc_{2,S}$. Their two-by-two state block is

$$\frac{\partial(f_{1,S}, f_{2,S})}{\partial(c_{1,S}, c_{2,S})} = \begin{bmatrix} -(d_1 + k) & d_1(r_I + r_R)/q_P \\ d_2 & -(d_2 + k) \end{bmatrix}. \quad (6.177)$$

The off-diagonal upper-right entry is created by recycle. Dropping it would treat the mixer as independent of the final reactor. For hypothetical values $d_1 = d_2 = 2$, $k = 1$, $r_I = 1$, $r_R = 0.5$, and $q_P = 2.5$, the block is $\begin{bmatrix} -3 & 1.2 \\ 2 & -3 \end{bmatrix}$. Its entries have units d^{-1} because the residual has concentration-per-day units and its derivative is with respect to concentration.

Finite differences permit an independent check of those entries. Increasing $c_{1,S}$ by 0.01 g m^{-3} changes $f_{1,S}$ by $-0.03 \text{ g m}^{-3} \text{ d}^{-1}$, giving $-0.03/0.01 = -3$. Increasing $c_{2,S}$ by the same amount changes m_S by 0.006 and $f_{1,S}$ by 0.012 , giving 1.2 . These values are exact for the hypothetical linear teaching equations. In a nonlinear residual, the check instead uses a small step h ,

$$J_{jk} \approx \frac{f_j(y + he_k) - f_j(y - he_k)}{2h}. \quad (6.178)$$

Here e_k changes only coordinate k . Near a non-negative boundary, a central step can leave the allowed state region. A second-order forward alternative is

$$J_{jk} \approx \frac{-3f_j(y) + 4f_j(y + he_k) - f_j(y + 2he_k)}{2h}. \quad (6.179)$$

The step has the physical unit of y_k . Scaling coordinates makes its relative size comparable across different component bases.

For three layers at fixed feed conditions, only neighboring concentrations enter each interface flux. Their layer Jacobian therefore has the pattern

$$J_{\text{layer}} = \begin{bmatrix} \star & \star & 0 \\ \star & \star & \star \\ 0 & \star & \star \end{bmatrix}, \quad (6.180)$$

where a star denotes an entry that may be nonzero. Feed solids and outlet reconstruction add connections to reactor coordinates. The zeros describe the declared dependency structure and are not approximations made to reduce computation.

The direct residual is sparse in its mechanistic state. A reactor balance depends on its own component vector and the preceding stage. Recycle reconstruction adds dependencies on final-reactor and underflow quantities. A layer balance depends on nearby layer concentrations, feed solids, and hydraulic controls. The inventory and reporting maps are linear aggregations. These dependencies provide a structured Jacobian rather than a dense arbitrary response map.

Scaling is applied before testing Jacobian conditioning or comparing residual sizes. Different component bases include oxygen, COD, nitrogen, phosphorus, solids, and charge equivalents. An unscaled norm could be dominated by a large solids balance and conceal a nutrient-balance defect. Analytic derivatives or differentiated smooth equations supply the optimization Jacobian. Independent finite differences are retained for derivative checks. Sparse linear algebra exploits the local reactor and layer connections without omitting recycle terms.

The direct search uses the interior-point filter method studied by Wächter and Biegler (2006). Its line search manages progress in objective and constraint violation. This numerical globalization controls acceptance of local steps. The surrogate route similarly separates numerical search progress from the final feasibility and stationarity audits.

6.3 Methodology

6.3.1 Case-study domain and fixed parameters

Topology and free controls

The case study has five equal-volume reactors and ten clarifier layers. Reactors 1 and 2 are unaerated. Reactors 3–5 have independent aeration controls. The free-control vector is

$$\vartheta = (H, a_3, a_4, a_5, r_I, r_R, w)^\top \in \mathbb{R}^7. \quad (6.181)$$

For every stage, $h_i = H/5$. The unaerated settings are $a_1 = a_2 = 0$, the aerated transfer coefficients are $k_L a_i = 47a_i \text{ d}^{-1}$, and $\xi_i = 0$. These are case-study conditions rather than structural restrictions on the connected formulation.

Table 6.3: Plant geometry and operating domain.

Quantity	Value or interval
Fresh-influent flow Q_0	10,000 m ³ d ⁻¹
Reactors n_r , layers L , feed layer ℓ_F	5, 10, 5
Reactor volume fraction f_i^V	1/5 for every stage
Clarifier area and total depth	1,500 m ² and 4.0 m
Clarifier total and individual layer volumes	6,000 m ³ and 600 m ³
Nominal total HRT H	6–36 h
Each of a_3, a_4, a_5	0–1 independently
Mixed-liquor recycle r_I	0–4
Return-sludge ratio r_R	0.25–1.25
Waste-sludge ratio w	0.001–0.05
Aeration reference E_A^{ref}	423,000 m ³ d ⁻¹
Wasted-solids reference X_U^{ref}	15,000 g TSS m ⁻³

The equal volumes provide a compact unaerated-to-aerated train. The layer depth is 0.4 m. The operating intervals preserve both positive clarifier outlet flow ratios. Together with 20 influent coordinates, the seven controls give 27 statistical inputs, 406 features, and 161 reduced-response coordinates. The mechanistic state has 110 coordinates. The projection has 76 equality rows and 12 physical inequality rows. Adding response non-negativity gives 173 inequality rows.

Changing the number of reactors or free controls changes the statistical dimensions. Changing the layer count changes the mechanistic dimension and inventory mapping but does not add coordinates to the aggregate surrogate response. In either case, physical and predictive validation must be repeated. The layer discretization enters through the mechanistic targets and inventory bounds.

Fresh-influent intervals

Table 6.4 defines independent intervals for the plant experiment. The nominal influent is their componentwise midpoint. The concentration bases and quality-map weights are those in Table 3.1. The plant alkalinity interval is 1.6–5.2 mol charge equivalents m^{-3} . This interval defines the connected-plant influent design.

Table 6.4: Independent fresh-influent intervals for the connected plant.

Component	Lower	Upper	Constituent basis
S_O	0	0.5	oxygen
S_F	20	180	COD
S_A	5	80	COD
S_{NH4}	12	55	nitrogen
S_{NO2}	0	3	nitrogen
S_{NO3}	0	8	nitrogen
S_{N2}	0	2	nitrogen
S_{PO4}	2	18	phosphorus
S_I	10	90	COD
S_{ALK}	1.6	5.2	charge equivalents
X_I	20	120	COD
X_S	60	280	COD
X_H	15	100	COD
X_{PAO}	5	60	COD
X_{PP}	2	20	phosphorus
X_{PHA}	1	30	COD
X_{AOB}	0.5	8	COD
X_{NOB}	0.5	8	COD
X_{MeP}	0	12	solids
X_{MeOH}	0	12	solids

All concentration values are in grams of the stated constituent per m^3 , apart from alkalinity. Numerically, g m^{-3} equals mg L^{-1} . Independent interval sampling distributes candidates across the declared experimental domain.

Stoichiometric, kinetic, and settling parameters

The constants in Tables 6.5–6.8 define the plant case. They are fixed parameters rather than fitted kinetic estimates. A parameter’s concentration basis follows the associated component. Symbols with several process subscripts on the same row have equal values. The two nitrifier populations remain separate, with different growth rates and oxygen half-saturation constants.

Table 6.5: Fixed yields, constituent contents, and conversion factors.

Parameter	Value	Unit or interpretation
Y_H, Y_{PAO}	0.625	g biomass COD per g substrate COD
Y_{PHA}	0.20	g stored COD per g phosphorus
Y_{PO4}	0.40	g phosphorus per g stored COD
Y_{AOB}	0.18	g biomass COD per g nitrogen
Y_{NOB}	0.08	g biomass COD per g nitrogen
f_{SI}	0	soluble inert fraction
f_{XI}	0.10	particulate inert fraction
$i_{N,SI}$	0.01	g nitrogen per g COD
$i_{N,SF}, i_{N,XS}$	0	g nitrogen per g COD
$i_{N,XI}$	0.02	g nitrogen per g COD
$i_{N,BM}$	0.07	g nitrogen per g biomass COD
$i_{P,SI}, i_{P,SF}, i_{P,XS}$	0	g phosphorus per g COD
$i_{P,XI}$	0.01	g phosphorus per g COD
$i_{P,BM}$	0.02	g phosphorus per g biomass COD
$i_{P,MeP}$	1/4.87	g phosphorus per g metal-phosphate solids
γ_{OH}	3.45	g metal-hydroxide solids per g phosphorus

Table 6.6: Fixed hydrolysis and heterotrophic rate and saturation constants.

Parameter	Value	Unit
K_H	3.0	d ⁻¹
$\eta_{hyd,NO3}, \eta_{hyd,NO2}$	0.6	dimensionless
$\eta_{hyd,fe}$	0.4	dimensionless
$K_{O,hyd}, K_{O,H}, K_{O,PAO}$	0.20	g oxygen m ⁻³
$K_{NO3,hyd}, K_{NO2,hyd}, K_{NOx,hyd}$	0.5	g nitrogen m ⁻³
K_X	0.10	g substrate COD per g biomass COD
μ_H	6.0	d ⁻¹
q_{fe}	3.0	d ⁻¹
b_H	0.40	d ⁻¹
$\eta_{H,NO3}, \eta_{H,NO2}$	0.9	dimensionless
K_F, K_{fe}, K_A	4.0	g COD m ⁻³
$K_{NO3,H}, K_{NO2,H}, K_{NOx,H}$	0.5	g nitrogen m ⁻³
$K_{NH4,H}, K_{NH4,PAO}, K_{NH4,NOB}$	0.05	g nitrogen m ⁻³
$K_{PO4,H}, K_{PO4,PAO}, K_{PO4,nit}$	0.01	g phosphorus m ⁻³
$K_{ALK,H}, K_{ALK,PAO}$	0.10	mol equivalents m ⁻³

Table 6.7: Fixed storage, nitrifier, and chemical rate and saturation constants.

Parameter	Value	Unit
q_{PHA}	5.0	g stored COD per g biomass COD per d
q_{PP}	0.60	g phosphorus per g biomass COD per d
μ_{PAO}	0.56	d ⁻¹
η_{PAO,NO_3}	0.07	dimensionless
η_{PAO,NO_2}	0.90	dimensionless
$b_{PAO}, b_{PP}, b_{PHA}, b_{AOB}$	0.20	d ⁻¹
$K_{NO_3,PAO}, K_{NO_2,PAO}, K_{NO_x,PAO}$	0.5	g nitrogen m ⁻³
K_{PS}	0.20	g phosphorus m ⁻³
K_{PP}	0.01	g phosphorus per g biomass COD
K_{max}	0.34	g phosphorus per g biomass COD
K_{IPP}	0.02	g phosphorus per g biomass COD
K_{PHA}	0.01	g stored COD per g biomass COD
μ_{AOB}	1.81	d ⁻¹
μ_{NOB}	1.52	d ⁻¹
b_{NOB}	0.17	d ⁻¹
$K_{O,AOB}$	0.74	g oxygen m ⁻³
$K_{O,NOB}$	1.75	g oxygen m ⁻³
$K_{NH_4,AOB}, K_{NO_2,NOB}$	0.5	g nitrogen m ⁻³
$K_{ALK,nit}, K_{ALK,PRE}, K_{ALK,chem}$	0.50	mol equivalents m ⁻³
k_{PRE}	1.0	m ³ per g solids per d
k_{RED}	0.60	d ⁻¹

Table 6.8: Fixed settling and oxygen-transfer parameters.

Quantity	Value
Maximum settling velocity v_{max}	250 m d ⁻¹
Theoretical settling velocity v_0	474 m d ⁻¹
Hindered-settling coefficient r_h	0.000576 m ³ per g TSS
Low-concentration coefficient r_p	0.00286 m ³ per g TSS
Nonsettleable fraction f_{ns}	0.00228
Receiver threshold X_t	3,000 g TSS m ⁻³
Oxygen saturation S_O^*	8.5 g oxygen m ⁻³
Maximum aerated-stage k_La	47 d ⁻¹

6.3.2 Mechanistic sampling, fitting, and prediction assessment

Simulation groups and accepted cases

The joint domain has seven control coordinates and twenty influent coordinates. Strength-one Latin-hypercube sampling places one point in each stratum of every coordinate. Independent random stratum orderings prevent a shared ordering from forcing artificial coordinate correlations. Open-unit jitter is a random fractional position strictly between zero and one. It places a candidate influent or control value inside each selected stratum, excluding its endpoints. The comparison of sampling designs by McKay et al. (1979) provides the basis for this coverage-oriented design.

The study design attempts 8,000 development candidates and 2,000 holdout candidates from separate seeded streams. Rejected candidates are retained in the accounting and are not replaced. The two-start physical acceptance procedure determines which candidates enter a statistical partition. Acceptance counts are recorded for both partitions.

Table 6.9: Randomization and fitting settings specified before assessment.

Setting	Value
Development Latin-hypercube seed	100042
Holdout Latin-hypercube seed	100043
Influent-scenario Latin-hypercube seed	314159
Grouped-fold permutation seed	271828
Number of fitting folds	5
Raw-response and logarithmic ridge grids	$\{10^{-8}, 10^{-7}, \dots, 10^2\}$
Standard-deviation divisor	fitting-set size
Near-constant coordinate cutoff	$10^{-12} \max(1, \max_i v_i)$
Low-overflow group	at or below development 25th percentile
High-overflow group	at or above development 90th percentile

The sampling arithmetic can be understood with four strata in one hypothetical coordinate. Each stratum has width $1/4$ in the unit interval. Number the strata from zero to three. Selecting stratum two and a within-stratum fraction of 0.6 gives $(2 + 0.6)/4 = 0.65$. For an illustrative physical interval from 10 to 20, this becomes $10 + 0.65(20 - 10) = 16.5$. Applying independent permutations to the other coordinates retains one observation in each marginal stratum without forcing the same combinations in every coordinate.

The exact jitter expression uses the same idea at a much finer resolution. The floor

operation $\lfloor v \rfloor$ means the largest integer no greater than v . Dividing U by 2^{11} and taking its floor yields an integer from zero to $2^{53} - 1$. Adding 0.5 selects the midpoint of the corresponding fine stratum. Division by 2^{53} places that midpoint inside the unit interval. In exact arithmetic, the extreme values are 2^{-54} and $1 - 2^{-54}$. Both are strictly inside the interval. Endpoint checks remain necessary when values are represented with finite numerical precision.

For an exact numerical randomization convention, each coordinate uses a descending Fisher–Yates permutation of its strata. The jitter from an unsigned 64-bit random integer U is

$$\zeta_U = \frac{\lfloor U/2^{11} \rfloor + 0.5}{2^{53}} \in (0, 1). \quad (6.182)$$

This definition excludes exact stratum boundaries. Accepted states retain their candidate ordering before the grouped-fold permutation is applied.

The fitting and prediction assessments use the states admitted by the integration, balance, branch, and stability checks. Rejection counts retain the corresponding outcomes from the attempted design.

Two initial states and the nonsmoothed steady solve

The first initial state assigns c_{in} to every reactor and fresh-influent TSS to every clarifier layer. The second retains the influent soluble concentrations and multiplies the particulate concentrations in reactors 1–5 by (1.5, 2, 2.5, 3, 3.5). Its final-reactor TSS sets a layer profile with multipliers

$$(0.002, 0.005, 0.01, 0.03, 0.10, 0.50, 1.25, 2.25, 3.25, 4.00) \quad (6.183)$$

from surface to hopper. The different solids distributions probe sensitivity to the initial biomass and storage state. Two-start state agreement is one of the numerical acceptance checks.

The positive-state transformation follows from the scalar chain rule. Treat y_f as the numerical concentration in its assigned native unit and introduce $\eta_f = \log y_f$ for $y_f > 0$. The logarithm therefore acts on the numerical coordinate, with one native unit as its implicit reference. Differentiating its inverse gives

$$y_f = \exp(\eta_f),$$

$$\frac{dy_f}{dt} = \exp(\eta_f) \frac{d\eta_f}{dt} = y_f \frac{d\eta_f}{dt}.$$

The original component balance requires $dy_f/dt = f_f(y)$. Equating the two expressions and dividing by positive y_f gives $d\eta_f/dt = f_f(y)/y_f$. The balance is transformed rather than replaced by another physical process.

For a hypothetical concentration of two native units and a balance rate of -0.4 native units per day, the transformed rate is $-0.4/2 = -0.2$ per day. A simple half-day

illustrative step in logarithmic coordinates gives

$$\eta_{new} = \log 2 + 0.5(-0.2), \quad y_{new} = \exp(\eta_{new}) = 2 \exp(-0.1) \approx 1.8097.$$

This example explains the algebra of a positive coordinate. The prescribed numerical integration retains the backward differentiation method and its error controls. The integration error controls and steady-state acceptance checks are applied separately.

Time integration uses the original nonsmoothed equations. A logarithmic state transformation keeps integrated coordinates positive. Before this transformation alone, an initial coordinate smaller than 10^{-8} is raised to 10^{-8} in its own physical unit. For $y_f = \exp(\eta_f)$, the transformed balance is $\dot{\eta}_f = f_f(y)/y_f$. This initialization safeguard is distinct from clipping a final predicted concentration.

The integration horizon contains two nested choices. First, the wasting coordinate is bounded below by 0.001 for this horizon calculation. Second, its reciprocal term is compared with a minimum horizon of 400 days. At a hypothetical $w = 0.01$, the inner maximum is 0.01, so the reciprocal term is $50/0.01 = 5000$ and the horizon is 5000 days. At $w = 0.20$, that term is 250, so the horizon is 400 days. At $w = 0$, the inner floor gives $50/0.001 = 50,000$ days. These arithmetic examples illustrate the declared rule and are not claims about actual convergence durations.

The values zero and 0.20 illustrate the branches of the horizon rule and lie outside the case-study wasting interval. They are not admitted operating candidates.

The backward differentiation method provides an integration route for stiff equations (Curtiss & Hirschfelder, 1952). The horizon is

$$T = \max\{400, 50/\max(w, 0.001)\} \text{ d.} \quad (6.184)$$

The calculation can stop early when its scaled residual falls below 10^{-9} . Dynamic relaxation supplies an initial state for algebraic steady calculation, consistent with the continuation rationale of Kelley and Keyes (1998) and its activated sludge use in Selerio (2026). A bounded least-squares polishing calculation is applied only if the integrated endpoint fails the physical audit. Solver convergence and physical acceptance use separate tolerances.

A residual first adds the signed physical terms in a balance. For hypothetical incoming, consumption, and outgoing contributions of 100, -60 , and -39.99999 in one consistent component-rate unit, the remaining imbalance is

$$100 - 60 - 39.99999 = 0.00001.$$

The largest absolute term is 100. Dividing the absolute imbalance by that scale gives $0.00001/100 = 10^{-7}$. The floor of one in the scale prevents division by a very small reference when all terms are near zero. The resulting dimensionless ratio assesses cancellation relative to the terms in that component equation. It is evaluated separately

Table 6.10: Steady-state generation and physical acceptance settings.

Setting	Value
Integration relative tolerance	10^{-7}
Transformed integration absolute tolerance	10^{-9}
Maximum integration step	$T/100$
Polishing evaluation limit	5,000
Polishing step, function, and gradient tolerances	10^{-9} each
Term-scaled balance tolerance	10^{-6}
State non-negativity tolerance	10^{-10}
Reaction-rate non-negativity tolerance	10^{-12}
Two-start scaled state agreement	10^{-6}
Rightmost reduced-state Jacobian eigenvalue	at most -10^{-8} d^{-1}
Eigenvalue agreement between two step sizes	10^{-6} d^{-1}
Matrix mass conservation and rank audit tolerance	10^{-10}

for each balance rather than obtained by adding balances with different constituent units.

For a balance with separately evaluated terms b_j , the acceptance residual is

$$r_{\text{bal}} = \frac{|\sum_j b_j|}{\max(1, \max_j |b_j|)}. \quad (6.185)$$

Reactor generation scales repeat $\max(1, c_{\text{in},f}^U)$ for each component. Layer scales use $\max(1, X_N)$. State agreement is tested in these scaled coordinates.

Local stability asks what happens to a small perturbation about an equilibrium. A first-order expansion of the dynamic equations has the form $\dot{\delta y} = J_{\text{dyn}} \delta y$, where entry (f, j) of J_{dyn} is the derivative of balance f with respect to state j . In a hypothetical independent two-coordinate example,

$$\begin{bmatrix} \dot{\delta y}_1 \\ \dot{\delta y}_2 \end{bmatrix} = \begin{bmatrix} -0.2 & 0 \\ 0 & 0.05 \end{bmatrix} \begin{bmatrix} \delta y_1 \\ \delta y_2 \end{bmatrix}.$$

The scalar equations are $\dot{\delta y}_1 = -0.2\delta y_1$ and $\dot{\delta y}_2 = 0.05\delta y_2$. Their perturbations evolve as $\delta y_1(0) \exp(-0.2t)$ and $\delta y_2(0) \exp(0.05t)$ when time is in days. One decays and the other grows. The two diagonal eigenvalues are -0.2 and 0.05 per day, so the rightmost eigenvalue is positive and this hypothetical equilibrium fails the stability condition.

For a coupled system, an eigenvector gives a combination of coordinates that behaves as one local mode. The real part of its eigenvalue determines growth or decay. The rightmost eigenvalue means the largest real part among all modes. This is a local test because the linear expansion describes small perturbations. Checking derivatives with two finite-difference steps tests numerical sensitivity of the estimated modes, while the stated negative threshold tests their local decay.

Reduced-state local stability is assessed by the rightmost eigenvalue of the dynamic Jacobian. Finite differences use normalized steps $\epsilon_{\text{mach}}^{1/3}$ and $2\epsilon_{\text{mach}}^{1/3}$. Central differences are used away from the non-negative boundary. Second-order forward differences are used near it. Both eigenvalue estimates must pass the upper-bound and agreement checks. These checks assess local stability of the reactor-and-aggregate-layer reference model.

Both starts must provide finite states, satisfy non-negativity and rate checks, pass component and invariant mass conservation checks, obey the operating domain and layer envelope, and meet the layer-flux residual tolerance. Generation also requires identical branch-classification tuples between the two starts. These tuples record the active kinetic and settling branches. A state that meets a residual test but fails another physical check is not admitted as a training target.

Fold-local fitting and two separately selected penalties

Every component and location from one accepted plant state stays in the same fold. This grouping prevents the same simulated plant from contributing training information and an assessment target within a fold. Fold sizes differ by at most one. Centers, input scales, quadratic-feature scales, and response scales are recomputed inside each fitting fold. The holdout contributes none of these quantities. The information-leakage analysis of Kaufman et al. (2012) explains why assessment information must be kept outside fitting choices.

The one-standard-error rule can be followed with a hypothetical five-fold comparison. Suppose the candidate with the smallest mean score has fold scores 0.40, 0.42, 0.38, 0.41, and 0.39. Their mean is 0.40. The sum of squared deviations from that mean is

$$0^2 + 0.02^2 + (-0.02)^2 + 0.01^2 + (-0.01)^2 = 0.001.$$

The sample standard deviation is $\sqrt{0.001/4} \approx 0.01581$. The standard-error summary is $0.01581/\sqrt{5} \approx 0.00707$. The admissible mean-score threshold is consequently $0.40 + 0.00707 = 0.40707$.

If a stronger hypothetical penalty has mean score 0.406, it lies within the threshold. A still stronger candidate with mean score 0.430 lies outside. The rule retains the largest candidate penalty within the threshold. It does not select the numerical score 0.40707 as a penalty. This hypothetical example explains the regularization choice.

Each candidate penalty is assessed by the standardized root mean square error of the complete raw response. The largest penalty within one standard error of the smallest mean score is selected. The rule favors stronger regularization when the score does not clearly distinguish the smaller penalty (Hastie et al., 2009). Projected errors do not choose the raw-response penalty. The logarithmic model uses the same folds, grid, and one-standard-error rule but is scored on its logarithmic target. Its penalty is selected separately.

The augmented ridge equations are solved with column-pivoted orthogonal-triangular factorization and audited using singular-value decomposition. A coordinate whose standard deviation meets the near-constant cutoff in Table 6.9 is rejected rather than assigned an arbitrary scale. Positive ridge penalties and the unpenalized intercept give a unique coefficient solution under the stated fitting design. The fitted penalties and coefficient matrices are retained as estimation quantities. The candidate grid defines the eligible penalties, and cross-validation selects their fitted values.

After selection, both models are refitted once on all accepted development states. Grouped out-of-fold predictions supply trust calibration and projection assessment. The overflow equality for an out-of-fold state uses the corresponding out-of-fold logarithmic model, not the final fit that has seen that state. The final models supply holdout predictions and optimization evaluations.

The overflow target must be strictly positive before logarithmic fitting. Predicted logarithms and their back-transformations must be finite. The development-set low-overflow and high-overflow strata are fixed by the percentiles in Table 6.9. Separate stratum assessment can reveal an error pattern concealed by a central concentration range. The imbalanced-regression analysis of Shahbazi et al. (2026) motivates attention to sparse or extreme response ranges. Separate concentration-stratum scores describe these response ranges.

Scientific admission and projection audits

Scientific admission is the decision to accept the fitted plant surrogate for operating assessment under the specified numerical criteria. Out-of-fold predictions are calculated for development cases excluded from the corresponding fit. They provide checks on new-case prediction within that development design before the surrogate is used to select controls.

Admission of the connected surrogate requires finite and independently audited out-of-fold projections for all accepted development states. The complete raw-response normalized root mean square error and the raw clarifier-inventory normalized error must each be below one. Because errors use development response scales, the threshold requires aggregate error below one scale unit. This threshold defines the case-study admission criterion.

Every projection is assessed for equality feasibility, physical inequalities, non-negativity, dual feasibility, stationarity, and complementarity. The overflow-closure residual is also reported separately. A failed admission criterion remains a failure rather than triggering an unreported refit or changed penalty selection rule.

Table 6.11: Projection solve and independent audit settings.

Setting	Value
Solver absolute and relative tolerances	10^{-12} each
Initial iteration limit	100,000
Polishing	enabled
Retry values of algorithm parameter ρ_{alg}	0.01 and 10
Retry iteration limit	200,000 each
Adaptive ρ_{alg} during retries	disabled
Scaled equality and inequality tolerances	10^{-8} each
Dual, stationarity, and complementarity tolerances	10^{-8} each
Physical non-negativity tolerance	10^{-10}
Active-set identification tolerance	10^{-7}

Prediction metrics and their denominators

The error calculations can be expanded with two hypothetical observations of one concentration coordinate. Let the reference values be 2 and 6 and the predictions be 3 and 4 in the same native unit. Subtracting reference from prediction gives residuals $3 - 2 = 1$ and $4 - 6 = -2$. Their different summaries are

$$\begin{aligned}
 \text{mean squared error} &= \frac{1^2 + (-2)^2}{2} = 2.5, \\
 \text{root mean square error} &= \sqrt{2.5} \approx 1.5811, \\
 \text{mean absolute error} &= \frac{|1| + |-2|}{2} = 1.5, \\
 \text{bias} &= \frac{1 + (-2)}{2} = -0.5.
 \end{aligned}$$

The mean squared error has squared concentration units. The other three quantities have concentration units. Bias retains the signs and can be small when positive and negative errors cancel.

For the same illustrative values, the reference mean is $(2 + 6)/2 = 4$. The squared error of the two predictions is $1^2 + (-2)^2 = 5$. Always predicting the reference mean instead would give squared error $(2 - 4)^2 + (6 - 4)^2 = 8$. Their ratio is $5/8$, so the coefficient of determination is $1 - 5/8 = 0.375$. The denominator describes variation of the assessment reference. It is not a training scale or a concentration limit. The general formulas record these same operations over all assessed observations.

For a response coordinate f and assessment state i , define the signed error $a_{if} = p_{if} - t_{if}$, where p is predicted and t is mechanistic. Physical-unit root mean square error, mean absolute error, and bias are

$$\left(\text{mean}_i a_{if}^2 \right)^{1/2}, \quad \text{mean}_i |a_{if}|, \quad \text{mean}_i a_{if}. \quad (6.186)$$

Normalized joint-response metrics replace a_{if} by $a_{if}/(D_\chi)_{ff}$ and state whether they pool errors across coordinates. Coordinatewise coefficients of determination use

$$R_f^2 = 1 - \frac{\sum_i (p_{if} - t_{if})^2}{\sum_i (t_{if} - \bar{t}_f)^2}. \quad (6.187)$$

A constant assessment target has an undefined denominator and is recorded accordingly. An arithmetic mean of coordinate scores is distinct from a score calculated on pooled observations.

Normalization changes the numerical comparison by assigning a denominator to each coordinate. In the preceding hypothetical error example, suppose the training response scale is two native units. Dividing residuals 1 and -2 by two gives 0.5 and -1 . Their standardized root mean square error is

$$\sqrt{\frac{0.5^2 + (-1)^2}{2}} = \sqrt{0.625} \approx 0.7906.$$

If a descriptive holdout report instead uses the observed range $6 - 2 = 4$, its normalized residuals are 0.25 and -0.5 . Their root mean square summary is approximately 0.3953. The predictions have not changed. The denominator has changed. This explains why a training-scale admission score and a holdout-range reporting score must be identified separately.

Location-quality summaries cover the mixer, five reactors, overflow, and underflow. Joint constraints link their errors, so the assessment retains each location alongside the plant aggregate. These reporting quantities connect prediction assessment to the treatment and resource measures used in operating decisions (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019). Outlet flows are divided by their own positive flow ratios before applying the quality map. For location ℓ and composite j , define the holdout reporting range and normalized error as

$$r_{\ell j} = \max_i t_{i\ell j} - \min_i t_{i\ell j}, \quad e_{i\ell j} = \frac{p_{i\ell j} - t_{i\ell j}}{r_{\ell j}}. \quad (6.188)$$

For positive ranges, an aggregate score pools the 8×4 location-composite coordinates. A location score pools four composites and a composite score pools eight locations. The displayed R^2 summaries average the corresponding coordinatewise scores. The holdout ranges serve only reporting. They do not enter fitting, admission, objective normalization, or trust calibration.

The overflow model receives concentration-scale and logarithmic assessment separately. The concentration report includes root mean square error, mean absolute error, bias, normalized error, and R^2 . The logarithmic report includes root mean square error and bias. Strict positivity of Equation (6.79) does not replace these assessments.

Exploratory simulation informs the specified response, closure, and acceptance rules. Development data supply input transformations, response scales, selected penalties,

overflow fitting, and trust calibration. The separate holdout stream evaluates the resulting fitted procedure. The operating evaluation uses independently solved connected responses at the selected controls. Fitted coefficients, preprocessing, priorities, trust thresholds, engineering safeguards, and numerical settings remain fixed throughout these assessments (Kaufman et al., 2012).

6.3.3 Operating searches and convergence evidence

A common baseline and an explicit basin-sensitivity experiment

A normalized control measures a fraction of its permitted interval. Consider an illustrative interval from 10 to 30 physical units. A control value of 15 lies five units above its lower bound in an interval of width twenty, so its normalized value is $5/20 = 0.25$. The midpoint 20 maps to 0.5, and the upper bound 30 maps to one. Conversely, a normalized value of 0.75 maps back to $10 + 0.75(30 - 10) = 25$. The general forward and inverse relations therefore use the same interval width,

$$\vartheta_j = \vartheta_j^L + (\vartheta_j^U - \vartheta_j^L)\eta_j.$$

This transformation allows operating variables with different units to use comparable numerical step sizes. It does not assign them equal importance in the engineering objective.

The baseline search for each route starts at the center of the normalized operating box. For each coordinate,

$$\eta_j = \frac{\vartheta_j - \vartheta_j^L}{\vartheta_j^U - \vartheta_j^L} \in [0, 1]. \quad (6.189)$$

The nominal influent is the midpoint of the fresh-influent intervals. The baseline explores one initial basin. The baseline records a local search from this common starting point.

For seven controls, the normalized center has seven entries equal to 0.5. Moving only the first coordinate upward by one quarter of its interval makes that coordinate 0.75. Moving it downward makes that coordinate 0.25. The six other entries remain 0.5 in both cases. Writing out the entries gives

$$\begin{aligned} \eta_{\text{first},+} &= (0.75, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5)^\top, \\ \eta_{\text{first},-} &= (0.25, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5)^\top, \\ e_1 &= (1, 0, 0, 0, 0, 0, 0)^\top. \end{aligned}$$

The unit coordinate vector e_1 records which coordinate moves. Two signs for each of seven coordinates give fourteen displaced starts. Adding the center gives fifteen. The compact expression below writes this same collection without listing every vector.

A planned multistart sensitivity experiment evaluates basin dependence separately from this baseline. It uses the same 15 starting control vectors for both routes,

$$\eta^{(0)} = \tfrac{1}{2}\mathbf{1}_7, \quad \eta^{(j,\pm)} = \tfrac{1}{2}\mathbf{1}_7 \pm \tfrac{1}{4}e_j, \quad j = 1, \dots, 7. \quad (6.190)$$

Each start receives the same route-specific budget and audit rules. A direct start obtains its mechanistic initial state from an independently accepted solve at those fixed controls. Failed initial states remain failures. The best independently validated feasible candidate is retained, with all starting-point outcomes and total search work reported. The reported baseline uses the single center start. The fifteen-start design defines a separate sensitivity analysis.

The distinction matters when interpreting a faster search. An algorithm that visits fewer basins can finish quickly while missing a better candidate. The expensive-function benchmarking study of Blik et al. (2023) motivates keeping objective quality, evaluation work, and computational time as separate comparison dimensions.

Surrogate search and derivative fallback

The initial surrogate outer method is sequential least squares programming, based on the constrained optimization approach of Kraft (1988). Each trial obtains a fully audited joint projection. The active-set derivative chain is used only after the rank, conditioning, multiplier, and independent derivative checks pass. An active-set change requires reconstruction of the projection and its derivatives.

When that chain cannot be validated, a deterministic value-only quadratic-approximation method continues within the normalized bounds (Ragoneau, 2022). It uses the same audited projection and does not substitute an approximate lower solve. Finite differences are used for independent derivative audits rather than as an unqualified derivative replacement. The best feasible visited point is solved again from an independent projection initialization before retention.

First-order evidence and finite-resolution polling

A candidate receives first-order stationarity evidence only when the derivative chain, lower projection certificate, and upper constrained-stationarity audit pass. This qualification concerns stationarity of the declared surrogate optimization problem at that candidate.

Coordinate and coupled directions can have different feasibility near an active constraint. Consider a hypothetical two-control problem with $x + y = 1$, non-negative controls, and objective $J = x$. At $(0.5, 0.5)$, any nonzero move in x alone or y alone violates the equality. A diagonal move that lowers x and raises y by the same amount preserves it. For a unit direction $(-1, 1)^\top / \sqrt{2}$ and illustrative step radius 0.1, the trial is

$$\begin{aligned} x_{trial} &= 0.5 - 0.1/\sqrt{2} \approx 0.4293, \\ y_{trial} &= 0.5 + 0.1/\sqrt{2} \approx 0.5707, \\ x_{trial} + y_{trial} &= 1. \end{aligned}$$

Table 6.12: Surrogate search, sensitivity, and finite-poll settings.

Setting	Value
Initial outer iteration limit	250
Initial outer function tolerance	10^{-10}
Active slack tolerance	10^{-7}
Required active inequality multiplier	greater than 10^{-8}
Domain-aware active-set perturbation	10^{-6}
Stationary-system condition number times machine precision	at most 10^{-8}
Sensitivity residual and scaled state reproduction	10^{-8} each
Independent derivative steps	10^{-6} and 5×10^{-7}
Derivative relative-error tolerance	10^{-5}
Upper feasibility and stationarity tolerance	10^{-6} each
Value-only fallback iteration and evaluation limits	250 each
Initial and final fallback trust radii	0.25 and 10^{-6}
Fallback feasibility tolerance	10^{-6}
Coarse and fine poll radii	10^{-3} and 10^{-4}
Coarse and fine direction counts	14 and 106
Absolute and relative sufficient-decrease tolerances	10^{-8} each
Direction-rank tolerance	10^{-10}
Certification evaluation limit	10,000
Ray growth factor and maximum probes	2 and 16
Objective relative tie tolerance	10^{-10}

The objective decreases by approximately 0.0707. An axis-only poll would contain no feasible nonzero trial in this example, even though a feasible improving direction exists.

In seven coordinates, each of the seven axes has a positive and a negative direction, giving fourteen. There are $7(6)/2 = 21$ unordered coordinate pairs. Each pair admits four sign combinations, giving $4(21) = 84$ pairwise directions. Eight regular-simplex directions supplement them. A regular simplex is the higher-dimensional counterpart of an equilateral triangle. Its directions distribute around a center rather than following only the coordinate axes. The fine direction count is $14 + 84 + 8 = 106$. These finite collections test coordinate and coupled moves.

If the derivative evidence is unavailable, the convergence procedure performs complete feasible-direction polls at normalized radii 10^{-3} and 10^{-4} . The coarse poll contains the 14 signed coordinate directions. The fine poll adds 84 signed pairwise diagonals and eight regular-simplex directions from a Helmert construction. The pairwise count is $4\binom{7}{2}$. Coupled directions can reveal feasible improvements that an axis-only test misses near a constraint.

Each complete poll must contain a feasible nonzero displacement. A rank calculation is diagnostic because active constraints can reduce the dimension of the feasible cone. An accepted improvement triggers a new complete poll. A fine-radius move also triggers revalidation at the coarse radius. After an improving step, up to 16 search-only probes grow along the same ray by successive factors of two. These probes may accelerate search but supply no convergence certificate.

Absence of sufficient decrease in both complete revalidated polls gives evidence only for the tested directions and radii. An incomplete poll, a failed calculation, or an exhausted certification budget leaves stationarity unresolved. The direction design follows the role of feasible polling discussed in the direct-search analysis of Audet and Dennis (2006).

Direct continuation, recovery, and endpoint checks

The direct route uses the interior-point filter algorithm of Wächter and Biegler (2006) through the three smoothing stages. Its endpoint audit independently checks all smooth reactor and layer equations, inequalities, non-negativity, multipliers, and stationarity. The full-state audit is separate from the final solver status.

If the initial direct search fails, one recovery may start from the same influent's certified surrogate controls and repeat the same smoothing sequence. Recovery is not attempted after an initially successful search or from an uncertified surrogate candidate. Because this recovery uses a surrogate decision, it is a dependent rescue calculation rather than independent basin exploration. Its status and computational cost are reported separately.

A validated feasible candidate can be retained when its search budget is exhausted or stationarity remains unresolved. Its convergence qualification remains visible. A

Table 6.13: Direct continuation and exact-evaluation qualification settings.

Setting	Value
Interior-point iteration limit per smoothing stage	2,500
Convergence and constraint tolerances	10^{-8} each
Dual and complementarity tolerances	10^{-6} each
Endpoint equality acceptance	10^{-8}
Endpoint inequality and stationarity acceptance	10^{-6} each
Endpoint active-set tolerance	10^{-7}
Receiver-switch ambiguity distance from X_t	1 g TSS m^{-3}
Other branch ambiguity distances	$10^{-6}S$
Two-start agreement under both prescribed state scales	10^{-6}
Engineering feasibility tolerance after exact evaluation	10^{-6}

case with no validated feasible candidate remains a search failure. A favorable success flag alone is insufficient for either route.

6.3.4 Independent mechanistic verification and fair comparison

The two searches use different response calculations but must be assessed on the same original-model basis. Figure 6.4 separates the search, native-candidate audit, independent physical evaluation, and comparison eligibility. Each stage retains its own outcome.

The common evaluation stage supplies treatment responses at fixed controls rather than returning information to the fitted models. A valid physical response can still violate a retained engineering safeguard and remain useful as a diagnostic. Separating these outcomes prevents numerical convergence, physical validity, and decision eligibility from being treated as the same property.

Evaluation at fixed selected controls

Every available selected decision is evaluated with its controls and influent held fixed. Both initial states are solved on the original nonsmoothed reactor and ten-layer clarifier equations. The independently solved state supplies the reference treatment outcome for either route. The role of an original-model check follows the feasibility and validation requirements discussed by Bhosekar and Ierapetritou (2018).

Exact evaluation uses the physical checks in Table 6.10. Two-start state agreement is tested under both generation scales and fixed state scales. The fixed scales are development-set coordinatewise median absolute deviations with a floor of one. A valid evaluation must pass mass conservation, non-negativity, layer residuals, the domain,

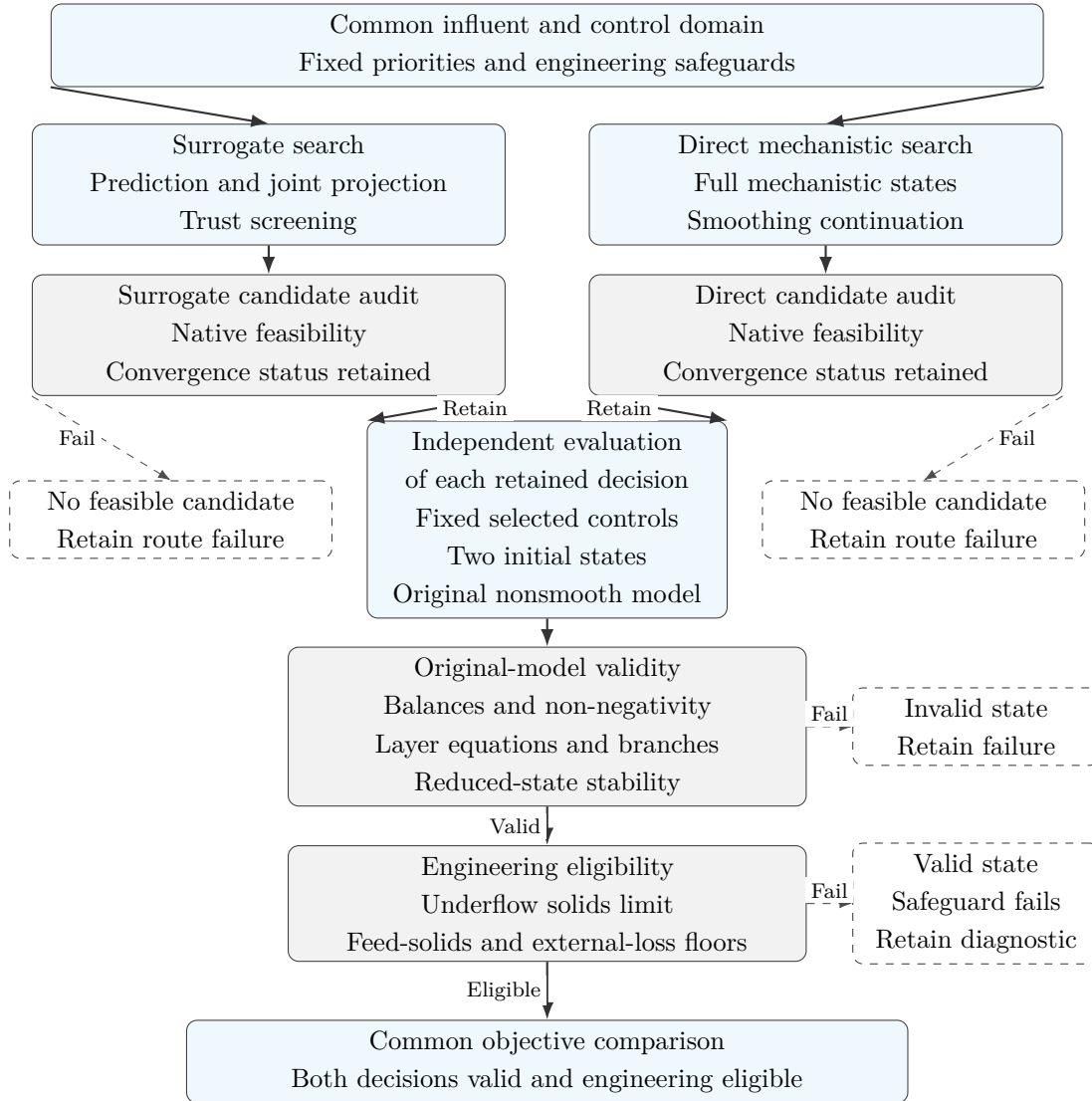


Figure 6.4: Conceptual paired-search and verification workflow. Each retained control vector receives its own independent original-model evaluation. The surrogate and smooth-direct native responses do not supply the reference treatment outcome. Original-model validity and engineering eligibility are distinct checks. Only eligible pairs enter the common-objective comparison, while failed or qualified cases remain in the accounting.

the endpoint envelope, and the reduced-state stability test.

Branch tuples must agree unless an evaluation start is boundary-ambiguous. Ambiguity bands cover the receiver threshold, zero storage capacity, zero settling offset, zero or maximum settling velocity, and equal adjacent gravitational fluxes. An ambiguity flag qualifies the state-level branch comparison. It does not prove that every mismatching branch falls within its own ambiguity band. Agreement of the physical states and all other audits is still required. Smooth stationarity or smooth-reference equivalence near such a boundary requires that qualification.

This verification is especially relevant near a prescribed nutrient limit. The aeration-scheduling framework of Ren et al. (2026) explicitly considers influent disturbances and effluent ammonium safeguards. This approach supports evaluating the particular effluent quantity and treatment requirement that governs an operating decision.

Native and common-reference objectives

Let route S denote the connected surrogate and route M the direct smooth mechanistic search. Their native responses are $\chi_{\text{nat},S} = \hat{\chi}$ and $\chi_{\text{nat},M} = \chi_{\text{sm},M}$. The independently solved responses are $\chi_{\text{ref},S}$ and $\chi_{\text{ref},M}$. Raw and projected surrogate predictions are also evaluated, where defined, at both retained control vectors.

Route-specific normalization can change an objective even when the physical response is identical. Consider an illustrative two-quality case with positive development standard deviations 0.5 and 2 in the corresponding native units. A scale with a floor of one gives denominators 1 and 2. The unfloored scale retains 0.5 and 2. For a hypothetical response of 0.8 and 4, their standardized coordinates are

$$\begin{aligned}\text{floored scale} &= (0.8/1, 4/2) = (0.8, 2), \\ \text{unfloored scale} &= (0.8/0.5, 4/2) = (1.6, 2).\end{aligned}$$

With equal quality weights, the means are 1.4 and 1.8. This difference arises entirely from the normalization. Writing the diagonal matrices makes the componentwise denominators explicit,

$$D_{T,S}^{\text{toy}} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad D_{T,M}^{\text{toy}} = \begin{bmatrix} 0.5 & 0 \\ 0 & 2 \end{bmatrix}.$$

These matrices illustrate the two general definitions. The comparison of selected decisions uses one common reference normalization so that such a numerical choice is not mistaken for better treatment.

For positive development-set effluent standard deviations σ_j , native quality normalization is

$$D_{T,S} = \text{diag}(\max(1, \sigma_j)), \quad D_{T,M} = \text{diag}(\sigma_j). \quad (6.191)$$

The unit floor is in the physical unit of each quality coordinate. The common reference objective uses $D_{T,M}$ for both decisions. Actual fitted standard deviations are estimation quantities and are not specified as theoretical constants.

A hypothetical comparison first assigns native objective values of 0.90 and 0.95 to two operating choices. Prediction favors the first choice by $0.95 - 0.90 = 0.05$. Independent mechanistic evaluation gives 1.10 and 0.98 at the same choices, reversing their ordering. Two distinct subtractions describe this result. For the first choice, the native-minus-reference difference is $0.90 - 1.10 = -0.20$. Its negative sign means that the native calculation presents a smaller objective than the independent evaluation.

For an illustrative direct-route decision with native objective 0.95 and reference objective 0.98, its native-minus-reference difference is -0.03 . Comparing the independently evaluated decisions instead gives $1.10 - 0.98 = 0.12$. On the shared objective basis, this positive difference means the surrogate-selected decision has the larger objective among these two choices. Both projected responses can satisfy the imposed balances while the independently evaluated objectives have this ordering. The arithmetic compares these two retained choices on a common reference basis (Alexandrov et al., 1998; Pedrozo et al., 2025). The compact expressions distinguish within-route prediction differences from between-route decision differences.

The native-to-reference and route-to-route differences are

$$\Delta J_{\text{nat-ref},k} = J_{\text{nat},k}(\vartheta_k, \chi_{\text{nat},k}) - J_{\text{ref}}(\vartheta_k, \chi_{\text{ref},k}), \quad k \in \{S, M\}, \quad (6.192)$$

$$\Delta J_{S-M} = J_{\text{ref}}(\vartheta_S, \chi_{\text{ref},S}) - J_{\text{ref}}(\vartheta_M, \chi_{\text{ref},M}). \quad (6.193)$$

The first can reflect both response approximation and route normalization. The second compares independently evaluated decisions using a common objective basis. The route-to-route difference is reported as a paired objective comparison.

An objective comparison and a control comparison also have different denominators. A hypothetical objective difference of 0.12 with reference direct objective 0.98 has direct-relative percentage $100(0.12)/0.98 \approx 12.24\%$. The symmetric scaled difference uses the denominator $\max(1, 1.10, 0.98) = 1.10$ and gives $0.12/1.10 \approx 0.1091$. The first uses one route as its reference. The second uses a symmetric magnitude scale with a floor.

For an illustrative control interval from 10 to 30, two selected settings of 25 and 20 differ by five physical units. Their normalized difference is $5/(30 - 10) = 0.25$. If this is the only differing coordinate among seven, the root mean square difference is $\sqrt{0.25^2/7} \approx 0.0945$, while the maximum absolute difference is 0.25. The average spreads one change over all seven coordinates. The maximum keeps the largest change visible.

Standardized response errors at each decision retain the raw, projected, and native quantities separately. A symmetric relative objective difference divides ΔJ_{S-M} by $\max(1, |J_S|, |J_M|)$. A direct-relative percentage divides $100\Delta J_{S-M}$ by J_M only when that denominator is nonzero. Normalized control differences are

$$d_j = \frac{\vartheta_{S,j} - \vartheta_{M,j}}{\vartheta_j^U - \vartheta_j^L}, \quad d_{\text{rms}} = \left(\frac{1}{7} \sum_{j=1}^7 d_j^2 \right)^{1/2}, \quad d_{\text{max}} = \max_j |d_j|. \quad (6.194)$$

These summaries distinguish similar objective values from similar controls.

Removal follows from the material entering and leaving on the stated concentration basis. For a hypothetical fresh-influent concentration of 200 mg L^{-1} and an effluent concentration of 2 mg L^{-1} , the concentration reduction is $200 - 2 = 198 \text{ mg L}^{-1}$. Dividing by 200 gives a fraction of 0.99, and multiplying by one hundred gives 99%. Algebraically,

$$100 \frac{z_{\text{in},j} - z_{E,j}}{z_{\text{in},j}} = 100 \left(1 - \frac{z_{E,j}}{z_{\text{in},j}} \right).$$

This step requires a positive influent denominator. It compares concentrations on one reporting basis. A component inventory balance additionally uses the appropriate flow rates and represented outlets.

Selected-decision removal for composite j is

$$\mathcal{R}_j = 100 \left(1 - \frac{z_{E,j}}{z_{\text{in},j}} \right), \quad (6.195)$$

when the fresh-influent denominator is positive. Predicted-to-reference removal differences use percentage points. Concentration percentage errors instead divide by the mechanistic effluent concentration. For a hypothetical influent of 200 mg L^{-1} and effluent values of 2 and 4 mg L^{-1} , the effluent prediction error is 100% relative to 2. The removal difference is only one percentage point. Reporting both makes their different denominators clear.

Validity, engineering eligibility, and failures

A paired objective comparison requires both routes to supply natively feasible candidates, valid exact evaluations, and satisfaction of the common retained engineering safeguards after evaluation. A valid physical evaluation that violates an underflow, feed-solids, or external-loss safeguard still retains its diagnostic objective and treatment response. It does not enter the eligible paired comparison. Solids retention time and loading measures are reported for every finite evaluation and do not add eligibility thresholds.

These status fields are not collapsed into one success label. Missing or nonfinite quantities, an undefined feature scale, a failed factorization, or an inconsistent numerical audit remain computational failures. A finite state that fails a scientific threshold remains a diagnostic state with the failed criterion identified. Generation rejection, optimization failure, projection infeasibility, and branch qualification have different causes and denominators.

Influent scenarios and time accounting

Ten influent vectors from a separate twenty-dimensional Latin hypercube define the scenario experiment. They use the seed in Table 6.9 and do not enter fitting,

Table 6.14: Distinct questions retained in failure and qualification accounting.

Assessment	Question answered
Generation acceptance	Did both original-model starts pass the required physical and numerical checks?
Projection feasibility	Did the fitted closure and physical constraints admit an audited joint response?
Native engineering feasibility	Did the search candidate satisfy its declared optimization constraints?
Convergence qualification	Did derivative audits or complete finite polls supply their stated evidence?
Exact-model validity	Did independent original-model evaluation converge to an accepted state?
Retained engineering feasibility	Did the exact response pass the common safeguards?
Paired eligibility	Were both decisions available and admissible on the common reference basis?

calibration, or tuning. Both routes are attempted for the nominal influent and every scenario. Available selected decisions receive exact evaluation even when the other route fails. These cases examine sensitivity to the specified influent vectors.

Optimization time is the duration of the route’s initial search. For the surrogate, it includes regression, projection, and any value-only fallback inside that search. For the direct route, it includes the initial three-stage smoothing sequence. Development sampling, fitting, convergence certification, conditional recovery, and exact-model evaluation are outside this interval. Failed initial attempts remain in the time accounting. Scenario aggregates use the ten influent cases and keep the nominal case separate. Both routes use the same processor allocation and thread count.

Search time is accompanied by evaluated points, iterations, projection retries, fallback use, completed smoothing stages, recovery status, and convergence qualification. The benchmark concerns identified by Blik et al. (2023) make these workload quantities useful when comparing expensive-function methods. Objective values and search qualification accompany the timing comparison.

A complete workflow account additionally includes development, certification, recovery, and verification. These tasks have separate cost terms. Repeated use can amortize development effort, but the relevant number of later decisions depends on the full per-decision cost and on acceptable mechanistic decision quality. The complete-cost calculation uses all of these terms.

The effect of repeated use can be shown with hypothetical total costs. Suppose development requires 100 seconds, and each later surrogate-assisted decision requires three seconds in total for search, qualification, and verification. Suppose a directly

evaluated decision requires six seconds for the corresponding complete task. For twenty decisions, the costs are $100 + 20(3) = 160$ seconds and $20(6) = 120$ seconds. For fifty decisions, they are 250 and 300 seconds. These hypothetical numbers illustrate amortization.

More generally, let n_{dec} be the number of later decisions, T_{dev} the development cost, and T_S and T_M the complete per-decision costs. Expanding the comparison gives

$$\begin{aligned} T_{dev} + n_{dec}T_S &< n_{dec}T_M, \\ T_{dev} &< n_{dec}(T_M - T_S), \\ n_{dec} &> \frac{T_{dev}}{T_M - T_S} \quad \text{when } T_M > T_S. \end{aligned}$$

The illustrative threshold is $100/(6 - 3) = 33\frac{1}{3}$, so at least thirty-four complete decisions are needed to cross it. If the complete per-decision cost is not smaller, a positive development cost cannot be recovered by this timing argument. The calculation also assumes that the compared decisions meet the required quality and feasibility conditions. Quality and feasibility are assessed through the common mechanistic evaluation.

The resulting evaluation keeps four types of evidence separate. Statistical assessment concerns approximation of accepted mechanistic states. Projection audits concern the imposed network and inventory constraints. Search audits concern local numerical evidence for the declared optimization problem. Original-model verification concerns treatment and feasibility at the selected controls. Together these assessments evaluate the connected surrogate as an operating decision aid.

6.4 Results

Treatment outcomes for Extended ICSOR and Smooth NLP are compared through independent evaluation with the original mechanistic equations. The nominal influent is denoted N, and the ten influent scenarios are denoted S1–S10. Optimization time is assessed separately from prediction and decision quality.

6.4.1 Accepted development and holdout states

The development design retained 7386 of 8000 attempted states. The holdout design retained 1845 of 2000 attempted states. Their acceptance fractions were 92.33% and 92.25%. Table 6.15 gives the corresponding rejection counts. Rejected states were not replaced. The accepted states supplied the fitting and holdout assessments.

The raw-response selection retained a ridge penalty of 0.1. Its mean grouped cross-validation normalized root error was 0.659. This score used the fitting response scales across the complete joint response. The holdout composite scores use the location-specific reporting ranges.

Table 6.15: Realized mechanistic samples for the connected-plant assessment. The separate development and holdout designs retained accepted states without replacing rejected candidates.

Design	Attempted	Accepted	Rejected	Accepted (%)
Development	8000	7386	614	92.33
Holdout	2000	1845	155	92.25

6.4.2 Aggregate and local holdout accuracy

Projection reduced the holdout nRMSE from 0.04690 to 0.04306. The relative reduction calculated before rounding was 8.19%. The nMAE decreased from 0.03044 to 0.02568, a reduction of 15.65%. Mean location-level R^2 increased from 0.925 to 0.938. Figure 6.5 separates the aggregate, location, and composite summaries. All eight location summaries and all four composite summaries had lower nRMSE after projection.

The composite nRMSE values decreased from 0.04741 to 0.04586 for COD, from 0.03724 to 0.03454 for TN, from 0.04170 to 0.03349 for TP, and from 0.05852 to 0.05476 for TSS. TP showed the largest relative improvement among these composite summaries. TSS retained the largest projected error. The location-specific holdout ranges define the weighting of these composite scores.

Projection reduced nRMSE in 30 of the 32 location-composite combinations. Figure 6.6 identifies the two exceptions. COD error increased by 0.47% in the final reactor and by 2.36% in the overflow. Overflow TP error decreased from 0.02948 to 0.01795, a reduction of 39.10%. The location-specific assessment identifies both the reductions and increases in prediction error.

6.4.3 Parity across the connected treatment system

Figure 6.7 pools the 14,760 location-observations for each composite while retaining location medians. Much of the density lay near the one-to-one line. The projected mean location-level R^2 values were 0.944 for COD, 0.963 for TN, 0.945 for TP, and 0.899 for TSS. These values average the location-specific scores.

The overflow remained an important source of error. Its projected location-average R^2 was 0.876, below the reactor averages of approximately 0.947. The projected overflow TSS nRMSE was 0.07392. TSS was underpredicted in 1008 of the 1845 holdout states, or 54.6%. The remaining states had exact or overpredicted TSS values.

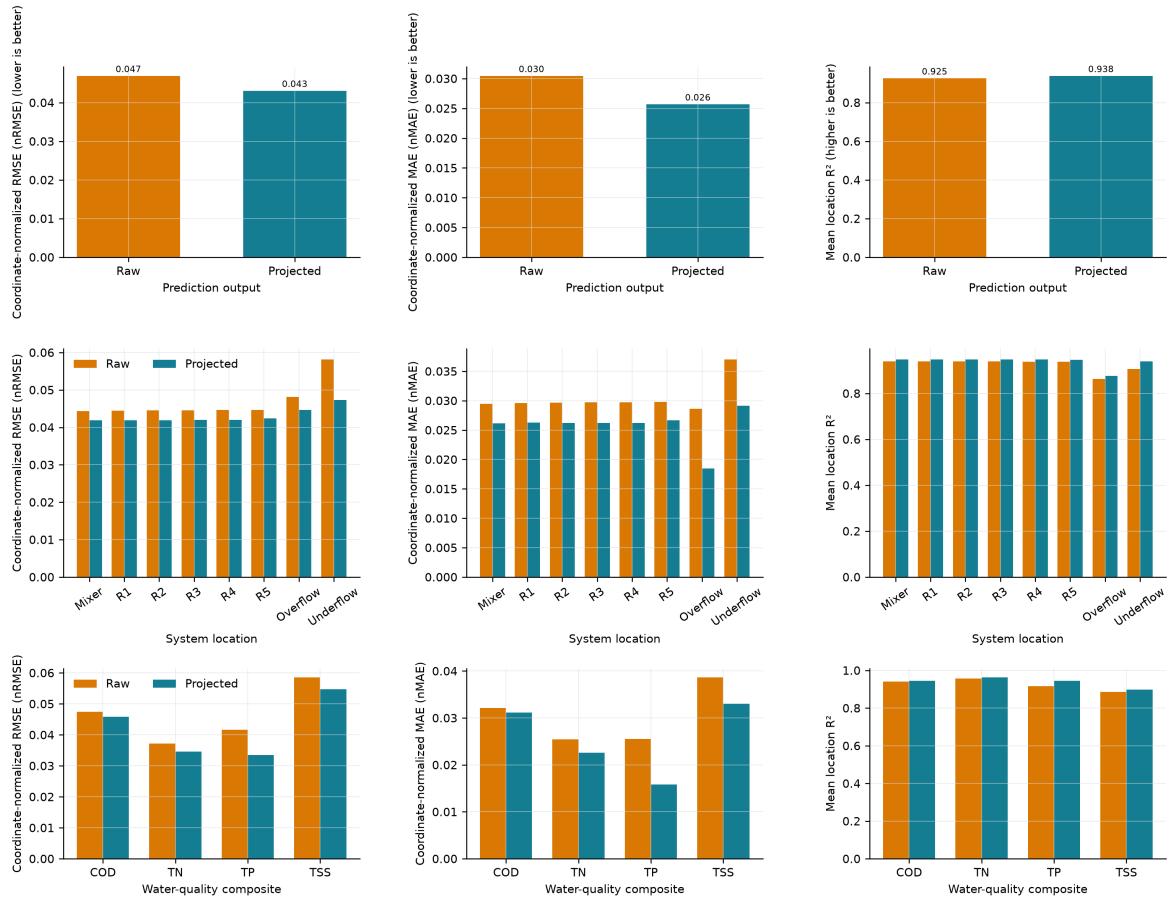


Figure 6.5: Raw and projected composite accuracy on 1845 holdout plant states. Rows show aggregate, location, and composite summaries. Columns show nRMSE, nMAE, and mean location-level R^2 . Errors are normalized by the mechanistic holdout range of each location and composite. The R^2 summaries average coordinate-specific scores rather than pooling locations into one score.

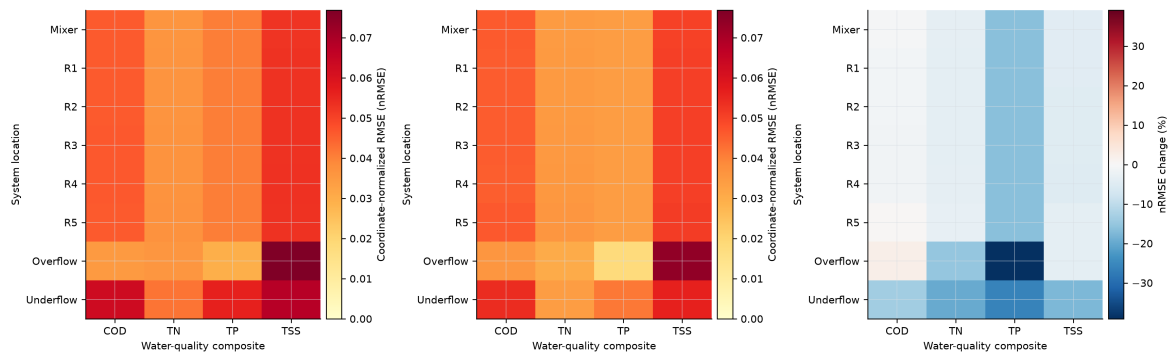


Figure 6.6: Holdout nRMSE by treatment location and composite. Raw and projected heatmaps share their color scale. The third panel shows percentage change relative to raw error. Negative values indicate improvement. The mixer, five reactors, overflow, and underflow are assessed separately.

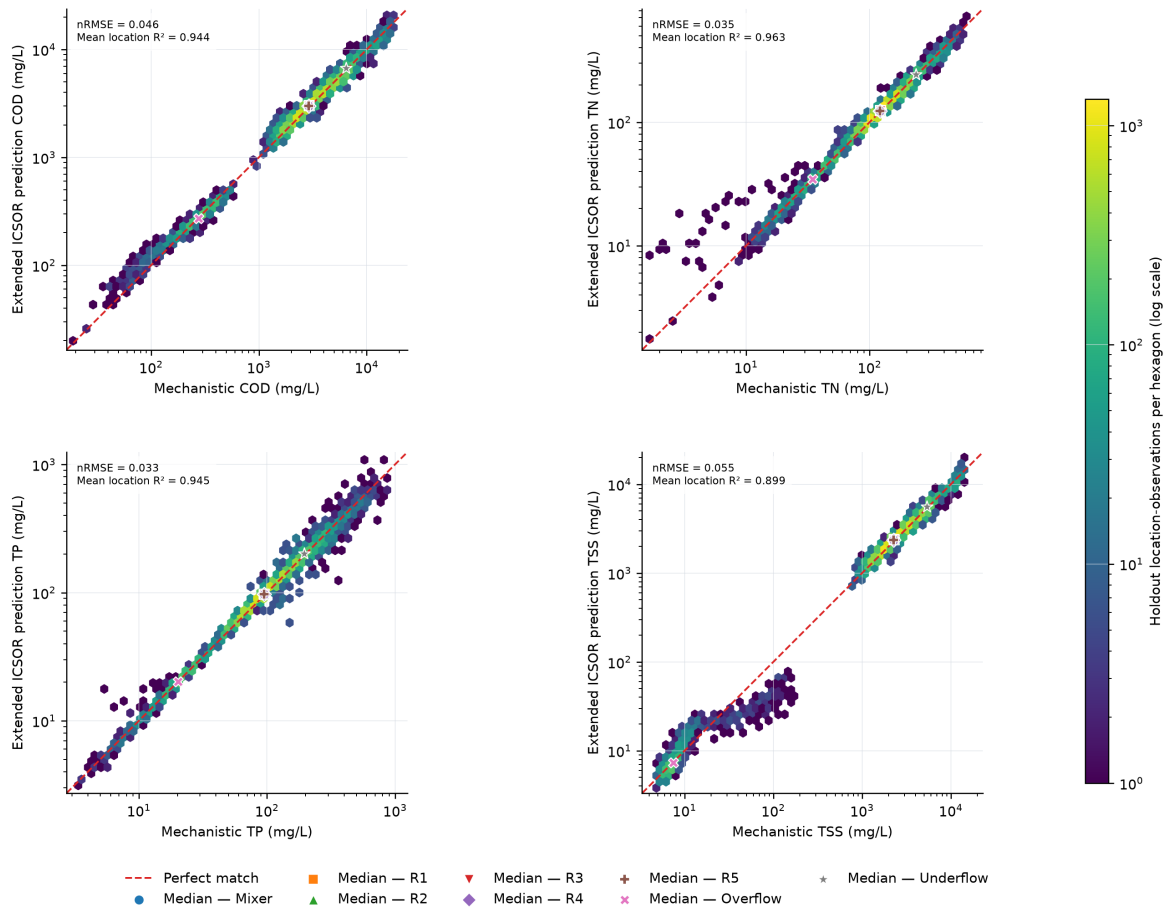


Figure 6.7: Projected composite parity across eight locations and 1845 holdout states. Both concentration axes are logarithmic. Hexagon color gives the number of location-observations, and labelled markers identify each location's median. The one-to-one line indicates exact agreement. Metric annotations retain location-specific normalization and average location-level R^2 .

6.4.4 Physical audits and search qualifications

Every raw holdout response had at least one mass conservation violation in the independent physical audit. Negative entries occurred in 1765 of the 1845 raw responses. The corresponding projected summaries recorded no mass conservation, non-negativity, or network-inequality violations under their stated audit definitions. These counts summarize the imposed numerical balance and inequality checks.

The surrogate route retained an initially feasible candidate for the nominal influent and all ten scenarios. The initial direct searches retained candidates for the nominal influent and nine scenarios. The S1 direct attempt did not retain a validated feasible candidate, and its permitted recovery did not resolve that search outcome. The two-route figures nevertheless include a separately evaluated valid direct decision for S1. This decision is included normally in the treatment and objective comparisons. It is not counted as a successful initial direct search.

Table 6.16 distinguishes these availability results from convergence evidence. Positive first-order qualification flags were recorded for the surrogate decisions at N, S6, and S10. Seven other surrogate decisions had finite-poll qualification. S4 remained unresolved. No retained initial direct candidate had a positive first-order certificate. The independent mechanistic responses supplied the reference treatment outcomes for these retained decisions.

Table 6.16: Qualification of the initial center-start searches. Each route was attempted in eleven cases. Finite-poll qualification uses the specified directions and radii. The separately evaluated S1 direct decision appears in the paired figures but does not alter these initial-search counts.

Outcome	Extended ICSOR	Smooth NLP
Initial searches	11	11
Retained feasible initial candidates	11	10
First-order qualification	3	0
Additional finite-poll qualification	7	0
Retained candidates with unresolved qualification	1	10
Cases without a retained initial candidate	0	1

6.4.5 Prediction fidelity at selected controls

Figure 6.8 compares Extended ICSOR predictions with independent mechanistic outcomes at its selected controls. Across the 44 case-composite comparisons, median absolute concentration error was 12.75% of the mechanistic effluent value. Mean absolute error on that percentage basis was 18.43%. Median and mean absolute removal

errors were 3.26 and 5.23 percentage points. These selected-decision summaries had different denominators and a different input distribution from the holdout scores.

In S6, predicted COD was 151.10 mg L^{-1} and the mechanistic value was 99.21 mg L^{-1} . In S10, the corresponding values were 232.23 and 167.78 mg L^{-1} . The predictions underestimated COD removal by 8.51 and 10.44 percentage points. These two COD predictions were conservative about treatment removal.

The nutrient errors in S4 had the opposite implication. Predicted TN was 2.67 mg L^{-1} against a mechanistic value of 7.47 mg L^{-1} . Predicted TP was 2.08 mg L^{-1} against 4.68 mg L^{-1} . TN and TP removals were overstated by 16.97 and 27.33 percentage points. These errors remained after projection and trust screening.

TSS was overpredicted in ten of the eleven selected surrogate decisions. Predicted values ranged from 5.53 to 8.05 mg L^{-1} , while mechanistic values ranged from 5.28 to 6.41 mg L^{-1} . In S1, values of 7.78 and 5.77 mg L^{-1} differed by only 0.61 percentage points in removal. A small removal error could therefore coexist with a substantial relative effluent-concentration error.

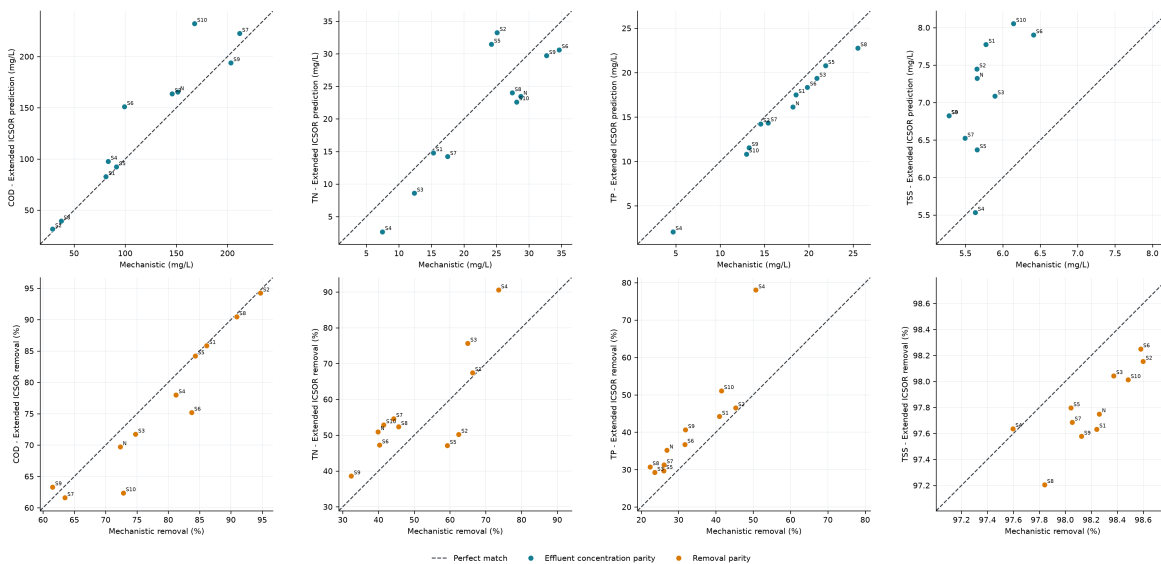


Figure 6.8: Extended ICSOR predictions at selected controls compared with independent original-model evaluation. The upper row shows effluent concentration parity in mg L^{-1} . The lower row shows removal parity in percent. Columns correspond to COD, TN, TP, and TSS. Teal identifies concentrations and orange identifies removals. Labels identify N and S1–S10. Dashed lines indicate exact agreement.

6.4.6 Treatment outcomes and operating choices

Figure 6.9 compares both routes on the mechanistic reference basis. Smooth NLP produced lower COD, TN, and TP in the nominal case and nine of the ten scenarios. Nominal COD was 151.26 mg L^{-1} at the surrogate decision and 57.60 mg L^{-1} at the

direct decision. In S9, values were 203.24 and 19.08 mg L⁻¹. Smooth NLP had lower TSS in six of the eleven comparisons. The TSS ordering therefore varied among cases.

Extended ICSOR selected the 6-h HRT lower bound at N, S6, S7, S9, and S10. Smooth NLP selected the 36-h upper bound in S5 and S10 within numerical tolerance. Since fresh flow was fixed, these HRT choices changed biological reactor volume. The comparison therefore combined reactor capacity selection with operating control.

Both routes selected the upper aeration bound in reactor 3 for every case within numerical tolerance. Aeration in reactors 4 and 5 also reached the upper bound in many cases. Both routes selected the return-sludge lower bound of 0.25. Smooth NLP selected internal recycle ratios of 4.00 in S2 and S10 and 3.44 in S9. Its other values were effectively zero. Extended ICSOR selected ratios between zero and 0.804.

The distinction between nominal HRT and residence time per passage was visible in S2. Smooth NLP selected $H = 31.40$ h with $r_I = 4.00$ and $r_R = 0.25$. The reactor-train flow ratio was 5.25, giving a residence time per passage of approximately 5.98 h. Recycle flow reduced the residence time for each passage through the train.

Wasting frequently reached its upper bound. Its resource contribution nevertheless depended on the selected underflow concentration through wX_U . Neither the wasting ratio nor an aeration setting alone described the corresponding material or resource measure.

6.4.7 Mechanistic treatment-train profiles

Figure 6.10 follows the original-model concentrations at each route's selected controls. All four composites increased from fresh influent to the mixer for both routes in all eleven cases. Recycle mixing brought concentrated material back into the biological train. The mixer increase reflected internal circulation.

Changes across the reactors differed from the concentration decrease across the clarifier. The former reflected the combined soluble and particulate state after biological conversion. The latter reflected particulate separation under the nonreactive clarifier assumption. A nearly constant composite concentration can accompany conversion among its component forms.

The large TSS decrease between reactor 5 and the final effluent showed the contribution of solids separation to treated-water quality. Species concentrations, outlet solids flows, and the reactor and clarifier inventories support interpretation of nitrification, denitrification, phosphorus uptake, and solids age.

6.4.8 Quality and resource trade-offs

Smooth NLP had the lower common-reference objective in the nominal case and all ten scenarios. Figure 6.11 separates the total objective from quality and weighted resource contributions. Nominal objectives were 0.8647 for Extended ICSOR and 0.7862

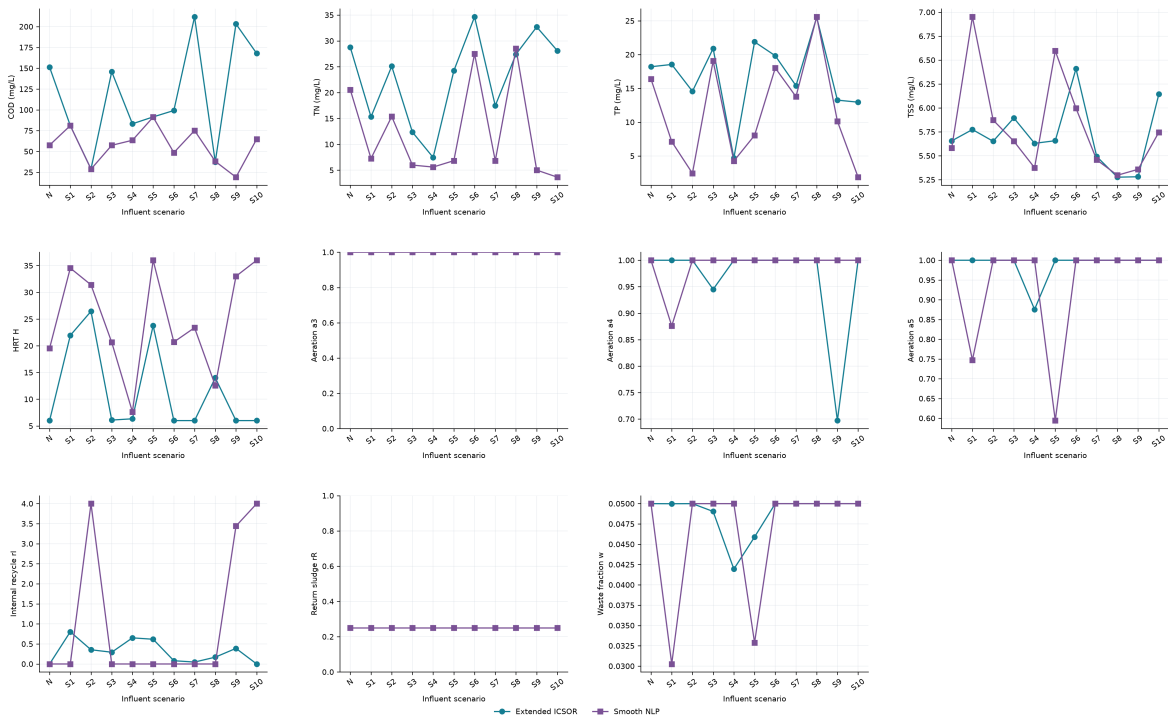


Figure 6.9: Mechanistic effluent outcomes and selected controls for both optimization routes. The top row shows COD, TN, TP, and TSS in mg L^{-1} . The remaining panels show HRT, aeration in reactors 3–5, internal recycle, return sludge, and wasting. The unused panel is omitted. N and S1–S10 appear in every comparison. The constant aeration and return-sludge panels use plain zero-to-one axes.

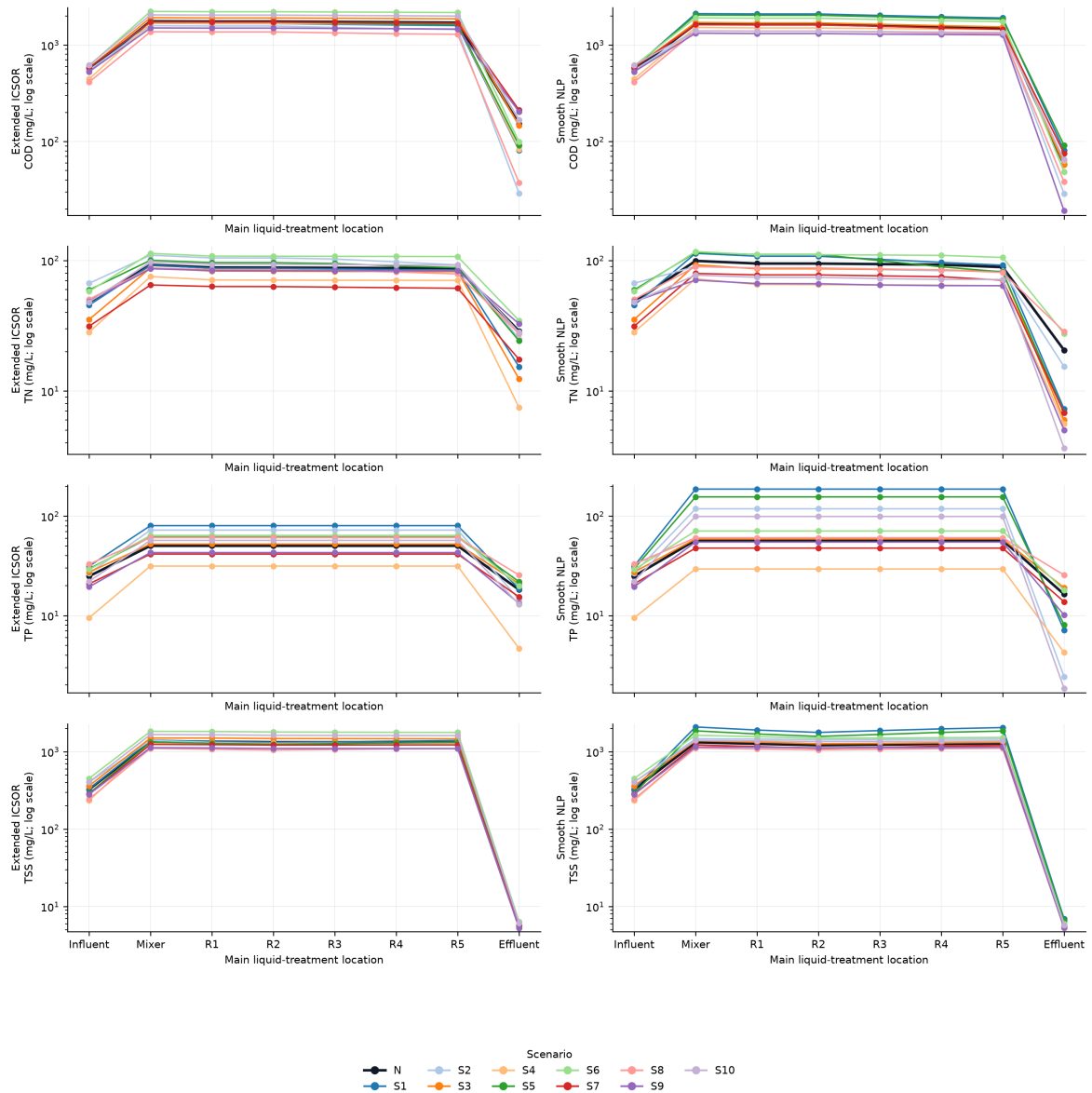


Figure 6.10: Mechanistic concentration profiles at controls selected by Extended IC-SOR and Smooth NLP. Rows show COD, TN, TP, and TSS from fresh influent through the mixer and reactors 1–5 to clarifier effluent. Each composite has identical logarithmic concentration limits for both routes. Colors identify N and S1–S10. Underflow is not placed in the main liquid-treatment path.

for Smooth NLP. The surrogate value was 9.99% higher relative to the direct value. S5 had the largest absolute difference, with values of 1.0101 and 0.7022. S8 had the smallest difference, with values of 0.9279 and 0.9234.

Smooth NLP had lower normalized quality cost in ten of the eleven cases. Its weighted resource contribution was higher in every case except S8. In S10, quality cost decreased from 1.4748 to 0.3699. Its weighted contribution decreased by approximately 0.5525. Weighted resource cost increased from 0.0557 to 0.4191. The resulting total objective decreased from 0.7931 to 0.6040. Better weighted treatment was obtained while accepting a larger capacity and resource contribution.

S8 illustrated the opposite trade-off. Quality cost increased from 1.5856 to 1.6110, but weighted resource cost decreased from 0.1351 to 0.1179. Smooth NLP therefore had a lower total objective even though every evaluated effluent composite was slightly higher. The resource reduction determined the lower total objective in this case.

The development quality scales were 106.61, 13.07, 7.11, and 17.38 mg L⁻¹ for COD, TN, TP, and TSS. All exceeded one. The two native normalization matrices in Equation (6.191) consequently coincided in this experiment. Both route objectives therefore used the same quality normalization.

The volume dependence of the aeration proxy was also important. When $H = 6$ h and all three aeration settings equal one, Equation (6.147) gives $j_A = 1/6$. Frequent low-HRT surrogate decisions thus received relatively small capacity and aeration contributions despite maximum aeration settings. The contributions measured normalized capacity and resource proxies.

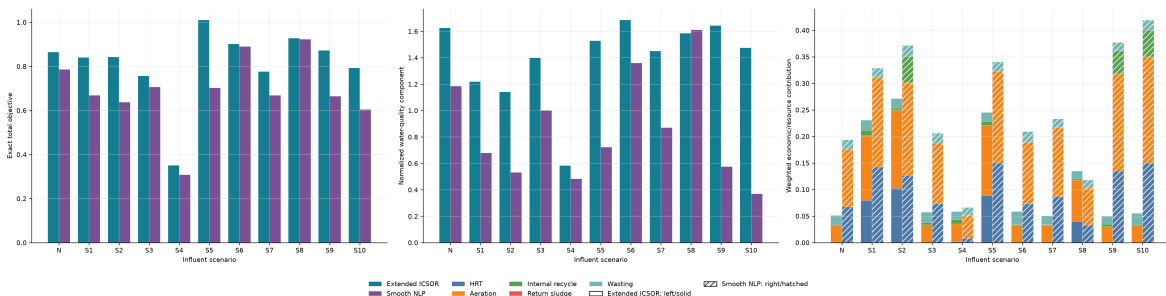


Figure 6.11: Common-reference total objective, normalized quality cost, and weighted resource contributions. Both routes are evaluated with the original mechanistic model and common quality scales. Resource stacks show HRT, aeration, internal recycle, return sludge, and wasting. Solid bars identify Extended ICSOR, and hatched bars identify Smooth NLP. The quality panel shows j_Q , whose contribution to the total objective is $0.50j_Q$.

6.4.9 Optimization time

Extended ICSOR had shorter recorded optimization time for N and every scenario. Figure 6.12 shows all eleven comparisons. Across S1–S10, times ranged from 0.758 to 2.702 s for Extended ICSOR and from 3.932 to 8.253 s for Smooth NLP. Means were 1.726 and 5.236 s. The ratio of these means was approximately 3.03. Nominal times were 0.985 and 5.062 s.

Scenario-specific direct-to-surrogate time ratios ranged from 1.90 to 7.99. The largest ratio occurred in S10. The comparison included initial failed attempts, including the S1 direct attempt. Candidate retention is recorded separately in Table 6.16.

The recorded times cover the initial optimization intervals defined in the methodology. Extended ICSOR had both a shorter search interval and a higher mechanistic objective in the paired comparisons.

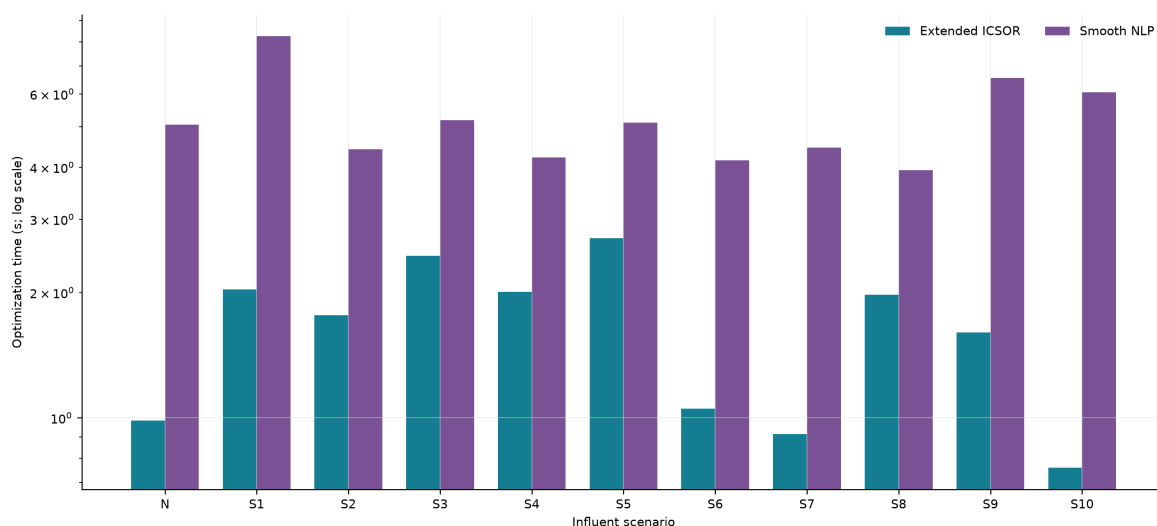


Figure 6.12: Recorded optimization time for N and S1–S10 on a logarithmic seconds axis. The interval includes the initial surrogate search with projection and fallback or the initial direct smoothing sequence. It excludes generation, fitting, certification, conditional recovery, and independent mechanistic evaluation. Initial failed attempts remain in the timing account. Scenario mean times use S1–S10 only.

6.5 Discussion

6.5.1 Connected enforcement and its statistical benefit

The joint response made the mixer, reactor train, clarifier outlets, and stored solids available in one prediction. Joint projection linked the material returned to the mixer with the material leaving the reactors and clarifier. It also reconciled reactor inventories with external effluent and waste-sludge flows. The observed reduction in holdout error

supported this connected formulation. Improvement across all eight location summaries and all four composite summaries showed that the statistical benefit extended across the treatment system.

The separate logarithmic model supplied a positive overflow solids endpoint. Joint projection combined that fitted endpoint with component balances, soluble continuity, particulate densification, and storage bounds. The two local COD increases and the remaining overflow errors describe how these simultaneous adjustments affected individual reporting quantities. Equation (6.122) relates weighted error reduction to feasibility of the reference, and Equation (6.129) includes its distance from the complete projection set. The location-specific results provide the corresponding assessment in the reported composite quantities. This connection between constraint choice, projection weighting, and prediction error follows the analysis of Sturm and Silva (2025) and Kircher and Votsmeier (2025).

6.5.2 Holdout accuracy and selected-decision fidelity

The operating search concentrates evaluations around preferred settings, so prediction errors at selected controls have a different distribution from errors across the holdout design. S4 illustrated the engineering importance of this assessment. Its projected TP prediction was 2.08 mg L^{-1} , whereas the independently solved value was 4.68 mg L^{-1} . For an illustrative treatment requirement of 3 mg L^{-1} , the predicted and mechanistic values fall on opposite sides of the threshold. The corresponding TN discrepancy also changed the estimated nutrient removal substantially.

Independent mechanistic evaluation supplied treatment outcomes from the biological kinetics and nonlinear clarifier equations at each selected control vector. Projection supplied network consistency, and trust screening supplied fixed checks on data support and model residuals. Their combination connected rapid candidate selection with a common process reference for judging the resulting decisions. The approximation-management work of Hameed et al. (2026) and the treatment-safeguard setting of Ren et al. (2026) support this emphasis on the effluent response that governs the operating choice.

6.5.3 Decision quality under declared priorities

Smooth NLP had lower common-reference objectives in all eleven paired comparisons. Extended ICSOR had shorter optimization times. These findings identify a trade-off between search computation and the quality of the retained decisions under the fixed priorities. Response approximation, trust restrictions, and the local search paths provide possible explanations for the selected controls. Each route is assessed through the same mechanistic response and quality scales.

The S8 outcome showed how resource priorities affect the preferred decision. Its

lower direct objective accompanied slightly higher effluent concentrations because the weighted resource contribution decreased sufficiently. In S10, the direct decision accepted a larger resource contribution to obtain much lower quality cost. Separate presentation of quality and resource terms made both choices understandable under the declared weights.

The free HRT coordinate connected reactor capacity with operating choices. At fixed influent flow, increasing HRT increased biological reactor volume. The aeration contribution then changed with both volume and the selected transfer coefficient. For application to an installed plant, fixed reactor volumes and specified treatment constraints define the corresponding operating problem. The component-resolved analysis of X. Zhang et al. (2026) and energy treatment of Miederer et al. (2025) support explicit accounting of hydraulic capacity, oxygen-transfer capacity, and resource demand.

6.5.4 Computational performance and operating assessment

The surrogate search varied seven controls and evaluated responses through second-order regression and joint projection. The direct search optimized controls together with reactor and clarifier states through a smoothing sequence. These computational structures were consistent with the shorter recorded surrogate search times. Development, fitting, qualification, recovery, and physical verification form the other terms in the complete-cost accounting described in the methodology. Retaining their costs alongside decision quality follows the benchmarking principles discussed by Blik et al. (2023).

First-order qualification and finite polling supplied distinct numerical evidence at the retained candidates. The finite polls tested specified coordinate and coupled directions at two resolutions, following the role of directional exploration in Audet and Dennis (2006). Their qualification results accompany the independently evaluated treatment outcomes and objective values. The resulting decision aid combines connected material accounting, rapid candidate evaluation, and a mechanistic assessment of the selected aeration, recycle, wasting, and capacity settings.

Chapter 7

Conclusion and Outlook

7.1 Synthesis of the research findings

This dissertation developed an integrated approach to physically constrained machine learning surrogates of activated sludge systems for process optimization. The central engineering problem was to obtain rapid treatment predictions while retaining a consistent account of the material that passes through biological reactors and solids separation. The research addressed this problem through three connected assessments. Chapter 4 compared statistical reactor predictors before and after physical projection. Chapter 5 developed an explicit regression whose operating, loading, and component relationships could be examined. Chapter 6 extended prediction and projection to a connected treatment plant and compared the operating decisions selected by surrogate and direct mechanistic searches.

The component state provided the common foundation. The adopted mechanistic model represented twenty component concentrations and twenty-eight reaction processes. Keeping the component response retained distinctions among organic substrate, nitrogen species, phosphorus storage, biomass, and solids. The fixed reporting map then produced COD, TN, TP, and TSS. This order connected the quantities commonly used to assess treatment with the detailed material accounting used to impose physical constraints. It also allowed the same predicted state to support balance checks, treatment reporting, and operating evaluation.

7.1.1 Physical enforcement across different reactor predictors

The reactor benchmark showed that projection could supply a common physical enforcement procedure across thirteen statistical learners. Every raw assessment state failed the specified mass conservation tolerance. Some predictors returned non-negative concentrations throughout the largest-design assessment, while others produced negative components frequently. Applying mass conservation and non-negativity together returned states that passed both checks in all 766,350 assessed projection instances.

The maximum invariant residual was 2.06×10^{-13} for the learning-domain assessment and 1.72×10^{-13} for the exterior assessment. The result established a practical numerical connection between statistical component prediction and the physical accounting of the reactor.

Projection also redistributed prediction error among components. At the largest design size, it improved 188 of the 260 predictor–component comparisons. The aggregate component nRMSE decreased for five predictors and increased for eight. Every predictor had lower TN and TP root error after projection, while COD and TSS root errors increased. These outcomes show why the effect of physical enforcement is most informative when examined at both component and reported-quantity levels. A single aggregate score compresses changes that have different implications for nutrient prediction, biomass, and solids.

The learning curves showed that the available simulation budget influenced predictor choice. XGBoost and LightGBM had the lowest observed raw errors at 500 cases. The multilayer perceptron led at 10,000 cases, with raw and projected component nRMSE values of 0.150 and 0.149. Its projected error increased to 0.251 under mild exterior inputs and 0.521 under severe exterior inputs. All exterior projected states still passed the physical checks. The benchmark therefore provided separate evidence about learning behavior, response to exterior inputs, and constraint enforcement. This separation gives engineers a clear basis for selecting a predictor for a particular data budget and operating domain.

7.1.2 Inspectable prediction through structured regression

ICSOR added an explicit representation of operating dependence and influent loading to the physical prediction framework. Its second-order driver separated baseline, operating, loading, curvature, and interaction terms. Learned component coupling connected these terms to the complete effluent state. Affine and constrained projections then returned the physically checked prediction. This structure made it possible to follow an operating input from its polynomial contribution through component coupling and output enforcement.

The model performed particularly well when fewer simulated cases were available. At 500 cases, its mean held-out pooled composite root error was 10.79, compared with 11.26 for XGBoost and 11.45 for LightGBM. It also had the lowest reported normalized root-error learning-curve area of 6.33. At 10,000 cases, the repeated-split error was 5.94 for ICSOR and 4.62 for the multilayer perceptron. LightGBM had the best mean rank across the broader assessment. The fixed partition gave a similar ordering, with pooled composite root errors of 5.98 for ICSOR and 4.38 for the multilayer perceptron. These results identify an engineering choice between an explicit approximation with favorable low-data behavior and more flexible predictors with lower full-size error.

The deployment assessment identified the role of each physical operation. Affine

projection returned non-negative states in 436 of the 2000 fixed-partition cases. The remaining 1564 cases required the constrained linear projection. Every final state passed the adopted numerical balance and non-negativity checks. The complete procedure thus combined structured regression with systematic output enforcement. The branch counts also quantified the additional calculation required to return the final state.

The coefficient analysis showed selective influent curvature in the raw COD mapping together with dense shared component coupling. Forty-four of 210 unique COD influent-curvature entries and 371 of 380 shared off-diagonal coupling entries exceeded their respective reporting thresholds. Named blocks and physical-unit coefficients made these fitted relationships available for inspection. Analytical expressions described the raw and affine sensitivities. The two-component illustration in Chapter 5 showed how conservation can reverse one raw slope and how an active non-negativity boundary can make the final local slope zero. Interpretation consequently follows the complete prediction calculation and the operating point at which it is examined.

7.1.3 Connected prediction and operating optimization

The connected plant formulation extended physical enforcement from one effluent vector to a joint response containing the mixer, five biological reactors, clarifier overflow, clarifier underflow, and stored solids. Hydraulic connections, stage invariants, flow-weighted separator balances, soluble-component continuity, and inventory bounds linked these outputs. The resulting formulation preserved information about both the liquid treatment path and the recycled solids that influence upstream conditions.

The development and holdout designs retained 7386 and 1845 mechanistic states. On the holdout, joint projection reduced aggregate nRMSE from 0.04690 to 0.04306, a relative improvement of 8.19%. Aggregate nMAE decreased by 15.65%. Projection improved 30 of the 32 location-composite nRMSE values. The largest relative composite improvement occurred for TP, and overflow TP error decreased by 39.10%. The two increases occurred for COD in the final reactor and overflow. These results support the value of a joint plant response while preserving the location-specific information needed to understand its treatment predictions.

Operating assessment introduced the controls selected by each search as an additional object of evaluation. Independent mechanistic calculations supplied the treatment outcomes at those controls. In scenario S4, the surrogate predicted TN and TP concentrations of 2.67 and 2.08 mg L⁻¹, while the mechanistic concentrations were 7.47 and 4.68 mg L⁻¹. This finding made nutrient prediction at the selected decision directly relevant to the operating comparison. It also showed the practical value of calculating treatment outcomes at the settings that would actually be chosen.

The direct route had lower common-reference objectives in the nominal case and all ten influent scenarios. Nominal objectives were 0.8647 for Extended ICSOR and 0.7862 for Smooth NLP. The direct route generally selected a larger treatment capacity

and accepted greater resource contributions to obtain lower quality cost. Scenario S10 illustrated this choice particularly clearly. Its quality cost decreased from 1.4748 to 0.3699, while the weighted resource contribution increased from 0.0557 to 0.4191. The total objective decreased from 0.7931 to 0.6040. In S8, the direct route instead achieved a slightly lower total objective through a lower resource contribution while all four effluent composites were slightly higher. Presenting the objective and its contributions together explained the treatment priorities embodied in each decision.

The connected surrogate shortened the recorded optimization interval. Mean times over the ten scenarios were 1.726 s for Extended ICSOR and 5.236 s for Smooth NLP, giving a ratio of approximately 3.03. This computational result accompanied higher mechanistic objectives for the surrogate decisions. The combined evidence identifies rapid candidate generation and mechanistic decision evaluation as complementary tasks. Their coordinated use is a direct route toward efficient process optimization with explicit treatment and resource priorities.

7.2 The integrated methodological contribution

The three assessments establish a progression from component prediction to operating choice. The reactor benchmark supplies a common comparison of learners and physical adjustments. The structured regression makes the fitted response and its sensitivities visible. The connected formulation preserves unit relationships and extends the calculation to plant decisions. Projection provides a shared link throughout this progression by converting a raw statistical response into a state that satisfies the specified physical relations.

Table 7.1 brings together the principal findings and their engineering uses. Each numerical comparison belongs to the assessment in which it was calculated. The reactor errors use training-based component scales, the structured comparison pools four native composite coordinates, and the plant holdout uses location-specific reference ranges. Together, these measures describe different aspects of the same engineering problem.

The integration also clarifies how process knowledge enters machine learning. Reaction stoichiometry supplies invariant relations. Flow and equipment boundaries supply transport and inventory relations. Statistical fitting approximates the remaining response over the sampled inputs. Interpretation examines the resulting mapping, and operating evaluation applies the engineering priorities. These contributions work together through defined quantities and checks. Their value lies in making the predicted treatment state and the selected decision open to examination.

For process optimization, the framework supports a sequence of screening, candidate selection, physical adjustment, and reference evaluation. Component-resolved predictions allow nutrient forms, biomass, and solids to be examined alongside the reported effluent quantities. The connected response accounts for recycle effects and

Table 7.1: Synthesis of the observed findings and their roles in activated sludge process optimization.

Assessment	Principal finding	Engineering use
Reactor projection	All 766,350 assessed instances passed both imposed physical checks	Return component states with consistent inventories and non-negative concentrations
Reactor prediction	Projection improved 188 of 260 component comparisons and five of thirteen aggregate scores at the largest size	Select learners and assess adjustments for the intended treatment quantities
Structured regression	Lowest mean composite error at 500 cases and explicit operating, loading, and coupling terms	Examine fitted treatment relationships with a limited simulation budget
Structured deployment	Affine projection sufficed in 21.8% of fixed-partition cases and constrained projection handled 78.2%	Follow the complete calculation that supplies the final checked state
Connected prediction	Holdout nRMSE decreased by 8.19%, with improvement in 30 of 32 local scores	Reconcile predictions across biological treatment and solids separation
Operating comparison	Mean scenario search times were 1.726 and 5.236 s, while all direct decisions had lower reference objectives	Combine rapid candidate generation with mechanistic treatment evaluation

solids storage. Independent evaluation then supplies treatment outcomes under the controls selected by the search. This sequence makes the time required to explore operating alternatives useful within a decision process that retains the original treatment objective.

7.3 Directions for future research

7.3.1 Adaptive sampling around operating decisions

Future studies can generate new mechanistic cases where the surrogate is most influential in the operating search. A first surrogate can screen the declared domain and identify candidate controls. Mechanistic calculations at those candidates can then supply additional training states. Sampling can concentrate on large nutrient discrepancies, strong projection displacements, and regions close to treatment requirements. Repeating fitting and operating assessment would connect the simulation budget to the decisions it is intended to support.

An informative comparison would hold the mechanistic evaluation budget fixed and compare uniform sampling, error-based sampling, and decision-based sampling. The outcomes should include effluent errors at selected controls, the common-reference objective, the number of accepted decisions, and the complete time required to deliver them. The S4 nutrient discrepancies provide a concrete target for such a study. Additional cases near those selected operating conditions could test whether focused sampling improves TN and TP prediction where the operating choice depends on them.

7.3.2 Projection aligned with treatment priorities

The reactor results motivate further study of projection weights and the quantities they emphasize. A projection that minimizes changes in component coordinates can redistribute error differently from an assessment based on reported nutrients or solids. Future formulations can compare component scaling, composite-sensitive weighting, and weights based on prediction uncertainty. Each formulation should retain the same mass conservation and non-negativity requirements so that changes in treatment accuracy can be attributed to the adjustment metric.

The connected plant offers an additional test of this idea. Weighting can distinguish internal state accuracy from final-effluent accuracy while preserving the unit connections. Studies can assess whether nutrient-focused weights improve selected-control predictions without producing large errors in biomass, solids inventory, or other treatment locations. This would make the relation between physical adjustment and the intended engineering decision more explicit.

7.3.3 Uncertainty-aware prediction and operating safeguards

Uncertainty can be developed alongside the projected point prediction. Repeated fits, model ensembles, or probabilistic component predictions can describe variation in the estimated treatment response. Each sampled state can be projected onto the same physical set. The resulting distributions can then be evaluated for their coverage of mechanistic reference concentrations, particularly for TN, TP, and overflow TSS.

Operating studies can use these distributions to formulate treatment safeguards. For example, a candidate setting can be selected to keep an upper predictive bound for TP below a specified treatment requirement. Mechanistic and plant-based evaluations can test the frequency with which those safeguards are met. Separating uncertainty associated with surrogate fitting, influent measurements, and mechanistic parameters would also show which information most improves the decision.

7.3.4 Stable interpretation of the complete response

Further interpretation work can examine the stability of coefficient groups across repeated training samples and operating domains. The present structure provides natural groups for operating terms, loading terms, curvature, and coupling. Resampling can show which groups remain prominent and which individual coefficients vary. Comparisons based on the complete predicted response can help identify different coefficient combinations that describe similar treatment behavior.

Sensitivity studies should also follow the final projection. Derivatives can be evaluated at representative operating points and checked against direct perturbations of the returned concentrations. Mapping the active constraints would show where a nutrient response changes smoothly and where a projection boundary alters its slope. Comparing these projected sensitivities with mechanistic sensitivities would connect model inspection more closely to aeration, recycle, and wasting decisions.

7.3.5 Dynamic prediction and control

A dynamic extension can replace steady-state targets with trajectories of component concentrations and stored solids. The governing balances would include the time derivatives of reactor and clarifier inventories. Training data can represent changing influent loads, aeration schedules, recycle settings, and sludge wasting. Physical enforcement can then account for the material transported and transformed during each prediction interval.

This extension would support studies of predictive control under daily influent variation and load disturbances. The evaluation can include transient nutrient peaks, recovery time, inventory changes, oxygen demand, and the computation required at each control update. The connected component representation provides a useful start-

ing point because it already retains the locations and inventories that influence delayed treatment responses.

7.3.6 Plant validation, topology, and resource models

Validation with measured plant data is a major next step. Studies can combine influent fractionation, calibrated mechanistic parameters, unit concentrations, clarifier solids measurements, and recorded controls. A prospective assessment can fix the fitting and operating procedure before collecting the final evaluation period. Testing across seasons and loading conditions would provide evidence about how the component surrogate transfers to observed treatment behavior.

Operating objectives can also be developed for specific facilities. An installed plant has defined reactor volumes, aeration capacity, recycle capacity, and discharge requirements. Electricity demand, chemical use, sludge handling, and operating costs can replace or supplement the engineering proxies used here. Multiobjective studies can present the attainable treatment and resource combinations before a preference is selected. Extensions to other reactor arrangements, attached-growth units, and resource-recovery connections can retain the same principle of defining component states and boundary streams before physical enforcement is imposed.

More extensive search comparisons can evaluate several initial controls, hybrid surrogate and mechanistic searches, and the fifteen-start basin-sensitivity design described in Chapter 6. Recording state generation, fitting, prediction, projection, search, and verification costs would establish when surrogate development becomes worthwhile for repeated decisions. The central assessment would remain the treatment outcome and common-reference objective obtained for a given computational budget.

7.4 Concluding perspective

Physically constrained machine learning can connect rapid activated sludge prediction with the material accounting needed for process optimization. The reactor benchmark demonstrated consistent numerical enforcement across diverse learners. The structured regression added an inspectable response with favorable performance at the smallest data budget. The connected plant formulation brought unit consistency, solids storage, and independently evaluated operating choices into the same framework.

The dissertation's integrated contribution is a defined way to use these capabilities together. Statistical models provide rapid approximations, process knowledge supplies physical constraints, and mechanistic evaluation supplies the treatment response at selected controls. Future development can strengthen each stage through focused sampling, uncertainty assessment, dynamic balances, and plant measurements. This progression supports activated sludge optimization in which computational efficiency

serves explicit treatment and resource objectives.

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